Philipp Spreer

# PsyConversion®

## 117 Behavior Patterns for an Improved User Experience and Higher Conversion Rates in E-Commerce

Springer

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Philipp Spreer
Hamburg, Germany

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# Preface

Effective, engaging, simple, and activating digital applications—which e-commerce professional doesn't have that on their professional wish list? However, the traditional methods of user experience and conversion optimization quickly reach their limits on the path to this goal. It's time for a fundamental expansion of the toolbox!

The fusion of psychology and behavioral science with conception, web design, and conversion optimization is how you approach and understand your customers as they are: how they think, decide, and act. PsyConversion® is the bridge between these largely separated disciplines. This book reveals the mechanisms underlying our decisions, describes the most essential cognitive "shortcuts", and provides hundreds of concrete ideas for implementing decision-optimizing design elements in digital business.

This book originated from a private collection. Over time, I collected behavioral anomalies, psychological mechanisms, research findings, and recurring decision patterns in a database whenever I encountered them in consulting projects or research and teaching activities. The initial goal was simple: to finally understand the often irrational behavior of users on the internet and use this knowledge to design the optimal website (or online shop)—in terms of user guidance, user experience, conversion rate, revenue, and environmental and social aspects. This collection repeatedly sparked great interest in the industry, proved its worth in projects, and grew steadily over time, including hundreds of A/B tests and usability studies.

Now, this knowledge is meant to be shared with the e-commerce community. I hope you will have a reading experience filled with "aha" moments and "wow" effects, and you can directly incorporate the collected suggestions into optimizing your digital projects. A positive side effect will undoubtedly follow: After reading this book, you will repeatedly catch yourself making logical mistakes or irrational decisions. These moments will make you a more conscious and, therefore, better decision-maker.

A request: Every collection is only as good as the feedback it receives! Due to the extreme breadth and dynamics of the behavioral sciences as a research field, this book cannot claim completeness even in its expanded current edition. If you have suggestions

for further behavior patterns, empirically proven design ideas, or corrections, please contact me anytime via www.psyconversion.de. There, you will also find empirical studies, examples of pages enriched with psychological principles, and materials you are welcome to use and share for free. Thank you very much!

Enjoy an enlightening reading experience.

Hamburg, Germany Philipp Spreer, Ph.D.

# Preface to the Second Edition

Over the past two years, many readers have provided valuable suggestions, constructive criticism, and optimization ideas, which have been considered in the second edition. I want to address one aspect briefly, as it has been mentioned repeatedly: the high proportion of technical terms. PsyConversion® is designed as a workbook and reference guide that directly transfers cutting-edge research findings into practice. I believe using widely established terms and designations is necessary to establish this connection without losses and allow for further research. The second edition has fundamentally adhered to this standpoint. Nevertheless, the entire text has been revised to use popular terms wherever they are equally established. Additionally, the language has been simplified to improve readability.

Furthermore, the book has been expanded in content: it has been updated to the latest state of research, which has kept its pace in the past two years. Seventeen new behavior patterns have been added, others have been supplemented, and two patterns have been merged.

Hamburg, Germany — Philipp Spreer, Ph.D.

# Reviews

Stefan Lein, director retail at Google:

> "The impressive collection of well-known and scientifically grounded behavioral patterns in PsyConversion® provides an excellent starting point for developing promising test scenarios and optimizations. Significant performance improvements are possible with this."

Dr. Thilo Pfrang, founder of behamics AG St. Gallen:

> "A brilliant book that systematically structures the groundbreaking insights of behavioral economics and makes them applicable to the e-commerce community. It will help you better understand your customers, yourself, and your decision-making process."

Dennis Esterl, head of onsite CX at Amazon:

> "A comprehensive overview and the best categorization of behavioral patterns in marketing—this book is guaranteed to find its place on my desk! It's a must-have for every marketer's toolkit and an excellent introduction to behavioral psychology."

Joachim Graf, futurologist and editor:

> "Human beings are fools, irrational fools. For those who already know this and want to apply psychological methods, PsyConversion® offers a wealth of tips, tricks, and methods. It's an indispensable reference when competing in the increasingly fierce attention economy."

Dr. Rolf Schulte-Strathaus, founder of the usability consultancy eparo:

> "After ten years of UX research, I am convinced that unconscious perception plays a central role in perfect digital services. PsyConversion® is a brilliant and incredibly comprehensive collection for this purpose. Despite my experience, I learned a lot from it. An indispensable reference!"

Dr. Matthias Wilken, head of channel management at Allianz:

> "If our online marketing were software, then Spreer's PsyConversion® would be the perfect update. It offers numerous concrete ideas for decision-influencing design elements in e-commerce. At Allianz, we constantly use these insights to improve user experience and conversion optimization."

Christoph Pschorn, founder & CEO of the digital agency DGTLS:

> "PsyConversion® makes psychological insights in e-commerce finally tangible and immediately applicable to everyone—even without prior knowledge and large budgets. It's essential reading and the new standard work for all e-commerce professionals!"

Dr. Kai Hudetz, managing director of the Institute for Retail Research in Cologne, Germany:

> "Spreer manages the feat of presenting scientifically grounded psychological effects in a practical and relevant manner. PsyConversion® is a compact reference work due to countless examples, excellent structuring, and comprehensive references to further literature."

Knut Polkehn, partner and consultant at artop—Institute at Humboldt University in Berlin and UXQB board member:

> "In the debate between marketing and UX on the question of 'customer or user experience,' it must be understood that it's about optimizing the benefits for both customers AND users. The thorough description of behavior patterns for genuine benefits rather than manipulation makes Spreer's book a must-read and highly recommended."

Kerstin Pape, Vice President Customer Excellence at Freenet:

> "I read the book with great attention and excitement and am impressed: simple and easily understandable patterns can significantly impact CRO (Conversion Rate Optimization). I look forward to working with this book!"

Jens Sievert, innovation manager at ERGO Digital Ventures AG:

> "The scientifically grounded collection of unconscious behavioral patterns allows us at ERGO to understand our customers much better. Additionally, PsyConversion® provides many useful, practical tips for adapting and optimizing digital products."

Prof. Dr. Hanno Beck, Professor at HS Pforzheim and author of "Behavioral Economics":

> "An extensive library of psychological behavioral patterns and an astonishing collection of implementation ideas in e-commerce. It's understandable, well-organized, entertaining, and highly precious!"

Prof. Dr. Philipp A. Rauschnabel, Professor at the University of the Federal Armed Forces Munich:

> "A perfect foundation for beginners who want to learn about psychological aspects of CRO. Simultaneously, it's a top-notch reference work for experienced online marketers searching for effective optimization ideas. In short: One of the few books that successfully bridges the gap between theory and application."

# Acknowledgments

Nearly two years after the release of PsyConversion®, the second edition you are currently holding is now available. Many thanks to thousands of buyers and readers of the first edition who made this possible. Since its launch, PsyConversion® has firmly established itself in the bestseller lists of usability and e-commerce literature, thanks to this support, for which I am humble and grateful.

I want to thank the PsyConversion® review team of e-commerce experts, psychologists, and neuroscientists who quality-assured the contents from their respective perspectives. My family also deserves the utmost respect and thanks for their understanding and love, which have accompanied me throughout the work on PsyConversion®.

Further, thanks go to the Behavioral Design Unit of elaboratum, where now more than twenty experienced consultants conduct outstanding PsyConversion® projects. The range extends from highly effective conversion optimization along entire customer journeys to designing digital products and portfolios, scaling persuasive content, and developing genuinely user-centric digital organizations. In addition to various e-commerce companies, the customers now also include telecommunications companies, manufacturers, utilities, and many insurance companies. The insights from around 150 consulting projects have also been incorporated into this second edition.

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# About the Author

**Philipp Spreer (Ph.D.)** is managing partner of elaboratum and supports organizations from medium-sized businesses to listed companies on their way to digital excellence. His focus areas are understanding customer behavior, striving for the perfect user experience in accordance with the conversion rate and managing complex digital projects.

Spreer has been researching customer behavior in the digital context for more than a decade. After studying in Kiel, Lyon and Göttingen, he completed his doctorate at the interface between marketing and psychology on the acceptance and effect of interactive service technologies. His research results and publications have been cited and further developed hundreds of times worldwide. At the same time, he was a long-time lecturer at renowned technical colleges and universities and is a regular speaker at leading digital conferences.

Based on current psychological and neuroscientific findings, PsyConversion® bridges the gap between usability, conversion optimization and online marketing and provides a standard work for practical work in the digital environment that no innovative e-commerce expert should be without.

**About Elaboratum**

Our conviction: Relationships are the critical success factor for your digital project. Because your customers, your prospects, and your employees have one thing in common: they are people. We help you understand them and use digitization in a way that creates long-term relationships based on trust.

This requires the right strategies, data, technologies, processes, products, and concepts. And the right people. That's why we founded elaboratum as a different consultancy: close, collaborative, trusting, and creative. This allows us to deploy experts with practical experience and top digital talents in your project, who can both shape the big picture with you and lend a hand in implementation themselves.

For more information: www.elaboratum.com



# Introduction

1


**Summary**

We all believe we have the freedom of will and are masters of our decisions. In most cases, however, this is not the reality. Evolutionarily, our behavior is primarily based on subconscious patterns of action – the **behavior patterns**. This disruptive insight changes how we view our existence, society, and purchasing and consumption decisions. This is precisely what PsyConversion® is all about: The book explains how we make decisions and what unconscious mechanisms underlie these processes. This knowledge is the key to a deeper understanding of the customer in two ways: E-commerce professionals can design more intuitive websites that customers can operate smoothly and effortlessly. As a result, previously unimagined conversion potential will be leveraged, and digital business key performance indicators will be significantly improved. These effects are measurable and are considered a **multi-billion dollar business** globally. It is obvious that the use of behavior patterns must occur within the framework of narrowly defined **ethical and moral guardrails**. PsyConversion® has become more than a source of inspiration; it is the standard reference for the daily optimization of all digital projects.


Let's face it: sometimes we wonder about our **own decisions.** Why did I order this shirt twice? What was I thinking when I chose that color? Many of my friends have this phone, but why wasn't there any other option for me? Such questions usually hit us like déjà vu: they suddenly appear on the mental scene but remain diffuse and rarely become actual thought processes. You could say that they are rather weakly cognitively verbalized feelings. Most of the time, we unconsciously suppress these feelings and return to business as usual. There is much more to it than that. These small moments suggest that we may have been influenced in our decisions. **Influenced? Yes, definitely.**

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We must say goodbye to the idea of "homo economicus"—the concept of human beings acting strictly rationally. We are by no means machines that operate according to a clearly defined schedule and rationally try to maximize their benefits. On the contrary, the vast majority of all decisions are made impulsively, spontaneously, and unconsciously. In science, "**Behavioral Economics"** has established itself as a discipline representing this idea. In contrast to Adam Smith's rationalist worldview of the so-called "classical economics", behavioral economics refers primarily to psychological findings. Accordingly, this book is based almost exclusively on psychological (and neuroscientific) research.

On closer inspection, the banal question "How do we make decisions?" turns out to be crucial for our entire existence. We live in a complex world in which we are constantly confronted with decisions and, in most cases, can decide freely. Our choices shape our actions, our togetherness, and our future. However, it is the case that freedom of will or **the ability to make free decisions** is, in most cases, an illusion—it only exists under controlled laboratory conditions. As soon as we go out into the world, we no longer make our decisions in isolation but always under the influence of external conditions. And these external conditions are primarily created by people. As a result, anyone who knows precisely which principles are used to make decisions has a **powerful instrument** in their hands. Not only can they make better decisions, but they can also influence other people in their decision-making. Obviously, this ability is accompanied by a great ethical and moral responsibility and is described in Sect. 5.7 of the book.

Thus, this book aims to shed some light on people's decision-making processes, explain what unconscious behavior patterns underlie our decisions, and show how this knowledge can be applied in e-commerce practice. Because one thing is sure: it doesn't matter how good your product is if no one wants to use it. The willingness and motivation to engage with your product need an **effective psychological trigger.** If you have internalized this, you will build websites and apps fundamentally differently. After all, only about 5% of all purchasing and consumption decisions are made predominantly rationally. For the remaining 95%, our feelings and unconscious decision-making patterns set the tone. Today, very few providers lack rational, factual arguments such as comparison tables, cost–benefit comparisons, statistics, and feature lists. Emotional needs, on the other hand, are hardly addressed. It's time to stop targeting only 5% of the customer's brain online!

If someone still asks you why it should be worthwhile to deal intensively with the decision-making processes of your users, present them with the following figure: Taking into account 37 individual studies, the Baymard Institute concludes that 69.2% of all users abandon the purchase after an initial favorable decision. The institute calculates a (theoretical!) optimization potential of US$ 260 billion (Baymard, 2017).

So, the art of convincing users (and that's what conversion optimization is all about) is a valuable skill in the truest sense of the word. And: It can be learned. **There is nothing mystical about being convincing** in digital business. It is the result of the structured application of good tools. At this point, the book provides helpful frameworks, an extensive collection of established behavior patterns, and **hundreds of immediately implementable**

**ideas**. What is particularly nice about it is that most of the adjustments cost no money and can be evaluated excellently.

At PsyConversion®, the business goals of e-commerce companies and the individual goals of users are always equally in the foreground. **You won't learn manipulation techniques** that degrade visitors to your website, online store, or app to externally controlled objects. Instead, the mechanisms described aim to get users to their destination with less cognitive effort and to empower them to make decisions more quickly. When this is achieved, there is inevitably a measurable positive impact on e-commerce success indicators, such as sales, customer lifetime value, or conversion rate. In addition, special attention is paid to measures that encourage users to behave in a socially desirable manner – e.g., not to produce unnecessary returns and pollute the environment or to commit fraud with customer loyalty programs.

> Persuasion must be honest and ethical to have an impact beyond a brief impulse. (Anders Toxboe, Usability Thought Leader, 2016).

PsyConversion® is, therefore, nothing less than the art of convincing users of one's cause (or product) in the long term through behavioral science mechanisms without using morally questionable techniques.

**Idea and Structure of the Book**

The book aims to distinguish itself from subjective opinions and present only tested fact-based knowledge. All behavior patterns that you will find in it have undergone a rigorous scientific peer review and, after an in-depth investigation, have been published in leading academic journals. This allows you to read all contributions in the original (the corresponding references are always indicated in the profiles of the behavior patterns). However, since the research in the initial studies is complex and highly time-consuming, this path remains blocked for most practitioners in everyday life. That's why PsyConversion® has done this work for you and researched **117 scientifically validated behavior patterns for the digital sector** and prepared them in an easy-to-understand way for daily use. This also includes hundreds of starting points for the **concrete implementation** of behavior patterns in e-commerce. This is because every behavior pattern needs a definite trigger to be activated – and these triggers must be conceptually developed and integrated into the interface of the website or online shop.

This book is thus

1. a practice-oriented guide through human decision-making processes,
2. a central reference work for psychological behavior patterns and
3. a comprehensive source of inspiration for your website's user experience and conversion optimization.

In addition, you will undoubtedly experience many "aha" moments, and you will not be able to resist a smile from time to time, for example, when you learn

- on which even Mother Teresa and Joseph Stalin agree (→ Sect. 4.1.3.3),
- why the "Sexiest Man Alive" sells best (→ Sect. 4.2.5.2),
- the detail that damaged the reputation of Michael Jackson (→ Sect. 4.2.1.14),
- what we can learn from the TV series "How I Met Your Mother" for the selection of images (→ Sect. 4.2.1.6) and
- why it can be worthwhile to drink a glass of wine in exam situations (→ Sect. 4.1.4.1).

Before these riddles are solved, chap. 2 presents the basics of decision-making. Building on this, chap. 3 introduces you to the behavior patterns mentioned above, the powerful shortcuts on the way to a decision. With this knowledge, you can familiarize yourself with the behavior pattern collection (chap. 4), find your way around quickly, answer the above questions, and derive promising starting points for optimizing your digital projects. How exactly all this can be implemented is then explained in chap. 5. Reading the first three chapters once in a row is recommended. Subsequently, you will have the prior knowledge to work with the behavior pattern collection in a goal-oriented manner and always when needed.

## Literature

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Toxboe, A. (2016). Beyond usability: designing with persuasive patterns. http://ui-patterns.com/blog/beyond-usability-designing-with-persuasive-patterns. Accessed: 4. Jan. 2019.




# Fundamentals of Decision-Making

# 2

**Summary**

After decades of economists assuming that people are rational utility maximizers, there has recently been a growing realization that we are not at all able to apply a rational decision-making process in many situations. This goes hand in hand with the renunciation of "homo economicus". Psychological and neuroscientific findings are increasingly woven into the economy's human image. Today, it is considered indisputable that we have an emotional decision-making system in addition to the rational one. The first is what the research branch of behavioral economics tries to understand and make usable for economic questions. Perhaps its most important point of reference is the "Prospect Theory", which was decisive in awarding the Nobel Prize in economics to psychologist Daniel Kahneman. Behavioral economics is primarily concerned with subconscious decision-making patterns (heuristics) and the associated errors of reasoning and biases. Together, these two groups of psychological phenomena can be described as behavior patterns. Research has identified and validated many of these patterns in the past twenty years.

We are the result of our choices. They shape our lives from day one. Turning left or right at a crucial moment can have drastic consequences. Therefore, when making decisions, people ask themselves several questions: Can I expect a profit or a loss? What is the probability of the respective events occurring? What is the relationship between risk and likelihood of profit? However, we do not ask ourselves these and other questions entirely consciously in the vast majority of decisions. Thus, if people are fundamentally interested in making optimal decisions, the question arises about how decision-making processes occur.

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> Cognition alone cannot produce action; to influence behavior, the cognitive system must work through the affective system. (Camerer et al., 2005, p. 13, one of the leading researchers in behavioral economics).

## 2.1 The Fairy Tale of the Rational Being

Economics has always been a reflection of the zeitgeist. Influenced by the leitmotif of industrialization, economists have long assumed that a decision-making process proceeds roughly as follows: We define a goal, research the available alternative courses of action to achieve this goal, collect information about the alternatives, then evaluate them about the expected degree of goal fulfillment, and finally select the action option with the **highest degree of goal fulfillment** (see Fig. 2.1).

This benefit maximization approach is at the core of the "**homo economicus"** idea: people behaved strictly rationally, researched, evaluated, and acted according to a predetermined script. Three characteristics describe the "homo economicus":

1. **Unrestricted rationality:** The economic person wants and can behave strictly rationally in all situations. He always weighs his decisions carefully and does not make any mistakes.
2. **Unrestricted willpower:** The economic person knows neither emotions nor does he ever lose control over his thoughts and actions. Once decisions have been made, they are implemented without hesitation.
3. **Unrestrained egoism:** The economic person maximizes only his own benefit and does not think in more significant social categories. The consequences of his behavior for those around him do not influence his decisions.

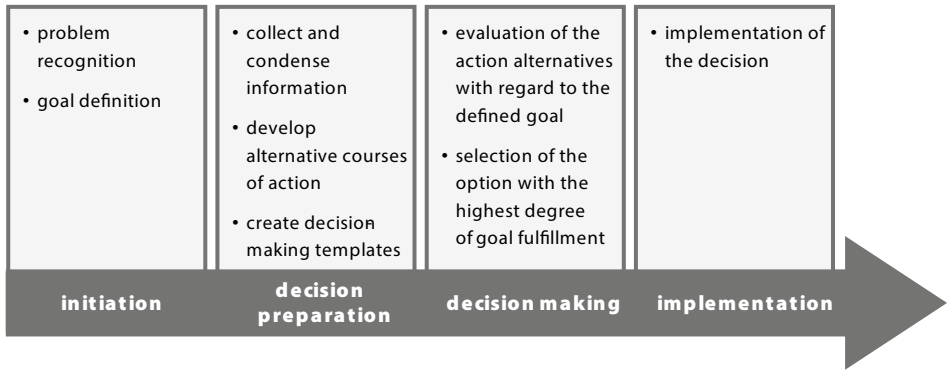



**Fig. 2.1** Rational decision-making process of "homo economicus". (Source: own illustration)

This view of humans has been increasingly criticized for quite some time. It is based on a fundamentally wrong idea of man – morally and regarding cognitive performance. For example, open-minded economists such as Herbert Simon interjected early on (1959) that people did not have the ability to maximize utility at all. This is plausible, as you would always have to know all available alternatives, have access to all information, and be able to correctly predict all consequences – an impossible task.

**Example**

Using the **example of buying a car**, the complexity of this process can be well understood. Dozens of vehicles fulfill the initial goal (e.g., comfortable mobility), each available in several lines and with various equipment features. The "homo economicus" would now have to create a list of all models, list the individual components (e.g., leather seats, alloy rims, heated windows, entertainment system, etc.) and measure them individually with regard to the goal with a quantitative value. Using a scoring procedure, the available options would then be evaluated as a bundle of characteristics, measured by the effort required and weighed against each other. Ultimately, the choice would inevitably have to fall on the model with the highest benefit score.◀

Simon calls the impossibility of this form of decision **"limited rationality"** (Simon, 1959). However, from the point of view of his critics, the other characteristics of "homo economicus" (willpower and egoism) are by no means unrestricted. In practical terms, limited willpower can be understood, for example, by the high number of failed diets or addiction cessations. Today, egoism is considered an evolutionary disadvantage rather than a strength because socially active communities have shown themselves to be more robust and capable of surviving.

**Background**

The well-known ultimatum game provides vivid proof of the thesis that we have an innate tendency towards fairness and are not unreservedly selfish (Camerer & Thaler, 1995). It works very simply: Player one receives a certain amount of money and then makes player two an offer on how this sum should be divided between them. Player two then has the option to accept the offer (in which case both players will receive precisely the proposed amount) or reject the offer (in which case both players will receive nothing). As a "homo economicus", player two would theoretically have to accept any offer from player one - because even 1 cent is better than no cent. In reality, however, it turns out that player two often rejects the offer if he thinks it is unfair. So, he waives money (or "pays") to punish player one and to encourage fairness.

The neuroscientific analysis of the game is also interesting: In the case of unfair offers, activations of the anterior insula occur, which is responsible for negative emotions such as pain, stress, hunger, or thirst (Sanfey et al., 2003). However, in the case of fair offers, activations are observed in the ventral striatum area associated with reward processes. Emotions, therefore, play a role in decision-making, especially in a social context.

With the objections of Simon and other leading scientists to the idea of a "homo economicus", the analysis of human decision-making behavior was revived, especially at the **intersection of economics and psychology.** More and more non-analytical strategies of decision-making have subsequently been described. The main driving force behind this development was that it was impossible to make a satisfactory prognosis of human behavior with the classical economic theory. Generally, theories must be measured by how reliably and accurately they can make predictions (Wong, 1973). To stay with the example of buying a car: Instead of "I will take the car with the highest predicted benefit score", car buyers worldwide are likelier to say, "I will take the black one." This begs the question: Why can people build rockets and fly to the moon but not make simple, logical decisions?

**Is man a cognitive failure?** Overestimated and become the dominant species on this planet only because all other living beings are even less cognitively powerful? This view falls short for two reasons:

1. **The "homo economicus" is a theoretical construct.** It is not a failure of reality but a model-like idea or a substantial escalation of a central assumption. It is precisely this simplification of reality that is why the idea of "homo economicus" has reflected the central image of man for so long. Simplification is a significant feature of all theoretical concepts; one could also say: the main strength of a theory. Only by reducing complexity can fundamental statements be made about interrelationships. The idea of the strictly rational human being thus makes it possible to work with economic decision-making models in the first place. By the way, the balancing act between simplicity and exactness is called the Bonini Paradox. The French poet Paul Valéry describes it vividly: "Everything simple is wrong, everything complicated is useless" (1937). This should always be considered when criticizing theoretical models due to erroneous or missing assumptions.
2. **Irrationality is a matter of perspective.** Behavior that contradicts rational theory does not necessarily have to be irrational. First, it can be worthwhile for people not to behave selfishly. Second, using supposedly irrational behavior patterns makes our decision-making process very fast and efficient – and efficiency is synonymous with rationality. Why we can be grateful to have a decision-making system that sometimes acts irrationally is explained ▶ in Sect. 2.2 in detail.

The fact that our decisions are not always particularly rational is undoubtedly also due to the sheer number of decisions we make daily. The German professor of medical psychology, Ernst Pöppel, once put their number at around 20,000 – per day (Pöppel, 2008). If we had to apply the complicated process described above to all these decisions, our brains would be permanently overloaded, and no life as we know it today would be possible.

In summary, to choose the correct option in a strictly fact-based and rational way, our brain has to expend unimaginable amounts of energy. As powerful as it may be, it is also

adept at reducing energy consumption. You can feel this first-hand: Most people feel great effort in decision-making and develop avoidance strategies – mostly unconsciously. If we cannot make our decisions strictly logically, we must turn away from the idea of a "homo economicus". Whether we like it (e.g., because it clearly distinguishes humans from cold-calculating machines) or not (e.g., because it makes it clear that we often decide and act "unreasonably"). The question now is: **How does the decision-making process work?**

Popularly speaking, we make most decisions "intuitively from the gut". By this, we mean they are cut down quickly and impulsively, without consciously thinking about them, and with great ease. What is surprising about this is that science confirms this popular description of the intuitive decision-making process quite accurately.

## 2.2 Two Decision-Making Systems

In Sesame Street, the serious Bert has a slightly crazy Ernie at his side; in the "Rosie Project" the structured scientist Don falls in love with the impulsive barmaid Rosie; in Steven Spielberg's animated series the emotional-intellectual laboratory mouse Pinky supports her brilliant partner Brain in the conquest of world domination: Life is full of unequal couples who need each other and harmonize with each other in their way. Rational and emotional complement each other well. The situation is very similar in our brains.

Psychologists and neuroscientists today largely agree that our brain works with two systems: A **rational system** we consciously control is slow but can solve complex problems. And an **intuitive system** that works impulsively without deliberate control but quickly and efficiently.

This differentiation into two systems is not entirely new; it can already be found, for example, in Freud's theory of the distinction between *unconscious, preconscious,* and *conscious* processes (Freud, 1923). However, the theoretical references of current studies are entirely different. One of the leading researchers and, to a certain extent, the new father of this idea is Daniel Kahneman, who, as part of his contribution to **dual-process theory**, distinguished the two decision-making systems and described them simply as *System 1* (intuition) and *System 2* (ratio) (Kahneman, 2011).

This view is now considered a scientific commons, not only since Kahneman was the first psychologist to be awarded the Nobel Prize in Economics in 2002. Similarly, the differentiation of *implicit* and *explicit* motives, which McClelland et al. (1989) consider to be the basis of our decisions, or the identification of *controlled* and *automatic* processes by Camerer et al. (2005) as well as *affective* and *cognitive* (Loewenstein & O'Donoghue, 2004). Table 2.1 illustrates the connection and the generally high degree of agreement between dual-process theories (for a more in-depth discussion, it is worth looking at Metz-Göckel, 2010).

At first glance, the designation of Smith (2008) seems to fall outside the very uniform framework of the other designations. However, his designation of emotional processes as

**Table 2.1** Designation of decision-making processes of selected dual-process theories

| Authors | Emotional process | Rational process |
|---|---|---|
| James (1890) | Associative thinking | True thinking |
| McClelland et al. (1989) | Implicit | Explicit |
| Loewenstein and O'Donoghue (2004) | Affective | Cognitive |
| Camerer et al. (2005) | Automatic | Controlled |
| Smith (2008) | Ecological rationality | Constructivist rationality |
| Kahneman (2011) | System 1 | System 2 |

"ecological rationality" expresses excellently that emotional processes are by no means irrational at their core. This is an important point: It is precisely the use of intuitive behavior patterns that allows our brain to work in a resource-saving manner – which can be equated with efficiency and rationality. Adding "ecological" clarifies that many behavior patterns arise from interaction with the environment, which triggers dynamic adjustments to our behavior.

The fact that the **two-system theory is also advocated by neuroscientists** (Lieberman, 2007) shows that the many automated decision-making processes do not follow the laws of logic but are evolutionarily anchored. The findings of the neurosciences are a clear plea for a more in-depth investigation of emotions in decision-making than has been the case so far in economic theory (well summarized by Camerer et al. 2004in the text "Neuroeconomics: Why Economics Needs Brains").

### System 1

"Experts are always right." This is a heuristic - an intuitive, simple, and superficial rule. Whenever we dispense with cognitive processing of information and instead resort to heuristics, System 1 is in the driver's seat. Kahneman describes it as an **intuitive decision-making system** based on automated behavior patterns. When we decide intuitively – i.e., our System 1 takes the lead – solutions come to us quickly, spontaneously, and without effort. In most cases, this works very well and flawlessly, as can be seen, for example, when driving a car – a situation in which we make dozens of decisions every minute and do not consciously think about them. Another handy example of System 1 decisions is the ball game: When an athlete catches a ball, he has not previously solved the equations to calculate the ball's trajectory and adjusted his position accordingly. He intuitively knows how to behave. The reason for the high speed of intuitive decisions lies in the good mental availability of heuristics. You guessed it: These kinds of intuitive decision processes are precisely what PsyConversion® is all about.

**System 2**

Please calculate the result of $24 \times 17$ now. Seriously, give it a try. $24 \times 17$. As you fight against the impulse to pull out a calculator, you notice an essential feature of System 2: it leads to exertion and physical stress. Therefore, thought processes in System 2 are noticeably different from those in System 1. They deal intensively with the available information and process it **rationally and consciously.** Based on the result, our attitudes and intentions for action are then formed in a structured manner. Kahneman mentions two core tasks of System 2:

1. "compute" (e.g., solve the above arithmetic problem)
2. "supervise" (System 1 and monitoring behavior or slowing down decision-making)

Often, our efficiency-optimized brain reacts to unwanted energy consumption by having the fast System 1 take over most of all decisions, and System 2 is only activated for challenging tasks. You have noticed that too: your first impulse was to skip the task and avoid the cognitive effort. In most cases, it remains the same: According to expert estimates, 75 to 98% of all decisions are made by System 1 (the majority assumes an order of magnitude of around 95%). Because we share almost 99% of all genes with chimpanzees and the genetic differences are the smallest in the brain (Junker, 2006), it is not difficult to believe that we only make strictly rational decisions in exceptional cases.

In his acceptance speech at the Nobel Prize ceremony, Kahneman describes the two decision-making systems as follows:

> The operations of System 1 are typically fast, automatic, effortless, associative, implicit (...), and often emotionally charged; they are also controlled by habits and are, therefore, difficult to control or modify. System 2's operations are slower, serial, laborious, more consciously monitored, and intentionally controlled. They are also relatively flexible and mostly rule-based. (Kahneman, 2003) (Fig. 2.2)

Also, from a neurological point of view, decisions are made in two different brain areas. In the **neocortex** (i.e., the "youngest" part of the cerebral cortex from an evolutionary point of view), conscious and rational thinking (system 2) takes place. Experiments have shown that rational thought processes activate the medial and lateral frontal brain areas (Lieberman, 2007). On the other hand, intuitive decisions (System 1) are triggered in the **amygdala** in the limbic system - i.e., precisely the part of the brain responsible for feelings and sensations. So neurologically, intuitive decisions hardly differ from emotions – a surprising finding, especially in a world that has long regarded emotionality as a disruptive factor in decision-making processes. Of course, the two areas of the brain do not work entirely separately from each other. The neocortex receives the signals from the amygdala and is thus influenced by its functioning. If this was not the case, we would be exposed to our impulsive decisions for better or worse. However, there is a two-way verification routine.

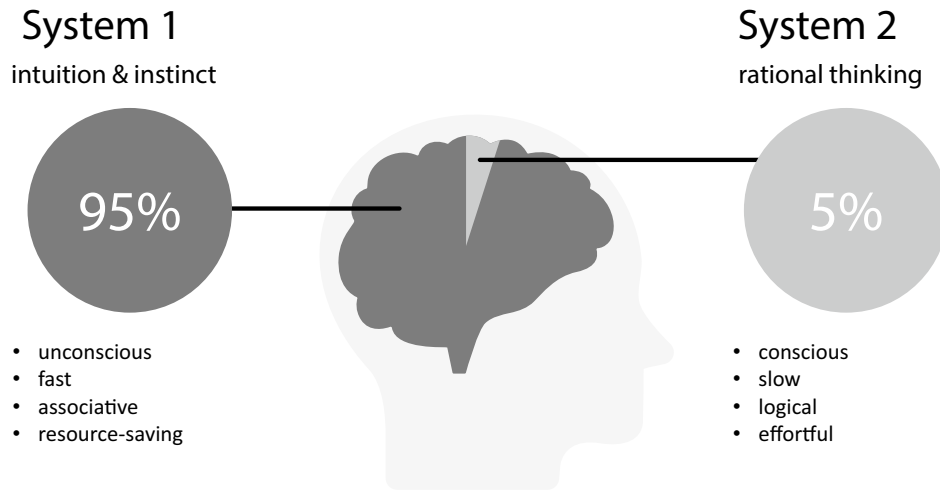



**Fig. 2.2** Features of System 1 and System 2. (Source: own illustration)

Neuroeconomic decision models are built using five components, shown in Fig. 2.3 (based on the description by Fehr & Rangel, 2011).

Have you ever heard of "Elliott"? He was the main character in a spectacular medical study by Antonio Damasio in the late 1980s (Damasio, 2014). The professor of neuroscience, neurology, and psychology at the University of Southern California wanted to explore the neurological connection between the two decision-making systems. To this end, he conducted studies with patients who had suffered brain damage in the course of a tumor disease. One was Elliott, who lost the ability to feel after brain surgery. Astonishingly, this was accompanied by his inability to make decisions from then on. As already indicated, the case shows that the old opposition, "mind vs. body" or "mind vs. feeling," falls far short of the mark. Damasio's research has thus proven a close connection between our neuro-anatomical structures and the human emotional world.

**Background**

**Neuroscientific Specialization: Damasio's Hypothesis of Somatic Markers** (1996).
Damasio and his colleagues were the first to claim that humans needed so-called "somatic" processes for decision-making in addition to cognitive ones. By this, they mean the perception of all processes occurring in the body. Somatic markers are generated in the limbic system and can evaluate options for action both consciously and unconsciously – especially when there is no time for rational analysis. The neural basis for the somatic markers is the connection between the prefrontal cortex and the limbic structures (e.g., amygdala, hippocampus, cingulate gyrus).

These processes were made measurable with a specially developed test for measuring skin resistance, the "Iowa Gambling Test". It was investigated whether patients with Elliot's impairment (ventromedial lesion) behave differently than healthy subjects when playing cards and whether this behavior can be predicted by skin resistance measurement. The result was impressive: the healthy participants could quickly see which two of the four decks of cards had a positive expected value

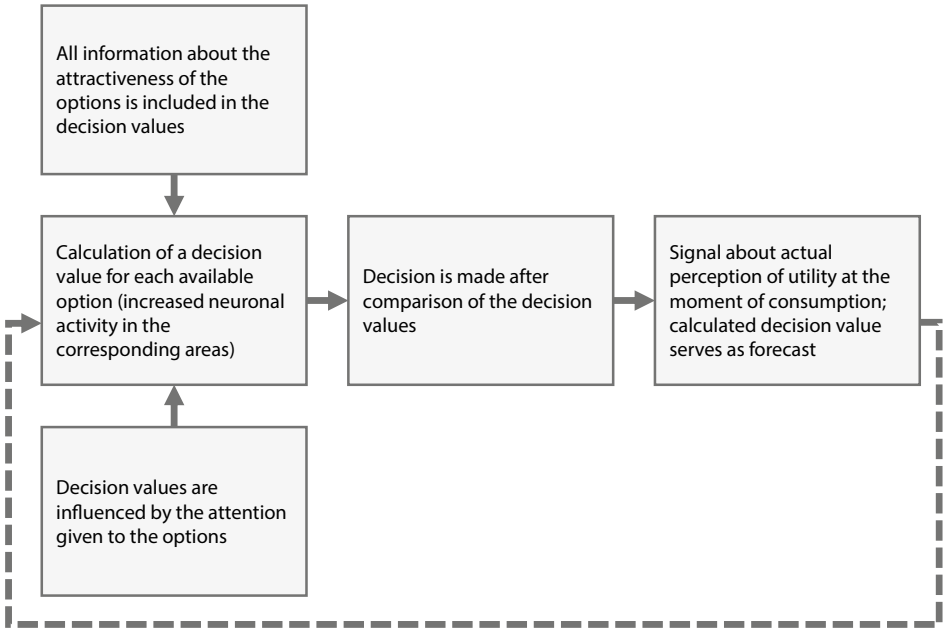



**Fig. 2.3** Decision-making process in neuroeconomic models

and which did not. Even before this finding could be consciously grasped and described by the test subjects, it was already possible to correctly predict whether a card from a deck would probably lead to a win or a loss via skin resistance. This was not the case for the participants with brain damage. In contrast to the healthy subjects, they ended the game with a loss. Damasio concluded that somatic markers represent an affective aspect of decision-making, i.e., **emotions shape our decisions**.

Generally, the two decision-making systems work together very efficiently: If we receive input from outside, it is usually processed in System 1, and a decision proposal is developed. If this proposal seems plausible to System 2 and there are no complications (e.g., based on our own experience), the proposal will be adopted and implemented. If, on the other hand, complications occur, System 2 boots up and "overwrites" the decision proposal of System 1, as shown in Fig. 2.4.

The **cooperation** between the two systems can also be felt firsthand, for example, with the "Stroop experiment" (Stroop, 1935). Take Table 2.2 and name the colors of the words as fast as possible (i.e., avoid reading the written words aloud) column by column. You will see that for every word, System 2 has to correct the first impulse of System 1, which is associated with noticeable concentration and effort.

Within the framework of this cooperation, System 1 and System 2 can support each other or lead to the same result. In the example of the ball player above, this becomes clear: both the intuitive decision-making system (System 1) and complex calculations

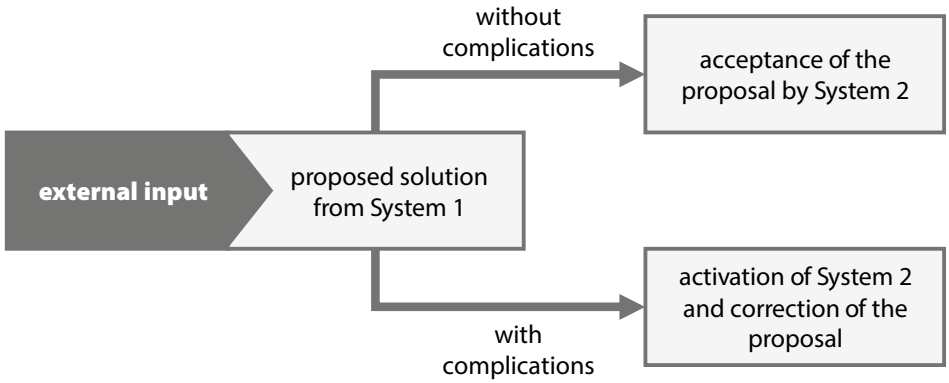



**Fig. 2.4** Cooperation between System 1 and System 2 (based on Kahneman, 2011)

**Table 2.2** The "Stroop effect" (own illustration based on Stroop, 1935)

| | | | | |
|---|---|---|---|---|
| Yellow | Green | Green | Blue | Black |
| Yellow | White | Blue | Black | White |
| Red | Black | Yellow | Yellow | Red |
| Black | White | Black | Green | Red |

(System 2) can be used to prepare for catching the ball. However, the two systems can also hinder each other: the capacity for mental effort is limited. That is why rational thought processes in System 2 usually interfere with each other. If a malfunction occurs, attention is withdrawn, and the task can no longer be solved satisfactorily. An impressive example of System 2's limited attention resources is the famous **"Monkey Business Illusion"**. The focus of attention is so strongly on a single task (in the experiment, count the ball changeovers of all players) that the test subjects even miss a gorilla moving across the picture. Intuitive processes in System 1, on the other hand, run effortlessly. They are not susceptible to malfunctions even if several tasks must be processed in parallel (Metz-Göckel, 2010). There is also neuroscientific evidence for the relationships between the two decision-making systems, emphasizing parallelism and possible competition (Loewenstein et al., 2008).

Incidentally, the "Monkey Business Illusion" makes it clear that **multitasking**, as a supposedly precious and desirable characteristic, is a demonstrably nonsensical construct: Performing several thinking tasks at the same time almost always leads to each task being performed with low quality or our brain prioritizing it independently and without regard to meaningfulness between these tasks. **Sometimes, 1 + 1 is less than 2.**

## 2.3 Better, but not Perfect

What is the capital of France?

Which compass direction is opposite to north?

How many animals of each kind did Moses take with him into the ark?

If you were able to answer these questions with "Paris", "south", and "two" without thinking twice, you have experienced first-hand how fast our brains can be – but also how error-prone! That's right: it was not Moses but Noah who built the biblical ship. The so-called "Moses Illusion" clarifies that we often only make decisions based on plausibility. Animals, Moses, ark – all of this comes from the bible, so it appears in a plausible context. In these situations, System 1 is the leading system; System 2 is only involved in situations where System 1 is uncomfortable. Obviously, sometimes, we make the wrong decisions when trusting System 1. By the way, it would have been different if Michael Jackson had been mentioned instead of Moses. This figure does not fit the given context, so System 2 would have been activated and noticed the error immediately.

Having two decision-making systems that can control each other is what allows us to live an everyday life as we know it. Without System 1, our rational decision-making System 2 would be chronically overloaded. Without System 2, on the other hand, we would act erratically, rashly, and often irrationally. So, we would lose the essential abilities that evolutionarily set humans apart from other species.

Nevertheless, the **duality of the two systems is by no means perfect**, as we have seen. We are not rational beings, nor can we entirely rely on our intuition. Instead, we switch back and forth between the two decision-making circles uncontrolled, unable to become aware of our thought processes or control them. Many cognitive biases and automated behavior patterns (heuristics) that have been ingrained over thousands of years are proof of this.

System 1 is usually blamed for **deceptions, distortions, and errors**, as it makes decisions even when the relevant influences and contextual factors are not sufficiently known. This is proven, for example, by the fact that we usually do not notice or even deny that we use automated behavior patterns (Pronin, 2009). Kahneman (2003) conducted extensive research on statistical intuition and was surprised that even experienced statisticians often come to spectacular intuitive misjudgments: "We were impressed by the persistent discrepancy between statistical intuition and statistical knowledge." (Kahneman, 2003, p. 697). Apparently, even experts cannot intuitively make good decisions in their own field of expertise. Almost even more astonishing: They are unaware of it and usually consider their intuitive assessments to be absolutely accurate. For us, this means: **Do not unquestioningly believe any expert**.

One of the best-known experiments dealing with the investigation of intuitive (thinking) errors is the so-called "**Linda problem"** by Sloman (1996). The test subjects were presented with the profile of a fictitious woman named Linda:

> Linda is 31 years old, single, straightforward, and very intelligent. She holds a university degree in philosophy. As a student, she was deeply affected by everyday discrimination and social injustice and participated in anti-nuclear demonstrations.

They were then asked to assess the probability of each of the following two statements:

a) "Linda is a bank teller."

b) "Linda is a bank teller and an active feminist."

Even if, from our current point of view, the question looks like an inappropriately deep reach into the cliché box: Which of the two statements would you consider more likely? Given the details from the profile, also statement b)? This is how most test subjects decide. However, they violate a simple conjunction rule: linking several statements can never be more likely than each statement alone. Statement a) must be more probable than statement b). In other words, the number of feminist bank tellers is wholly included in the number of bank tellers. Intuitively, however, we discover a plausible connection between Linda's social commitment during her student years and her affiliation with the feminist movement. Trusting our intuition and plausibility makes us make a decision that is obviously wrong from a rational point of view. System 2 would have avoided this error but was not involved in the decision of the affected test subjects because the information was presented in a plausible context, similar to the Moses illusion.

People have a similar experience when it comes to **purchasing decisions** – especially when uncertainty is involved. A great example is the conclusion of a life insurance policy. In the experiment by Johnson et al. (1993), the willingness to pay for two variants of this insurance was determined:

1. Payment of $100,000 in case of death
2. Payment of $100,000 for death by terrorism

Astonishingly, people are intuitively willing to pay more for variant 2, although it only pays out the amount in a fraction of cases compared to variant 1. This is due to the activation potential of feelings such as fear of terrorism. You could also put it this way: People are more afraid of dying in a terrorist attack than of dying at all. This (admittedly drastic) example once again illustrates the susceptibility of intuition itself to errors and, at the same time, the influence of public reporting on our perception of probability. Interestingly, there is a solid cultural impact here: While the fear of terrorism is irrationally high

among Americans, the same effect has been demonstrated in Canada using the example of bird flu (Sunstein, 2005).

As already mentioned, however, one should by no means conclude from this that our (at least quantitively) dominant decision-making system 1 is irrational overall. The opposite is true: using heuristics also leads to a good result in most cases and makes our brains extraordinarily fast and efficient. Alternatively, to put it another way, **heuristics make the decision-making process economical.** So, the calculation "heuristics = irrational" does not add up.

## 2.4 Decision Theory 2.0: Behavioral Economics

If the classic economic theory were software, it would urgently need an upgrade. Their assumptions no longer seem up-to-date and conflict with current research findings, especially from psychology and neuroscience. In its pure doctrine, therefore, it is hardly represented today. However, this does not mean that all economic ideas should be discarded. Instead, they will be expanded and updated to bring them closer to today's reality and to be able to make better forecasts.

Beck (2014, p. V) logically formulates the primary concern of behavioral economics with the aim of actually "weaving the psychological foundations of human action into economic theory." Therefore, behavioral economics is not an independent theory but an **application-oriented "upgrade" of established approaches.** This "upgrade" shows where psychological explanations of human behavior can be used in business. About the early origins of economic theory, some authors also speak of a **reunification of psychology with economics** (Camerer, 1999). However, it is also clear that its application becomes more complex when a theory is expanded. This is acceptable as long as the gain in accuracy more than compensates for the loss of simplicity (one could also say: user-friendliness).

In contrast to the core assumptions of classical economic theory (1. unlimited rationality, 2. unlimited willpower, 3. unlimited egoism), behavioral economics describe the following **foundations of decision-making:**

1. Imperfect markets: The availability of information is limited, as is the transparency of existing options. As a result, people do not always behave according to a market-efficient expectation. So, if we assume that individuals do not (or cannot) behave rationally indefinitely, perfect markets cannot emerge at a higher level.
2. Framing: How a problem or decision is presented influences the decision. Despite some recent criticism from the academic community, strong evidence of this are the impressive Milgram experiments (Milgram, 1974) or the prison experiment of Zimbardo (Haney et al., 1973). Here, enormous influences of the situation and the instructions on the behavior were revealed. In the prison experiment, the test participants in the role of

a prison guard even went so far as to punish the alleged inmates with life-threatening electric shocks due to the corresponding framing.

3. Use of heuristics: People often make decisions based on a simple, fast, and stable rule of thumb, not just an analysis of all possibilities. Heuristics are thus shortcuts on the way to decision-making. Beck (2014) illustrates a heuristic with the Gordian knot: a complex problem (knotted rope) is solved with a simple approach (cutting with a sword). This solution is simple and efficient but can lead to undesirable side effects (rope is destroyed).

Although classical economics already assumed that people can make wrong decisions, it never regarded them as systematic. Frequently cited arguments for limiting the occurrence of errors are the following three aspects: arbitrage, survival of the fittest, and learning.

- **Arbitrage** refers to the opportunities that open up to someone when someone else behaves economically irrationally. The stock market illustrates this well: If investors pay an irrationally high price for a stock, it will encourage rational stock owners to sell. The increased supply due to the opportunistic exploitation of this arbitrage leads to a price decline, and the irrational price tendencies are eliminated. As a result, according to classical economics, there should be no irrational behavior in the long term.
- **Survival of the fittest** means that only rational market participants will establish themselves in the long term. On the other hand, irrational people and institutions make mistakes that remove them from the market. Therefore, an entrepreneur who does not calculate rationally would be displaced by rationally acting competitors.
- **Learning** is the third argument for the absence of irrational decisions in economic theory. Accordingly, anyone who makes mistakes can learn from these mistakes and avoid them in the future. Thus, irrational behavior cannot occur in the long term.

None of these arguments of classical economics is free of criticism: Can't seemingly irrational stock prices also be rational in a sustained upward trend? Does not a look at nature show that even physically suboptimal life forms can be successful in the long term? Is there any possibility of learning from behavior due to a lack of feedback? Nevertheless, these arguments illustrate the basic assumption of economic theory that irrational behavior is not systematic and, thus, merely a temporary market distortion. To cut a long story short, this is a significant difference from the behavioral economics approach: they regard irrational behavior as a natural side effect of heuristics and, therefore, require the respective economic theories to include these anomalies in behavior modeling.

The basic assumptions of behavioral economics listed above always (but not exclusively) apply when we make **decisions under uncertainty.** This is the case when we do not know with certainty the probability of an event occurring – i.e., in almost all purchasing and consumption decisions.

## 2.5 Prospect Theory as a Point of Reference in Behavioral Economics

This raises a question: If classical economic theory is no longer up-to-date, but behavioral economics is not a theory in its own right and, therefore, cannot be used as an alternative explanatory model, which theory do we build on our understanding of decision-making?

The Prospect Theory is favored by many experts for this purpose. It was presented by Kahneman and Tversky in 1979 and is "the basis for awarding Daniel Kahneman the Nobel Prize in Economics in 2002. It explains how people make decisions under uncertainty. In its earliest form, the two scientists investigated uncertainty using the example of lotteries with different maximum winnings and winning probabilities, which is why it was initially called the lottery theory".

It is well suited for use in digital business because all information and purchase decisions for customers are uncertain. A complete assessment of all risks is impossible. In addition, digital decisions usually take place alone (i.e., without social influences), which once again emphasizes the uncertainty and scope of one's decisions. Therefore, precisely the conditions described in the theory can be found in the online environment – without the fathers of the theory even remotely thinking about the Internet.

The core idea of the Prospect Theory is that people do not always behave in the same way in decision-making situations (as the strictly rational utility maximization approach in classical economics assumes) but that the assessment of the probability of occurrence of an event and its evaluation are highly individual. For example, people tend to weigh losses (with the exact probabilities of occurrence) more than gains. So you are annoyed about losing $1,000 more than happy to win the same sum. It quickly becomes comprehensible if we imagine these two scenarios in our own lives. As a rule of thumb, avoiding a loss is about twice as motivating as making a profit. From a strictly economic point of view, this leads to the irrational behavior of many investors on the stock market: profits are realized too early, while loss-making stocks are held on for too long due to loss aversion.

#### Background

Prospect theory describes **two phases of decision-making:**

1. **The editing phase**: The available alternatives are preliminarily evaluated and sorted. To be able to do this quickly and resource-efficiently, the other options are usually simplified. Six mechanisms are available for this purpose, which already fundamentally distinguish prospect theory from classical expectation utility theory: *coding* (results are not perceived absolutely, but in relation to a reference point), *combination* (alternatives with the same result are summarized and their probabilities are added), *segregation* (safe components of an alternative are excluded, the expected value of an event multiplied by the high likelihood is then 1), *cancellation* (equal components of other options are not taken into account, i.e. the decision is only made based on the distinguishing features), *simplification* ("crooked" probabilities are mentally rounded, very unlikely events are entirely excluded) and finally *elimination* (alternatives that are subject to other options in all components are not further taken into account in the decision-making process).

2. **The evaluation phase**: The decision between the now-edited alternatives occurs in the evaluation phase. This can be modeled with a simple function consisting of two elements: a) the subjective value of an alternative and b) the individually weighted probability of occurrence of this alternative.

When evaluating alternatives based on a value and its probability of occurrence, prospect theory does not seem to differ much from the expected utility theory, which also argues with expected values and probabilities. The central difference is the subjective (i.e., not necessarily rational) assessment of the elements of this function. In other words, **the objective expected benefit is irrelevant to a decision**. This is illustrated very vividly by the coin toss game, in which you lose one dollar with "heads" and one dollar with "tails". The expected utility theory would now say that both events (heads and tails) have the same probability of occurrence with 0.5 each and that the possible win and loss are the same – the game's expected value would thus be 0. The Prospect Theory comes to a different conclusion: As described above, people attach more weight to losses than to gains. So if a decision-maker estimates the loss of a dollar to be higher than the gain of a dollar and both occur with the same probability, his expected value is negative – he would not make a bet (unlike the "homo economicus").

#### Background

A bird in the hand or two in the bush?

A brief anticipation of the collection of behavior patterns described in chap. 4: The unequal treatment of possible profits and losses is known from the "**endowment effect".** It explains that people value things from their possessions more than things they could possibly receive because of the high fear of loss. Let us give it a try: You have the choice between two options:

1. Loss of $30,000 with a probability of 0.1%
2. Certain loss of $30

The expected value is identical in both cases ($30). Why do the majority of you still spontaneously opt for option b? Because just the idea of the possible loss of $30,000 creates so much "pain" that the probability is hardly taken into account anymore. Incidentally, this banal example can also explain the business model of insurance companies: You have certainly insured your car worth $30,000 with a monthly premium of $30 (probably significantly more) and probably overestimated the probability of occurrence of an insured event.

For the decision on participation in such games, one must additionally emphasize the reference value from the editing phase of the Prospect Theory. It clarifies that the willingness to participate should not be statically modeled: Someone with a wealth of $100,000 will be hurt by the possible loss of one dollar less than someone who only has a wealth of $10. These two people would, therefore, probably not come to the same decision to participate – completely independent of the expected value.

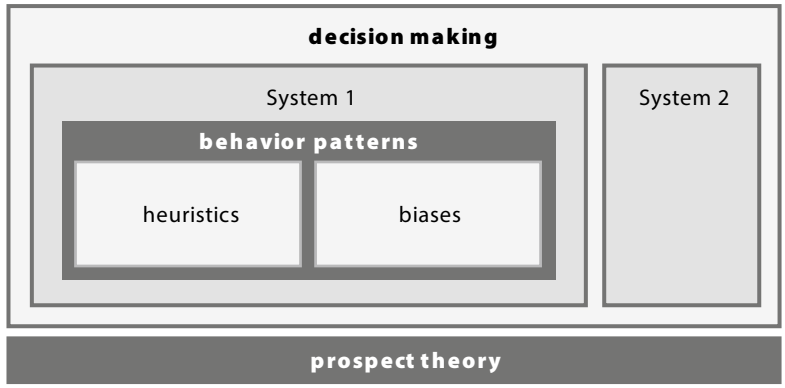



**Fig. 2.5** Context of the most important terms. (Source: own presentation)

## 2.6 The Most Important Findings Summarized

- Humans possess **two decision-making systems,** one for rational and controlled thinking and one for impulsive and intuitive thinking. The majority of all decisions are made intuitively.
- Intuitive decisions are based on simple rules, the **heuristics.** These are universal and anchored in evolutionary biology. They can be detected in almost all people.
- While heuristics are efficient, they are not necessarily conducive to the quality of decisions. Their use often entails **cognitive distortions** or biases.
- **Behavioral economics** is the school of thought that considers the heuristics and biases in behavioral modeling. Their main point of reference is prospect theory.
- Many heuristics are now known in detail. This opens the possibility of **influencing** unconscious decision-making processes by activating specific **behavior patterns**.

The most important terms of this introductory chapter and their interrelationships are described in Fig. 2.5: In PsyConversion®, we are thematically on the left side of the figure, the intuitive decision-making System 1, in which the behavior patterns unfold their effect.

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# Behavior Patterns: Foundations of Our Decisions

3

**Summary**

Behavior patterns are standardized processes that underlie the majority of our decisions. They can be classified in the field of "persuasive communication". Their use in digital business improves the user experience and increases the conversion rate (as well as other relevant performance indicators). Prerequisites on the company side are a marketable and competitive product and a functioning technical infrastructure. If this is the case, behavior patterns can be used to redesign or optimize websites, apps, online advertising media, etc. At the same time, their effect should not be overestimated; they certainly do not represent the legendary "buy button" in the brain. Although the effectiveness is mainly universal, the potency is not. As a result, the better a company knows and classifies its users, the more accurate the selection of behavior patterns can be. In this context, the real-time dynamization or personalization of the website along the identified user groups offers further conversion potential. For the identification of suitable behavior patterns, this book provides several frameworks. In addition, it is advisable to combine mutually reinforcing behavior patterns in many cases.

After discussing the fundamental interrelationships of decision-making processes in Chapter 2, we now go one level deeper and deal with the individual behavior patterns that can be observed repeatedly in System 1 decisions. These behavior patterns are nothing less than the basis of our decision-making ability.

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## 3.1 What are Behavior Patterns?

To say it right at the beginning, behavior patterns are neither good nor bad. And they certainly should not be used as evidence of human stupidity or cognitive inadequacy. This becomes clear when one considers what these behavior patterns are: First and foremost, behavior patterns are, as already mentioned in chap. 2, mainly about **heuristics, shortcuts, or trails on the way to a solution**. By using these rules of thumb, our decision-making system becomes fast and efficient, which is fundamentally positive in the first place. From an evolutionary point of view, the usefulness of heuristics also becomes clear: In primeval times, early humans depended on being able to decide immediately whether a rustle in the bush was a danger or not. Standardized behavior patterns provide the necessary speed for this**.** The fact that there was no saber-toothed tiger behind every rustle (i.e., there was a risk of a wrong decision) was gladly accepted. Without heuristics, our ancestors would probably not have fled from the noise early enough, and none of us would be here today. This indicates that the **gain in efficiency goes hand in hand with a loss of accuracy**. For example, heuristics do not consistently deliver the correct result and can, therefore, be the basis of cognitive biases – the second source of behavior patterns (see also Fig. 2.4). In summary, this means: Yes, people are intelligent and good decision-makers. And no, they are not always right.

Behavior patterns are thus the **basic schedules for automated decisions** that we all carry (to a greater or lesser extent) within us. Interest in these patterns has steadily increased over the past four decades as the realization of the potential of their knowledge has spread: Whoever holds the schematic in his hands can quickly determine which levers need to be set in motion to trigger a specific behavior.

> Persuasive communication is any message designed to shape, amplify, or change someone else's reactions. (Gerard Miller, 1980, p. 11, psychologist and author of numerous books on persuasion)

Somewhat less externally determined is the understanding of the author, Nathalie Nahai. She describes behavior patterns as a means of "reducing the distance between divergent points of view and working together towards a mutually positive outcome" (2017, p. 60) – whether it's about convincing children to eat their broccoli, preventing hostile states from declaring war, or turning website visitors into customers. Gerard Miller's definition is essential as a basis because it classifies the targeted activation of behavior patterns as **persuasive communication**. However, adding Nathalie Nahai is necessary to clarify that persuasion must not be made to the detriment or against the customers' interest.

Behavior patterns almost always work together. They embed themselves in the context of the environment and merge with all other influences presented. This shows the importance of properly designing a functioning visual trigger to activate the corresponding pattern (see section 5.2 for more details). Strictly speaking, not just one trigger is usually used, but several. These can address the same or different behavior patterns. A

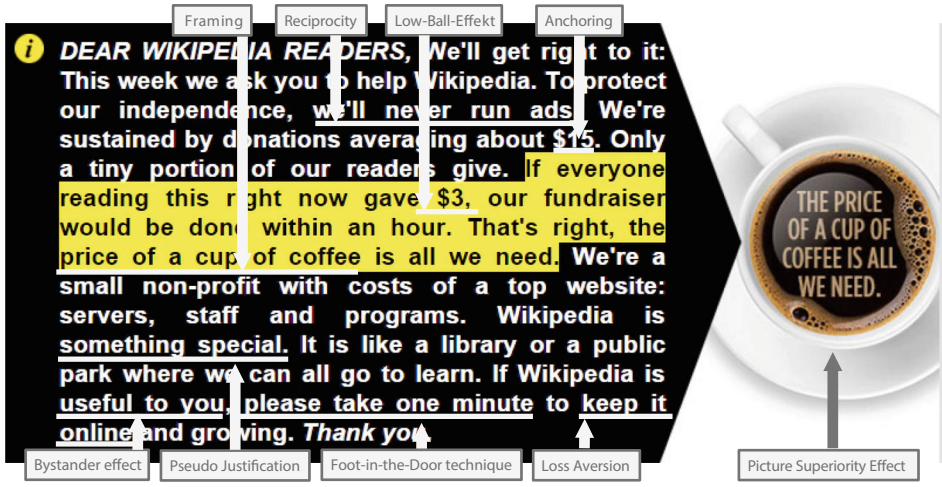



**Fig. 3.1** Example: Fundraising on Wikipedia. (Source: own illustration with a screenshot of wikipedia.org)

combination of several behavior patterns is generally used in critical sections or when there is no possibility of meaningful personalization of the page. An exciting example of both is Wikipedia's call for donations. The non-profit online encyclopedia is primarily financed by donations raised with advertisements on the website. Therefore, the success of these reports directly impacts the portal's existence. Accordingly, the text is peppered with many persuasion mechanisms, all described in chap. 4 (see Fig. 3.1).

## 3.2 Why Should Behavior Patterns be Used?

The use of behavior patterns (or more precisely, the use of triggers in the interface to activate behavior patterns) is primarily suitable for two central motives: E-commerce providers want to increase their **conversion rate and sales** – obviously, or to put it more precisely: actively shape the growth curve of their company. In addition, behavior patterns also offer the possibility of a substantial improvement in **usability and user experience.** The user experience (UX) can be most easily described as the sum of all experiences of a website visitor and includes all "emotions, ideas, preferences, perceptions, physiological and psychological reactions, behaviors and services that arise before, during and after use" (DIN ISO 9241–210). Usability is an excerpt of the user experience and describes the effective, efficient, and satisfactory achievement of a user's goals (DIN ISO 9241–11). Loosely based on Steve Krug (2014, "Don't make me think!"), customers strive to achieve their goals on a website with the least possible cognitive effort. Knowledge of automated decision-making processes can contribute significantly to this and make online shopping

a pleasurable experience. Therefore, optimizing the **user experience is not an altruistic end but a business requirement**.

Consequently, only at first glance is there a conflict of objectives between the business perspective (i.e., increase in sales) and the customer perspective (i.e., improved user experience). Considering the disproportionately higher costs of acquiring new customers than keeping existing customers, it becomes clear that the entrepreneurial goal must be a sustainable relationship with loyal customers. This is only the case if customers are satisfied with their decisions in the long term, i.e., if an intuitive decision was not made against rational will.

## 3.3 What Requirements Must be Met?

For behavior patterns to be activated by appropriate triggers in a website, several factors must be given. A distinction can be made between factors in the sphere of the customer and factors in the sphere of the company.

About the **requirements that customers must bring with them**, the following characteristics are mentioned often (e.g., Fogg, 2009, see in more detail also Sect. 4.2):

1. Motivation: There must be a concrete reason to engage in behavior.
2. Capacity to act: Users must have the practical ability to engage in behavior.
3. Resources: Time and money must be available in sufficient quantities.
4. Limited effort: The physical and cognitive effort must be low compared to the benefit of the behavior.
5. Social acceptance: The user's environment must not reject the behavior.

The first thing to mention about the **requirements on the company side** is the freedom from technical errors in the shop. Drastic violations of UX conventions and habitualized behaviors learned in e-commerce hinder the flow of the user and ensure that System 2 is permanently reactivated and the control over the decision no longer lies within System 1. In addition to many other conditions, good content with meaningful product descriptions or the availability of all information relevant to decision-making is essential. In addition, selection processes and digital order forms must be designed and presented in a manageable manner. Finally, a product that is not marketable is challenging to sell even with the best combination of high-performance behavior patterns.

## 3.4 How Strong is the Effect of Behavior Patterns?

When you read this headline, it was probably already clear that this question can hardly be answered precisely. This is true, but some clues can help to come to an assessment.

First, the **effect of behavior patterns is enormously glorified** in many (more or less reputable) practice publications and sold as the key to the "active management" of customers. This is exaggerated and misses the basic idea of working with behavioral scientific findings: Website visitors are not suddenly controlled by an appeal to their deep-seated behavior patterns and unthinkingly buy random stuff in the next best online shop. There is simply **no "buy button" in the brain,** which amateur neuroscientists often invoke. John-Dylan Haynes, brain researcher and director at the Bernstein Center for Computational Neuroscience at Charité in Berlin (Germany), clarifies: "Manipulating people at the push of a button doesn't work. […] There is no way to stimulate the brain to make a purchase directly" (Socaciu, 2017, n.p.). This is because very few online ordering processes are completely System 1-based. This is already ensured by process steps such as entering the address and selecting the shipping method, which are mainly processed consciously and rationally. The payment method is also a reliable activator of System 2: Paying is always associated with loss and risk, which usually leads to System 2 being called in by System 1 for help and performing its control function. Nevertheless, the general direction can certainly be set by System 1. Thus, the purchase decision can be made here, while System 2 is responsible for working out the details of the procurement process.

In addition, behavior patterns are often most likely to unfold their effect in the last steps of the purchase decision. Suppose all technical requirements are met (see ▶ Sect. 3.3) and there are no objective-rational arguments against a purchase (e.g., exorbitant prices, long delivery times, non-availability). In that case, they can **"nudge" in the right direction**. It is not for nothing that the psychologists Richard Thaler and Cass Sunstein simply call their book on the subject, which is well worth reading, "Nudge" (2009) (see also the list of recommended literature in the appendix). How strong the effect is also depends on how products are purchased. In the case of hedonistically purchased products (e.g., luxury food, toys, cosmetics), System 2 has less decision-making authority. Conversely, this means that System 1 is leading the decision – and it is precisely this that is controlled by the behavior patterns.

The effect is thus **overestimated and underestimated simultaneously,** which also expresses the still immense need for research: Behavior patterns cannot (fortunately) force customers to buy against their will. They can only address needs that are already predisposed in us. Used correctly, they can tip the scales when deciding whether to purchase. For the sake of completeness: Dealing with behavior patterns can not only lead to incorporating conversion boosters into the website. Due to the more profound understanding of the decision-making process, conversion killers are just as often revealed on the current page. Eliminating them is the second conversion lever of working with behavior patterns.

## 3.5 With Whom do Behavior Patterns Work?

In short: **With all people and in (almost) every situation.** Behavior patterns or the heuristics and biases they contain are the basis of how our brain works and most decisions. However, ► Sec. 3.4 has already shown that the strength of the effect always **depends on many contextual factors**. The relative strength of the two decision-making systems depends on the product, but also the customer journey (i.e., the process from the first impulse through the information phase and the purchase to the post-purchase phase) plays a significant role. Some steps of this journey are more System 1-based and thus particularly well suited for the effectiveness of behavior patterns. Anyone taking one of these steps (e.g., product evaluation; see ► sec. 3.6) is more receptive to the effect than someone in a System 2-dominated phase (e.g., entering the payment information).

In addition to the customer journey, individual **personality traits** are also decisive for whether behavior patterns are effective. Bargain hunters, for example, are more likely to be influenced by patterns from the area of behavioral pricing (see ► Sect. 3.6) compared to security-oriented, all-around carefree customers. Customer segmentation, e.g., based on historical behavioral data, personas, or psychological-neurological profiles, is helpful in classification. A famous example of the latter is the seven limbic© types of the Nymphenburg Group (*Harmonizer, Bon Vivant, Hedonist, Adventurer, Performer, Disciplined, Traditionalist*; Gruppe Nymphenburg, 2018).

## 3.6 How do you Find Suitable Behavior Patterns?

A vital note right at the beginning: The selection of behavior patterns presented here does not claim to be complete. There are two reasons for this: Given the exponentially increased enthusiasm for automatic decision-making processes in research and practice, exciting new findings occur almost weekly, repeatedly leading to the discovery of previously unknown behavior patterns. Some have been presented for the first time in this edition of PsyConversion®. Thus, this collection can only ever be a snapshot of the current state of research. The second reason lies in the application focus of the book: Not all behavior patterns can be used in the digital environment, for example, because a personal conversation with a salesperson is a central component of the mechanism of action. To keep the scope of the behavior pattern library manageable, such behavior patterns without immediate application potential have not been listed.

The central challenge for working with behavior patterns is identifying suitable patterns for realizing the goal (e.g., building trust, increasing perceived value for money, preference for higher-quality products, etc.). For this reason, the book provides a **framework for structuring behavior patterns.** It structures the behavior patterns in a distant way

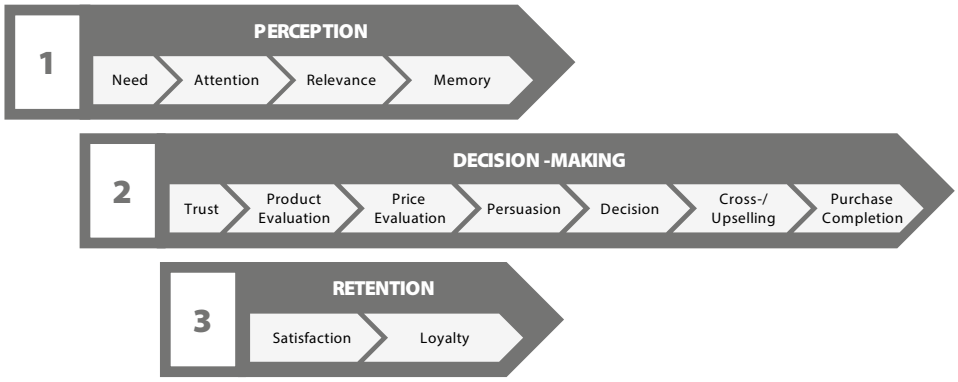



**Fig. 3.2** Idealized decision-making process in e-commerce

based on Karimi's "Decision-Making Process Model" (2013) along an idealized decision-making process. A rough distinction is made between the three phases: 1) perception, 2) decision-making, and 3) retention (see Fig. 3.2). It is important to note that this phase model is only used to structure the behavior patterns and does not reflect the actual decision-making process of all users. To assume this would be contrary to the non-linear operation of System 1 and would not be compatible with the nature of behavior patterns described in chap. 2.

The **perception** phase refers to bringing an online shop or e-commerce provider into the customer's consciousness. It includes the awakening or discovery of the basic need, the generation of attention for the shop, the determination of the relevance of the provider, and ultimately, the memory of the provider since there are usually several (marketing) contact points with a customer before the start of the actual purchase process. The second phase is called **decision-making** and represents the core of the process considered here. For this reason, it is subdivided more delicately and includes seven dimensions: building trust with the customer, evaluating the product as suitable, evaluating the price as appropriate, the direct persuasion of the customer (as a push process), indirectly influencing his decision (as a pull process), the use of cross-selling or up-selling opportunities and finally the avoidance of abandonments in the process.

However, the purchase does not end the e-commerce decision-making process. It only ends with the **retention** phase (the "post-purchase phase"). This includes, on the one hand, satisfaction with the purchase and, on the other hand, loyalty to the provider as the basis for subsequent purchases. Due to the better possibilities for addressing customers and the resulting lower activation costs of existing customers, the phase plays a crucial role.

Most of the 117 behavior patterns described in this book can be used in several phases or dimensions. Therefore, the framework distinguishes between **primary and secondary patterns** in each of these areas. A primary pattern shows in which area (e.g., attention) the

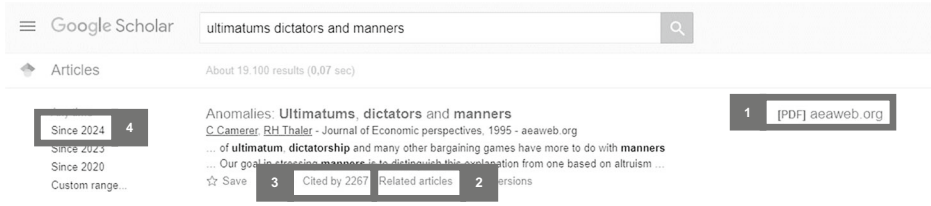



**Fig. 3.3** Google Scholar results page. (Source: Google)

behavior pattern has its historical origin and usually also the highest efficiency. Secondary patterns describe behavior patterns that can be used in other areas beyond the primary application context or whose effect has also been proven in studies there.

Chapters 4 (perception), 5 (decision-making), and 6 (retention) present the behavior patterns along the e-commerce decision-making process.

▶ When working with this book, you will often come across exciting behavior patterns but need more in-depth information – for example, to find application examples in your industry, to research the exact effectiveness in experiments, or to deal with the criticism of the behavior pattern. For this, "Google Scholar" is recommended as a research platform. Here, Google lists papers from academic journals and conference papers, almost all of which have undergone a scientific review. Under scholar.google.com, enter the source mentioned in the profile. Thus, you will often find not only the freely available original texts (1) but also related articles (2) or articles that refer to the original source and usually further develop the core idea (3). In addition, you can use filter functions to understand the significance of older behavior patterns (4) (see Fig. 3.3).

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# Behavior Patterns for the Perception Phase

# 4

**Summary**

Scientific research across disciplines (especially psychology, neuroscience, and economics) has identified and validated many behavior patterns. This library summarizes the most effective and most suitable for e-commerce use. It is structured along the e-commerce process and is divided into 1) the perception phase (chapter 4), 2) the decision-making phase (chapter 5), and 3) the retention phase (chapter 6). These three chapters are the core of this book. The collection of behavior patterns has been built up, qualified, cleaned up, and described over the years. The result is not a complete collection of all identified behavior patterns but a **practical selection** to provide quick and immediately actionable answers and inspiration for e-commerce professionals who want to increase their communication and sales success.

The **perception phase** presented in this chapter is divided into the sections need, attention, relevance, and memory (see Fig. 4.1). It contains a total of 24 patterns that have their primary application context in one of these four blocks. The **need** describes the initial impulse that leads users to deal with the possibility of a purchase in the first place. **Attention** clusters behavior patterns that can be used to claim users' scarce attention resources and be noticed online. The **relevance** section goes one step further and bundles approaches that place an online shop in the "relevant set" so that it is considered when making a decision. The fact that not all products are purchased at the first contact is expressed in the **memory** section. Many purchasing decisions require several (in extreme

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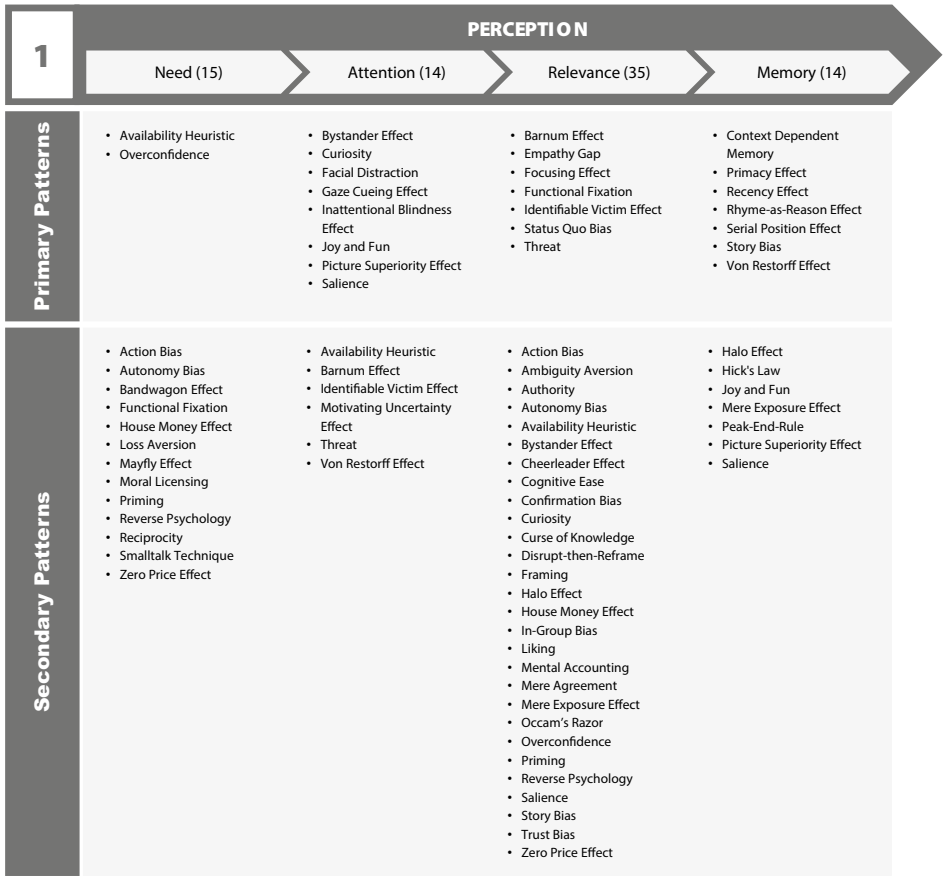



**Fig. 4.1** Overview of primary and secondary patterns for the perception

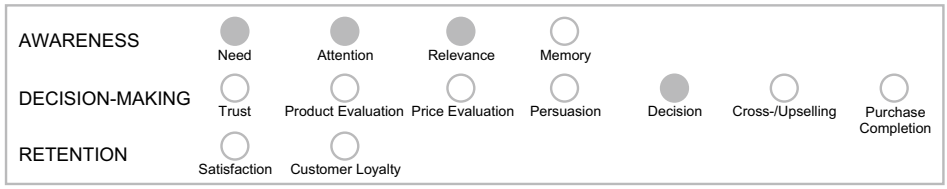
cases, dozens) contact points to form a clear preference. Therefore, online shops need to stay in the minds of potential customers about what can be achieved with these behavior patterns.

## 4.1 Need

### 4.1.1 Availability Heuristic

Availability stands for the cognitive ease with which we find a suitable example of an event in our memory with System 1. When we think about buying a product, we look for situations where we would have needed it. If it is easy for us to find such examples (i.e., with high availability), we consider the product relevant, make an intuitive purchase decision, and refrain from activating System 2. If, on the other hand, System 1 finds it difficult to find a solution, System 2 jumps to its side. The availability heuristic is an illusory correlation: From the ease of finding an example, we infer the intensity of demand, although these two variables are usually not causally related. Several factors influence availability: repeated experience, clear presentation, priming (more on that later), good mood, or the subjective effort in the search for examples. If the conditions are right, we feel a flow experience: Everything feels good, simple, familiar, and effortless.

Simply put, what we easily remember seems probable or relevant. This also explains, for example, many people's fear of flying: (statistically extremely rare) plane crashes are intensively reported in the media, leading to high mental availability of such events and, thus, to an overestimated probability.




**Implementation in e-commerce**

- Products should have familiar-sounding names (and not "cx45b-d_v2").
- The formatting of text (especially the font's legibility, strong contrasts, and bolding) influences availability. Conversely, however, the following also applies: If you want users to deal deeply with a topic, System 2 can be activated using difficult-to-read typography (e.g., "Mistral" or "Freestyle Script").
- Key statements can be repeated at decision-making points, as well as the necessity of the product. Take insurance, for example: After a flood, more homeowners and natural hazard insurances are bought in the affected areas because the mental example of "flood" is easier to find.
- Products can be linked to events from the media that have a high level of recognition. Although the recognition applies to the media event, it is transmitted through the joint presentation (spill-over), also called the "news illusion". After significant media events

related to the product, advertising expenditure should be increased to take advantage of the (temporary) effect.

- Low-probability events may seem more relevant if the time horizon is changed (e.g., decades instead of years). In the insurance example, the risk of experiencing a "flood of the century" in the next 20 years that destroys one's house is significantly higher than the annual risk.
- Availability Heuristic can even be used to influence the outcome of satisfaction surveys (e.g., NPS surveys) astonishingly: If you ask customers to name two aspects a company still needs to improve, experience has shown that this is very easy. The Availability Heuristic concludes that there is great potential for improvement – the rating is correspondingly poor. But if you ask for ten points of criticism instead of two, the respondents quickly start to ponder and can only fill the list with incredible difficulty. The rating is correspondingly better. Whether this makes strategic sense must be decided on a case-by-case basis.




**Source:** Tversky A, Kahneman D (1973) Availability: A heuristic for judging frequency and probability. Cognitive Psychology 5(2):207–232.

**See also:** Cognitive Ease, Anchoring, WYSIATI Effect, Picture Superiority Effect, Empathy Gap, Curse of Knowledge, Priming.

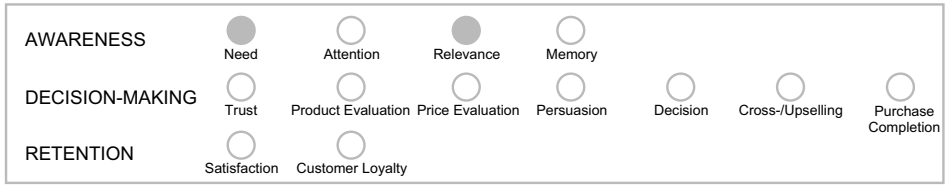
**My Notes**

### 4.1.2 Overconfidence

Also: Over-Optimism, Overestimation of Oneself

People tend to overestimate themselves and their abilities, knowledge, or achievements. Well-known manifestations are the incorrect assessment of probabilities, the Illusion of Control (see: Illusion of Control), and the self-serving bias. Interestingly, as the complexity of the task increases, so does the level of over-optimism. This effect also applies the other way around: In the case of simple tasks, confidence in one's performance dwindles. This is called the "hard-is-easy effect" (Fischhoff et al., 1977).

This effect has been empirically proven in lawyers, physicians, investment bankers, and business managers – i.e., professional groups that are generally not assumed to have the problem of a lack of self-confidence. Self-confidence and overconfidence are closely correlated: the more self-confident you are, the more vulnerable you are. Especially at management levels, parallel worlds of self-image often develop due to a lack of feedback and strong isolation. Before the question arises: Beyond all gender stereotypes, there is robust empirical evidence that men succumb to Overconfidence Bias far more often and more severely than women. Moreover, it hits less intelligent people harder. The two psychologists, Justin Kruger and David Dunning (1999), explain this aptly: "*If someone is incompetent, then he cannot know that he is incompetent.*"




**Implementation in e-commerce**

- Overconfidence can prevent people from recognizing a need (e.g., from guidebook literature, insurance, vitamin supplements). This effect can be corrected by giving immediate feedback and explicitly pointing it out, for example, by using dropout surveys. In this case, the user signals by moving the mouse toward the browser's input line that he wants to leave the page or does not consider the product relevant. This triggers a pop-up window describing the threat scenario and the underestimated danger (e.g., "You also think vitamin deficiency does not affect you? Far from it, 60% of the population lack valuable vitamins and are impaired as a result.").
- This corrective measure can be reinforced by providing users with extensive in-depth information, whereby overconfidence becomes weaker and weaker as the amount of information increases. Monetary incentives (e.g., a rebate) also mitigate the effect.




**Sources:** Oskamp S (1965) Overconfidence in case-study judgments. Journal of consulting psychology 29(3):261–265; Fischhoff B, Slovic P, Lichtenstein S (1977) Knowing with certainty: The appropriateness of extreme confidence. Journal of Experimental Psychology: Human perception and performance 3(4):552–564; Kruger J, Dunning D ((Kouchaki et al., 2013)) Unskilled and unaware of it. How difficulties in recognizing one's incompetence lead to inflated self-assessments. Journal of Personality and Social Psychology 77(6):1121–1134.

**See also:** Self-Efficacy, Illusion of Control.

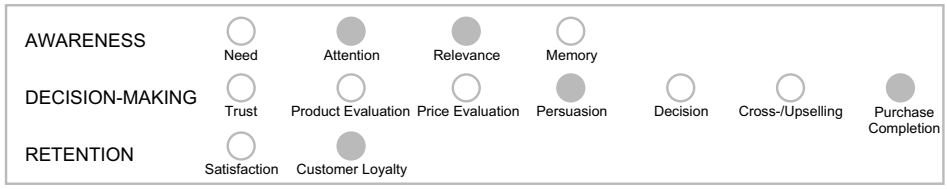
**My Notes**

## 4.2 Attention

### 4.2.1 Bystander Effect

The presence of other people inhibits the impulse to help, the willingness to help decreases steadily as the number of people present increases. There is no active action because no one wants to be the first to be ridiculed (intervening without any real cause) or take a risk alone. This trend is self-reinforcing: a situation is not perceived as an emergency because others have not yet intervened. Here, it is assumed that everybody else can better assess the situation and that the non-activity is based on a well-founded assessment (see Unity). The effect got its name because of its discovery in the context of accidents and criminal assaults, where it quickly became apparent that people usually stand around uninvolved in the face of tragedy – even if this contradicts their inner beliefs.

However, there are ways and means to overcome the effect: The probability of helping increases as soon as public attention is created. In an experiment, the subjects were asked to participate actively in an online forum. The bystander effect caused this request to fizzle out without effect. This changed abruptly when the names of the participants were highlighted in red, and a webcam was used to create publicity. The Bystander Effect works when there is a lack of clear responsibilities in a group. If the responsibility is spread over many shoulders, everyone bears only a tiny share, which can easily be overlooked. The discoverers of the effect refer to this as "pluralistic ignorance". The effect can be overcome if individual group members' actions are brought into focus. Given social visibility, people want to demonstrate to the group that they are up to the challenge and are perceived as proactive, decisive, and capable of acting.




**Implementation in e-commerce**

- Instead of addressing the clientele broadly, individual users (e.g., on social media) can be addressed directly, for example, to respond to criticism.
- In your own platforms, it makes sense to distribute digital status badges and (supposedly) stand out users from the crowd.
- In content marketing, the direct customer approach ("you should…") works better than general, impersonal formulations ("it is recommended…").
- The need for intervention must be verbalized: "We still need information from you here.", "You have to decide now.", "Please write a short review about this product you just bought."

- The fear of negative consequences of shopping (e.g., no guaranteed anonymity) can be taken away from users: “Do not worry, you can change your decision at any time.“, “Your details are safe with us; we will never share information about you on social networks.“




**Source:** Darley JM, Latane B (1968) Bystander intervention in emergencies: diffusion of responsibility. Journal of Personality and Social Psychology 8(4p1):377–383.

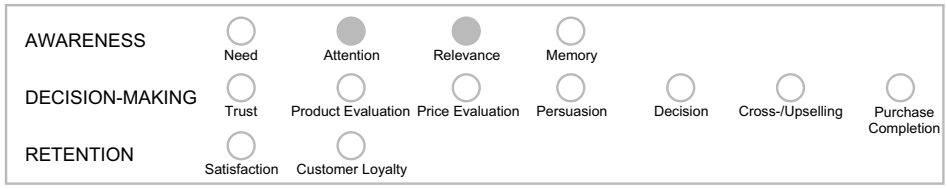
**See also:** Unity, Trust Bias, Social Proof, Inequity Aversion.

**My Notes**

### 4.2.2 Curiosity

Functional magnetic resonance imaging has shown that curiosity is associated with strong amygdala activation, the region where the most profound human emotions are located. It is, therefore, not surprising that curiosity is a potent driver of behavior. According to the "Desire Theory of Motivation" (Reiss, 2004), it is one of the most vital motivational factors.

Nevertheless, "Curiosity" is vastly underestimated as a motivator. Even in the scientific community, simple curiosity is often forgotten as an influencing factor in modeling human behavior. It is now clear that five different forms of curiosity can be decisive as drivers of our behavior (Kashdan et al., 2018): 1) joyful exploration, gaining new knowledge and rejoicing in learning and growing; 2) loss sensitivity, fear and tension outweigh joy; 3) stress tolerance, doubt, confusion, fear acceptance, which is achieved through the exploration of unknown events arise; 4) social curiosity, desire to know what other people think and do; 5) thrills, willingness to take physical, social and financial risks to have intense experiences.




**Implementation in e-commerce**

- A newsletter subscription can be advertised with "exclusive insider information" and surprises.
- Caution: There is a fine line between arousing curiosity and rightly discredited clickbaiting techniques (i.e., lurid headlines that create a maximum urge to read with subsequent disappointment). Marketing agencies repeatedly rely on this way of lead generation and later blame shop usability if the (completely unqualified) traffic does not convert sufficiently.
- Curiosity only works if a minimum amount of information on the topic has already been transmitted, so a short outline with a build-up arc of suspense is a good idea before placing the trigger. The conversion point should be at the peak of the curiosity curve. As soon as so much information is provided that the most pressing curiosity is satisfied, the arc of suspense quickly drops.



**Source:** Loewenstein G (1994) The psychology of curiosity: A review and reinterpretation. Psychological Bulletin 116(1):75–98.

**See also:** Motivating Uncertainty Effect, Ambiguity Aversion, Status Quo Bias, Story Bias, Reverse Psychology.

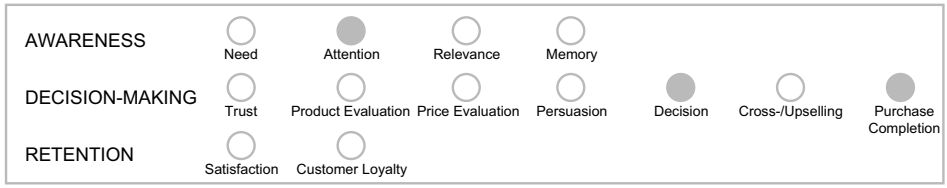
**My Notes**

### 4.2.3 Facial Distraction

Also: people-look-at-people phenomenon

We cannot escape direct eye contact. When a person looks directly at us, we look them in the eye and "waste" many resources trying to decipher facial expressions to classify our counterpart as a friend or foe. This is not without consequences. The resources used are then lacking elsewhere. For example, the content intended to be supported by the image is less noticeable. Interestingly, the facial distraction phenomenon leads to entirely different results offline and online: In a face-to-face situation in brick-and-mortar retail, direct eye contact increases the willingness to perceive the verbal message of the other person. The opposite is the case online: Since information is usually not available verbally but in text or diagram form, this transfer does not occur; attention sticks with the person. This connection is more robust the more familiar a face is (see: Authenticity).

From a neuroscientific point of view, facial recognition is also very different from object recognition in terms of visual processing. It occurs in the so-called "Fusiform Face Areas" and requires considerably more resources. People with certain neurological disorders (such as "prosopagnosia") can no longer recognize faces. On the other hand, they have no problems with objects because they are processed in different areas.




**Implementation in e-commerce**

- In your own shop or website, people in pictures (e.g., testimonials) should not look the user in the eye. According to the Gaze Cueing Effect (see ► Sect. 4.2.4), it is instead a good idea to direct the user's gaze with the eyes of the person depicted in the picture – for example, in the direction of the button (call-to-action).
- Outside of your platform, the situation is different: Here, it makes perfect sense to let the user look directly at the testimonials used to draw their attention to a display banner ad or similar. The eyes are a central component of the effect, which should be particularly emphasized (keyword: childish scheme).




**Sources:** Friesen CK, Kingstone A (1998). The eyes have it! Reflexive orienting is triggered by nonpredictive gaze. Psychonomic Bulletin & Review 5(3):490–495; Frischen A,

Bayliss AP, Tipper SP (2007) Gaze cueing of attention: visual attention, social cognition, and individual differences. Psychol Bull 133(4):694–724.

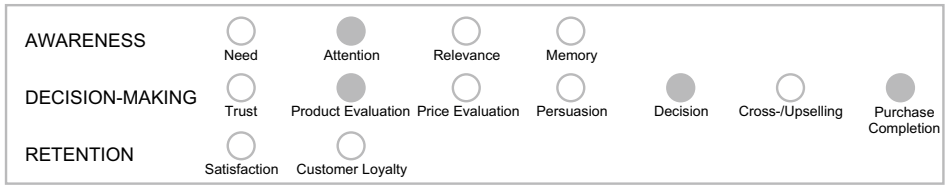
**See also:** Gaze Cueing Effect, Authenticity.

**My Notes**

### 4.2.4 Gaze Cueing Effect

Unconsciously, when we come to a new environment, the first activity is to analyze the faces of the people in that environment. This begins in the first minutes after birth. With our looks, we already show toddlers the world, while they could not follow a verbal request for a long time. So, our attention is focused on where other people are looking or pointing. This can be felt particularly well in everyday situations, such as crowds of people that form during accidents or when looking at birds in the sky.

The Gaze Cueing Effect dulls only surprisingly slightly: Even if we have followed the gaze of our fellow human beings several times without recognizing an explanation for their increased interest in this direction, we hardly stop doing so. This is because this orientation towards others occurs unconsciously; test subjects often deny it in post-experience interviews. An evolutionary explanatory approach suggests that a concrete danger was often to be expected from the direction in which our fellow human beings look. Given the possible consequences, it is understandable that the effect also often works repeatedly.




**Implementation in e-commerce**

- When selecting images, depending on the context of use, it should be noted that the people depicted support the Gaze Cueing Effect in favor of the most crucial text block or the call-to-action, i.e., look or point in the appropriate direction. Therefore, images in which the depicted persons (or animals) guide the user out of the picture are unsuitable.
- Conversely, gaze cueing can also divert attention from critical zones, e.g., prices. If these areas are placed in the areas facing away from the view, they play a more minor role in the decision-making process.
- A unique feature is the "Visual Cueing Effect". It refers to the same phenomenon, but the user's gaze is not directed with the eyes or hands of the people depicted but with graphic elements. Examples are arrows (especially with one-pagers) or triangles, to the top of which our gaze is directed. Extremely effective (albeit relatively trivial at first glance) are arrows that point from a USP communication to the buy button: This arrangement is linearly processed in the brain as "This is a good product, I want to buy it." An alternative application of the Visual Cueing Effect is the arrangement of centrally placed text with lines that narrow downwards. This creates an arrow-like shape that points to the call-to-action placed underneath.




**Source:** Frischen A, Bayliss AP, Tipper SP (2007) Gaze cueing of attention: visual attention, social cognition, and individual differences. Psychological Bulletin, 133(4):694–724.

**See also:** Facial Distraction, Salience.

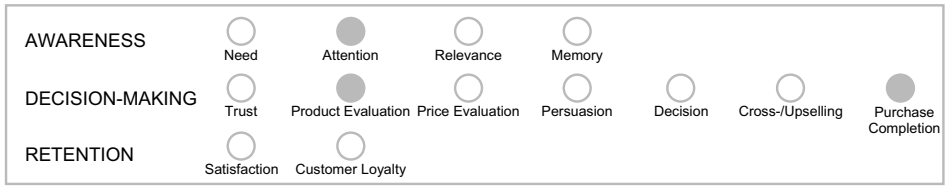
**My Notes**

### 4.2.5 Inattentional Blindness Effect

Also: Change Blindness Effect

People do not perceive unexpected events when engaged in an elementary observation task. This is true even if these events are long-lasting, conspicuous, and placed right in the visual center of our attention. The extent of inattentional blindness depends on the difficulty of the observation task: the more cognitive resources are required, the more "holey" the perception of the surrounding context becomes.

The Inattentional Blindness Effect was proven by the selective attention test of Simons and Chabris, which later became famous as the "Monkey Business Illusion" (Simons & Chabris, 1999). If you do not know this test yet, be sure to give it a try: youtu.be/IGQmdoK_ZfY.




**Implementation in e-commerce**

- If something happens in the page's background, this activity should be visualized visually (e.g., with a waiting spinner for processes with a processing time of more than one second).
- If changes take place in the interface, they must be recognizable or prominently pointed out (e.g., with an input effect, a disturber, or a highlight). This is currently well solved in many shops in the context of ad-hoc stock reports and error communication in the ordering process.
- Another well-known application context is banner blindness. This refers to the phenomenon that areas that do not receive our cognitive attention (e.g., because we have learned that advertising without functional added value is usually shown there) are no longer considered. This is important for optimizing the online marketing mix, selecting placements, media buying, etc.
- You should check whether attention-grabbing elements in the shop take up a lot of cognitive resources or for which elements this applies. Suppose these are not desired (e.g., due to an application of the "disrupt-then-reframe technique"); in that case, these elements presumably represent barriers to attention, the removal of which has a positive effect on conversion.



**Source:** Simons DJ, Levin DT (1997) Change blindness. Trends in cognitive sciences 1(7):261–267.

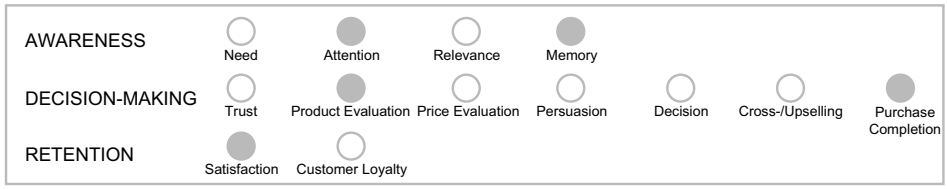
**See also:** External Reference, WYSIATI Effect, Gaze Cueing Effect, Disrupt-then-Reframe-Technique, Salience, Empathy Gap.

**My Notes**

### 4.2.6 Joy and Fun

Playful and entertaining elements make uncomfortable or unloved tasks more bearable and support the processing through intrinsic motivational incentives. Example: Volkswagen's "The Fun Theory" campaign, where annoying everyday activities such as climbing stairs became favorite thanks to entertaining elements. As part of the campaign, the car manufacturer installed pressure-sensitive surfaces on the steps, which emitted a sound on contact, and with these "piano stairs" managed to ensure that the escalator right next to it was hardly used anymore. The unpleasant option became the favored alternative using "Joy and Fun".

Along with "Perceived Usefulness" and "Perceived Ease of Use", "Perceived Enjoyment" is one of the central drivers of acceptance, also in e-commerce (Moon & Kim, 2001; Van der Heijden, 2004). The "Joy and Fun" pattern is aimed at "Perceived Enjoyment".




**Implementation in e-commerce**

- A good approach is micro-interactions to make everyday and necessary actions on the website more attractive. Explicit interactions, such as verbal feedback on the user's actions (such as overlays of dialogue speech bubbles), or implicit interactions, such as user-leading anchors, zoom effects, or fade-outs, can be considered. These micro-interactions are often used, for example, when adding products to the shopping cart, reviewing products, or initiating contact.
- The pattern can also be deeply rooted in content marketing and shape the tone of dialogue across the page. Entertaining or surprising texts, images, and videos are just a few examples.
- Ultimately, the pattern is also suitable for avoiding abandonments, for example, if the site has a humorous approach to the users' negative experiences. Examples are links to pages that no longer exist, out-of-stock articles, or technical interruptions in the ordering process.
- In all cases, it is essential that the entertaining elements do not distract from the user's conversion goal but support it. UX designer Ian Armstrong describes this as "creating multiple relevant experience funnels for people to explore, and ruthlessly eliminating distractions in the funnel itself", with the critical addition that "the goal-oriented shopper still needs their shortcuts" (Armstrong, 2015). In addition, offensive jokes should be avoided at all costs, and the congruence with your brand presence should always be checked.




**Source:** Liu C, Arnett KP (2000) Exploring the factors associated with Web site success in the context of electronic commerce. Information & Management 38(1):23–33.

**See also:** Aesthetics Heuristic, Availability Heuristic, Cognitive Ease, Curse of Knowledge.

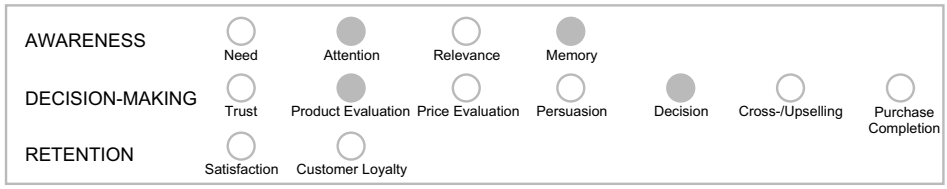
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### 4.2.7 Picture Superiority Effect

The Picture Superiority Effect is based on the realization that humans have two separate systems for representing verbal (text) and non-verbal information (image). Learning – and nothing else is a search for information in an online shop – works particularly well when information is visually and verbally encoded ("multimedia principle"). The reason for this is that although both codes work independently of each other, they have an additive effect on memory.

In concrete terms, adding the two encodings means that if you read a text without images, you will remember about 10% of the content the next day. If the exact text is reinforced with images, the proportion of remembered content increases to an astonishing 65% after three days (Versteege, 2017).

The effect can also be seen in the speed of perception: Images are processed much faster in the brain than texts (approx. 0.01 s are sufficient for a visual scene), which can be a decisive advantage in highly competitive e-commerce with steadily decreasing user attention spans. However, this also means outstanding care should be taken when selecting images. If an image does not support the text optimally, the superior perception quickly leads to cannibalization of verbal attention and thus to misinterpretations on the part of the user. Image selection is, therefore, not a task for a temporary worker with access to a stock photo platform but an essential part of editorial work or content marketing.




**Implementation in e-commerce**

- In the early phase of the customer journey, providers must stand out to get into the relevant set of users, e.g., with impressive or controversial images. In later (i.e., more decision-making) phases, on the other hand, images should support the message or even speak for themselves. Using generic stock photo material usually offers little benefit, but vivid visualizations do.
- Content that requires explanation (e.g., a pension gap related to the sale of retirement savings policies or the construction of a high-performance membrane in outdoor clothing) should always be presented in a combined text-image presentation.
- In many places, images could be used sensibly, but many websites and shops still use non-visual interaction elements in their place. A typical example is category drop-down menus, which can usually be replaced 1:1 by an illustrated or illustrated category overview. The better the product images are, the more likely a grid should be chosen for display; in the case of weak images, on the other hand, a listing usually proves its worth.

- At least three high-resolution product photos should be used for product presentation, preferably supported by interactive 360°representations or videos.
- Representations that show customers using the product are particularly effective. This is also known as the "Visual Depiction Effect" and describes the increased willingness to engage with a product when visualizing the future usage scenario.




**Source:** Paivio A (1990) Mental representations: A dual coding approach. Oxford University Press, New York.

**See also:** Availability Heuristic, Identifiable Victim Effect, Von Restorff Effect.

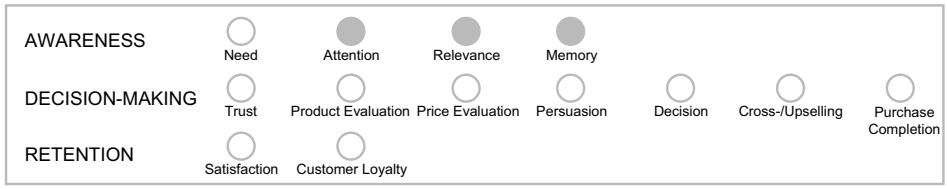
**My Notes**

### 4.2.8 Salience

Salience means that a visual stimulus stands out clearly from its surroundings. As a result, more perceptual resources are focused on this stimulus, which increases the likelihood that we will perceive this stimulus. A frequently cited example of an object with high salience is the yellow rubber duck in a white bathtub. Brake lights at night also illustrate the behavior pattern well.

The salience of an object can be calculated precisely: In the salience model of Itti and colleagues (Ishizu & Zeki, 2011), the deviation of the object from its environment in terms of color, luminance, and orientation is measured. The individual deviations are added up, creating a map with salience values. The model has been expanded several times in recent years, and visual movement (e.g., animations) is now counted among the dimensions. So, salient objects guide our gaze and captivate our attention – independent of our intention or conscious action.

The stimulus-driven focus of attention can be increased if it matches a person's current needs or motivation. Example: If you consider buying a Mini Cooper, you will suddenly notice such cars everywhere. The attention effect is maximum if the stimulus is unknown or occurs for the first time.




**Implementation in e-commerce**

- Obviously, Salience is particularly relevant in the early phase of the customer journey. At the beginning of almost all buying processes is the attention to a product, a service, or a company. All classic online marketing channels are struggling with limited user attention resources. Advertising media that contain objects with high salience are perceived particularly well and record significantly higher click-through rates.
- Salience aspects also come into play in the design of buttons, e.g., for the primary action of the page, the contrast should be as great as possible to the environment of the button on the one hand and the text within the button on the other. In the case of secondary actions, it often makes sense to reduce this inner and outer contrast to achieve a clear decision-making hierarchy. For this purpose, "ghost buttons" are usually used with a transparent fill – i.e., no external contrast.
- Instead of visually strengthening individual elements, the Salience effect can be achieved by visually attenuating less essential environmental elements. As a result, more attention is paid to the more salient elements. From the user's perspective, this is sometimes the

more sensible option because it reduces "optical noise" and the application is perceived as calmer and less blatant overall.




**Source:** Itti L, Koch C, Niebur E (1998) A model of saliency-based visual attention for rapid scene analysis. IEEE Transactions on Pattern Analysis & Machine Intelligence 20(11): 1254–1259.

**See also:** Focusing Effect, Inattentional Blindness Effect, WYSIATI Effect, Gaze Cueing Effect.

**My Notes**

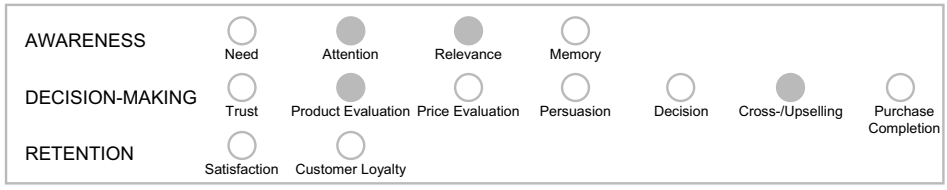
## 4.3 Relevance

### 4.3.1 Barnum Effect

Also: Forer Effect, Personal Validation Fallacy

The Barnum Effect describes the tendency of people to interpret vague and universal statements about themselves as accurate descriptions and to assume that they have been understood very precisely. The effect explains, for example, why horoscopes, future predictions, and personality tests are so popular and widespread.

The pattern is effective because the statements are so general and, therefore, impossible to refute. Similar to the base rate phenomenon (see the section on the Linda problem), the joy of the supposedly accurate statement dominates the criticism of arbitrariness. General characteristics or universal desires (for example, security or wealth) are often used as a basis. In addition to the appropriate language (a lot of subjunctives and relativizations, personal salutation), the framing of the statements is also important: If you present people with a personality description with the note that it is taken from the horoscope of a woman's magazine, it is (unsurprisingly) not evaluated nearly as accurately as the same description, which is supposedly the result of a complex psychological test (Forer 1949).




**Implementation in e-commerce**

- Where segmentation does not allow for a more precise needs-oriented approach, it is best to work with vague, flattering statements ("You work hard for your money.", "You deserve it.") or to address general states of mind (e.g., "Would you also like to look more fashionable sometimes?"). In this way, you can give users the feeling of speaking to them 1:1, even though you are addressing millions with a broad media campaign.
- A very interactive way to implement the Barnum Effect is quizzes and tests in social media ("Take the test: Which ball sport suits you?"). Experience has shown that such mechanisms generate much interaction and are well-suited for lead generation.
- The cross-selling area can also be charged accordingly: "Especially for your needs, we recommend you…"



**Source:** Meehl PE (1956) Wanted–a good cook-book. American Psychologist 11(6):263–272.

**See also:** Framing, Overconfidence, Base Rate Fallacy.

**My Notes**

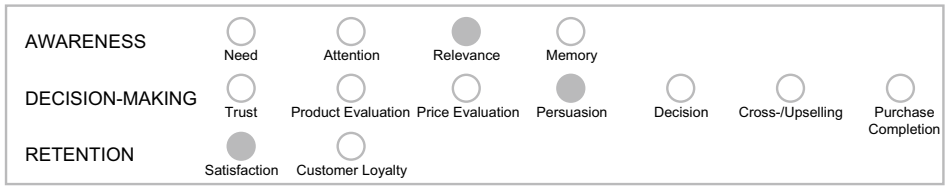
### 4.3.2 Empathy Gap

Also: Hot–cold Empathy Gap

The Empathy Gap is one of George Loewenstein's most significant contributions to the development of behavioral economics. He and his colleagues have recognized that people massively underestimate the influence of their visceral urges on their decision-making behavior. Visceral urges are understood as influences such as hunger, thirst, sexual arousal, an addict's craving for drugs, physical pain, and intense emotions. If we act under the influence of such drives, we are in a "hot" state and (irrationally) do everything we can to appease the influences. As a result, we act almost exclusively intuitively and impulsively based on behavior patterns – but this is usually not perceived by humans. On the contrary, we think that satisfying short-term urges aligns with our long-term goals.

At its core, the empathy gap describes that our empathy is state-dependent: When we are angry, it is difficult for us to understand at that moment what it is like not to be angry. It is hard to imagine what it is like to be in love when we are not in love. This noticeable empathy gap – which we are certainly all familiar with – is challenging to minimize.

In practice, the empathy gap often appears in connection with or in the form of the False Consensus Effect: The simple assumption that all other people agree with us serves as a heuristic for not having to deal with the needs of others. This avoids the high cognitive effort required to empathize empathetically and saves cognitive resources.




**Implementation in e-commerce**

- The empathy gap is probably partly responsible for the fact that debates and discussions on the Internet – especially in social media – are dominated by an often harsh tone. In the case of a written point of view, we lack the indicators for the context of the contribution, which drives the necessary empathetic effort for an appropriate response to almost unattainable heights. This can be counteracted with a code of communication, moderated discussions, and continuous reminders to consider others' points of view.
- A hot–cold empathy gap also exists when we receive negative customer reviews and feedback. Customers hand them over in highly emotional moments, i.e., they are viscerally driven ("hot"). This often leads to exaggerations and exaggerations. As recipients (in a "cold" state), we need to have the necessary empathy to respond appropriately. This is particularly challenging in telephone disputes because of the real-time character.
- When content is created, we must anticipate precisely the emotional state in which the content will be consumed: Are customers insecure? Have you had an experience of loss?

Are they euphoric? This strongly impacts which activation patterns work and at what speed ("pacing") users can be led to the conversion goal.

- The content itself can also activate visceral urges while reading: depictions of bare skin, appealing videos of steaming food, images of people suffering from pain that resemble us – all this leads to a "hot" state of users. However, it should not be trivially deduced from this that higher emotional activation leads to a better response to behavioral economic triggers. The context of the product must be given so that the termination point is not merely shifted to the back of the decision-making process.




**Source:** Van Boven L, Loewenstein G, Dunning D, Nordgren LF (2013) Changing places: A dual judgment model of empathy gaps in emotional perspective taking. Advances in experimental social psychology 48:117–171.

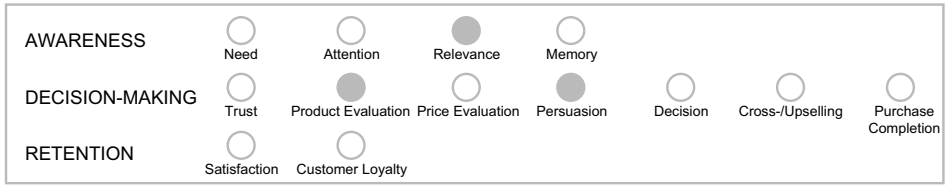
**See also:** Availability Heuristic, Context-Dependent Memory, Inattentional Blindness Effect, Mere Exposure Effect, False Consensus.

**My Notes**

### 4.3.3 Focusing Effect

We can only focus on a limited number of things. The Focusing Effect, therefore, describes that the human brain attaches a disproportionately high weight to a few aspects of the decision-making process. As a result, mainstream positions or sensationalized units of information are usually given preferential treatment, while other (possibly more important) factors are neglected in decision-making.

In a high-profile study, Schkade and Kahneman (1998) asked their subjects whether they believed people in California or the Midwest lived happier lives. The result was as clear as expected: the vast majority considered Californians to be the happier people. In their assessment, the test subjects focused on depictions often shown in the media (beach, palm trees, and laughing successful people in the sunset). The Midwest is not known to have such clichés to focus on. However, there is no difference between the people's happiness in both regions. Therefore, The assessment is distorted by focusing on a few plausible factors. It ignores aspects that would provoke a conflict with the coherent picture (in the example of California, for example, crime, natural disasters, hard work in agriculture, dependence on the weather).




**Implementation in e-commerce**

- Customers are looking for products that make their lives better in some way. Due to the Focusing Effect and the Belief Bias (i.e., people want to believe), products are evaluated based on the prominently highlighted properties. For digital marketing strategies, this leads to the recommendation to focus on a few of the most widely recognized and differentiating factors across channels. Empirically, three to four factors or USPs have been proven.
- In addition, a practical application of the effect is to describe the significant change in people's lives that they can expect after purchasing the product or digital service.
- Due to the Focusing Effect, the importance of money for personal well-being is also systematically overestimated. It is, therefore, an excellent idea to provide one of the focused marketing aspects with a financial aspect (e.g., "Save much money with this product.").
- In terms of design, white space can support the effect. Empty spaces can be an effective persuasion tool and are certainly one of the most underrated approaches to conversion optimization. They guide users through the page and reinforce key messages more effectively than typographic highlighting.




**Sources:** Schkade DA, Kahneman D (1998) Does living in California make people happy? A focusing illusion in judgments of life satisfaction. Psychological Science 9(5):340–346; Cherubini P, Mazzocco K, Rumiati R (2003) Rethinking the focusing effect in decision-making. Acta Psychologica 113(1):67–81.

**See also:** Representativeness Heuristics, Belief Bias, Halo Effect, Salience, Ockham's Razor, Zero Risk Bias.

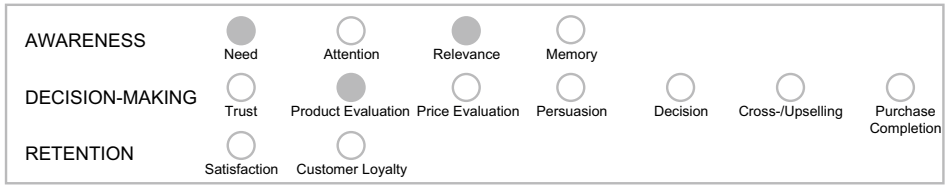
**My Notes**

### 4.3.4 Functional Fixation

Functional Fixation restricts people to using an object only as it is traditionally used. This insight comes from Gestalt psychology, which emphasizes the holistic cognitive processing of stimuli. Accordingly, we build a mental block against the novel use of an object required to solve a problem. Surprisingly, this distortion is not innate; it only manifests itself at the age of about seven years. Until then, children show no signs of Functional Fixation.

Someone looking for a paperweight but only has a hammer forgets that the hammer is not only suitable for hammering nails into the wall. The Functional Fixation thus prevents the available resources from effectively finding solutions. In other words, MacGuyver would never have escaped prison with a garter, chewing gum, and paper clip if he had only used these objects for traditional use.

The most famous experiment on Functional Fixation was conducted by Duncker (1945): he gave participants in his study a candle, a box of thumbtacks, and a matchbook. He asked them to attach the candle to the wall so that it did not drip onto the table. Most people tried to attach the cores to the wall with the thumbtacks or by melting them. Only a few considered pinning the empty box to the wall and using it as a candlestick. Apparently, they were too fixated on the traditional way of using a box as a container.




**Implementation in e-commerce**

- Functional Fixation can be used as a fundamental law of usability: All visual objects are associated with a specific type of use: You click on buttons, you swipe an image gallery from right to left. When designing elements, it is necessary to pay strict attention to the fact that the standard type of interaction corresponds to the behavioral goals of the designer. Since users have a Functional Fixation, they will not click on buttons that do not correspond to the learned design.
- The principle can be used by memorable staging (strictly speaking, often superfluous) objects in which no traditional type of use is anchored. For example, the best way to pit a pomegranate is to hit the peel from the outside with a blunt object - anything from a wooden spoon to a rolling pin. However, people often do not recognize this alternative use of objects present in every household due to the functional fixation, so they can feel that they urgently need a hand-flattering clapper made of walnut wood with brass applications for 200 dollars.




**Source:** Duncker K (1945) On problem solving. Psychological Monographs 58(5): full issue.

**See also:** WYSIATI Effect, Diderot Effect, Commitment and Consistency.

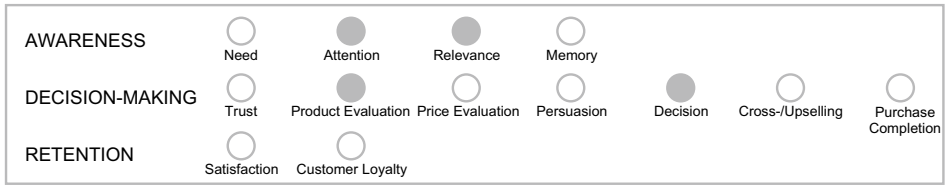
**My Notes**

### 4.3.5 Identifiable Victim Effect

The fate of individuals touches us much more than the fate of an anonymous group of people. The Identifiable Victim Effect works both ways: We feel a more vital impulse to support individuals in need, but we are also more willing to hold individual offenders accountable and punish them. Abstractly, the connection means that images and representations of individual people activate us more than mere statistics and graphs of larger groups.

The effect is probably the only example of Joseph Stalin's and Mother Teresa's agreement. The Soviet dictator is associated with the phrase, "A single death is a tragedy; a million deaths are a statistic." The Nobel Peace Prize laureate once said, "If I looked at the masses, I would never act."

Psychologist Keith Payne summarizes humanitarian catastrophes: "We feel the least when needed most." He thus points out the connection that our emotional sympathy paradoxically decreases with the increasing number of victims. This paradox is explained by the fact that we have a self-protection mechanism that is supposed to prevent us from being overwhelmed by our feelings. So, we distance ourselves emotionally and reduce the sympathy as the number of victims increases.




**Implementation in e-commerce**

- The effect is often used at Christmas time by charitable organizations that want to raise funds (e.g., for starving children in Africa) or win supporters for the common cause (fundraising).
- Statistics can be very practical in the right place (see: Pseudo Justification) but fail to have an effect elsewhere. Regarding emotionalizing users and shifting decision control in favor of System 1, images of individuals are far more effective.
- The impact of testimonials is based on the same effect. Individual fates usually activate us more strongly (e.g., when taking out insurance) than abstract hazard potentials and accident statistics.



**Source:** Jenni K, Loewenstein G (1997) Explaining the identifiable victim effect. Journal of Risk and Uncertainty 14(3):235–257.

**See also:** Picture Superiority Effect, Bystander Effect, Commitment, Pseudo Justification.

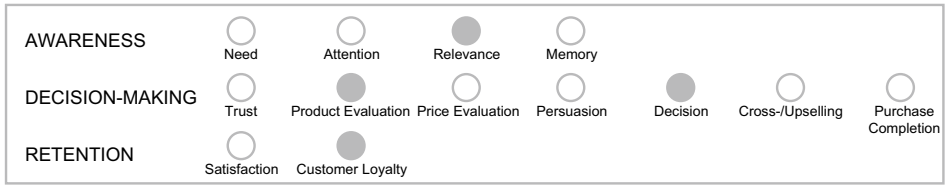
**My Notes**

### 4.3.6 Status Quo Bias

Also: Omission Bias, Default Effect

People tend to want to preserve the current state. This is regularly evident in politics: in political elections, for example, the incumbents usually have a measurable office bonus, and there are regular protests against reforms. As the saying goes: "Man is a creature of habit". Habits and routines are reluctant to be abandoned, they create security, scale search efforts, and reduce complexity. In this respect, the Status Quo Bias is not per se a distortion in the negative sense but can also be understood as an efficient heuristic.

It is sometimes explained by the endowment effect, sometimes by the loss aversion of the prospect theory (see: Loss Aversion). Just as often, the bias is also considered an independent effect, which is explained by the attachment to a decision once made: If one were to revise a decision, one would have to painfully admit a mistake, which people strongly try to avoid (regret aversion). The longer the status quo exists, the stronger the Status Quo Bias usually is. It is concluded that the current state must be perfect if it has lasted so long – we assume evolutionary superiority. As a result, the effect continuously reinforces itself.




**Implementation in e-commerce**

- Shop operators can promote customer loyalty by emphasizing the convenience of the current state of affairs with loyal existing customers. People tend to be loyal and buy the same brands or shop in the same stores. This is particularly effective when combined with functional lock-in effects (e.g., a delivery cost flat rate or a bonus program).
- In the case of regular purchases (e.g., consumer goods), customers are relatively insensitive to price fluctuations. Discounts are usually not worthwhile here. On the other hand, experimental price increases can be used to determine how the consumer surplus can be optimally skimmed off.
- Another implementation approach would be to offer free trials of products. Once you get used to it, you will be reluctant to return the product, especially if it involves increased handling effort (e.g., returning a couch or mattress).
- The desired behavior can be triggered so that no action is required on the user's part: For example, when choosing a product configuration, you can preselect the bestseller as "default selected".




**Source:** Samuelson W, Zeckhauser R (1988) Status quo bias in decision making. Journal of Risk and Uncertainty 1(1):7–59.

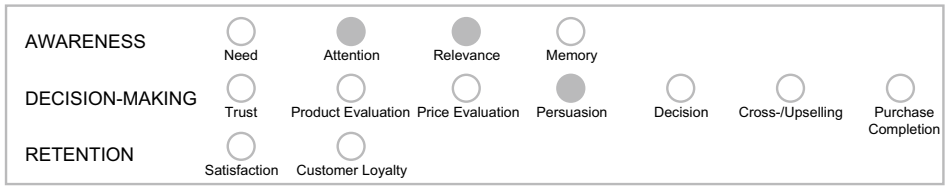
**See also:** Action Bias, Endowment Effect, Confirmation Bias, Loss Aversion, Anchoring.

**My Notes**

### 4.3.7 Threat

Fear is one of the emotions with the highest activation potential for evolutionary reasons. We are programmed to act quickly out of a stressful or anxious situation, which could make the difference between life and death in primeval times. This shifts the decision-making power to System 1. Products that appear in this context can be well staged as a friend or a savior. Coupled with the high willingness to act, this is what makes the threat pattern so effective.

However, this correlation only applies to fears we consider real and specific. Kahneman (2011) cites climate change as an example, which has little activation potential for most people because we neither know for sure when and to what extent it will affect us nor do we understand the immediate consequences for our lives. If one considers the relationships between the pattern "time inconsistency" (see ► Sect. 4.2.4.19), after we are geared towards immediate rewards and hyperbolically discounting, it quickly becomes clear why such long-term threat scenarios irrationally seem to have no effect on our lives.




**Implementation in e-commerce**

- Insurance companies and manufacturers of safety-oriented products (e.g., climbing equipment, baby care, pharmaceuticals, cars, etc.) often work subtly with fear-oriented representations of damage risk to increase the perceived probability of a claim.
- Needless to say: The law against unfair competition prohibits fear advertising if it influences the purchase decision in an unobjective way and exploits the resulting feeling of helplessness (§ 3 and § 4, number 2 UWG). This boundary is inviolable and must never be affected by the threat pattern.




**Source:** Tversky A, Kahneman D (1983) Extensional versus intuitive reasoning: The conjunction fallacy in probability judgment. Psychological Review 90(4):293–315.

**See also:** Affect heuristics, time inconsistency, hyperbolic discounting.

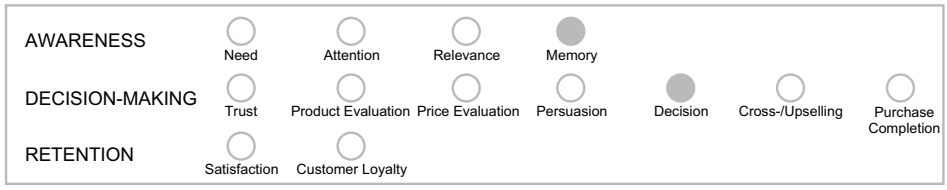
**My Notes**

## 4.4 Memory

### 4.4.1 Context-Dependent Memory

People tend to forget things outside the environment's current context. Crucial for this are cues that link memories to a specific context. In everyday life, we encounter "Context-Dependent Memory" repeatedly: You go from the living room to the kitchen, and – once you arrive – you do not know what you want there. Back in the living room, the memory comes back immediately. Memories are, therefore, often tied to an environment with "contextual cues", which can also be defined locally (in this case, the living room).

This is another explanation from everyday life for all those who have often treated themselves to a glass of alcohol while studying during their studies: Presumably, you then wrote the exam soberly. What sounds reasonable does not necessarily have to be conducive to the result. Studies on state-dependent memory have shown that physical and emotional status is also a contextual cue. Surprisingly, those who have learned drunk also deliver better results when drunk compared to someone who has learned drunk and has been tested sober (Goodwin et al., 1969).




**Implementation in e-commerce**

- Features that define a context should be kept consistent across all channels. This is not only a question of branding and corporate communication but also a question of conversion, as the pattern shows.
- A concrete example: In retargeting, users interested in a product usually find themselves in a completely different context when the ads with the corresponding product advertisement are played out to them. To ensure that the remembering context is not lost, it makes sense to reuse as many "contextual cues" as possible, e.g., the icon, the claim, the color scheme, and the testimonials. All this increases the recall of the brand and product and promotes the effect of retargeting.
- For returning users, cues from their last session, e.g., the last product viewed, can be shown.



**Source:** Godden D, Baddeley A (1975) Context-dependent memory in two natural environments. British Journal of Psychology 66(3):325–331.

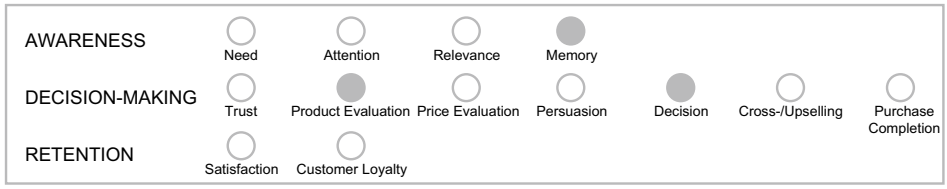
**See also:** WYSIATI Effect, Zeigarnik Effect, Recency Effect, Empathy Gap.

**My Notes**

### 4.4.2 Primacy Effect

People form an opinion early in decision-making, which is difficult to change later. Colloquially speaking: "The first impression remains" or "There is no second chance for the first impression". This bias can be explained by Confirmation Bias: we are reluctant to change our minds and instead prefer to look for affirmative arguments. As a result, for example the first contact with a dealer is so important as the basis for forming an opinion. In addition, there is an explanatory approach from memory research: It is apparently easier for information to find its way into long-term memory if no other information has yet been stored on this topic or if there are no disturbing interferences. In this way, the first day of vacation is usually remembered more strongly than the following days.

Symbolically, a golf analogy is well suited: The first shot determines the direction and proximity to the target, so it should be the strongest. Translated, the most potent argument should be put forward first.




**Implementation in e-commerce**

- Optimization of the homepage (for users who enter the provider's ecosystem via direct type-in) or landing pages (for campaign traffic) with regard to positioning: If you are perceived as an award-worthy provider (e.g., due to price-oriented campaign routes), not so many positioning arguments are needed later in the purchase process, as the primacy effect, the primary perception is already predisposed and users later ignore contradictory arguments or give them less weight.
- In print advertising (highly relevant for many e-commerce companies), the effect shows that a magazine's first and last pages are particularly effective for ad placement.
- Lists of benefits and USPs should start with the strongest argument. However, this does not mean that the arguments should be sorted in descending order of strength because the last point of a list is also remembered intensively (see: Recency Effect).




**Source:** Asch SE (1946) Forming impressions of personality. The Journal of Abnormal and Social Psychology 41(3):258–290.

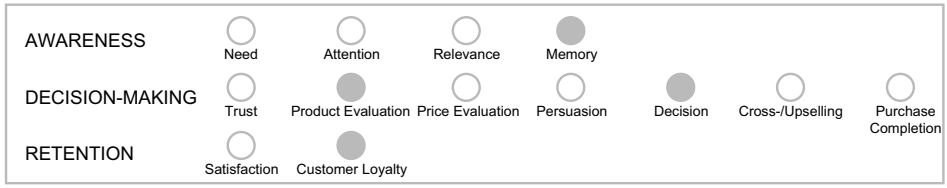
**See also:** Confirmation Bias, Halo Effect, Recency Effect, Peak-End Rule.

**My Notes**

### 4.4.3 Recency Effect

The counter-effect to the primacy effect or.dem overarching confirmation bias: Not the first, but the last experience you have is better remembered and dominates the formation of opinion and decision-making. The recency effect occurs when information is stored in short-term memory, while the primacy effect is more aimed at long-term memory. The explanation for the recency effect is that the last information is not overwritten and, thus, better memorized. In addition, we always assume a chronological order for information received. Accordingly, the latest information would be the most up-to-date and, therefore, the most relevant (novelty effect). How strong the effect is can be seen from the simple attribute "new" in product launches: Even though many iPhone generations have produced few genuine innovations, the sheer novelty has made customers queue up in droves.

The primacy and recency effect are strongly interdependent, they are often combined into one effect. The probability of memory follows a U-curve, i.e., the first and last pieces of information are most likely remembered. The fact that there does not have to be a contradiction in this can be expressed with the mnemonic sentence: "The first impression counts, the last impression remains."




**Implementation in e-commerce**

- Wow effects in fulfillment and after-sales create a positive event at the end of the ordering process, positively impacting the evaluation of the entire buying experience.
- The Recency Effect provides a positive conclusion by congratulating and summarizing all the benefits customers have secured with the purchase (see: Post-Purchase Rationalization).
- In case of delays in fulfillment, a negative recency effect can be counteracted with a voucher (for the next order) or a small gift as an excuse. This turns critics into enthusiastic promoters (NPS) and creates the basis for a follow-up purchase.




**Source:** Atkinson RC, Shiffrin RM (1968) Human memory: A proposed system and its control processes. Psychology of Learning and Motivation 2:189–195.

**See also:** Halo Effect, Primacy Effect, Post-Purchase Rationalization, Peak-End Rule, Neomanie.

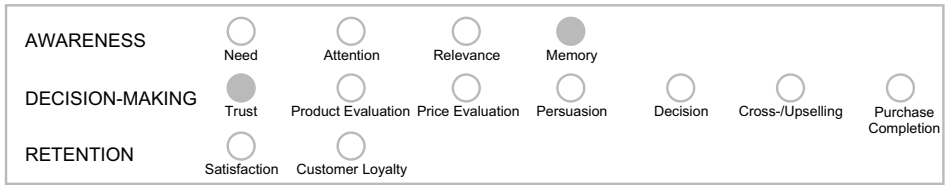
**My Notes**

### 4.4.4 Rhyme-as-Reason Effect

Also: Eaton Roses Phenomenon

Rhyming statements and aphorisms are perceived as more truthful than inconsistent statements. There are two possible explanations for the Rhyme-as-Reason effect. First, rhyming statements have a more aesthetic effect; Aesthetics, on the other hand, is used as a heuristic for assessing credibility (Keats heuristics). Secondly, rhymes are processed faster in the brain, which gives the statement a higher value and is, therefore, preferentially memorized.

Well-known examples and partly real classics are Pringles ("Once You Pop, You Can't Stop") or "Jaguar ("Grace. Space. Pace.").




**Implementation in e-commerce**

- Enhance language with rhymes wherever possible (e.g., claims, banner ads, product descriptions, headlines, etc.). Attention: Applying the pattern must continue to fit the brand identity and not lead to forced snarling.



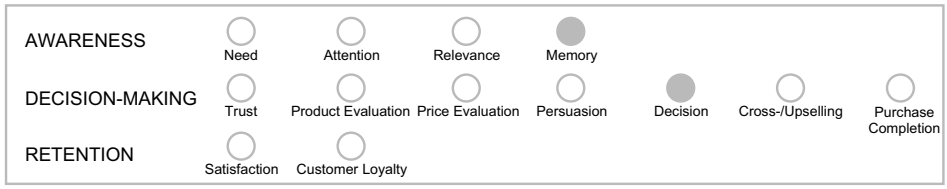

**Sources:** McGlone MS, Tofighbakhsh J (2000) Birds of a feather flock conjointly (?): Rhyme as reason in aphorisms. Psychological Science 11(5):424–428; Filkuková P, Klempe SH (2013) Rhyme as reason in commercial and social advertising. Scandinavian journal of psychology 54(5):423–431.

**See also:** Aesthetics Heuristic, Story Bias.

**My Notes**

### 4.4.5 Serial Position Effect

Memory (and thus the number of decision alternatives considered) depends significantly on the order of all options presented. These are most likely to be remembered if they are at the beginning (Primacy Effect: all cognitive resources for memorization are still available) or at the end (Recency Effect: no displacement from short-term memory by subsequent information units) of a list. Options in the middle, on the other hand, are in a difficult sandwich position that is not supported by any cognitive mechanism. Beyond the extreme positions, memory decreases in lists from the top or bottom towards the middle (see: Extremeness Aversion). The second option has an even stronger perception than the penultimate one, and so on.




**Implementation in e-commerce**

- The Serial Position Effect is used, for example, in online advertising, where it can be used to determine and purchase the best (i.e., most attention-grabbing) placements for advertisements.
- Knowledge is also helpful for the structure of content pages: content at the top and bottom of the page is best remembered. Therefore, the essential messages of the page (often presented as a summary or repetition at the end) and the primary call-to-action should be placed here. The same applies to content production in newsletter marketing.
- When listing unique selling points (USPs) or advantages of products or the provider, the most convincing arguments should be mentioned first or last.
- The same applies to the structure of the primary navigation: On the left and right are the top positions where the product categories with the highest click, turnover, or margin (depending on the strategic orientation) should be positioned.




**Source:** Murdock Jr BB (1962) The serial position effect of free recall. Journal of Experimental Psychology 64(5):482–488.

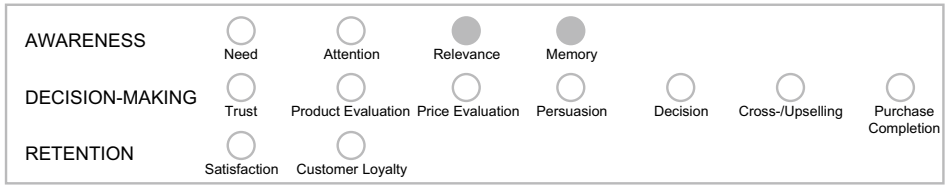
**See also:** Recency Effect, Primacy Effect, Context-Dependent Memory.

**My Notes**

### 4.4.6 Story Bias

Facts presented in the form of stories catch on better with the reader. It is easier to remember connections and details of a narrative, and metaphors often used in this context reinforce the effect. It is not without reason that memory artists work with exactly these methods. In the case of processing and memorization, however, it is completely ignored whether the connection between the partial events is factually logically justified or artificially constructed. The author Rolf Dobelli (2011) cites as an example the description of the imminent death of two monarchs: "The king died, and then the queen died of grief." A sequential sequence of two unconnected events becomes a story that feels right and meaningful. In other words, stories frame facts with meaning.

We are attracted to well-told stories but bored by sober facts. From a social point of view, this carries the risk of an emotional artificialization of factual content, to which many media companies have succumbed for quite some time. From a marketing perspective, however, Story Bias is a sharp weapon in the battle for people's scarce attention resources. The most expensive commercials on the planet have perfected this: Every year, the who-is-who of the advertising industry meets at the Superbowl and presents the most emotional creative outpourings in the marketing world. What better proof of the power of Story Bias could there be than the fact that leading companies fill their 30 s, which cost up to five million US dollars, predominantly with emotional stories (Kantar Media 2017)?




**Implementation in e-commerce**

- Regardless of whether they are entirely logical or fragmented, users are more likely to engage with product descriptions in narrative form. Working with personas, which is common in most companies, is already a suitable parenthesis, and the storytelling repertoire from content marketing provides even more impulses here.




**Source:** Fico F, Richardson JD, Edwards SM (2004) Influence of story structure on perceived story bias and news organization credibility. Mass Communication & Society 7(3):301–318.

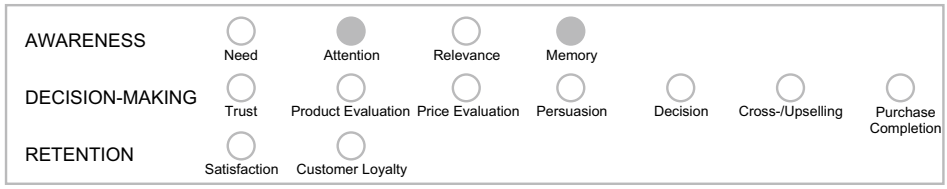
**See also:** Rhyme-as-Reason Effect, Joy and Fun.

**My Notes**

### 4.4.7 Von Restorff Effect

Also: Bizarreness Effect, Perceptual Incongruence, Distinction Effect

Surprising content is much more entrenched than uniformly designed content. The surprising moment can be crazy, funny, bizarre, or just different from the context of the environment and refer to texts, images, products, or communication messages. A single word written in red in the middle of black text is, therefore, more noticeable and better memorized. The Von Restorff Effect was first demonstrated by the German psychologist Hedwig von Restorff. It showed that our eyes and brain are constantly looking for elements that break with the expected value (which is why the effect is also called the "distinction effect").




**Implementation in e-commerce**

- The use of the effect is versatile. Good application scenarios are the language used (use of unusual terms and typefaces), call-to-action designation, and the physical product experience (e.g., with unusual packaging).
- The Von Restorff Effect can also be elevated to a communicative leitmotif. For example, the body care manufacturer Old Spice has provided a collection of all conceivable applications of the effect in its commercials for years.
- The effect can also be interesting for the placement of ads. A few years ago, Dahlén et al. (2008) investigated how the congruence between an advertisement and the platform in which the advertisement was embedded affects the recall of the ad. Unsurprisingly, low congruence (i.e., ad and platform do not match) increases the likelihood of the ad being noticed.
- The bounce rate (i.e., the proportion of users who leave the website immediately) can be reduced if the page performs an unexpected action directly after loading.
- A small English car leasing company has implemented the effect in its purest form. Once you have visited Lingscars.com, you will never forget the website – good proof of the impact of the effect. By the way: Contrary to the first intuitive assumption, the company is financially very successful; Newsweek calls the site "one of the best websites in the internet" (Veix, 2016).



**Source:** Von Restorff H (1933) On the effect of area formations in the field of traces. Psychology Research 18:299–334.

**See also:** Inattentional Blindness Effect.

**My Notes**

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# Behavior Patterns for the Decision-Making Phase

# 5


## Summary

The "decision-making phase" is not surprisingly at the heart of the whole decision process (Fig. 5.1 and Fig. 5.2). Consequently, this is also where the most significant proportion of primary behavior patterns can be found. In total, 84 patterns have their central application context in one of the seven sections of this phase. **Trust** describes the first step in decision-making: a provider is evaluated as a reliable partner for the upcoming purchase. If this test is positive, the **product evaluation** follows**.** This is where a decision is made as to whether an offered product meets the functional and emotional requirements of the user. If this is the case, the **price evaluation** follows, in which the price charged is set in relation to the individually evaluated performance of the product, and a price-worthiness assessment is derived. If a user accepts the product and the price, it often requires more substantial external incentives to force the purchase decision actively. This takes place in the **persuasion** section. If, on the other hand, a user is to be influenced less explicitly, but instead, the choice is to be made easier, the behavior patterns from the area of **decision** come into question. Once all decision-inhibiting factors have been overcome, the challenge for shop operators is to increase the shopping cart volume with **cross-/up-selling**, whereby behavior patterns can also support. Ultimately, however, all this must not lead to the purchase process being interrupted or the **purchase completion** as a central goal being jeopardized. With the help of suitable behavior patterns from this area, it can be ensured that users can circumvent the typical termination points in the closing process (such as entering payment data or legal instructions).


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**Fig. 5.1** 1 Overview of Primary and Secondary Patterns for the Decision Making phase, Part 1

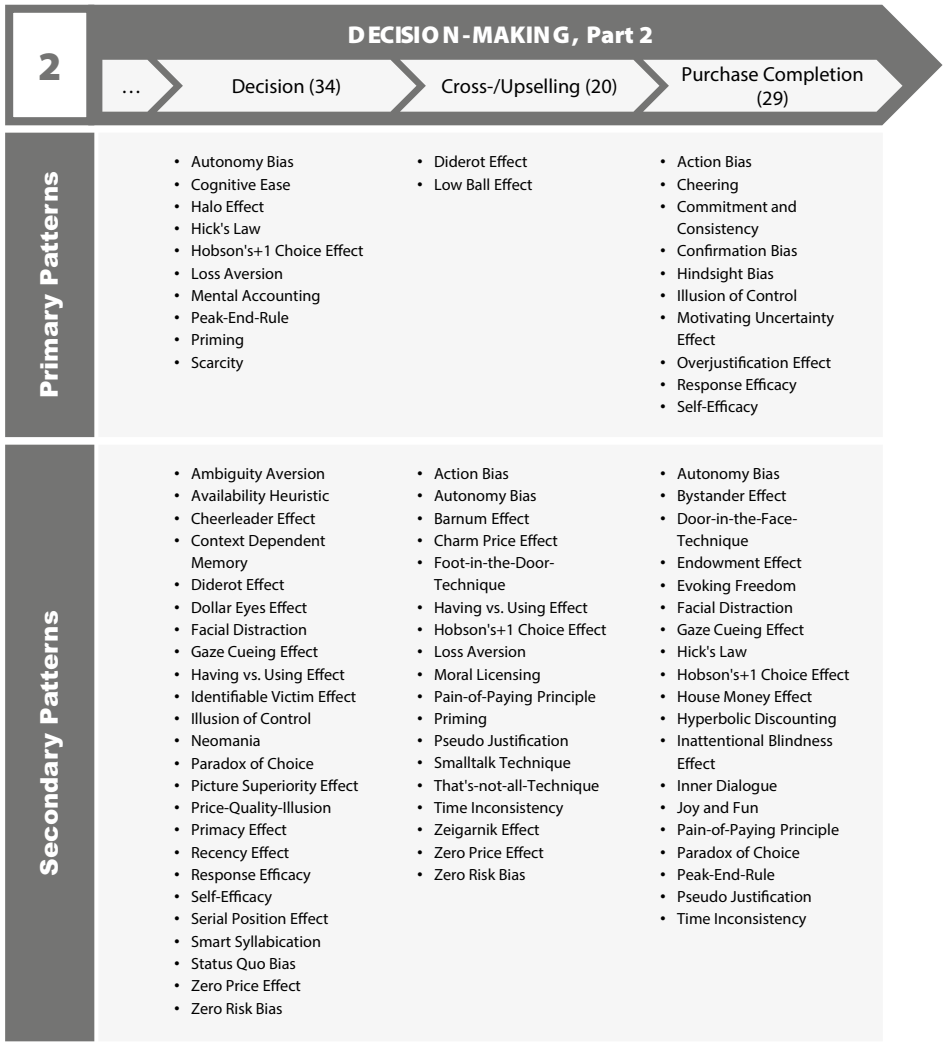



**Fig. 5.2** Overview of Primary and Secondary Patterns for the Decision Making phase, Part 2

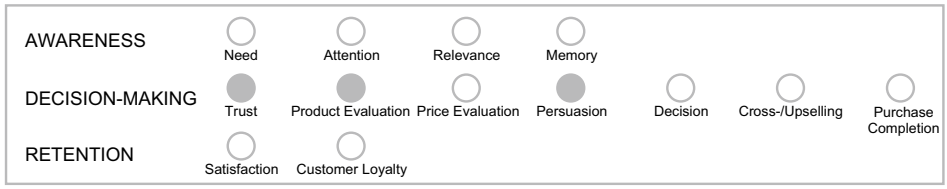
## 5.1 Trust

### 5.1.1 Affect Heuristic

Opinions and decisions are often based solely on emotional affection or aversion to the alternatives. The Affect Heuristic replaces the (difficult-to-answer) question "What do I think about this?" with the (easy-to-answer) question "What do I feel about it?" and then uses the answer to it as an answer to the original question. The processing mechanism is often trivial: a distinction is only made between good and bad feelings, directly leading to a positive or negative attitude towards the alternatives.

The Affect Heuristic leads (through processing in System 1) to the fact that probabilities are systematically misjudged: Probable events are underestimated, and improbable events are overestimated.

When people are later confronted with this connection, they often deny that their decision-making was emotionally biased and react by rationalizing the decision-making (emotional) arguments – another vivid example of the occasional lack of link between System 1 and System 2.




**Implementation in e-commerce**

- Users find it difficult to be convinced of opposing points of view with rational arguments. However, you can set emotional processes in motion and hand over control to System 1. This can be well illustrated by the example of insurance companies, where customer reports of dramatic damage (e.g., water damage, lightning strikes, fire) overestimate the probability of such events and increase the perceived necessity and willingness to pay.
- It is not for nothing that many commercials appeal to deeply stored memories from childhood, be it for Nintendo, Kellogg's, Hershey's, or Coca-Cola. Activating pleasant emotional sensations can be a far shorter way to a purchase decision than persuasion with good rational arguments.
- Interestingly, a low risk and a high benefit have the same effect in the Affect Heuristic. Instead of focusing on the benefits, emphasizing the low risks for some products (e.g., children's toys) may be a good idea.



**Source:** Finucane ML, Alhakami A, Slovic P, Johnson SM (2000) The affect heuristic in judgments of risks and benefits. Journal of Behavioral Decision Making 13(1):1–17.

**See also:** Belief Bias, Zero Price Effect.

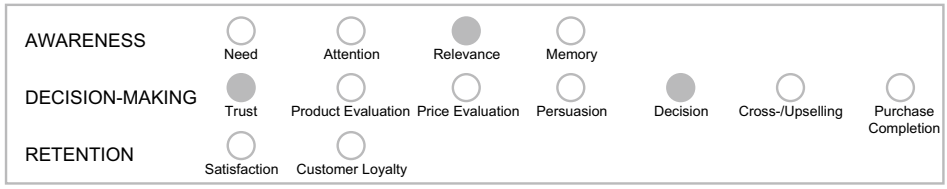
**My Notes**

### 5.1.2 Ambiguity Aversion

Also: Ellsberg Paradox

People tend to avoid decisions as soon as there is a need for more information (or ambiguity). As a result, in decision-making situations, it is often not the most advantageous alternative, but the one with the highest known probability is chosen because we try to avoid uncertainties. Thus, the pattern is similar to the loss aversion.

The effect was discovered in the following experiment: The test subjects were asked to pull balls out of a container and guess the ball's color. The balls are either red or green. Participants could choose between two containers: the first contains 50 red and 50 green balls. In the second, red and green balls are in an unknown ratio. Most people choose the first container, although there is no way of knowing whether the second container has a higher or lower probability of the selected color and thus has the same expected value as the first container. The first option provides a predictable result or more security.




**Implementation in e-commerce**

- Statistics, number-based statements, customer reviews, etc., must be backed up with supporting documents. Otherwise, ambiguity aversion takes effect in the event of ambiguity, leading visitors to avoid the decision.
- In the case of links, it must always be clear where they lead. This is often not the case when using a URL shortener because the link is not in an understandable structure (e.g., www.website.com/folder/page) but is shown in a cryptically truncated form (e.g., bit.ly/xyz). Links should, therefore, consistently be named descriptively and displayed on mouseover.
- Metaphors can be used to describe complex things and reduce uncertainty.
- When users choose between two providers, they are likelier to choose the more familiar one, as it offers more security. Therefore, an unknown website in a prominent place should introduce itself briefly at the first call and, if necessary, show the creators behind it (who, what, why, where).



**Source:** Ellsberg D (1961) Risk, ambiguity, and the Savage axioms. The Quarterly Journal of Economics 75(4):643–669.

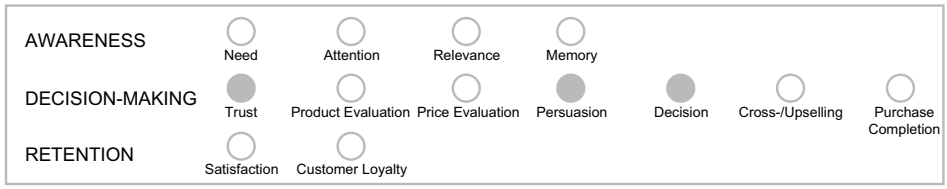
**See also:** Status Quo Bias, Loss Aversion, Action Bias, Zero Risk Bias.

**My Notes**

### 5.1.3 Authenticity

We are more open and positive towards people we perceive as genuine and authentic. The connection can be justified relatively trivially: We have good antennae for real and fake feelings of others and can (not always consciously) reliably recognize feigned affection, a disingenuous recommendation, or half-assertions. From an evolutionary point of view, authentic behavior over thousands of years brought with it the necessary reliability, sincerity, and predictability and represented a resilient basis for social structures.

In the digital environment, authenticity is perceived as a permanent scarce commodity: It allows a puzzle of all (existing and desired) facets of our personality and enables target group-specific self-marketing. On Facebook and Instagram, you present yourself as a hedonistic bon vivant, and on LinkedIn, simultaneously, as a serious business person. Given this, the question constantly arises: How real is what I see here? Is it a decal of my counterpart optimized to my ideas, or is it authentic? Is a representation descriptive or marketing?




**Implementation in e-commerce**

- Against this background, a plea for more authenticity in advertising and digital communication can be derived from authenticity: Neither top models nor superstars sit in front of the computer. So why is this type so stressed on the screen? An indication of the sensible return to normal (also in terms of business indicators) is provided, for example, by the successful cross-channel campaign "Natural Beauty" of the cosmetics brand Dove.
- Real photos are usually more persuasive than generic catalog material from stock photo exchanges. However, it is also important not to conflict with the Aesthetics Heuristic. Authentic photos must also meet the aesthetic demands of the customers.




**Source:** Taylor C (1992) The ethics of authenticity. Harvard University Press, Cambridge.

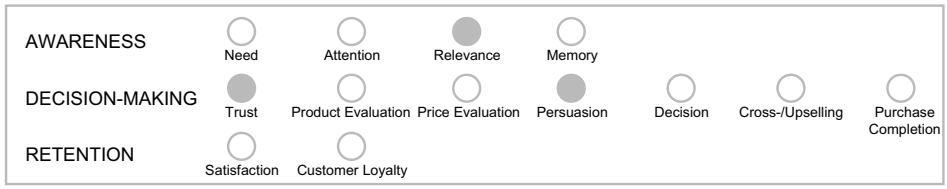
**See also:** Picture Superiority Effect, Aesthetics Heuristic, Mirroring.

**My Notes**

### 5.1.4 Authority

People tend to trust authority. This begins in childhood when we are taught respect and obedience to parents, teachers, police officers, etc. Later, professional incentive systems worked on the same principle: he who obeys makes a career; the number of subordinates measures success. Authority knows many faces in everyday life: expertise, experience, popularity, uniforms, titles, wealth, physical characteristics, or even self-confidence. Since we cannot cognitively afford to evaluate all information regarding the trustworthiness of its source, we use the principle of Authority as a heuristic.

The Milgram experiments are the most famous studies and remarkable proof of the effectiveness of Authority. Here, subjects were put in the role of a teacher who was supposed to punish his student (an actor) for poor performance with electric shocks. As soon as another actor attended the subject in a medical coat (= authority), the "teachers" followed the instructions of the supposed physician largely uncritically to the point of life-threatening electric shocks due to the legitimization by an authority.




**Implementation in e-commerce**

- Own expert status: "We were honored for outstanding market expertise."
- Recommendation of recognized experts: "Dentists recommend this toothbrush." or "Professional athletes prefer this fitness product."
- Professional groups with high trust, e.g., doctors and pharmacists, are often used in advertising. Their portraits adorn shampoo bottles and insurance advertisements, and in television commercials for dietary supplements, they walk thoughtfully past glass walls on which unlabeled axes show curves. Although these stylistic devices are relatively simple and are rejected by customers as unconvincing in surveys (i.e., System 2-based), in fact, the message still catches on subconsciously (System 1-based), which can be explained by the loss of contextual information and the Mere Exposure Effect.
- Seal of approval as proof of authority of (supposedly) neutral test institutes and authorities: Various studies show that heuristics are almost independent of the logical relationship of the seal to the product. For this reason, it is essential to use seals; what these are, on the other hand, is secondary (within the spectrum of trustworthy institutions).
- Media: Well-known phrases such as "known from radio and television" still have their raison d'être today. Thus, it is assumed that what is mentioned in the media has been reviewed and approved by an authority.




**Sources:** Milgram S (1963) Behavioral Study of obedience. The Journal of abnormal and social psychology 67(4):371–378; Cialdini RB (1984) Influence: the psychology of persuasion. Harper Collins: New York.

**See also:** Trust Bias, Mere Exposure Effect, Social Proof, Commitment and Consistency.

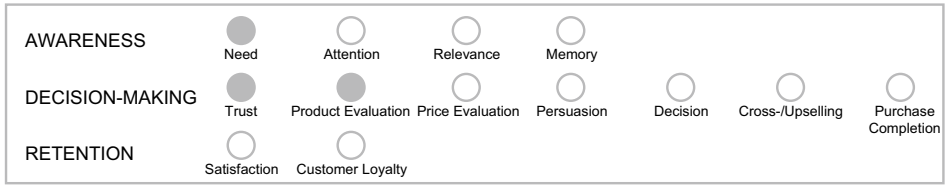
**My Notes**

### 5.1.5 Bandwagon Effect

Also: Herding, Follower Effect

People are more inclined to buy a product when others have already bought it. Thus, the Bandwagon Effect is a cognitive bias in which a decision is not made based on functional utility characteristics but on the popularity of a product in the peer group and on perceived conformity and the desired sense of belonging to a group. In addition, there is also an evolutionary explanation here: Humans learn evolutionarily strongly by observing their fellow human beings and can thus avoid dangers and accelerate learning processes.

The Bandwagon Effect is well summarized by its discoverer Harvey Leibenstein (1950, p. 184) as "the desire of people to wear, buy, do, consume, and behave like their fellows; the desire to join the crowd, be 'one of the boys'". In short, "If everyone has it, I want it too."




**Implementation in e-commerce**

- Similar to the social proof trigger, formulations such as: "Already sold 123 pieces", "Already 100,000 enthusiastic customers", and "Bought particularly often" are suitable.
- Based on anchoring mechanisms, purchase-independent vital figures can also be used as triggers, e.g., the number of page views or video views.
- The effect also works powerfully in social media (e.g., a need is felt when friends' products appear in one's timeline). It thus forms the basis of successful influencer marketing.




**Source:** Leibenstein H (1950) Bandwagon, Snob, and Veblen Effects in the Theory of Consumers' Demand. The Quarterly Journal of Economics 64(2):183–207.

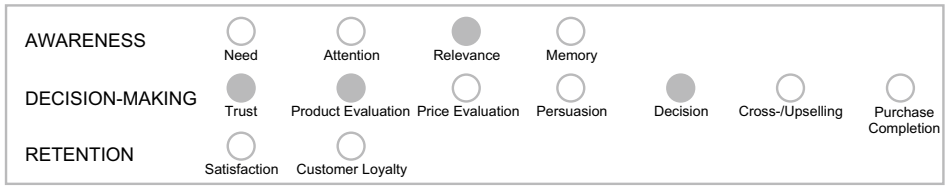
**See also:** Social Proof, Anchoring, Commitment and Consistency, Trust Bias, In-Group Bias, Moral Licensing.

**My Notes**

### 5.1.6 Cheerleader Effect

The Cheerleader Effect describes that individuals within a group are perceived as more attractive than each individual on their own. This applies to both men and women. More precisely, the unattractive facial features of individuals in the group are compensated by (complementary) features of the other group members. Groups are thus processed as average values of phenotypic traits – and average faces are generally perceived as sympathetic (see: Liking or Langlois & Roggman 1990). The size of the group, the duration of the observation, and the context of the group presentation do not play a role in the effect.

The TV series How I Met Your Mother has given the effect wide recognition (for those interested: season 4, episode 7).




**Implementation in e-commerce**

- Products and testimonials should always be presented in a group to make them look more attractive (the average smoothes out unattractive features). In addition, in the case of groups, there is a higher probability that the user will be able to identify with at least one of the people depicted. In this case, the pattern mirroring also applies.
- The effect works better for homogeneous groups (such as the eponymous cheerleaders) than for inhomogeneous groups (such as a family of four), which are more challenging to process than averages. This should also be considered when choosing testimonials and images.




**Source:** Walker D, Vul E (2014) Hierarchical encoding makes individuals in a group seem more attractive. Psychological Science 25(1):230–235.

**See also:** Liking, Mirroring, Picture Superiority Effect.

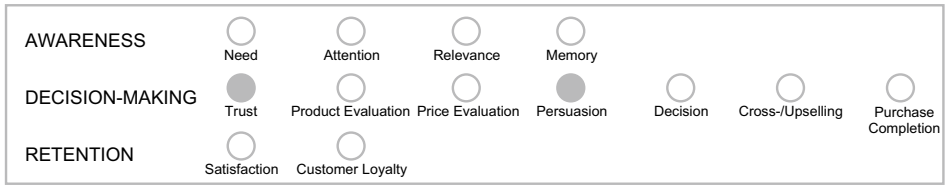
**My Notes**

### 5.1.7 Equality Attraction

Our willingness to interact increases significantly when we meet someone with the same name. The following effect gradation can be seen: Same first name < same surname < same first and surname. Similar results can be measured in the case of matches between the birthday or the city of origin. This connection can be seen in the proverb "birds of a feather flock together".

Equality Attraction can be explained as follows: The names of people who are similar or even the same as our name naturally seem immediately familiar to us – after all, we hear this name dozens of times every day. In this way, the effect of the Mere Exposure Effect (repeated presentation leads to increased trustworthiness) is transferred to the other person. Trust is an indicator of security, which shuts down System 2 and relinquishes control to System 1. According to the available studies, the effect is higher in women, as they are more likely to be relationship-oriented.

It should be noted, however, that the effect has been predominantly demonstrated in the Anglo-American cultural area, where the direct salutation with the first name is familiar and accustomed. In Central Europe, conversely, a more distant tone is usually maintained with customers, which could negatively impact the effect strength.




**Implementation in e-commerce**

- With the help of personalization approaches, a (fake) employee in the call center or customer service can have the same name as logged-in users. Be careful, however, with the same first and last name; this can quickly create reactance or uncertainty (especially with rare names).
- Given the low use of the effect in the Western cultural area so far, exact testing before introduction is once again strongly recommended concerning possible customer reactance or other undesirable side effects.



**Source:** Gamer R (2005) What's in a name? Persuasion perhaps. Journal of Consumer Psychology 15(2):108–116.

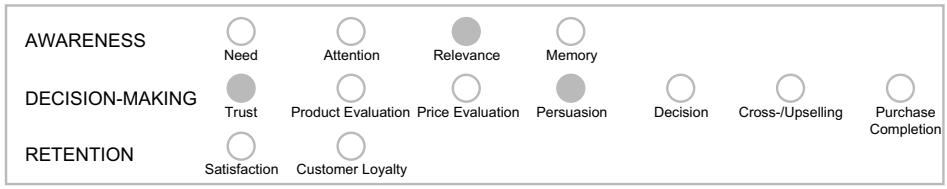
**See also:** Liking, Mere Exposure Effect, Reactance, Reverse Psychology.

**My Notes**

### 5.1.8 In-Group Bias

As early as the beginning of the twentieth century, it was recognized that people felt safer and more comfortable in social groups and tended to perceive their group as superior to other groups. Our social identity and self-confidence are essentially fed by belonging to groups. Consequently, we attach great importance to the opinions of members of our group, regardless of the actual competence of these people.

In his experiments, Henri Tajfel and his colleagues impressively demonstrated how quickly such social structures develop and what power they unfold – even if they have arisen randomly for all participants. In the experiment, for example, a random division into two groups broke up the friendly relationships between 14 boys built up over the years within a very short time. They began not only to stand up for their own group but also to deliberately harm the other group, even though the latter (outside the experiment) was composed exclusively of their friends. The thought of an in-group victory was just as motivating as an out-group defeat.




**Implementation in e-commerce**

- In-Group Bias is often found among highly brand-loyal customers ("evangelists"), such as those found among buyers of Apple products. Owning the same products creates a commonality that establishes the basis for social bonding and draws a sharp dividing line with users of other smartphones.
- The effect can be used through testimonials that the members of the target group would like to know in their in-group and where customers hope to establish a commonality through the purchase. Often, one will realize that it is not necessarily prominent testimonials but other target group members who serve this purpose.
- Verbally, this feeling can also be generated. Terms such as "we" and "our" are completely underestimated: they powerfully express that product users are not opportunistic buyers but members of a movement.




**Sources:** Brewer MB (1979) In-group bias in the minimal intergroup situation: A cognitive-motivational analysis. Psychological Bulletin 86(2):307–324; Tajfel H, Turner

JC (1979) An integrative theory of intergroup conflict. In: Austin WG, Worchel S (eds) The social psychology of intergroup relations, Brooks/Cole, Monterrey, pp. 33–47.

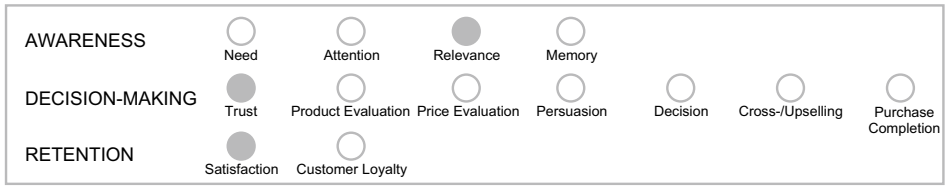
**See also:** Social Proof, Bandwagon Effect, Not-Invented-Here Syndrome, Moral Licensing.

**My Notes**

### 5.1.9 Liking

We tend to be persuaded more quickly by people we like. There are four different approaches to generating sympathy: similarity, attractiveness/familiarity (see: Halo Effect, attractiveness suggests competence), contact/cooperation (in achieving common goals), and praise/recognition (see: reciprocity, we want to "give back" to the seller).

It is difficult to escape the effect of these mechanisms around liking. Even if the other person's motives are known and even in the case of apparent dishonesty, we perceive people who flatter us as sympathetic and want to make decisions together (see: Commitment and Consistency). This impression is strengthened if these people also build a rapport with us, e.g., by mirroring our body language and becoming similar to us (see: Mirroring).




**Implementation in e-commerce**

- The target image of the salesperson (whether virtual or physical on the sales floor) should be to become the customer's best friend. Sentences like "We're pulling in the same direction" help here.
- Affection can be built by showing the people behind the platform or the shop. The message is: "We are real people, just like you. We are similar. We like you, and you like us." When introducing employees (e.g., in service or call centers), one possible application would be to intersperse personal information (e.g., "Peter loves dogs and has a soft spot for sneakers.") that often appears in the target audience so that customers can recognize similarities.
- E-commerce brands should also be touchable in real life, e.g., with road shows, customer meetings, open days, events, etc.
- Ultimately, shared goals with customers can be highlighted: "We are committed to the perfect wrapping experience."



**Source:** Cialdini RB (1984) Influence: the psychology of persuasion. Harper Collins: New York.

**See also:** Reciprocity, Equality Attraction, Halo Effect, Commitment and Consistency, Mirroring, Cheering.

**My Notes**

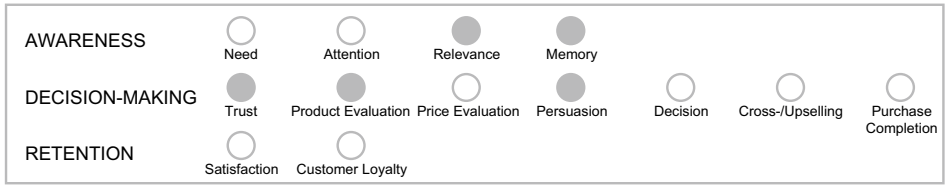
### 5.1.10 Mere Exposure Effect

Also: Illusion-of-Truth Effect, Repetition Effect

The Mere Exposure Effect describes that we are more likely to consider the information we have encountered frequently true. The mere repetition of messages creates familiarity and, under uncertainty, reduces the perceived risk of a decision. When we recognize something familiar, the bonding hormone dopamine establishes a sense of reward. The fact that we opt for alternatives that we have encountered more often can also be proven neurologically. Measurement of brain waves shows that the brain first evaluates which alternative seems most familiar in the first fractions of a second. This comparison can even predict later decisions (Rosburg et al. 2011).

Surprisingly, it does not play a significant role in how the information is embedded: Long-term memory associates complex information with similar previous information. However, in this association process, the context (e.g., a consumer protection warning) and the specific characteristics (e.g., a low level of source trustworthiness) of the information are usually lost, leaving only the essential information.

However, the decision based on familiarity can easily be misleading. Shares of well-known companies, for example, are only in greater demand due to their high media coverage, without considering the share's actual value.




**Implementation in e-commerce**

- The Mere Exposure Effect can be applied, for example, in the context of retargeting. With this marketing tool, interested users are repeatedly confronted with advertising messages after a visit to the online shop (cookie-based) until they get caught. However, for technical reasons, it is often impossible to measure the actual effect, as the mere exposure effect has a prolonged impact cycle that often goes beyond the usual capping in retargeting (i.e., messages are only played out with a limited frequency).
- In customer service, the same employee can be positioned as the face of the provider over a long period: Through frequent encounters with this person (or even a company or an object), we find them increasingly sympathetic and trustworthy. Examples are known from companies such as Ford, Vanguard, and Verizon.
- If companies rely on prominent testimonials in their external presentation, spill-over effects usually occur: The celebrity leads to a basic familiarity with the person, who is then transferred to the advertised brand as part of the advertising.

- Call-to-actions are clicked more easily if placed on the page with the same label multiple times.




**Source:** Zajonc RB (1968) Attitudinal effects of mere exposure. Journal of Personality and Social Psychology 9(2, Pt.2):1–27.

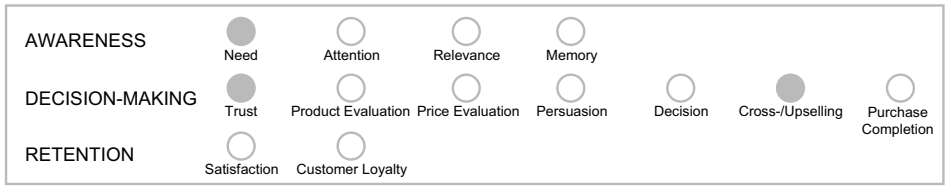
**See also:** Liking, Availability Heuristic, Empathy Gap.

**My Notes**

### 5.1.11 Small Talk Technique

A unique form of framing is achieved through small talk. This means (despite all sales targets) not to rush to topics but to first get into the conversation away from the topic. In the experiments that scientifically investigated the effect of small talk, small talk phrases about sensitivity, opinions, or questions about directions were used. Other empirically proven well-suited topics are sports, film and culture, food, or hobbies.

Now, the question arises about what the Small Talk Technique does with conversion optimization and usability in e-commerce. The effect of such approaches works similarly in all (even impersonal) contact channels: On the one hand, small talk prevents immediate blocking of the customer, and on the other hand, it activates a behavioral script typically used when dealing with friends. Therefore, thinking about transfer approaches in the digital world can be worthwhile.




**Implementation in e-commerce**

- In the case of telephone interaction or chats, the (virtual) agent can first thank you for contacting them and ask how the customer is doing. Chatbots can even do this, which establishes a superficially friendly relationship.
- For small talk to work, no actual interaction process is needed. The corresponding script must be activated; this can also be achieved, for example, by introducing an employee who takes care of the respective order at the beginning of the closing process. The screening includes several personal details (e.g., family, hobbies, favorite movies). The effect can be enhanced if combined with the Unity Pattern, e.g., if the employee makes a personal cross-selling recommendation ("I also recommended this to my family.").



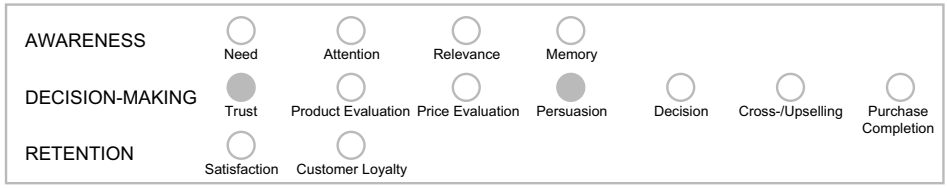

**Source:** Dolinski D, Nawrat M, Rudak I (2001) Dialogue involvement as a social influence technique. Personality and Social Psychology Bulletin 27(11):1395–1406.

**See also:** Framing, Liking, Unity, Mere Agreement, Foot-in-the-Door Technique.

**My Notes**

### 5.1.12 Social Proof

When we don't know how to decide, we are guided by other people's actions. In such situations of uncertainty, we believe that other people (who have already made a decision) have more specific knowledge about the situation and have been able to make informed decisions based on this. This lowers the perceived risk of one's own decision. In addition, a parallel to the Commitment and Consistency Pattern can be seen here: We try to remain true to ourselves and make decisions in conformity with our peer group.




**Implementation in e-commerce**

- Individuals (experts, influencers, testimonials): Besides a real photo and name, authentic details (e.g., profession, purchased product configuration, place of residence) are particularly important when presenting individual customer advocates. Also, the provider's team members can be considered as social proof. They show that the shop is genuine and trustworthy.
- Collectives (other users or customers): Evidence of the collective appreciation of the company or a product should be documented with the number of interactions (likes, views, tweets, subscribers, etc.) and further demanded with social sharing buttons. In addition, highlighting a "most popular product" or "bestseller" has proven its worth. Procedural savings can also be achieved: For example, the most favorable payment method for the provider can be strengthened ("How do you want to pay? 72% of our customers use PayPal.").
- Institutions (test institutions, authorities): A particular form of social proof dealt with in more detail in the "Authority" pattern.
- For the placement of all social proof elements, proximity to the call-to-action and under-the-product presentation has proven effective. They are only needed at the top of the page in exceptional cases (e.g., young providers with credibility issues).



**Sources:** Sherif M (1935) The psychology of social norms. Harper & Row: New York; Cialdini RB (1984) Influence: the psychology of persuasion. Harper Collins: New York.

**See also:** Authority, Commitment and Consistency, Trust Bias.

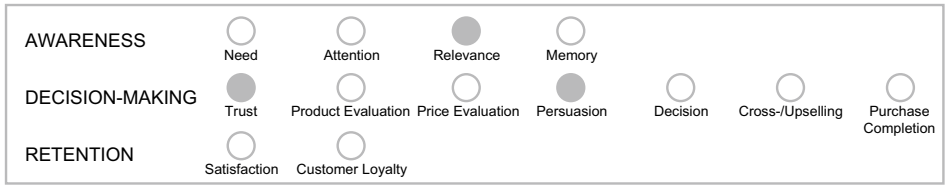
**My Notes**

### 5.1.13 Trust Bias

Trust is the basis for even considering shopping in a shop. Some Western countries have a particular culturally determined case: Statistically, countries such as the USA or Germany are among the most anxious and cautious online users worldwide. This underlines the importance of trust-building elements along the entire customer journey and, thus, the Trust Bias itself. Particularly critical points are online advertising (since, for example, in the case of display banners, the exact link target is usually not known before the click), the start and product pages (since the purchase decision is always made here), and the application process or the ordering process (since sensitive data such as address and bank details have to be provided).

The great importance of trust elements also applies to developing a sustainable customer relationship as a whole. This illustrates the timely horizon that should be applied when using behavior patterns: In addition to short-term conversion optimization, there is always a long-term relationship level. Trust can be built using "sedative pills" for System 2: The site should send enough signals to reduce stress and turn off the ratio. This is where the old usability wisdom of Steve Krug (2014) comes in: "Don't make me think."

A brief digression into science: It is now clear that the complex construct of "trust" has several dimensions, all of which must be fulfilled for a sufficient calming effect (Mason et al., 2013): perceived competence, integrity, and benevolence (i.e., ultimately, the assumption that action is being taken in my interest). If you honestly ask yourself which of these dimensions of trust users might find deficient, you usually quickly come up with an appropriate reaction.




**Implementation in e-commerce**

- Trust-building elements serve as heuristics for assessing the store's trustworthiness.
- Probably the best-known examples of trust elements are seals of approval such as Trustpilot, Google Trusted Stores (Google Customer Reviews), Better Business Bureau, and many more.
- The company's social and ecological commitment lends itself to a prominent presentation: This is where the representativeness heuristic comes into play and reinterprets the promotion of sustainability as "If you are good to the world, you will also be good to me."

- Technical trust marketing is another implementation approach: Common examples are safe listing in tools such as "Web of Trust" (WOT), SSL encryption, sending passwords separately from usernames, 2-factor authentication, double opt-in for newsletter subscriptions, and the renunciation of non-transparent distributions.




**Source:** McCornack SA, Parks MR (1986) Deception detection and relationship development: The other side of trust. Annals of the International Communication Association 9(1):377–389.

**See also:** Social Proof, Authority, Bandwagon Effect.

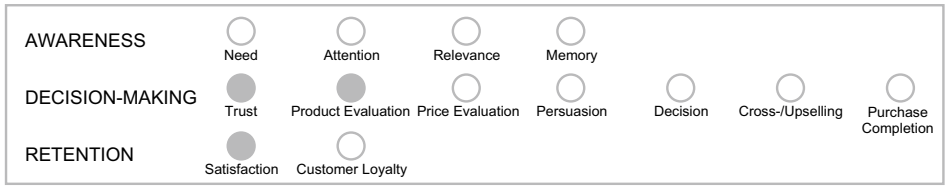
**My Notes**

### 5.1.14 Uncanny Valley Effect

The Uncanny Valley Effect can be easily described as "incredibly realistic." It refers to the paradox that artificial or simulated figures (e.g., avatars and robots) are not necessarily more accepted by us when they become more human-like. First of all, however, this connection exists: to a certain extent, we like similarity in artificial figures. At a certain point, however, the increasing resemblance creates discomfort, and the acceptance of these figures sinks or turns into rejection.

The acceptance gap between a human-like robot and an actual human is known as the "uncanny valley." For example, highly humanoid robots often lay in this valley of rejection. The effect is also often used to explain why we have acceptance problems with people who have undergone severe cosmetic surgery (a prominent example in literature is Michael Jackson towards the end of his life). Due to limited facial expressions and the lack of typical human features (such as wrinkles or birthmarks), these are perceived as robot-like people similar to human-like robots - in other words, precisely in the Uncanny Valley.

The unwanted animation of non-living figures can explain the observed connection. We fear we will be unable to tell the difference between a human and a robot in the near future, but by nature, we see humans as part of a different group. To upgrade our group (in-group), we systematically devalue the group of humanoid robots (out-group).




**Implementation in e-commerce**

- If digital assistants are to be used on the website, these findings must be considered. We don't like perfectly animated alter egos but non-human-looking figures with human traits. Good examples are the robot "Wall-E" from the well-known Disney movie or one of the most commercially successful robots currently, "Pepper" from SoftBank (for its acceptance, see Volland and Meyer 2018). Both utilize human looks (e.g., childish schema) and human behavior (e.g., emotions such as astonishment and joy).
- The same applies to multi-channel providers who use robots on the sales floor.
- The effect can also occur without a human–robot appearing visually: For example, Google has long struggled with the acceptance of many of its personalized offers (e.g., unsolicited slideshows and video animations of one's own holiday photos, evaluation of movement and fitness data with a derivation of recommendations for action). Despite the power of modern automation algorithms, the boundary between man and machine should never

be blurred. Customers must always be able to understand how personalized services are created and, ideally, request them explicitly.




**Source:** Seyama JI, Nagayama RS (2007) The uncanny valley: Effect of realism on the impression of artificial human faces. Presence: Teleoperators and virtual environments 16(4):337–351.

**See also:** Not-Invented-Here Syndrome, Unity, Illusion of Control.

**My Notes**

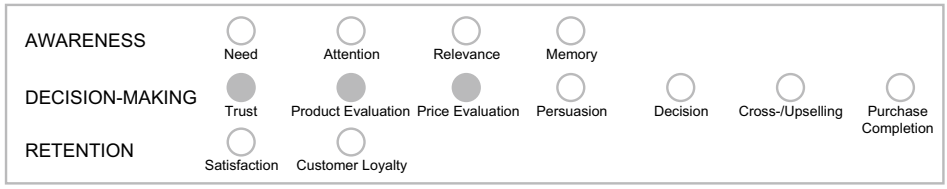
## 5.2 Product Evaluation

### 5.2.1 Aesthetics Heuristic

Good visual design significantly increases the perceived persuasiveness of functional aspects of products. These results are from the PsyConversion® study by elaboratum (Klein et al., 2018). In addition, experiences with an aesthetic product are described as more exciting, and there is an increased sense of pride in owning the product. In this heuristic, the beauty and appearance thus act as a substitute for the functional benefit. Essential: The willingness to pay also increases significantly.

Aesthetics is a fluid term. This becomes clear when you look at the ideals of beauty of past decades or even centuries. The once-popular Rubens figurines have as much in common with today's zero-size models as fries have with celery. Applied to e-commerce, all the drop shadows, gradients, and 3D effects from before the turn of the millennium seem entirely out of time compared to today's flat or material design. And this design trend will also pass shortly. The Aesthetics Heuristic (like almost all other patterns) requires continuous adaptation along the current zeitgeist.

In the meantime, it has become a scientific commons that, despite substantial temporal and regional differences, there are some universal standards of evaluation for beauty that the research branch of neuroaesthetics is trying to find. Brain scans found that the presentation of beautiful content activated the medial orbitofrontal cortex, while ugly content activated the amygdala (Ishizu and Zeki 2011). Examples of motifs universally perceived as attractive are children playing or bluish landscapes. Even depictions that have a mystical aspect and leave the interpretation partly to the imagination of the viewer function universally. Regarding portraits, general recommendations can also be made: faces should be symmetrical, average, with prototypical shapes, and rather curvy contours to be perceived as aesthetic.




**Implementation in e-commerce**

- High-quality web design is not just a visual hobby but a benchmark for product evaluation and desirability. It often turns out to be significantly cheaper to improve the aesthetics of the virtual product presentation than to revise the product itself – with comparable effects. The decisive factor here is the orientation towards objective and measurable design criteria and standard UX conventions. The orientation towards established design guidelines is thus also a measure to reduce business risks.

- Symmetry is crucial for the perception of aesthetics. This can be explained with the help of an evolutionary heuristic: A symmetrical face could be used as a reliable indication of intact genetic material and thus helped in the choice of reproductive partner. A website should, therefore, also be symmetrical. This can be achieved, for example, by storing a symmetrical grid for all page templates to which the content objects are aligned.
- When selecting images, those mentioned above universal aesthetic standards should be taken into account (e.g., symmetrical faces, children playing, bluish landscapes).




**Source:** Chitturi R (2015) Good Aesthetics is Great Business: Do We Know Why? In: Batra R, Seifert CM, Brei DE (eds) The Psychology of Design: Creating Consumer Appeal. Taylor & Francis Group, Routledge, pp. 252–262.

**See also:** Joy and Fun, Halo Effect, Unity, Rhyme-as-Reason Effect, Affect Heuristics.

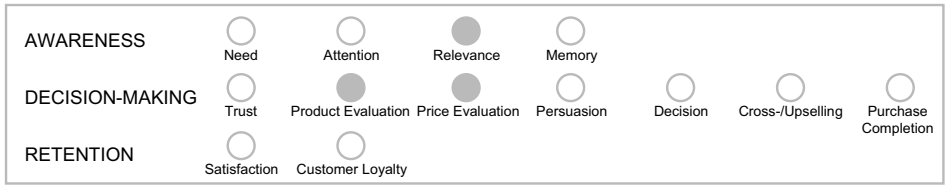
**My Notes**

### 5.2.2 Curse of Knowledge

The Curse of Knowledge is a cognitive bias that causes people who are well-versed in a topic to find it impossible to look at that topic from the point of view of a less informed person. In practice, this leads to many concepts and ideas not being presented and explained clearly enough because the audience is assumed to have too much knowledge and understanding. The problem is that only in the rarest of cases do interested parties freely admit that they could not follow the provider's explanations – which is why the effect is usually clearly underestimated. This makes Curse of Knowledge the perfect alternative to information asymmetry, in which it is generally assumed that the professional superiority of a salesperson leads to customers reliably joining his recommendation.

In 1990, psychologist Elisabeth Newton conducted an impressive experiment on the Curse of Knowledge: she separated the participants in the investigation into two groups. One group tapped well-known songs, while the other listened and was supposed to recognize the songs. The group of "knockers" estimated the rate of correctly guessed songs to be at least 50% - in reality, it was only 2.5%. Knowing which song was meant made the solution obvious, and a realistic estimate of the success rate became impossible (Newton 1990).

For once, the Curse of Knowledge is a behavior pattern affecting e-commerce managers, not customers. Nevertheless, it is indispensable for designing effective digital interfaces, as it often stands in the way of actual customer orientation. If people with in-depth knowledge of the product's features and benefits set a price, uninformed prospects will likely perceive it as too high – especially in high-priced segments.




**Implementation in e-commerce**

- Product features and benefits should be considered from the customer's point of view. The old logic of USPs ("Unique Selling Propositions") deals far too much with the arguments the provider considers relevant for a successful SALE – but rarely with how the respective product can improve the users' lives.
- Curse of Knowledge is a plea to test new pages and products with users before publication constantly. Standard analysis questions are: What is being sold here? What are the benefits? Is the product relevant to me? The choice of methodology and the design of the test structure are of great importance (see Chapter 7).
- Surveys are an excellent idea to understand the users' level of information. This way, it can be clearly understood (preferably packaged as a service initiative) how large the

information gap between the provider and the interested parties is and at what point they must be picked up.

- Product names should be catchy and self-explanatory and give even the uninformed prospects an instant sense of what they are all about. Which TV do you find more grippy at first glance? A "TX-40GXW804" (Panasonic) or "The Frame" (Samsung)?
- Especially in B2B business, Curse of Knowledge is crucial: Here, an average of five people are involved in a purchase decision. If the advantage argumentation for a product is too complex for only one of these people, the conclusion will likely not occur.




**Sources:** Camerer C, Loewenstein G, Weber M (1989) The curse of knowledge in economic settings: An experimental analysis. Journal of Political Economy 97(5):1232–1254; Newton L (1990) Overconfidence in the communication of intent: heard and unheard melodies. PhD thesis, Stanford University.

**See also:** Cognitive Ease, Availability Heuristic, Joy and Fun.

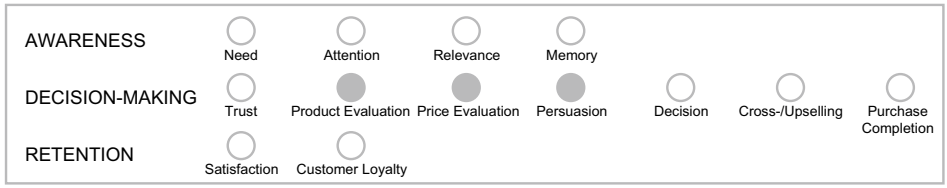
**My Notes**

### 5.2.3 Decoy Effect

The Decoy Effect refers to the phenomenon that many people cannot decide between two options, A and B (or choose the cheaper alternative, A). However, if you add a third "asymmetrically dominated option" C, the decision is suddenly straightforward. Deciding between an attractive (B) and an unattractive option is much easier (= decoy/C). The option that prevailed in this comparison (B) is also perceived as better than the original comparison value (A) and is predominantly chosen.

In the experiment, this was proven, for example, by buying a car. Customers were faced with choosing between a simple car at a low price (A) and a well-equipped car at a higher price ( B). The Decoy Effect comes into play when the seller adds a third option (C): This has slightly better features than A (but significantly fewer features than B) and is almost as expensive as B. By comparing these three options, B appears to be by far the best choice. The original decision was difficult because we cannot think and decide in terms of absolute utility. Options A and B offer entirely different benefit dimensions (A: Price; B: Performance), which are difficult to compare. An option is preferred if contrasted with a comparable, unattractive option.

The behavior pattern does not contradict the assumptions of the Paradox of Choice or Hobson's +1 Choice Effect since the additional alternative is not considered, but only the number of theoretical options has increased. Nevertheless, care should be taken not to let the number of comparison options with a decoy become too large (i.e., > 5).




**Implementation in e-commerce**

- Product comparison tables: If two products (e.g., smartphones or insurance) are compared, there is a tendency towards the cheaper product. Suppose you now add a decoy ("bait") to the comparison, which is even more expensive than the more expensive product but not necessarily better (i.e., it makes no economic sense for most customers). In that case, the middle product tends to be preferred. Like the car example above, a decoy can be designed as a medium product.
- The Decoy Effect is also often used in the context of real estate viewings: Many brokers first present an acceptably cheap option, then an unattractive, expensive, and finally an attractive, expensive option, which is also chosen in most cases. Newsletters can be designed according to the same principle in e-commerce: With three products presented, the decoy should be in the middle and the high-margin (usually most expensive) product at the end.




**Source:** Huber J, Payne JW, Puto C (1982) Adding asymmetrically dominated alternatives: Violations of regularity and the similarity hypothesis. Journal of Consumer Research 9(1):90–98.

**See also:** Extremeness Aversion, Paradox of Choice, Hobson's +1 Choice Effect.

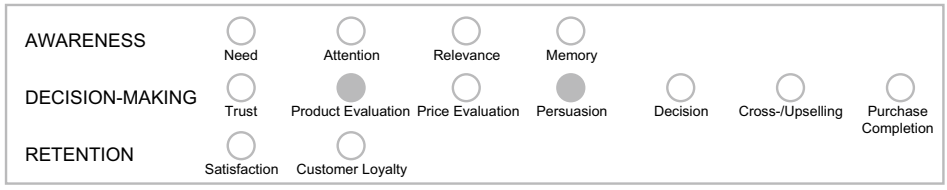
**My Notes**

### 5.2.4 Extremeness Aversion

Also: Center-Stage-Effect

In uncertain situations, people avoid extreme decision alternatives and prefer a compromise (often a middle option). In cases with three products, the cheapest one appears unsuitable because it is associated with poor quality (see: Price-Quality-Illusion); the most expensive, on the other hand, seems superfluous or insignificant features. Consequently, the middle one is chosen as a compromise with sufficient quality and an acceptable price. Thus, the premium product serves as a reference value for the election decision and emphasizes the advantages of the medium alternative over the low-cost one.

Due to loss aversion, the loss of individual features when choosing the cheap alternative appears severe and worth avoiding. There is also a social component: In the case of uncertainty, the middle position can best be justified to oneself and third parties, if necessary.




**Implementation in e-commerce**

- The effect can be used in every decision with a comparison of alternatives. The tendency towards the mid-range product is powerful when the premium product is only slightly better in a feature but disproportionately more expensive (see: Decoy Effect). Typical application contexts are:
  - Design of product comparison tables
  - Product finder tools
  - For non-physical products: contract levels (e.g., bronze, silver, gold)
- Following these insights, the product portfolio must not be optimized based on sales figures of individual products. It must explicitly include shifts in demand: Even a supposed poor dog with a low contribution margin and low rotation or sales speed can be an elementary component for the portfolio if it triggers a tendency to compromise towards the middle as a "decoy" (bait).
- In addition, the middle product can be graphically highlighted. Then, Loss Aversion reinforces the lever of Extremeness Aversion and increases the likelihood that customers will follow the recommendation.



**Source:** Simonson I, Tversky A (1992) Choice in context: Tradeoff contrast and extremeness aversion. Journal of Marketing Research 29(3):281–295.

**See also:** Decoy Effect, Loss Aversion, Anchoring, Price-Quality-Illusion.

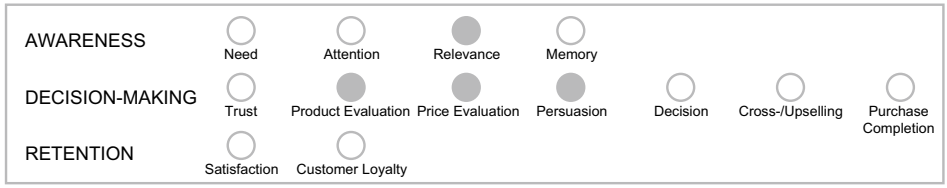
**My Notes**

### 5.2.5 Framing

Framing states that the formulation of a problem influences the decision, even if the logical statement is identical. This would be impossible if classical economic theory assumptions were valid. Framing is, therefore, a frequently cited example of the renunciation of homo economicus.

Positive framing tends to have an activating effect (as it relativizes existing risks), and negative framing tends to inhibit action (as it highlights existing risks). Therefore, negative framing is rarely used in e-commerce, except in presenting the dire possible consequences of not buying (e.g., insurance or retirement products). It is now known that we react more strongly to positive framing than negative framing because it triggers positive thoughts and creates a mental synergy with the message sender.

Well-known examples are margarine (75% fat-free vs. 25% fat), meat (25% lean vs. 75% fat), diseases (10% die vs. 90% survive), and drinks (half full vs. half empty). These examples are called attributive framing. In addition, there is action framing, which compares the positive consequences of an action with the negative consequences of non-action (e.g., when advertising preventive medical check-ups).




**Implementation in e-commerce**

- Language is a powerful tool in product descriptions: To develop persuasive power, all quantitative characteristics should be formulated positivistically (e.g., for fitness or diet products: "90% success rate" instead of "only 10% failures").
- Strong retailer brands can define the framework for the perception of product brands and products if they are placed again in the environment of the product description (spill-over).
- In out-of-stock situations, it is better to speak of "soon available" (positive) than "sold out" (negative). Amazon, for example, uses the addition "More is on the way" when stocks are scarce.
- Savings (given fit to the product) can be symbolized by translating them into everyday units customers often consume (e.g., "Buy this product and save enough for three delicious coffees!").
- Framing always happens, whether intentional or not. It is, therefore, strongly recommended that you examine your website for unwanted Framing effects to increase the conversion potential through correct framing.




**Source:** Tversky A, Kahneman D (1986) Rational choice and the framing of decisions. Journal of Business 59(4):251–278.

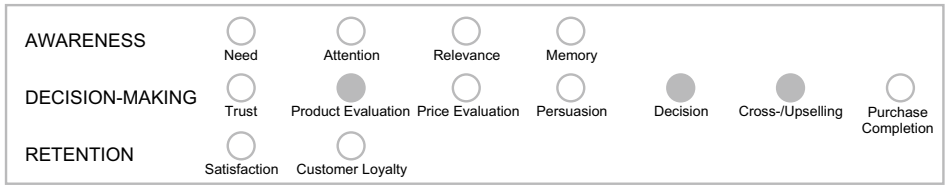
**See also:** Anchoring, External Reference, Rule of Hundred, Priming, Cheering.

**My Notes**

### 5.2.6 Having vs. Using Effect

People prefer things that offer a wide range of features – regardless of whether they use them. Having vs. Using Effect, for example, explains the triumph of mobile phone contracts over prepaid offers: When making a purchase decision, we weigh extensive guaranteed data volume, telephony packages, and services beyond a rational level and regardless of our actual usage behavior. In the case of contract extensions, loss aversion is exacerbated, which leads us to continue to hold on to the functions (despite demonstrable superfluity).

The more technically complex the functions are and the less prior knowledge you have in the product category, the stronger the effect. The reason: A high number of functions are used heuristically to indicate a high-quality product (system 1) if one cannot formulate the actual demand and compare it with the available alternatives (system 2). The pattern also clearly expresses the materialism and consumerism of our society, so the effect also has a social impact component: Social status is defined more by the ownership of the most extensive product than by the proportion of actively used functions or the correspondence between actual needs and purchased parts. This is visible in expensive districts of all major cities, where off-road-capable SUVs are often used for shopping.




**Implementation in e-commerce**

- A tendency towards a higher-quality product can be triggered if the number of functions of product alternatives is trivially compared. A significantly higher number of features will tend to encourage preference for the premium product.
- The superiority of the top product can also be represented by aggregation. With attributes such as “most powerful in its class” or “new and even better,” the effect is used globally without going into detail about the individual functions. The effect is well implemented in the current iPhone 15. This was simply introduced as “Pro. Beyond.” without giving any further reasons for buying.
- All product functionalities should be shown on product pages; concerns about the users’ “information overload” are inappropriate here. However, the prerequisite is an easy-to-consume display and grouping of the same elements within the page (e.g., through pagination, horizontal sliders, or section-by-section display with accordion menus).




**Source:** Goodman JK, Irmak C (2013) Having versus consuming: failure to estimate usage frequency makes consumers prefer multi-feature products. Journal of Marketing Research 50(1):44–54.

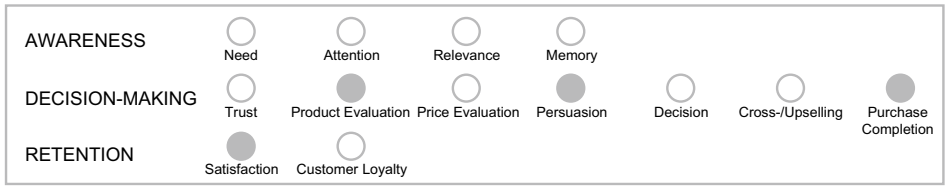
**See also:** Price-Quality-Illusion, Loss Aversion, Decoy Effect.

**My Notes**

### 5.2.7 Inner Dialogue

Users are constantly in an inner dialogue that resembles a soliloquy. Every step of the decision-making process can be coded as a conversation with a stringent syntax ("Should I buy this product? What are the arguments in favor of this? What's wrong with that?"). The course of this Inner Dialogue is crucial for forming the later intention to act. If, for example, the brain receives negatively colored impulses from the inner dialogue, it searches the "archive" (memory) for evidence of this impulse (see Confirmation Bias). The documents found are returned immediately, and the corresponding essential attitude is reinforced.

Positive framing of critical thought processes (e.g., "Can I afford this?") is crucial for a message's persuasiveness. The inner dialogue is designed as a self-fulfilling prophecy, so controlling this dialogue is an effective conversion tool.




**Implementation in e-commerce**

- Initiating thought processes in a targeted manner and leading them into dialogue paths that lead to persuasion or conversion can be achieved by asking the user questions in the content.
- Relevant (i.e., common and meaningful) questions should be picked up as FAQs and prominently placed. In this way, a classic content dead end is turned into an effective persuasion tool.
- Scenarios where customers may find themselves are described concretely to "externalize" and explicitly reinforce internal dialogue processes.
- Positive reinforcement at points with a high drop-out rate, for example, when entering an application process for a complex product (e.g., Financial products, insurance, customization), acts as a motivator: "Don't worry, you can do it."
- Language should be used in the present tense, and subjunctives should be avoided (creates uncertainty). The same applies to vague terms such as "maybe", "possibly", "perhaps", and negations should also be avoided, as they are often mentally classified incorrectly. Unambiguous instructions, on the other hand, trigger purposeful inner dialogues.



**Source:** Schulz von Thun F (1998) Talking to each other 3 – The 'inner team' and situational communication. Rowohlt, Reinbek.

**See also:** Confirmation Bias, Framing, Post-Purchase Rationalization.

**My Notes**

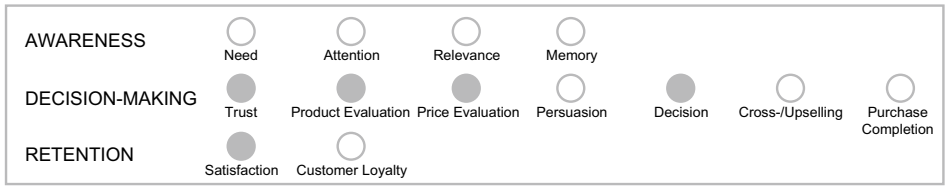
### 5.2.8 Neomania

Also: Appeal to Novelty, Argumentum ad Novitatem.

The term "Neomania" was coined by Nassim Nicholas Taleb, one of our time's most influential thought leaders and book authors. He describes the behavior of many people to prefer things only because of their novelty, the constant search for something new, and the greed for innovation. The complex question of whether it makes sense to buy a product is replaced by the simple question of whether a product is new. The answer is then related to the original question. The effect is thus similar to the Halo Effect. As a reminder, this describes that one dominant feature can outshine the perception of all other features.

Taleb views Neomania as a negative cognitive bias. He understands innovations and new products like a species in nature: what has proven itself in centuries of innovation will likely remain so. Old technologies have stood the test of time because of their "inherent logic"; new technologies have yet to prove this – which, in most cases, will fail.

Nevertheless, the effect is of considerable strength, which can also be seen in the product policy of large manufacturers: While there used to be ten years between two generations of the VW Golf, it is now only around five. In China, the product life cycle of Lenovo's smartphone models often ends after only four months – then the successor model follows.




**Implementation in e-commerce**

- The large group of "early adopters" is especially receptive to the attribute "new" in e-commerce. Actual and wannabe trendsetters can be effectively activated with appropriate hints. This includes stickers on category pages, corresponding filter options, and your landing pages, which present a current collection and aim at the self-perceived pioneering role of the users. The desire for individuality and standing out from the crowd also plays a role: Only new products can ensure that individuals will be perceived as something special with the product (at least for a limited period).
- To be honest, most products do not have a unique market position (consider consumer goods or cars). So, suppose a direct comparison with competitors' products cannot be processed into a clear purchase recommendation. In that case, a product can still be distinguished by the novelty feature in the context of campaigns. The often-voiced accusation that go-to-market campaigns fail to meet user needs may be correct rationally, but they can make sense from a behavioral economics perspective.

- An exciting side effect: Framing as a "novelty" effectively removes a product from the user's expectation to make a bargain. It has been learned that products are not reduced at the beginning of their life cycle, which measurably increases the willingness to pay of individual target groups.
- Neomania – as the name suggests – also has something compulsive about it. Instead of working on absolute novelties, many companies spend time marketing pseudo-innovations to satisfy customers' greed for updates. This leaves room for a countertrend: In the age of sustainability, emphasizing the longevity and durability of products is now the more substantial selling point for a constantly growing target group. It is essential not to warn about the longer product life cycles as a necessary renunciation (this triggers loss aversion for us) but to focus on the financial savings, the ecological-social value, the resource savings, and the efficiency gain (gain framing).




**Source:** Taleb NN (2012) Antifragile: Things That Gain from Disorder. Random House, New York.

**See also:** Halo Effect, Recency Effect, Loss Aversion.

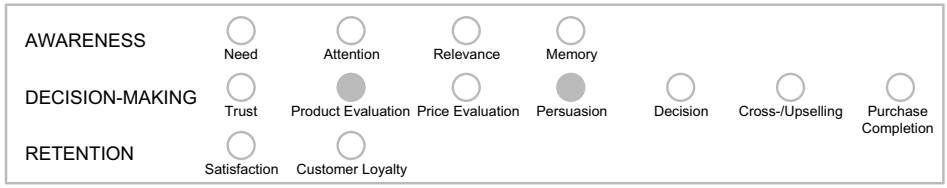
**My Notes**

### 5.2.9 Not-Invented-Here Syndrome

Also: Self-Generation Affect Effect

What was not conceived or produced within one's own (extended) peer group is dismissed as inferior. Instead, one's ideas are exaggerated and irrationally favored over other alternatives. The effect is becoming massively more critical with the development from an industrial to a knowledge society: Those who refuse to use existing knowledge in other parts of the world or other sectors today will never realize their full potential and will be at a massive competitive disadvantage.

Evolutionarily, the effect stems from the (once vital) skepticism towards everything foreign and is part of groupthink: by distinguishing it from the out-group, the in-group is upgraded. Therefore, there is always a lot of vanity, status thinking, ignorance, protectionism, selfishness, and lack of decision-making.




**Implementation in e-commerce**

- Even e-commerce organizations are not immune to the Not-Invented-Here Syndrome: In the case of a make-or-buy decision (e.g., when introducing a shop system or developing a private label), it causes the make option to be systematically overestimated and sometimes generating high follow-up costs. This connection can often be seen within organizations, resulting in good ideas being ignored and discarded as soon as they stem from other departments ("Why does controlling interfere in our marketing strategy? They don't have a clue!").
- This can only be broken with intensive work on corporate culture. Good starting points are incentive systems (ideas should never be rewarded more than the implementation) and the brainstorming culture (specifically asking colleagues outside the subject and department for input). The former director of innovation at Procter & Gamble, Chris Thoen, describes the vision with a simple label that he would like to see more often in his company: "Proudly found elsewhere" (Kelley 2009).
- If you turn your attention back to customers in e-commerce, it makes sense to bring some patriotism and local color into the presentation of the company and its products. Apple has been implementing this with "Designed in the USA" for years and benefits from an increased commitment to its home market. In an increasingly globalized e-commerce world, where Asian vendors (especially the ubiquitous Alibaba and the up-and-coming eBay clone Taobao) are beginning to shake up Western markets, this pattern will become more important there.




**Source:** Katz R, Allen T.J. (1982) Investigating the Not Invented Here (NIH) syndrome: A look at the performance, tenure, and communication patterns of 50 R&D Project Groups. R&D Management 12(1):7–20.

**See also:** Unity, Labor Love Effect.

**My Notes**

### 5.2.10 Ockham's Razor

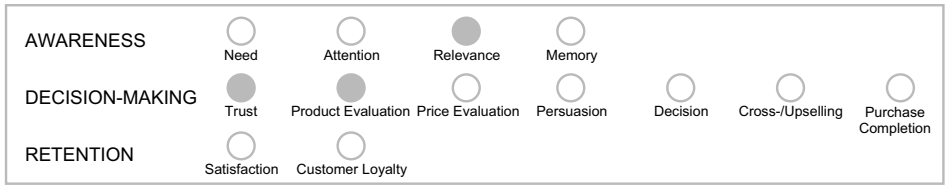
Also: Principle of Parsimony, Lex Parsimoniae, Principle of Economy

The English Franciscan monk and philosopher William of Ockham (c. 1287–1347) used his penchant for simplicity to defend the idea of divine miracles. He evaluated the following: "If there are several possible explanations for a phenomenon, then the most simple one is most likely the correct one." This philosophical razor was used repeatedly in selecting hypotheses as explanatory approaches for observed behavior, constructing theoretical models, and dramaturgy and literature.

The focus on the essentials, pragmatism, and logic has steadily developed as a paradigm. The principle of "Chekhov's Gun" is an example: The famous short story author Anton Chekhov put it in a nutshell:

> "Remove anything that has no relevance to the story. If you say in the first chapter that a rifle is hanging on the wall, it must go off in the second or third chapter. If it's not going to be fired, it shouldn't be hanging there." (Thaler, 1981)

Even in Ockham's time, the idea of focusing on the essential and straightforward was only partially new. Many great thought models of the recent past also take the same line: the 80/20 principle of Pareto, the law of diminishing marginal utility of Gossen, or the image of "low-hanging fruits."




**Implementation in e-commerce**

- In the digital context, the principle is increasingly being strained, too: In 2017, for example, digital marketing guru Avinash Kaushik recommended that companies radically stop all social media activities aimed at organic reach, as there was no acceptable proportion between effort and return. The blog post was simply titled "Ockham's Razor".
- In fact, the flood of arbitrary, interchangeable, and often poorly produced messages can also be found on the websites and shops of many e-commerce companies. This ignores the limited receptivity of users. In addition, the described complexity-reducing heuristics (see: Focusing Effect) lead to the fact that the relevant content is more likely to be lost among the irrelevant content.
- Each page of a portal should have a clear conversion goal. All content elements that do not directly or indirectly contribute to this goal must be removed. Experience has shown that this is extremely difficult for marketers – because they, too, succumb to cognitive

distortions such as the Endowment Effect, Status Quo Bias, and Loss Aversion and are attached to existing content. Therefore, it is an excellent conceptual principle to start with a blank page and only include elements that meet the relevance criterion. The KISS principle ("Keep it short and simple") also goes back – not coincidentally – to Ockham's Razor.




**Source:** Mole P (2003) Ockham's Razor cuts both ways: The Uses and Abuses of Simplicity in Scientific Theories. Skeptic 10(1):40–47.

**See also:** Focusing Effect, Paradox of Choice, Hick's Law.

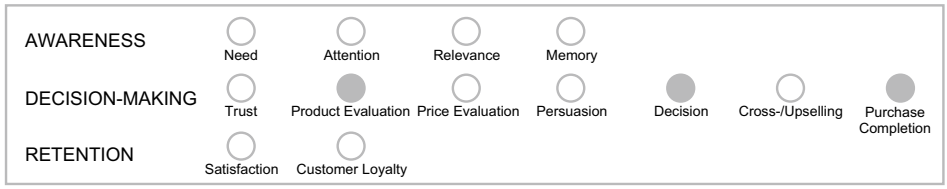
**My Notes**

### 5.2.11 Paradox of Choice

It sounds counterintuitive initially: the more choices we have, the worse our decisions are. Many options do not increase our decision-making capacity; they lead to uncertainty and fear. So, even if we believe that a wide range of decisions makes us free, happy, and satisfied, the opposite is true.

There are several reasons for this effect: the more alternatives are available, the greater the effort required to evaluate and weigh them up. However, our brain is programmed to use heuristics to reduce effort – this is how conflict arises. In addition, many options disproportionately inflate our expectations, so the chosen product will often be subject to expectations – buyer's remorse is the logical consequence.

As a result, the paradox of choice states that for many options, we will either not decide at all or choose an option we will not be satisfied with afterward.




**Implementation in e-commerce**

- The paradox of choice is a significant challenge, especially for online shops that offer many long-tail products (which are rarely purchased): A large assortment is a supposed unique selling point (USP) but can massively complicate decision-making and lead to users abandoning their purchase. A wide range should, therefore, be organized rather deeply and divided into smaller categories.
- It also makes sense to reduce the number of products on overview pages and display them in list format rather than with a two-column grid.
- Product comparison tables can present the alternatives' respective optimal values. This is highly context-dependent and often industry-related (e.g., insurance companies and telecommunications companies usually have three different product alternatives). Depending on the most relevant KPI, it is either optimized so that users stay in the shop in the first place or a correspondingly smaller part makes a purchase decision.
- It should also be possible to select products with checkboxes and compare them directly with each other. In this way, users can determine the number of options themselves and receive the comparison in a simplified manner.
- Intensive use of filter criteria and sorting algorithms reduces the number of options displayed to the ideal amount (identified in the A/B test).




**Source:** Schwartz B (2004a) The paradox of choice: Why more is less. Ecco, New York.

**See also:** Hobson's +1 Choice-Effect, Buyer's Remorse, Hick's Law, Ockham's Razor.

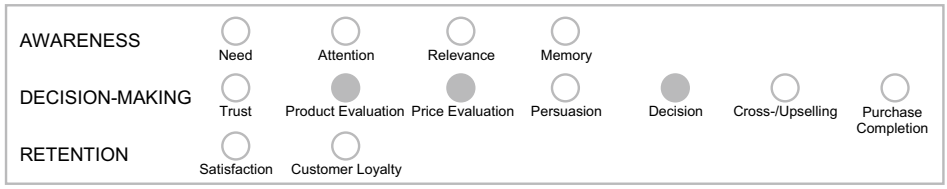
**My Notes**

### 5.2.12 Price-Quality-Illusion

The benefits people see in a product depend not only on the primary characteristics (such as taste, quality, design, etc.) but also on the expectations regarding the quality of the product. The price plays a decisive role in this: it changes the utility coding in the brain but does not affect the perception of primary properties. Simply put, the price deceives the brain but not the senses (Beck 2014).

In the experiment, the subjects conducted a wine tasting. The remarkable thing about it was that the wine was always the same; only prices differed. The participants rated the wine with the higher price indication as better regarding quality and taste. Using a functional magnetic resonance imaging (fMRI) scanner showed that the higher price of wine activated the medial orbitofrontal cortex, which is associated with evaluating taste, smell, and music. Thus, the expectation of a better wine activated other brain regions, influencing the actual benefit experienced.

It is interesting to note that customers have long since stated in surveys that they no longer believe in a stable relationship between the price and quality of a product, as a GfK study has shown (GfK, 2013). This makes it clear that the pattern works subconsciously (System 1) and does not penetrate the buyer's consciousness (System 2).




**Implementation in e-commerce**

- To increase the perceived benefit of a product, it is not necessary to change the product itself. It may be enough to adjust product attributes, such as the price. This works particularly well for products where customers do not have a precise idea of the value for money or cannot evaluate the quality independently (e.g., insurance, wine, or software).
- The effect of the price-quality illusion is subject to strong fluctuations. There seem to be threshold values anchored in customers' consciousness due to learning processes used as a benchmark. At these thresholds, the effect weakens, and in between, it becomes stronger. Suppliers usually have a good knowledge of the price elasticities of their products and should incorporate this into the design of the corresponding trigger.



**Source:** Plassmann H, O'Doherty J, Shiv B, Rangel A (2008) Marketing actions can modulate neural representations of experienced pleasantness. Proceedings of the National Academy of Sciences 105(3):1050–1054.

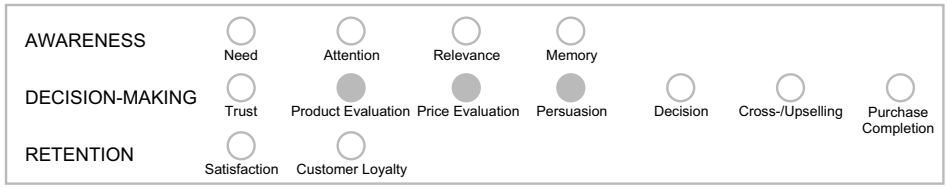
**See also:** Framing, Anchoring, External Reference.

**My Notes**

### 5.2.13 Time vs. Money Effect

The Time vs. Money Effect, which is still relatively young, describes how people make purchase decisions and subsequent satisfaction dependent on whether a product has been advertised with time- or price-oriented arguments. Products mainly defined by time-related variables perform significantly better than products with price-related classification. This finding can be explained as follows: Time intensifies the thoughts of the positive use of the product and creates an emotional bond between the customer and the product. On the other hand, money focuses on the transactional part of the purchase decision and emphasizes payment (and the pain associated with it; see: Pain-of-Paying Principle). The effect is also powerful because time (just like money) is not only a scarce resource but also far less replaceable than money – once invested, time is lost without any possibility of recovering it.

But be careful: In the case of products whose benefits are mainly defined by possession (e.g., in the form of increased social status or distinctly materialistic groups), the effect is precisely the opposite.




**Implementation in e-commerce**

- The effect offers the opportunity to stimulate preferences beyond financial arguments and can be an essential support on the way out of the price pressure trap. So, instead of relying on financial benefits (e.g., discounts), the time saved by using a product can be used as a convincing argument in marketing.
- The same applies to value propositions such as "hours of fun with this product": Everyone evaluates their free time and entertaining activities differently and thus comes to a different willingness to pay. However, this is usually averaged over universal criteria expressed via the price.
- On a deeper level, the effect can also be used to support the "fast lanes" of the conversion (e.g., express check-out or 1-click order).




**Source:** Mogilner C, Aaker J (2009) The time vs. money effect: Shifting product attitudes and decisions through personal connection. Journal of Consumer Research 36(2):277–291.

**See also:** Framing, Pain-of-Paying Principle.

**My Notes**

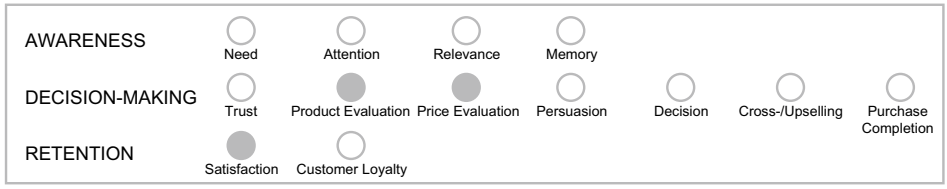
### 5.2.14 WYSIATI Effect

Also: Presence Effect

When making a decision, people only consider the information provided or available and, therefore, often come to hasty or incorrect conclusions due to inaccurate or incomplete data (WYSIATI = "What You See Is All There Is"). There are also cases where we consciously or unconsciously only retrieve some of the available information. This phenomenon is called the illusion of attention: we think we are getting everything when we look at everything in our field of vision. But in fact, we only see what we focus on. Attention is no longer enough for unexpected events. In other words, what is there (presence) can be evaluated much more quickly than what is not (absence). Example: If we are healthy, we think it is unlikely that we can also get sick.

System 1 is entirely insensitive to the quality and quantity of information; a critical examination does not occur. This can have extreme consequences: People allow information to flow into their decision unfiltered even if the low reliability of the source is explicitly pointed out.

The WYSIATI effect can explain mechanisms such as Framing, Overconfidence, and Base Rate Fallacy (i.e., the suppression of all information about the frequency of a trait within the population).




**Implementation in e-commerce**

- There is no explicit to-do for this since the effect always works based on available information, and its effect is not linked to clear triggers.
- Nevertheless, to reduce return rates, it is recommended to critically examine whether users can always make meaningful decisions based on core target groups/personas.




**Source:** Kahneman D (2011) Fast thinking, slow thinking. Siedler Verlag, Munich.

**See also:** Framing, Inattentional Blindness Effect, Base Rate Fallacy, Availability Heuristic, Salience, Functional Fixation.

**My Notes**

## 5.3 Price Evaluation

### 5.3.1 Anchoring

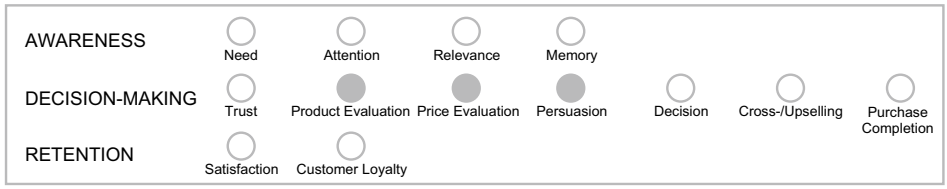
Also: Contrast Principle

Anchoring can be well explained figuratively: When a ship has dropped anchor (= "anchor"), it can no longer move far away from this point and stops in the anchor chain's radius. Price anchors work similarly: Once a numerical value has been perceived (regardless of whether it makes sense or is helpful), it is always used as a reference value when processing a price. This comparison distorts price perception in the direction of the anchor value.

Empirically, the effect was proven as follows: The participants in the experiment were asked to spin a manipulated wheel of fortune with 100 sections. Due to the manipulation, it stopped at either 10 or 65. The subjects were then asked whether the percentage of African countries in the UN was above or below the value of the wheel of fortune (i.e., 10 or 65). In the second step, the concrete proportion should be estimated as a number. The result: If the wheel of fortune showed a value of 10, the average estimate was 25%. If, on the other hand, the wheel of fortune stopped at 65, the estimate was an astonishing 45%. As a reference or anchor point, the wheel of fortune has, therefore, significantly influenced the estimate – even though one obviously has nothing to do with the other.

There are various explanations for the mechanism: For example, an anchor can function as a social orientation value, but it can also work due to insufficient adjustment of the estimate (the further one adjusts one's estimate away from the anchor value, the more certain one must be). In addition, some assumptions assume a numerical imprint and an exceptionally high accessibility (see: Availability Heuristic) of the anchor-forming information.

This effect is so strong that neither high financial incentives for an exact estimate (to activate System 2) nor the explicit indication that a bias is likely to occur through the anchor led to better estimates (Chapman & Johnson, 2002). Anchoring is thus one of the "superstars" of behavior patterns.



**Implementation in e-commerce**

- In general, high values (i.e., above the absolute digit level of the price) can be scattered on product pages, e.g., in the number of available products, satisfied customers, logged-in visitors, reachable employees, etc.
- In the context of home or category pages, high-priced premium products should be placed first (instead of special offers and cheap bestsellers).
- When communicating the recommended retail price, the old (canceled) price must first be mapped, followed by the current price to trigger an Anchoring effect.
- In addition to the product, more expensive products can be displayed (e.g., in the cross-selling area). Amazon uses this intensively in the presentation of marketplace offers.
- In addition, a limited dispensing quantity per person can be defined. As a result, customers set their demand close to the maximum amount.
- In the case of contract products, it is sometimes a good idea to display the monthly (instead of annual) price to suggest a low-price level compared to a higher anchor.
- Processing or shipping costs should always be added only after the purchase decision, and the product price in the shopping cart should be displayed more prominent than the shipping costs.
- It sounds paradoxical, but it is effective: the prices of obsolete products should not be lowered but increased. Customers usually do not evaluate prices based on value for money (because the absolute value of products can hardly be determined) but as reference values to each other. Current bestsellers are thus promoted, while a devaluation of the old products (hardly bought anymore) is unproblematic.




**Source:** Tversky A, Kahneman D (1974) Judgment under uncertainty: Heuristics and biases. Science 185(4157):1124–1131.

**See also:** Availability Heuristic, Framing, External Reference, Precision Heuristic, Rule of Hundred, Priming.

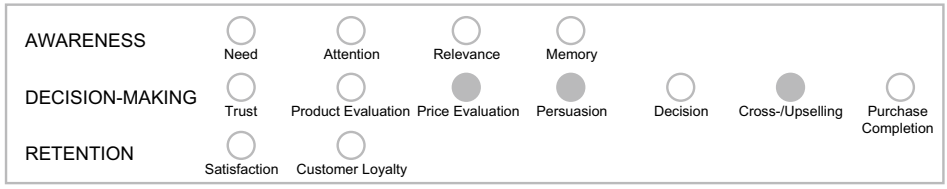
**My Notes**

### 5.3.2 Charm Price Effect

Also: Odd Price Effect, Left-Digit Effect

Fractional prices with the number "9" at the end are perceived as cheaper than smooth or rounded prices. However, this only applies if the first digit of the price also changes during the increase of one cent (e.g., when jumping from $ 1.99 to $ 2.00) and the comparison with an alternative product is strongly influenced by whether the first digit is higher or lower by the value 1 (Thomas & Morwitz, 2005). Investigations of scanner data prove this hypothesis based on actual purchases. However, there is an exciting differentiation here: In the case of hedonistic purchases, a smooth price has proven itself (the decision then feels "round" in the truest sense of the word), while the Charm Price Effect is confirmed in functional purchases (Wadhwa & Zhang, 2015).

There are two primary explanations for this phenomenon. A neuroscientific analysis suggests that our brain processes numerical information from left to right (in reading direction) and that the first digit defines a mental reference value (see: Anchoring and Primacy Effect). On the other hand, from a sociological point of view, it is assumed that we were brought up to associate 99 prices with a favorable price perception.




**Implementation in e-commerce**

- For prices (whether as part of promotions, strike prices, or regular MSRPs), you should use fractional 99 prices if a product is not bought predominantly hedonistically or emotionally.
- In the case of vouchers and discounts, on the other hand, you should round up to whole values, as these should not be perceived as low or the Charm Price Effect is not desired.
- Important: Due to the high follow-up costs of a price change, it should be tested quantitatively before widespread use, as the effect has not been found in all industries, and other factors influence the price-worthiness assessment as well (see: Smart Syllabication).



**Sources:** Bader L, Weinland JD (1932) Do Odd Prices Earn Money? Journal of Retailing 8:102–114; Blattberg RC, Neslin SA (1990) Sales Promotion: Concepts, Methods and Strategies. Englewood Cliffs, Prentice Hall, pp. 349–350.

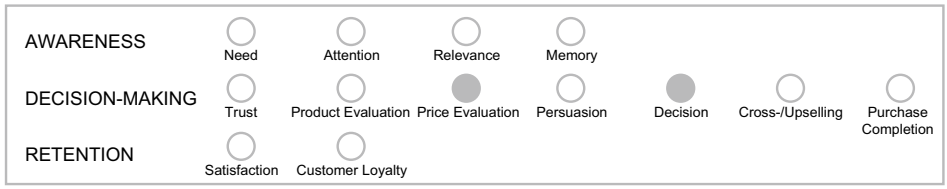
**See also:** Smart Syllabication, Anchoring, Primacy-Effect, House Money Effect, Dollar Eyes Effect, Precision Heuristic, Rule of Hundred.

**My Notes**

### 5.3.3 Dollar Eyes Effect

From a neuroscientific point of view, money creates a strong self-benefit and activates the same brain areas as food, cocaine, or sports cars. This refutes economists who assume that money has only a functional exchange function but does not generate independent utility.

This goes so far that even the sight of money ("dollar eyes") can influence our behavior. Vohs et al. (2006) found that people switched to a more market-oriented decision-making mode in contexts where concrete incentives and meritocracy (or value for money) shape the decision. People mentally shaped in this way behave less socially, build distance, are less willing to be helped, and are less helpful. They increasingly think of categories such as "business" or "profit." On the other hand, the market-oriented decision-making mode also increases the ability to concentrate, motivate, and focus. Recent studies on the Dollar Eyes Effect have shown that even the mere thought of money can be enough to behave unethically or cheat in an experiment (Kouchaki et al., 2013).




**Implementation in e-commerce**

- Classic purchase psychology reinforcers (e.g., discounts, time or quantitative shortages, bonuses, etc.) do not work equally for all target groups. These amplifiers can be used even more effectively with the simple depiction of money-associated motifs (e.g., piggy bank, bundle of bills, stack of coins). Someone who has received such triggers may search directly for coupon codes to improve their "deal" once again. As a provider, you should satisfy this need and, for example, offer a newsletter subscription.
- Without price discounts, however, the depiction of money tends to hurt shop conversion, as users become increasingly stingy and no longer willing to take financial risks. The willingness to help other users altruistically (e.g., by writing reviews or product ratings) is also lowered by depicting money motives without financial incentive.
- Currency signs can also trigger the Dollar Eyes Effect, and their omission can avoid the effect. For example, prices are perceived as cheaper if displayed without currency signs. This means users switch less strongly into market-oriented decision-making mode and behave more intuitively.



**Source:** Vohs KD, Mead NL, Goode MR (2006) The psychological consequences of money. Science 314(5802):1154–1156.

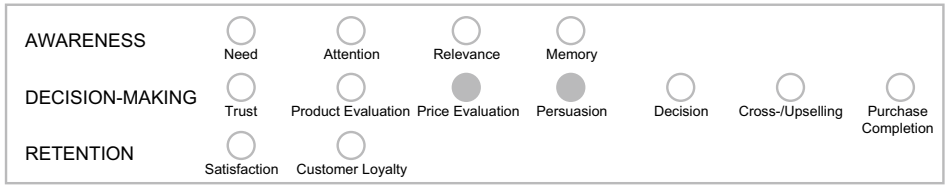
**See also:** Pain-of-Paying Principle, Rule of Hundred, Priming.

**My Notes**

### 5.3.4 External Reference

Also: Adaptation Level Theory, Context

Helson's adaptation level theory is one of the essential reference points of Kahneman and Tversky's later work on anchoring. It states that when evaluating prices, people not only look at the price of the relevant product but also subconsciously incorporate the prices in the environment. As an "external reference," these act as an anchor of judgment. This means that the price is set in relation to the surrounding prices and processed depending on them. The environmental information, together with the price knowledge from previous buying experiences, results in the level of adaptation. Higher prices lead to a higher level of adaptation and a correspondingly higher willingness to pay. The same connection is sometimes described without the price reference as an independent behavior pattern "context."




**Implementation in e-commerce**

- On product overview or category pages, individual high-priced items prove their worth and define the fundamental willingness to pay, especially among customers with low brand preference and product knowledge. This is also achieved, for example, by setting the sorting logic of the products to "sorted by relevance" by default, and this algorithm plays out many high-priced products in the first places.
- This effect can also be used on product pages' cross-selling area. Contrary to the widespread assumption that low-priced items are to be preferred, high external reference prices can be used as anchors. This may not fully exploit the deadweight effect, but it supports the more relevant decision, namely whether a product is bought.



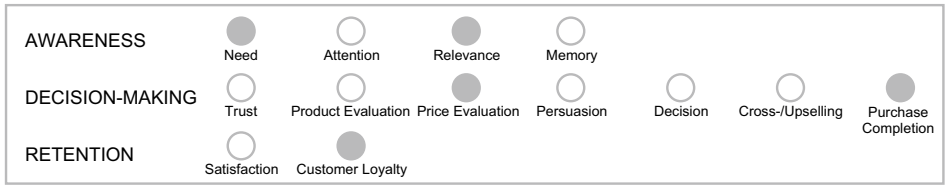

**Source:** Helson H (1964) Adaptation-level theory: an experimental and systematic approach to behavior. Harper and Row, New York.

**See also:** Anchoring, Framing, Magnitude Priming, Rule of Hundred.

**My Notes**

### 5.3.5 House Money Effect

How money (or its equivalent) has been obtained influences how wealth is handled. This can often be observed in the casino: If you win money first, you will be more careless. The money won is considered the "money of the house," hence the name of this behavior pattern. The loss of such foreign money produces incomparably less pain. Similar correlations have been demonstrated in the case of found and inherited money. In the case of speculative or passively obtained profits, there is, therefore, a higher willingness to take risks than in the case of hard-earned money. This can be easily reconciled with the Labor-Love Effect.




**Implementation in e-commerce**

- In online settings, the House Money Effect can be used, for example, in the context of voucher communications ("$50 for free"), which is often used at online casinos or betting providers. This amount is spent without significant reservations and serves as an entry into a customer relationship or lowers the inhibition threshold for a valuable purchase.
- Credit card providers and banks are happy to offer an account already filled with credit as part of their new customer acquisition.
- Customer loyalty programs do the same with virtual currencies and offer first points when joining the loyalty program (e.g., airline loyalty miles).
- Another application idea is aimed primarily at users who plan to abandon the webpage: "Surprising" pop-ups show time-limited vouchers for users, which are often redeemed immediately due to the higher willingness to take risks.




**Source:** Thaler RH, Johnson EJ (1990) Gambling with the house money and trying to break even: The effects of prior outcomes on risky choice. Management Science 36(6):643–660.

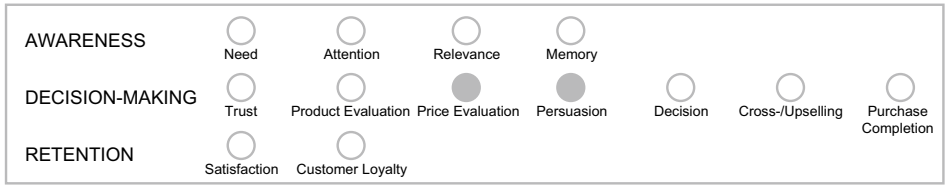
**See also:** Labor Love Effect, Mental Accounting, Loss Aversion.

**My Notes**

### 5.3.6 Magnitude Priming

People are influenced in their judgment by reference values in the environment of the valuation object. We already know this from the Anchoring and External Reference patterns: A high strike price acts as a reference value for the reduced price and makes it appear extremely cheap. The research team led by Daniel Oppenheimer has shown that not only do numerical values trigger an Anchoring effect, but that physical or visual influences also succeed in doing so. So, people cannot free themselves from being shaped by various influences when making decisions.

Numerical expressions are processed and encoded in the brain as size representations, i.e., evaluated depending on their relative size. The experiment showed that subjects who were asked to draw short lines as part of a first task estimated the length of the Mississippi River (or the mean temperature in Hawaii) to be significantly lower than subjects who had drawn long lines. Magnitude priming thus proves visual stimuli have a comparable anchor effect to numerical values.




**Implementation in e-commerce**

- The central use case of this pattern is the representation of prices: If prices are perceived as low, keeping the font size and line thickness of the numerical value low is recommended. A shade of gray instead of a deep black font color can also trigger this effect.
- At first glance, the following implementation of the effect seems unusual: the price should be placed at the bottom left of the page. This placement is almost universally superior in experiments because humans process numerical values as horizontal lines that ascend from left to right (much like a timeline). The lower left value is thus always perceived as the lower end of a fictitious scale.
- In the case of reductions, the strike prices (e.g., recommended retail prices) should be displayed correspondingly more prominent than new prices. The basic prerequisite for implementing any Anchoring Effect is that the anchor is perceived before the numerical value whose perception is to be influenced.



**Source:** Oppenheimer DM, LeBoeuf RA, Brewer NT (2008) Anchors aweigh: A demonstration of cross-modality anchoring and magnitude priming. Cognition 106(1):13–26.

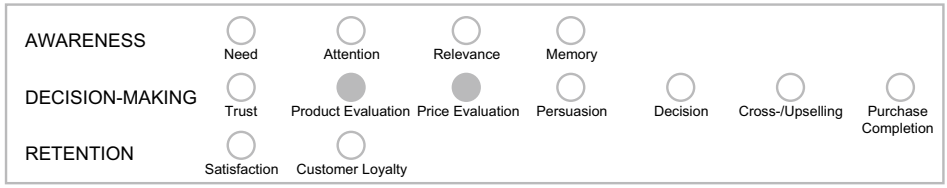
**See also:** Anchoring, Priming, Smart Syllabication, External Reference, Precision Heuristic.

**My Notes**

### 5.3.7 Money Illusion

The Money Illusion describes the cognitive error of non-perception of inflation. In their decisions, people are under the illusion that money always has the same value. In the case of long-term investment products or asset-forming insurance products, this can entail a massive risk in the form of currency devaluation. Even when it comes to real estate financing, people usually do not start from their actual income but from their numerical income and thus underestimate the effect of inflation.

The Money Illusion also applies in a second related context: people perceive amounts of money according to their absolute amount. As a result of the changeover to the Euro shortly after the turn of the millennium in many European countries, people in the euro area subjectively felt poorer than before. The reason for this is that the nominal value of the euro is lower than that of most of the former currencies, such as the Deutschmark, the Franc, or the Lira. Inflation was massively overestimated in this course: instead of the actual 2%, it was perceived as 9% on average (Wertebroch et al. 2007). This finding puts parts of the research on the Money Illusion into perspective and must be considered when using the pattern.




**Implementation in e-commerce**

- Costs should be presented as a percentage, while revenues should be presented as absolute amounts. For example, the administrative costs for a pension contract can be communicated as "effective 2%", which sounds much more bearable than expressing the costs over the term of 35 years in absolute terms ("costs over the term: approx. $100,000").
- The pattern is particularly effective when concluding contract products, e.g., electricity, gas, telecommunications, pay-TV, streaming services, insurance, financial services, and banking products.



**Sources:** Simon HA (1986) Rationality in psychology and economics. Journal of Business 59(4):209–224; Shafir E, Diamond P, Tversky A (1997) Money illusion. Quarterly Journal of Economics 112(2):341–374.

**See also:** Hyperbolic Discounting, Time Inconsistency.

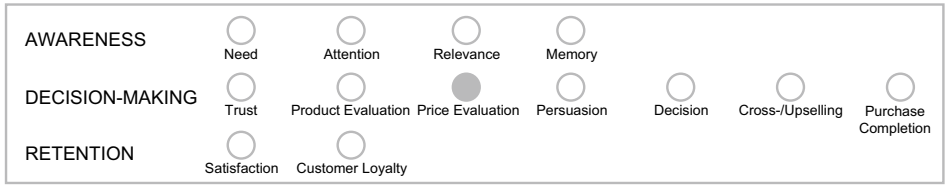
**My Notes**

### 5.3.8 Pennies-a-Day Effect

Also: Temporal reframing.

Prices represent the losing side of any trade and create latent pain in clients. As a result, providers of contract products (e.g., insurance companies, banks, telecommunications, energy suppliers, and software providers) have, in some cases, switched to showing prices not on the usual market metrics but on a smaller basis. In concrete terms, a low daily price is communicated instead of a high annual price – i.e., a considerably lower numerical value. Strictly speaking, this contradicts the prognosis of Kahnemann and Tversky's Prospect Theory, which suggests that we aggregate losses, i.e., combine them into a value – true to the motto: I'd rather have a lot of pain once than small pinpricks every day.

Nevertheless, the approach enjoys some popularity in practice: If you get a mobile data plan bundled with a smartphone at AT&T, the extra costs for the phone will be calculated per month instead of the total cost. Similarly, meal kit delivery services such as BlueApron or Hello Fresh display prices per serving instead of the monthly subscription fee. Scientifically, the "pennies-a-day" strategy has also proven to be effective in various experiments (Gouldner, 1960). In concrete terms, an iPhone 6 s was perceived as cheaper in six out of seven image dimensions if the price was shown at 0.71 $ / day (over 730 days) instead of 519 $. Apparently, the reframing of the price presentation can relativize the high perceived price level of Apple products. We remember: People can hardly imagine prices in the future (see: Hyperbolic Discounting), so adding up a small amount over a long period remains abstract. On the other hand, the daily equivalent of mobile phone use as a constant companion is present, so the effect is hardly surprising.




**Implementation in e-commerce**

- Prices of term products should be presented as a monthly payment amount and not as a sum over the entire term. The experiment shows a deterioration in price perception due to such term prices of about 15%, which is quite significant. However, this corresponds to the current industry standard and is only handled differently by a few providers (mostly on a test basis).
- The decomposition of the price with the pennies-a-day strategy is a good idea when the daily benefit of a product is perceived as very high – as in the example of the iPhone mentioned above. For products we do not use daily (i.e., the daily benefit is regularly zero),

this strategy can have a negative effect because the cost–benefit ratio is then perceived as poor.

- If the price presentation breaks with the customers' expectations, the focus of the decision is naturally strongly directed towards the price as a criterion. If, on closer inspection, this proves to be unmarketable, the strategy is not advisable.
- In conclusion, the effect has been shown to improve price perception in some cases but should always be closely examined using appropriate measurement methods.




**Sources:** Gourville JT (1998) Pennies-a-day: The effect of temporal reframing on transaction evaluation. Journal of Consumer Research 24(4):395–408.

**See also:** Loss Aversion, Precision Heuristic, Zero Price Effect, Hyperbolic Discounting, Disrupt-then-Reframe, Precision Heuristic.

**My Notes**

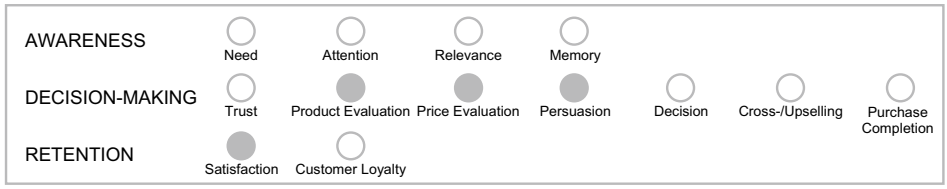
### 5.3.9 Precision Heuristic

The Precision Heuristic is almost a classic of price distortions, which we encounter in many facets in research and practical everyday life. This effect is also based on the core insight that our product and price evaluation also depends on the specific price level, which we can intuitively agree with the product's perceived value. In concrete terms, the more precise a price appears, the more willing we are to pay it.

In a large-scale study of 27,000 real estate transactions, Thomas, Simon, and Kadiyali (2007) found that people paid significantly more for a house when the price was crooked or supposedly precise (e.g., $325,425 vs. $325,000). They explained this connection with two mutually reinforcing phenomena:

- With crooked prices, we are confronted with lower digits, which our brain uses individually as low price indicators. Consequently, the Precision Heuristic works particularly well in addition to the Charm Price Effect if the numbers 1, 2, or 3 are used instead of "*0.99".
- We interpret crooked prices as a signal that the seller has derived the price very precisely and conclusively (and thus rationally). From this, we implicitly conclude that there is little room for negotiation and are inclined to accept the price without further ado.

What is exciting about the connection is that it was detected in the experiment almost independently of the absolute level of the respective prices. So, whether we want to sell a house or a liter of milk plays only a subordinate role. At the same time, the effectiveness is even stronger in areas where negotiations are common (such as buying real estate) because of the second lever (and because these tend to be purchases we make less frequently – i.e., we have less price knowledge).




**Implementation in e-commerce**

- Whether dominant-fractional prices with a 0.99 extension (see: Charm Price Effect) bring the highest selling leverage should be investigated experimentally with A/B testing. The Precision Heuristic may help to make price communication even more effective.
- Especially in the context of time-limited offers, precise prices are attractive. This circumstance can also be used in communication to reinforce the effect on the verbal level ("as low as we can go").

- "Precise" prices also result when all price components are integrated: Adding processing and shipping costs to the shopping cart value and showing them as an amount almost naturally results in a crooked amount. By the way: In the context of Prospect Theory, this is also perceived as particularly bearable and painless (integration of losses; see Loss Aversion).
- It has not yet been conclusively clarified whether the Precision Heuristic can also be used effectively outside of price perception – e.g., in the case of quantities on a package ("1,012 g" content vs. "1 kg"). This can create an exciting field for new A/B test cases.
- In the context of emotional purchasing decisions, in contrast to the core hypothesis of the Precision Heuristic, it can pay off to bet on round prices to bring as little attention as possible to the price and not disturb the user's flow moment.




**Sources:** Thomas M, Simon DH, Kadiyali V (2007) Do consumers perceive precise prices to be lower than round prices? Evidence from laboratory and market data. Evidence from Laboratory and Market Data (September 2007). Johnson School at Cornell University Research Paper 9(7); Mason MF, Lee AJ, Wiley EA, Ames DR (2013) Precise offers are potent anchors: Conciliatory counteroffers and attributions of knowledge in negotiations. Journal of Experimental Social Psychology 49(4):759–763.

**See also:** Magnitude Priming, Charm Price Effect, Anchoring, Loss Aversion, Pennies-a–day Effect, Mayfly Effect.

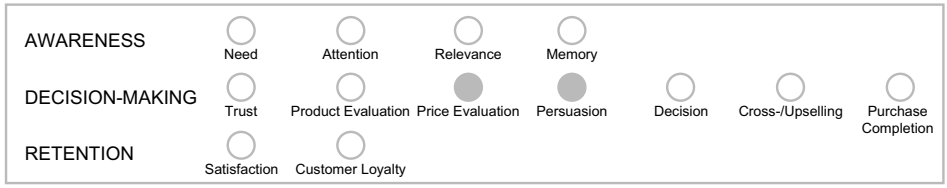
**My Notes**

### 5.3.10 Rule of Hundred

We are bombarded daily by discount offers: Save VAT at the furniture store, get a $ 10.00 discount with the supermarket's customer card, and get 20% off everything from the hardware store. This form of price communication should be tested, but the first thing to do is to understand why some discount campaigns work and others don't.

If a t-shirt is reduced from $20.00 to $15.00 and advertised with "$5 off", the additional demand effect is usually low and can even negatively affect sales (as a product of price and quantity). However, if a "25% discount" is granted on the product, we often see much stronger effects in practice. The final price of both promotions is identical, so why is there this difference?

First, we must note that percentage discounts are not generally more effective than absolute savings. On the contrary, there is a concrete threshold at which the effects are reversed. For amounts below $100, a percentage discount has a stronger effect, while for amounts over $100, an absolute discount. This is the statement of the "Rule of Hundred".




**Implementation in e-commerce**

- In short: If product prices are below $100, discounts should always be expressed as a percentage. For more expensive products, on the other hand, it is an absolute amount ("Save $500 on this stereo for $1,500" instead of "Save 33%").
- The percentage disclosure of a discount also works so well for small amounts because the cognitive effort to calculate the "real" savings (in dollars) is shied away from. Some retailers are experimenting with an even more complex discount structure (e.g., percentage discounts on individual components of a product bundle). It is essential to ensure that the focus is not on concealing the actual prices.




**Source:** Berger J (2013) Contagious: Why Things Catch On. Simon & Schuster, New York.

**See also:** Anchoring, External Reference, Framing, Dollar Eyes Effect, Charm Price Effect.

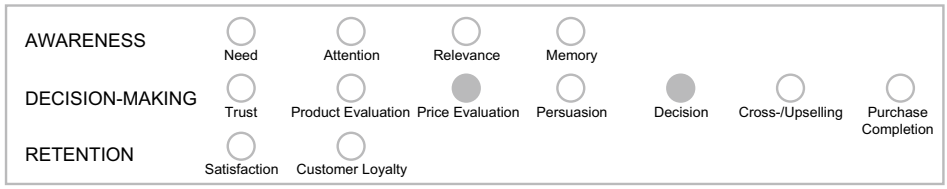
**My Notes**

### 5.3.11 Smart Syllabication

The number of syllables of a prize influences the perception of the price level. The more syllables a prize has, the higher it is perceived. A price of $7.79 ("seven seventy-nine" = six syllables) would, therefore, be classified as more expensive than a price of $8.12 ("eight twelve" = two syllables), although it is cheaper.

The explanatory approach of Smart Syllabication is that the brain processes the numerical information in an auditory-linguistic format. Visual stimuli are translated and "read aloud" in the brain. The more syllables a price has, the longer the cerebral processing procedure takes, or the more resources are consumed. The amount of resources consumed is then used as a benchmark for evaluating the price level.

The same effect occurs (but more through visual processing) when the thousands separator is removed at a four-digit price.




**Implementation in e-commerce**

- First, test: The discovery of the effect is still relatively recent and has not yet been clearly proven outside an experimental setting. Thus, the effect is a promising basis for an A/B testing hypothesis but not yet sufficient evidence for a change in the pricing structure, which usually entails high administrative follow-up costs.
- A relatively risk-free application is the removal of the thousand's separator at all prices.
- In several of our A/B tests in the context of financial products, the effect could not be clearly isolated. There is a high probability of interaction effects with other price-related behavior patterns, such as the Charm Price Effect, which requires further investigation.




**Source:** Coulter KS, Choi P, Monroe KB (2012) Comma N'cents in pricing: The effects of auditory representation encoding on price magnitude perceptions. Journal of Consumer Psychology 22(3):395–407.

**See also:** Charm Price Effect, House Money Effect, Money Illusion, Dollar Eyes Effect, Anchoring.

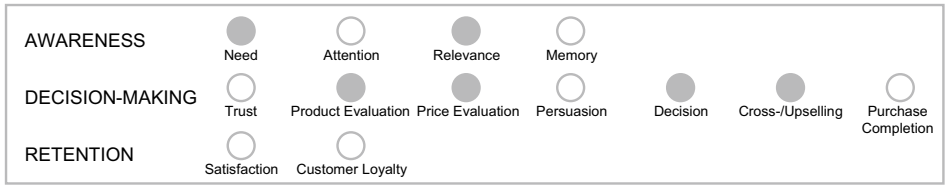
**My Notes**

### 5.3.12 Zero Price Effect

If a product is free, it has a significantly higher attractiveness than if it was only minimally more expensive – this is what the Zero Price Effect teaches us. The effect was first described by Daniel Ariely and colleagues (2007). In the first part of the experiment, subjects were given a choice between a (low-quality) chocolate for 1 cent and a (high-quality) chocolate for 14 cents. 8% chose expensive chocolate, 30% chose cheap chocolate, and 62% chose no chocolate at all. In the second part of the experiment, a distinction was made between the low-quality chocolate for 0 cents and the high-quality chocolate for 13 cents. The result: 31% chose the free one, 13% the expensive one, and 56% resisted the temptation. So, it turned out that many people were willing to deviate from their actual preference for a free product and switch to an inferior alternative.

The fact that the absolute savings – i.e., the rational argument – were identical in both cases is irrelevant to the observed decision. It is important to note that even if only a single component of a larger bundle is available free of charge, the effect affects the entire bundle.

Recent studies have shown that the Zero Price Effect is particularly effective in hedonic goods, i.e., things that bring us joy or pleasure. The explanation for the effect goes back to Affect Heuristics: options without disadvantages trigger a stronger affective response. Conversely, stress is reduced when you don't have to deal with costs when deciding. A decision without risk does not have to be rationally questioned, i.e., it takes place entirely within the intuitive decision-making system and accordingly saves cognitive resources, which the brain considers fundamentally beneficial.




**Implementation in e-commerce**

- Freemium models: If a product is available for a limited period for 0 dollars, users are much more willing to engage in a test. This applies, in particular, to contractual products or software.
- Detach individual components from the total price and show them separately for 0 dollars, e.g., price batteries in the electronic product with 0 dollars or highlight free shipping.
- Customer loyalty programs: Virtual currencies can also trigger the effect. If free points are credited for each purchase, the free component increases the attractiveness of the overall bundle.

- Free bonuses or additional products: In addition to virtual bonuses such as loyalty points, free physical supplements such as additional products also work. These can be presented as gifts, thus triggering additional reciprocity among customers.
- A reverse effect is possible: undesirable or environmentally harmful behavior can be sanctioned by charging prices as "penalties." For example, the number of plastic bags issued has fallen massively since a small amount (usually 5 – 10 cents) is charged for them.




**Source:** Shampanier K, Mazar N, Ariely, D (2007) Zero as a special price: The true value of free products. Marketing Science 26(6):742–757.

**See also:** Affect Heuristics, Foot-in-the-Door Technique, Pain-of-Paying Principle, Reciprocity, Pennies-a-day Effect.

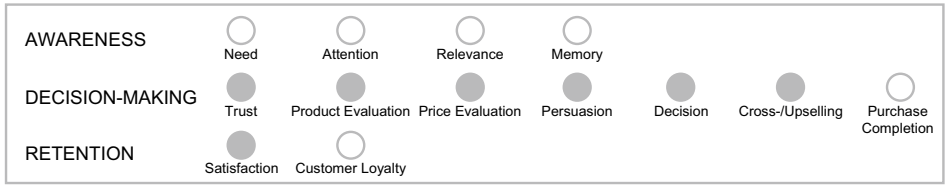
**My Notes**

### 5.3.13 Zero-Risk Bias

We are not masters at realistically estimating probabilities; we have already seen this in various places. Another example of Zero-Risk Bias: we tend to pay significantly more (or overestimate the benefits) for the complete elimination of a risk than for an identical (relatively speaking) risk reduction with a terminal value that does not correspond to zero. This is illustrated by a well-known experiment conducted by Viscusi and colleagues in 1987 (Ueberweg, 1868): They interviewed customers of a hardware store after their purchase and found that reducing the health risk of potentially dangerous products (specifically toilet cleaners and insecticides) significantly increased the willingness to pay. Surprisingly, however, this effect was not linear: the reduction of the risk from 10 to 5% had a considerably weaker effect (toilet cleaner: +19 cents, insecticide: +54 cents willingness to pay) than the reduction of the risk from 5 to 0% (toilet cleaner: +83 cents, insecticide: +571 cents).

We try to focus strongly on decision-relevant aspects. You don't have to consider the possible negative consequences of a zero-risk product. This saves cognitive resources and allows faster decisions – both desirable from an evolutionary point of view, even if we buy this effect at a high price.

Zero-Risk Bias is particularly common when people are concerned with health, safety, and the environment. This is probably familiar to each of us from the inner urge to reduce the deductible for an insurance policy to zero - whatever the cost. The otherwise hardly explainable success of bonds can also be traced with the effect: Although there are very few potential returns, the risk is also close to zero – the loss aversion (see: Loss Version) sends its regards.




**Implementation in e-commerce**

- In e-commerce, we are always dependent on users perceiving the offers as low-risk to be considered as a provider at all. However, the explicit statement "You don't take any risks" has additional power here, relieving customers of the strenuous evaluation of the risk associated with the purchase.
- This can be underpinned visually (see: Picture Superiority Effect) with seals of approval, guarantees, and customer reviews.
- In addition, customers may experience several seasonal risks, which should be explicitly addressed (see: Inner Dialogue). This can be the shipment of heat-sensitive goods in

summer ("Will my goods arrive intact?"), or an order shortly before school enrollment, Easter, or Christmas ("Will my goods arrive on time?").

- In addition, perceived risks can be reduced functionally. Examples are payment after receipt of the goods ("Buy now, pay later"). Amazon now also offers its customers the option of paying via a monthly statement. IKEA's experiment with a lifetime return guarantee, which could not be realized financially, is undoubtedly extreme.
- Size guides and meaningful product descriptions and images also reduce the perceived risk – although never to zero.
- In some cases, digital companies also use Zero-Risk Bias as part of their customer loyalty programs: Regular customers can benefit from extended return periods or special insurance policies (e.g., always insured shipping, reimbursement in the event of illness).
- It is no coincidence that warranty extensions are a best-seller in cross-selling. Especially with expensive electronic products, which are always tainted with the flaw of a risk of failure, this is an effective way to increase the size of the shopping cart.




**Source:** Viscusi W, Magat W, Huber J (1987) An Investigation of the Rationality of Consumer Valuations of Multiple Health Risks. The RAND Journal of Economics 18(4):465–479.

**See also:** Zeigarnik Effect, Focusing Effect, Loss Aversion, Ambiguity Aversion.

**My Notes**

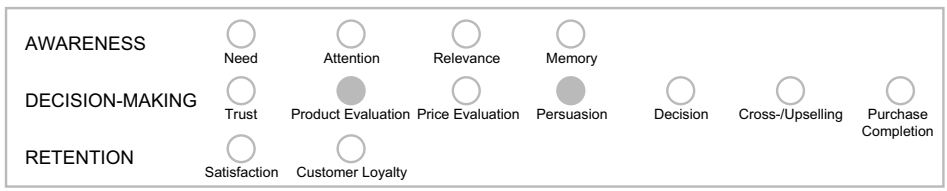
## 5.4 Persuasion

### 5.4.1 Base Rate Fallacy

Also: Prevalence Error

We often make our decisions based on poor consideration of base rates, i.e., we estimate probabilities detached from the number of occurrences of a trait in the population. What sounds abstract is illustrated by Tversky and Kahneman with a simple example: In a city with 85% green taxis and 15% blue taxis, an accident happens. A witness then sees a blue taxi fleeing. Its reliability is stated by the police at 80%. What is the probability that a blue taxi was actually involved in the accident? Most of the participants in this experiment gave this probability as 80% – ignoring the base rate. In fact, the problem can be solved with the Bayesian formula: For example, the witness will recognize the color incorrectly in 20%, i.e., identify 17 as blue in 85 green taxis. Conversely, out of 15 blue taxis, he will perceive three as green. In another twelve cases, he correctly recognizes a taxi as blue. This means the witness will see a blue taxi $17 + 12 = 29$ times but will only be correct in 12 cases. The answer to the question of the probability of a correct statement is, therefore, only $12/29 = 41\%$.

In the case of the Base Rate Fallacy, System 1 again seduces us into a quick but wrong assessment. According to the Recency Effect, the latest information is of particular importance. Therefore, the way of describing this problem influences the result (see: Framing).




**Implementation in e-commerce**

- The fact that the base rate is not sufficiently perceived and processed can be taken advantage of in the presentation of statistics by changing the reference value. For example, satisfaction scores or recommendation rates can be specified as "99% of our active customers" instead of "85% of our customers". The higher value shapes the perception, while the lower base rate is wholly neglected as a counter-effect.
- Conversely, in the case of negative correlations: "0% of our active customers have returned the product" works much better than "18% of our customers have returned the product".



**Source:** Tversky A, Kahneman D (1982) Evidential impact of base rates. In: Kahneman D, Slovic P, Tversky A (eds) Judgment under uncertainty. Cambridge University Press, Cambridge, pp. 153–160.

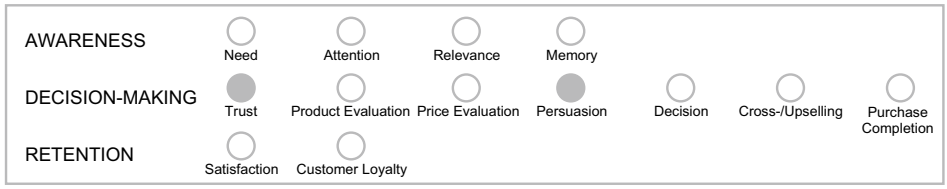
**See also:** Representativeness, Recency Effect, Framing, Barnum Effect.

**My Notes**

### 5.4.2 Belief Bias

Similar to Confirmation Bias, Belief Bias describes the tendency of people to believe rather than doubt information - the sources' authenticity (see: Authenticity) and validity are usually not adequately considered. Arguments are evaluated exclusively based on their plausibility (massively influenced by one's opinion, see: Confirmation Bias) instead of their truthfulness and considered in the decision-making process.

Often, feedback takes place in which the method or source of the information is subsequently classified as reliable if the information fits into one's belief pattern.




**Implementation in e-commerce**

- Despite limited statistical validity, information can be useful in decision-making or persuasion of users. However, the information provided must not be misleading. A vivid example is the presentation of product reviews based on only a few opinions. A 5-star rating based on a customer's opinion has a comparable effect to the same rating based on hundreds of votes. It is recommended to use the Belief Bias only temporarily, for example, when the validity of arguments (e.g., customer reviews) is still under construction.
- It can also be useful to describe how a product helps users in real life. The positive effects are classified as desired, the argument is thus believed more strongly, and the sender (= the shop) is classified as credible.




**Source:** Wilkins MC (1928) The effect of changed material on the ability to do formal syllogistic reasoning. In: J. Winawer (ed.): Archives of Psychology, Vol. 16, No. 102.

**See also:** Confirmation Bias, False Consensus Effect, Availability Heuristic.

**My Notes**

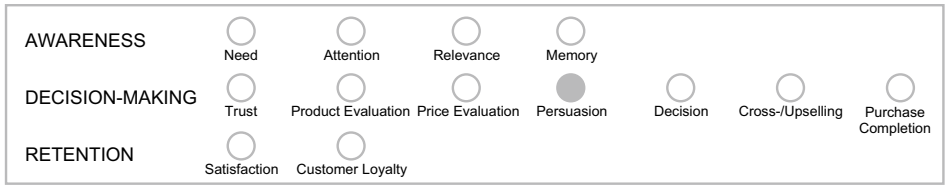
### 5.4.3 Black-and-White Fallacy

Also: False Dilemma, False Dichotomy, Polarized Thinking

If the space of possible solutions is to be described, only the extreme (e.g., the best and worst) options are mentioned. Grey areas do not exist or are systematically underestimated. This can be explained, for example, with the Availability Heuristic (see: Availability Heuristic): We try to describe the solution space with as little cognitive effort as possible and fill it with options. The extreme positions receive far more attention in the media than the "normal" positions in between, which means that we can think of an example for them more and more quickly.

For example, "In the U.S., you're either a Democrat or a Republican. If you're not a Democrat, you have to be a Republican." The conclusion seems logical and convincing – but it is not. Finally, the statement creates an artificial dichotomy of options that ignores the fact that one can also be a liberal, anarchist, socialist, or completely apolitical.

We often use the pattern as a figure of speech to generate persuasion and decisiveness: "If you're not with us, you're against us," and "If you're not a winner, you're a loser." Differentiated neutral positions are cognitively more difficult to grasp and are not modeled as options.




**Implementation in e-commerce**

- Various insurance products are often sold with a combination of the Threat and Black-and-White Fallacy patterns: For example, long-term care insurance is sometimes launched on the market with the message "It affects every second person in your country." The implicit conclusion is: "If I can't rule it out for myself (due to the high probability), it will probably hit me."
- Product positioning: By focusing on extreme positions, a product portfolio does not appear complete until the extreme positions are covered or at least their names sound like extreme positions (e.g., economy and premium gold). If this is the case, there is no longer any need to research elsewhere because the current provider seems to offer all relevant products.
- Preference for a product can be generated in two steps: 1) presentation of two very different options, A and B, of which A should be chosen by the user; 2) Explanation of the negative consequences of the other option B, so that in the case of a supposedly logically binary solution space (A or B), only one solution A remains, which seems very

plausible. In practice, option A could, for example, be purchasing a particular car, and option B could be the prospect of living unhappily and with a low social status.




**Sources:** Ueberweg F (1868) System of Logic and History of Logical Teachings. Adolph Marcus: Bonn (3rd ed.); Solove DJ (2011) Nothing to Hide: The False Tradeoff Between Privacy and Security. Yale University Press: New Haven.

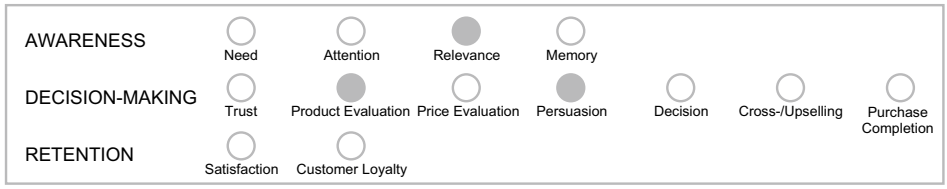
**See also:** Hobson's +1 Choice Effect, Threat, Availability Heuristic.

**My Notes**

### 5.4.4 Disrupt-then-Reframe

The disrupt-then-reframe pattern could be summed up as "confuse first, then explain." The idea: The conscious mind (System 2) is occupied with confusing and unexpected content to clear the way to the subconscious mind (System 1). In doing so, cognitive resources are specifically claimed, and a load or overload of System 2 is brought about. With this (sometimes questionably used) technology, it is unable to recognize the clumsy attempt to influence and leaves the control system to System 1. The prerequisite for the effect is the triad of DISRUPT (confusion: "This candle costs 300 cents"), REFRAME (explanation: "That's $3"), and SUGGESTION (suggestive classification: "This is an excellent offer").

An exciting experiment was carried out at Stanford University in the 1990s: The test subjects had to memorize different numbers of digits – sometimes two, sometimes seven. Then, as if by chance, they were led past a buffet with fruit and pie. The participants were much more likely to reach for the cake if they had to remember many digits. The explanation of the scientists: If the mind is busy, the impulsive behavior is not controlled, and individuals are more likely to give in to one's ill-considered impulses.




**Implementation in e-commerce**

- When presenting the product, an application would be to first show complex technical gibberish (DISRUPT), then explain the resulting advantages for customers (REFRAME), and then give a recommendatory rating (SUGGESTION).
- Differentiation from the competition: "That's how complex the others handle it (DISRUPT), that's how simple we handle it (REFRAME), that's why you should buy from us (SUGGESTION)." Start-ups with disruptive aspirations, such as digital insurers and insurtechs, like to position themselves according to this principle to differentiate them from traditional insurance companies.
- The initial confusion (DISRUPT) can also be used with surprising linguistic variations, which is useful in e-commerce. In one experiment, for example, cupcakes were referred to as "halfcakes"; this unusual and counterintuitive designation was already sufficient to create openness to the explanation (REFRAME) among customers.
- The disrupt-then-reframe technique is one of the strongest (the average uplift of all published studies is about 100%) but also one of the riskiest. The spam industry has massively used it for years and has a clear negative connotation regarding seriousness among customers. To avoid creating a negative impact, the approach must be exceptionally well thought through and tested.




**Source:** Carpenter CJ, Boster FJ (2009) A meta-analysis of the effectiveness of the disrupt-then-reframe compliance gaining technique. Communication Reports 22(2):55–62.

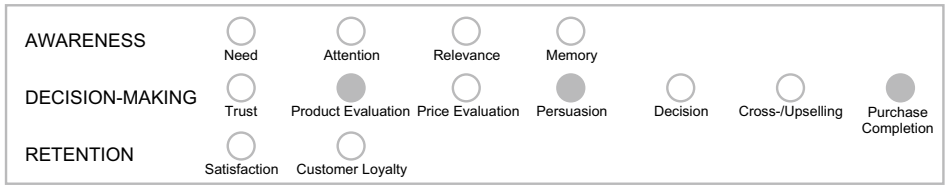
**See also:** Framing, Inattentional Blindness Effect, Pennies-a-day Effect.

**My Notes**

### 5.4.5 Door-in-the-Face Technique

Instead of preparing a big favor with a small one (Foot-in-the-Door Technique), the door-in-the-face technique goes the opposite: you place a large entrance request, which will likely be rejected. However, this provoked refusal increases the willingness to agree to a subsequent second request (i.e., the desired result). It is essential that the proverbial door must be slammed shut first: If both requests are presented at the same time instead of one after the other, the willingness to agree to the second request decreases significantly.

On the one hand, this can be explained by the latent feeling of guilt of the rejecting party and the perception of the second request as a cooperative concession on the part of the questioner. This triggers the reciprocity pattern and requires consent, which makes it possible to further correspond to one's self-image as a helpful person (see: Commitment and Consistency). On the other hand, the first offer "anchors" the second and acts as a reference value so that the "costs" of the second offer then have a correspondingly lower effect. This is also known as the "contrast principle." At the same time, there is always the risk that the first offer will be perceived as excessively exaggerated and destroy the personal relationship between the negotiating partners. After all, you don't like to slam the door in anyone's face.




**Implementation in e-commerce**

- The principle is used, for example, in promoting "shop the look" approaches. A complete outfit is offered here, which is usually rejected by customers (at least one part does not please, similar shoes are already available, etc.). An offer placed below to buy individual products from the look is then preferred within the technology framework.
- The effect can also be used in marketing magazine subscriptions and contract products by first offering long contract terms for expensive products and provoking a rejection. Subsequently, trial subscriptions or short-term contracts can be marketed more easily, suitable as entry-level products in a long-term customer relationship (companionship).



**Source:** Cialdini RB, Vincent JE, Lewis SK, Catalan J, Wheeler D, Darby BL (1975) Reciprocal concessions procedure for inducing compliance: The door-in-the-face technique. Journal of Personality and Social Psychology 31(2):206–215.

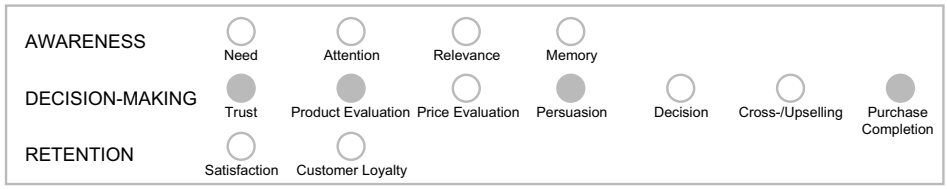
**See also:** Reciprocity, Anchoring, Commitment and Consistency, Foot-in-the-Door Technique.

**My Notes**

### 5.4.6 Evoking Freedom

An offer or request (an offer to sell is, in fact, nothing more than a request to buy a product) is formulated with the addition "But you are free to decide whether you want to buy it or not" or the note "But do not feel obligated." The effectiveness of this technique, which at first glance seems banal, is based on two effects: On the one hand, it anticipates the reactance reaction of the interested parties and avoids a feeling of oppression, to which interested parties would typically react with a defiant reaction (rejection or post-purchase dissonance; see: Buyer's Remorse) to restore their freedom. On the other hand, an increased degree of friendliness, politeness, and submissiveness is documented, which makes the provider appear more sympathetic (see: Liking).

The Evoking Freedom effect has been proven in various contexts (e.g., selling pastries, bumming cigarettes, getting money for a bus from strangers, acquiring participants for studies, etc.). The technique works best in the context of face-to-face interactions but has also been proven in the context of online communication (Guéguen et al., 2013).




**Implementation in e-commerce**

- This technique can be observed at regular intervals in Wikipedia's fundraising activities (see the annotated screenshot in Sec. 3.1). The online dictionary, which relies heavily on voluntary collaboration, thus successfully addresses the altruism of its active users.
- In the context of product presentations, text additions such as "Your free decision" or "Decide for yourself" (e.g., which tariff of ours you choose). The request to make comparisons with competitors also works with Evoking Freedom. The underlying message is: "You can compare us, and then you will find that we are the best."
- In the ordering process, the choice of creating a customer account or ordering as a guest is a sensible implementation. Similar to Hobson's +1 Choice Effect, the decision is shifted – from "Do I really want to order here?" to "Should I create a customer account when ordering or not?".




**Source:** Gueguen N, Pascual A (2000) Evocation of freedom and compliance: The 'but you are free of…' technique. Current Research in Social Psychology 5(18):264–270.

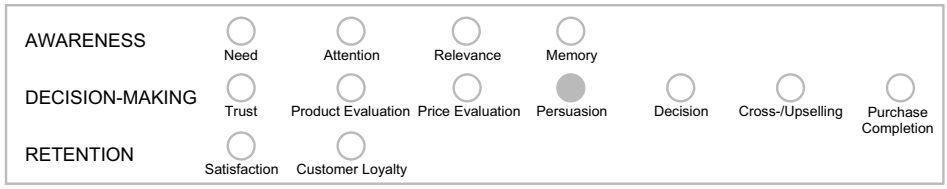
**See also:** Liking, Hobson's +1 Choice Effect, Buyer's Remorse, Reactance, Autonomy Bias.

**My Notes**

### 5.4.7 False Consensus

We tend to overestimate the correspondence of other people's opinions with ours. This is based on the egocentric assumption that everyone thinks, feels, and prefers the same way as we do. This leads to a feedback loop that gives us confidence or strengthens our decision. This decision (which is self-confident and well-founded from the outside) can, in turn, encourage other decision-makers with uncertainty to make the same decision as we do (see: Social Proof).

The phenomenon can be explained by the Confirmation Bias, the Barnum Effect, and the Availability Heuristic: We primarily look for confirmation (not for refutation) and interpret leeway accordingly in our favor or in favor of a clear and dissonance-free decision.




**Implementation in e-commerce**

- The False Consensus effect naturally plays into the hands of e-commerce companies, as users (with a qualified interest in buying) are more likely to look for affirmative elements of their predefined decision anyway. The use of the Social Proof toolbox can reinforce this tendency.
- The effect is particularly strong in people with a high level of individual trust. Using the Trust Bias pattern, work should, therefore, be done on the website's general trustworthiness to trigger the False Consensus effect.




**Source:** Ross L, Greene D, House P (1977) The "false consensus effect": An egocentric bias in social perception and attribution processes. Journal of experimental social psychology 13(3):279–301.

**See also:** Social Proof, Confirmation Bias, Barnum Effect, Availability Heuristic, Trust Bias, Empathy Gap.

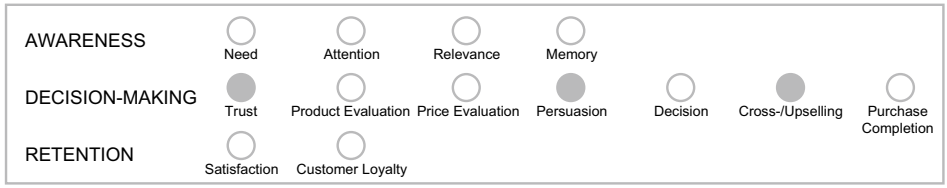
**My Notes**

### 5.4.8 Foot-in-the-Door Technique

Also: Ben Franklin Effect

The Foot-in-the-Door Technique breaks down the effort of persuasion into several small parts: You first ask your counterpart for a small favor that they practically cannot refuse. With the consent, your proverbial foot is in the door, and the actual concern is revealed later. With the acceptance of the first small request, the acceptor creates a self-image of himself and demonstrates composure towards the inquirer. Due to one's striving for consistency (see: Commitment and Consistency), one will most likely agree to further favors. In addition, the fact that you have already trusted someone once and done a favor creates affirmative feedback, which even leads to an increase in sympathy for the profiteer, which in turn is the basis for the following favor. The request does not have to be explicitly formulated but can result implicitly from the context.

The technique is also known as the "Ben Franklin Effect," who described the following apt maxim in his biography: "He that has once done you a kindness will be more ready to do you another, than he whom you yourself have obliged." There is no better way to summarize what psychologists discovered almost 200 years later.




**Implementation in e-commerce**

- The customer journey should be structured as a buying process that leads from broad (non-binding favor) to pointed (binding favor). From the first click on the page to reading free content and providing the e-mail address for the purchase, you can gradually develop the customer relationship from a tiny request to completion.
- The request to use filters to display suitable products (e.g., sneakers in size 10) can also be used as a small request to prepare the later purchase.
- The effect can also be used in a cross-selling context. Instead of "other customers also bought," the note can be formulated as "complete the look" in a more activating way. However, since this request is not always clearly related to the original product, the pattern does not have the same effect.
- It is widespread to find small requests for Facebook likes or a newsletter subscription to prepare big requests (like purchases).
- The technique is used in almost all freemium models: A free trial acts as a small request, and the purchase of the full version as a big one. The same applies to music in the years before streaming came along: The single (small request) is bought quickly, resulting in a higher probability of purchasing the whole album (big request).

- The successful customer development paradigm "Companionship" (the company as a partner of the customer), which is common and successful in e-commerce, is also primarily based on this technique.




**Source:** Freedman JL, Fraser SC (1966) Compliance without pressure: the foot-in-the-door technique. Journal of Personality and Social Psychology 4(2):195–202.

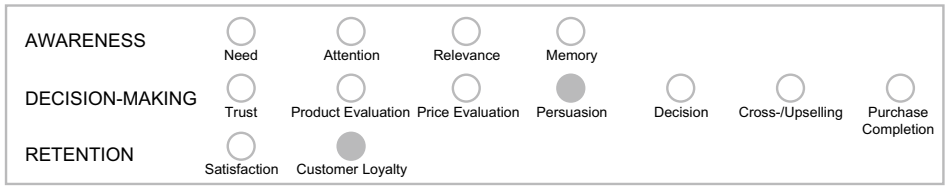
**See also:** Commitment and Consistency, Reciprocity, Foot-in-the-Face Technique, Small Talk Technique, Zero-Price Effect.

**My Notes**

### 5.4.9 Foot-in-the-Face Technique

After demonstrating that both the sequence of small requests/large requests (foot-in-the-door) and the sequence of large requests/small requests (door-in-the-face) can positively influence consent, the scientists wondered about the sequence of multiple medium-sized requests. The psychologist aimed to develop a method of persuasion with less potential for conflict or rejection than the door-in-the-face technique. Surprisingly, Dolinski's experiments also found an increased willingness to participate in situations where two requests of approximately equal complexity were formulated – the foot-in-the-face technique was discovered (Dolinski 2011).

The effect is highest when two to three days between the two requests pass. However, even then, the effect size in most experiments is not as high as in the foot-in-the-door and door-in-the-face techniques.




**Implementation in e-commerce**

- Effectively, large parts of the existing customer marketing toolbox are based on the principle. The willingness to make a follow-up purchase is usually higher than a first purchase. In other words, existing customers convert better than new customers. However, this is not big news; it is common knowledge in e-commerce. Since the technique has proven less effective than the two approaches presented before (foot-in-the-door and door-in-the-face) in most comparative studies (albeit all outside the e-commerce context), it should be used mainly when these alternatives cannot be used, e.g., for image reasons.




**Source:** Dolinski D (2011) A Rock or a Hard Place: The Foot-in-the-Face Technique for Inducing Compliance Without Pressure. Journal of Applied Social Psychology 41(6):1514–1537.

**See also:** Foot-in-the-Door Technique, Door-in-the-Face Technique, Mere Agreement.

**My Notes**

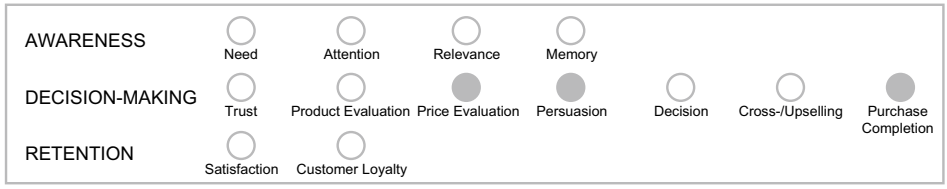
### 5.4.10 Hyperbolic Discounting

Also: Immediacy Effect, Instant Gratification, Present Focus Bias

People do not evaluate events consistently but always depend on temporal factors. For example, if people have the choice between $50 in six months and $60 in seven months, they will most likely opt for $60 in seven months. However, suppose you ask this person again after six months whether they want to get $50 immediately or $60 in one month. In that case, the decision usually favors an immediate reward ($50) – which is why the effect is also called instant gratification. Although the two options are identical and only the timing has been shifted, the preference has changed. Thus, people behave in a time-inconsistent manner (see: Time Inconsistency).

Hyperbolic discounting thus describes the phenomenon that people are impatient in the short term and forego an immediate reward only for a high premium. Conversely, they are patient and charge far lower waiver premiums in the long run. So, mathematically speaking, we discount in the form of a hyperbola – hence the pattern's name.

This explains why many people go on a diet (the future benefits seem high, the renunciation is not yet noticeable), but only a few successfully enforce it (at the moment of confrontation with a reward such as a high-calorie dessert, the benefit no longer seems so high and the renunciation is painful).




**Implementation in e-commerce**

- So, people react much more strongly to immediate rewards than to perspective incentives, even if they are greater. Therefore, mechanisms such as "buy now, pay later" have proven successful.
- Customer loyalty tools can also be supported in this way: "Join the customer club now and secure benefits immediately."
- Fast shipping options (overnight, same-day delivery) shorten the waiting time for the product and also correspond to a timely reward. They should, therefore, be offered, even if they cost significantly more.
- Supply bottlenecks and out-of-stock situations should not be communicated as "currently unavailable" but rather as "more is already on the way; order now." In this way, the temporal focus of the contemplation is shifted from the future, which would be hyperbolically discounted, to the present.

- Postponements of the payment time also rely on hyperbolic discounting (e.g., payment on account with a payment term of 30 days). If the "pain" of payment arises later, it is discounted accordingly in its intensity.




**Source:** Laibson D (1997) Golden eggs and hyperbolic discounting. The Quarterly Journal of Economics 112(2):443–478.

**See also:** Time Inconsistency, Money Illusion, Pennies-a-day Effect.

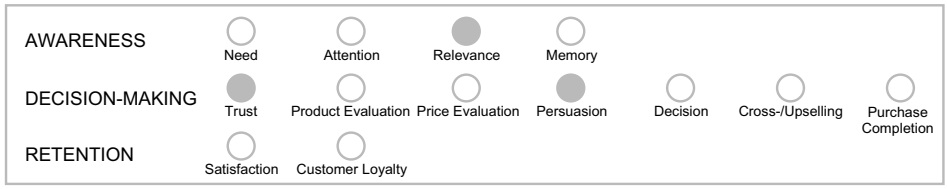
**My Notes**

### 5.4.11 Mere Agreement

Also: Yes Street

When someone asks us several questions in a row, all of which we answer "yes" (however insignificant or obvious they may be), this creates a sense of similarity or rapport to the questioner and makes us increasingly helpful (i.e., more willing to buy in the context of sales). This effect is based on the liking pattern. The building blocks are generally supported statements such as "I think animal cruelty should be banned, what about you?" (Common Sense), but also phrases like "A wonderful day, right?".

The Foot-in-the-Door Technik (or the Small Talk Technique) suggests that questions answered with "no" also have this effect. However, the effect is even higher with a series of "yes" answers (Mere Agreement). This is because in the case of a "Yes Street," the consistency pattern also applies (see: Commitment and Consistency).




**Implementation in e-commerce**

- When introducing the customer, such statements can be placed in the upper third on product pages or newsletters, which customers intuitively agree with. If the decisive call-to-action follows, the sense of similarity resulting from the Mere Agreement effect leads to an increased click-through rate on the button.
- This pattern was developed for the personal sales pitch and works best there. In e-commerce, the closest thing to this real-time interaction is a dialogue with a chat (bot). Here, the customer service agent or chatbot should not get down to business immediately but first place some small talk phrases that the customer can easily agree to.



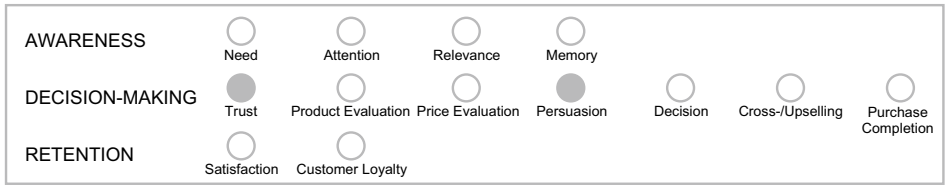

**Source:** Pandelaere M, Briers B, Dewitte S, Warlop L (2010) Better think before agreeing twice: Mere agreement: A similarity-based persuasion mechanism. International Journal of Research in Marketing 27(2):133–141.

**See also:** Foot-in-the-Face Technique, Liking, Unity, Commitment and Consistency, Small Talk Technique.

**My Notes**

### 5.4.12 Mirroring

Mirroring is a persuasion technique in which the other person's behavior is mirrored to appear more convincing. In neuro-linguistic programming (NLP), this is also referred to as "pacing." Just like mirroring, this means aligning body language, facial expressions, voice, language, and strategies. Expressing emotions that reflect the other person's emotional life reinforces a message's persuasiveness. The basic idea of mirroring (also behavioral mimicry) is the artificial construction of a feeling of emotional connectedness (see: Affect Heuristics, Liking).




**Implementation in e-commerce**

- Since direct face-to-face interaction is missing in e-commerce, only parts of mirroring can be used, which weakens the overall effect compared to other application contexts.
- Nevertheless, a typical application can be found in the selection of images: The visual language should correspond to the emotional world of the users. A blatant example: You shouldn't use cheesy vignette photos of happy teenagers on the beach when selling funeral supplies. Ideally, pictures that are slightly more positive than the emotional world of the users (determined with user-centered qualitative methods) are presented.




**Source:** DeSteno D, Petty RE, Rucker DD, Wegener DT, Braverman J (2004) Discrete emotions and persuasion: the role of emotion-induced expectancies. Journal of Personality and Social Psychology 86(1):43–56.

**See also:** Barnum Effect, Authenticity, Affect Heuristic, Liking.

**My Notes**

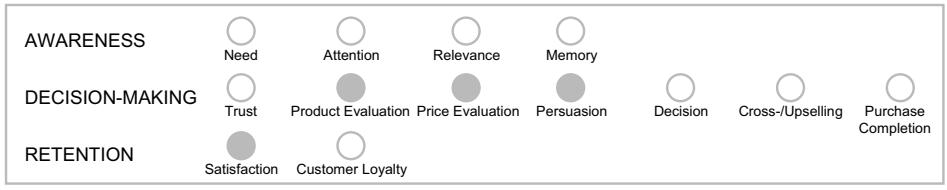
### 5.4.13 Money Omission

Two forms of incentives, monetary and social, function completely differently and cannot be combined. While social incentives are strongly System 1-based (e.g., Reciprocity, Unity, Inequity Aversion), monetary incentives are strongly System 2-based. The latter leads a purchase decision to a contract-like situation in which the rational flow of the decision-making process overrides the vast majority of all behavior patterns. An example by psychologist Bart Schutz: A social paradigm becomes a contractual relationship when a kindergarten decides to introduce a penalty for picking up children too late (e.g., $10 for every quarter of an hour late). Parents arriving late derive from the regulation a right to pick up their children at any time ("After all, I'll pay you for it!") and thus discard socially established behaviors. The punishment would then trigger exactly the opposite of the intended effect: parents would increasingly pick up their children late.

In the original experiment of the discoverers of Money Omission, Heyman and Ariely (2004) asked participants the following question: "Imagine you need help to carry your couch into the car. Which option would you choose?".

1. Just ask someone
2. ask someone AND offer a candy
3. ask someone AND offer a candy AND mention that it's worth $0.50
4. ask someone AND offer $0.50

It turned out that the first two options worked best (there was no significant difference between them). When money comes into play, the subjects behaved less socially or were less helpful. This effect was later replicated in various other situations.




**Implementation in e-commerce**

- When customers receive a gift (e.g., as a bonus), the gift's monetary value should not be put in the foreground – surprisingly. Nevertheless, almost all corresponding marketing activities work this way ("You also receive… worth…!"), which would therefore have to be questioned.
- When it comes to attracting customer reviews, financial incentives have not proven themselves and are hardly found in e-commerce today. Far more effective are social appeals to the altruism and helpfulness of customers. These should be formulated with a reference

to the social values and norms of the customer ("Are you a helpful person?") and in personal form (see: Bystander Effect).




**Sources:** Heyman J, Ariely D (2004) Effort for payment: A tale of two markets. Psychological science 15(11):787–793; Ariely D (2016) Payoff: The Hidden Logic that Shapes Our Motivations. Simon & Schuster, New York.

**See also:** Reciprocity, Commitment and Consistency, Social Proof, Bystander Effect, Unity, Inequity Aversion.

**My Notes**

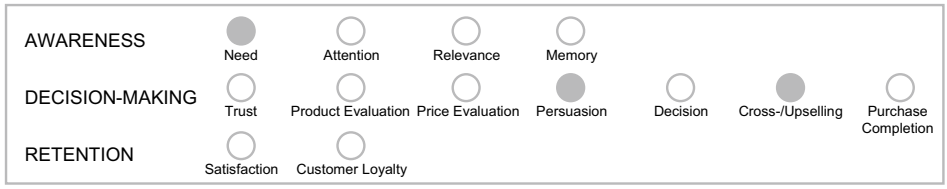
### 5.4.14 Moral Licensing

Also: Self-Licensing, Free Pass, Licensing Effect

If we have done something good recently, we are more inclined to do something bad later. We use morally desirable behaviors as legitimation to act immorally later on. A surprising number of studies confirm what sounds contradictory: Those who shop in organic markets get there by SUV. You can reward yourself for being a vegetarian with a long-distance trip to New Zealand. If you went for groceries for your elderly neighbor, you don't have to let someone cut in line at the checkout. Burgers and fries do not contradict the diet plan if you drink a Coke Light with it.

The explanation for Moral Licensing is as follows: We have a mental morality account. We can top up this account with good deeds. If the account is well-filled, we succumb to the temptation to act selfishly and not to see our own or society's moral principles too narrowly. "Nobody is perfect; you're already doing so many good things," says the voice in our heads. Feelings of guilt or a guilty conscience, therefore, do not arise.

Interestingly, the social weight of good and bad deeds hardly plays a role: We decide impulsively and often dichotomously (is my moral account filled: yes/no) whether we can "indulge" ourselves in a violation of principles or not. Socially, this can be risky because it encourages prejudice, discrimination, selfishness, and waste.




**Implementation in e-commerce**

- Ecologically aware customers (i.e., hopefully, everyone!) can consider whether the order is appropriate and covered by their own moral account, especially in the case of orders with a small value or large volume. This is because the $CO^2$ emissions associated with shipping are particularly striking here. Moral licensing could increase the willingness to complete the purchase, e.g., by emphasizing to existing customers how much $CO^2$ they have already saved with the provider's climate-friendly shipping policy.
- For new customers, it can be decision-making to emphasize how the company is committed to environmental protection or how durable the products in the shopping cart are. Sustainability engagement can, therefore, have a direct effect on conversion.
- Recent research suggests that moral licensing also works when we strongly identify as part of a group: Even the good deeds of our peer group are recorded by our brain in our moral account. This moves the effect in the direction of the Bandwagon Effect. So, you could use the socially desirable activities of other customers to dispel moral concerns about the purchase. Some sustainable fashion companies, for example, show

how many climate protection projects have already been supported with the purchase of other customers.




**Sources:** Mazar N, Zhong CB (2010) Do green products make us better people? Psychological Science 21(4):494–498; Miller DT, Effron DA (2010) Psychological license: When it is needed and how it functions. Advances in Experimental Social Psychology 43:115–155; Mullen E, Monin B (2016) Consistency versus licensing effects of past moral behavior. Annual Review of Psychology 67:363–385.

**See also:** Commitment and Consistency, Mental Accounting, In-Group Bias, Bandwagon Effect.

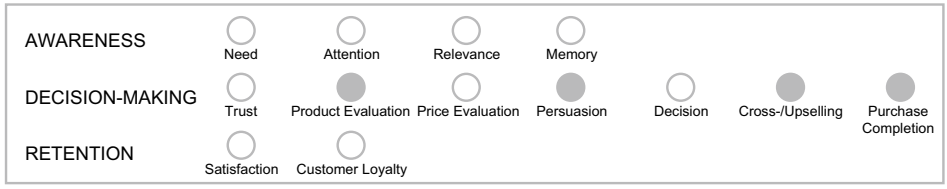
**My Notes**

### 5.4.15 Pseudo Justification

Also: Pseudo Math, Nonsense Math

A request is much more likely to be agreed upon if you justify the necessity. Surprisingly, however, it hardly matters how plausible this justification is. In the experiment by Key et al. (2009), subjects were asked whether they would be allowed to enter the copier. One justification significantly increased the willingness of the respondents. However, there was little difference between the justifications "because I am in a hurry" (plausible justification) and "because I have to make copies" (implausible justification).

The mode of action of pseudo-justification can be described as follows: Our brain hardly questions the meaningfulness of the justification (especially in the case of small requests). It simply registers, "Would you let me go to the copier because blablabla." The heuristic is that in the past, almost everyone has provided a plausible justification after the keyword "because." System 1 assumes this will probably also be the case now. System 2 is only activated if the request is so large that the consent would have substantial negative consequences (e.g., a long waiting time if you let someone cut in who wants to make several thousand copies).




**Implementation in e-commerce**

- The effect is sometimes described as its own pattern named "Pseudo Math" or (not quite apt in terms of content) "Nonsense Math." Accordingly, any statistical or mathematical justification leads to the product being considered relevant and purchased. Curves on unlabeled axes or histograms with distorting sections of the data set are particularly popular. Incidentally, the less mathematical education the users have, the stronger the correlation is in the experiment.
- An important addition: For the effect to occur, the statistics shown do not necessarily have to be nonsensical ("pseudo" or "nonsense"). It's just that, conversely, meaningfulness is not a necessary prerequisite.
- The pattern proves that justifications in the form of advantageous arguments are necessary in e-commerce to trigger purchase intention.
- Call-to-actions should always be accompanied by a justification, even if it does not add value in terms of content ("Click now so you can buy the product" or "Click to go to step 2").




**Sources:** Langer EJ, Blank A, Chanowitz B (1978) The mindlessness of ostensibly thoughtful action: The role of "placebic" information in interpersonal interaction. Journal of Personality and Social Psychology 36(6):635–642; Key MS, Edlund JE, Sagarin BJ, Bizer GY (2009) Individual differences in susceptibility to mindlessness. Personality and Individual Differences 46(3):261–264.

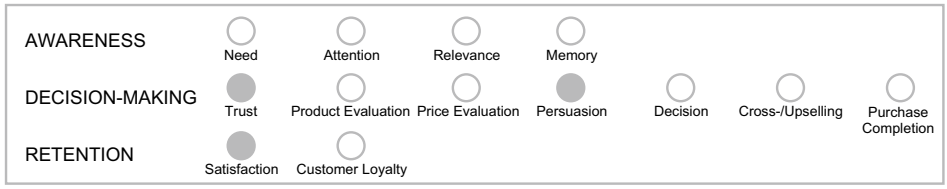
**See also:** Availability Heuristic, Post-Purchase Rationalization, Confirmation Bias.

**My Notes**

### 5.4.16 Reactance

Reactance is a psychological defense strategy used when people are pressured to perform a certain action. It is a reaction to the attempt to limit personal freedom (see Evoking Freedom) and often has extreme manifestations, e.g., the immediate termination of an ordering process. In this way, it acts like a boomerang of non-conformity: To regain our freedom, we represent exactly the position from which we are to be dissuaded or take the exact opposite position. The process occurs largely unconsciously; only the result becomes perceptible, usually as an abrupt rejection.

When working professionally with behavior patterns – usually driven by the goal of motivating people to take a specific action – the reactance pattern is undoubtedly one of the most important. It shows the limits of influencing and underlines that humans are not easily manipulated creatures, even when activating unconscious behavioral patterns, but have evolutionarily proven defense strategies for this purpose. From this, the following imperative can be derived: Reactance can only be avoided if you never lose sight of the customers' needs during conversion optimization.




**Implementation in e-commerce**

- Users should never be offered just one choice. Such pseudo-decisions without freedom of choice create reactance and a trade-off between buying and not buying, as Hobson's +1 Choice Effect has shown. Instead, a second option should always be offered, shifting the trade-off to "buying A vs. buying B."
- Even (moderately) critical customer reviews may be allowed to demonstrate that they are genuine opinions and that no regulating censorship authority monitors the activation of reviews.



**Source:** Brehm JW (1966) A theory of psychological reactance. Academic Press, Oxford.

**See also:** Evoking Freedom, Hobson's +1 Choice Effect, Reverse Psychology, Autonomy Bias.

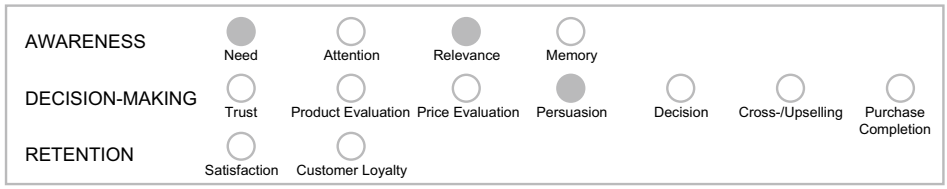
**My Notes**

### 5.4.17 Reverse Psychology

Sometimes, it doesn't help to demand or ask for a certain behavior. Some people feel harassed and pressured by this – they react with the opposite behavior. This reaction pattern uses Reverse Psychology. By telling a person to do the opposite of the desired action, one achieves the behavior they hoped for.

The principle is known, for example, in the dating sector ("You want what you can't have"), in raising children ("Please don't make your bed until I'm there so I can help you"), but also in classical literature (for example in Shakespeare's "Julius Caesar").

The principle is quite strong, but not equally for all people: While people who tend to resist are very susceptible to it, reverse psychology can also have a negative effect: If you accidentally apply it to people who tend to be accommodating, they will perform the required (but not desired) action. People who resist their character increasingly try to maintain their autonomy and not allow themselves to be corrupted. That is why they react with reactance to demand and do the opposite.




**Implementation in e-commerce**

- The application of Reverse Psychology in a digital context is risky and complex. As we've seen, it only works for people with certain personality traits you usually can't identify online. Therefore, caution is always required, and good planning is necessary. Also, in order not to favor manipulative applications, only use cases with borrowings from Reverse Psychology are presented below.
- Reverse Psychology works well in conjunction with scarcity principles (see: Scarcity). For example, a newsletter can start with a request to unsubscribe from the recipient list immediately because not enough products are in stock for all recipients. Positive side effect: Users who unsubscribe had no intention to buy anyway and improve the quality of the recipient list through the cleanup.
- Reverse Psychology also works in combination with the "Curiosity" behavior pattern: a call-to-action labeled "don't click" exerts an almost magical attraction.
- The outdoor brand Patagonia ran ads urging people not to buy a new jacket to conserve resources ("Don't buy this jacket"). The message, which at first glance seemed irritating, had two effects: Patagonia strengthened its brand perception as a sustainable manufacturer and triggered a firm purchase intention among many of its customers (+40% during the campaign period (Seyama & Nagayama, 2007).




**Source:** Fowler S (2014) Why motivating people doesn't work… and what does: the new science of leading, energizing, and engaging. Berrett-Koehler Publishers, San Francisco.

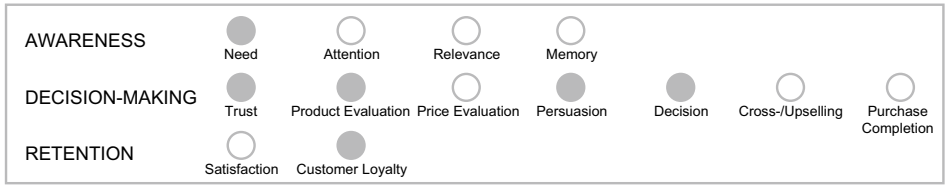
**See also:** Reactance, Equality Attraction, Inequity Aversion, Scarcity, Curiosity.

**My Notes**

### 5.4.18 Reciprocity

Reciprocity is the principle of moral obligation: if someone does something good for us, we feel obliged to return the favor – according to the motto "give and take."

However, this sense of obligation applies both ways. If someone violates the norms of cooperation, we tend to punish them for it. In social structures, reciprocity leads selfish persons to turn to the common cause and stop their uncooperative behavior – not out of altruism, but because it also allows better results to be achieved for the individual.




**Implementation in e-commerce**

- Offering gifts, free samples, and free downloads can trigger a willingness to buy as a reciprocal action. Under certain conditions (e.g., hedonistically purchased products), the effect can still be coupled with the Curiosity pattern and marketed as a surprise gift.
- Beyond a certain sales threshold, customers can be offered free shipping on orders.
- Product upgrade: Hotels offer a free upgrade to a larger room, and car rental companies want to convince with a higher vehicle class. Often, these upgrades are only played out with an exit-intent pop-up window, i.e., they only appear when a user moves the mouse toward the "Close Tab" button or the browser input line, thus signaling an intention to abort.
- Freemium models, in which companies initially provide customers with part of the service free of charge and rely on customers to feel that they have committed themselves to buying it later as revenge, are also based on reciprocity.
- During the holiday season, calendars for the new year or small gifts can be sent together with the order catalog, which increases the probability of ordering in the catalog or the online store.




**Sources:** Gouldner AW (1960) The norm of reciprocity: A preliminary statement. American Sociological Review 25:161–178; Cialdini RB (1984) Influence: the psychology of persuasion. Harper Collins: New York.

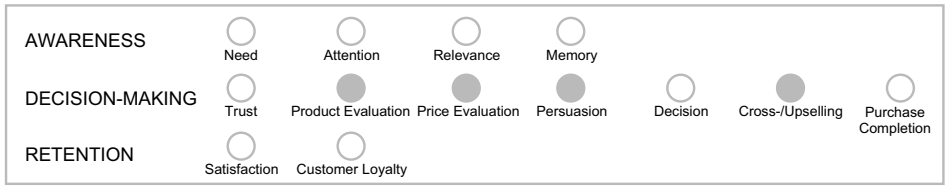
**See also:** Curiosity, Liking, Money Omission, Foot-in-the-Door Technique, That's not-all Technique, Zero Price Effect.

**My Notes**

### 5.4.19 That's not-all Technique

This pattern begins like the effective but often socially problematic door-in-the-face technique: a large request (or an expensive offer) that would most likely be rejected is presented. However, the offer is constantly being improved until the customer accepts it. Thus, by gradually improving the deal, acceptance is built.

Two aspects come together in effect: First, you trigger the reciprocity norm and put people in a situation in which they want to return the favor for the friendly courtesy of the seller with a purchase (see: Reciprocity). In addition, the initial offer functions as a price anchor and creates the contrast effect: Compared to the high starting price, products with the new price (or an extended service bundle) appear relatively cheap (see: Anchoring).




**Implementation in e-commerce**

- The approach is familiar from teleshopping but can also be used in e-commerce, e.g., with price reductions communicated differently: "Today not 100 $, not 90 $, not 80 $, but only 60 $" (instead of "~~MRSP 100 $~~, now 60 $").
- An Expanding the service portfolio is also suitable: "That's not all: In addition, you will receive [bonus 1]. And that's not all: you also get [Gift 2]."
- 2-for-1 campaigns and bundling approaches also rely on this principle and put the initial price into perspective by improving the performance side.
- The principle also appears promising in the insurance context. Up to now, products have been built from the baseline and have become more expensive with each customer's information. For example, car insurance costs "from $ 89 per year" and becomes more expensive with each parameter changed (the insured vehicle is not from the smallest category, the deductible should not be at the maximum value, the mileage is higher, you do not have the lowest no-claims class, etc.). This approach continuously creates moments of frustration and uncertainty for the customer. Astonishingly, the principle has never been reversed: The That's-not-all Technique suggests that products should not be built from the baseline but from the topline and become cheaper with each indication of the user. Instead of punishments, this approach would work with rewards, which seems to make far more sense in the context of this book.



**Source:** Burger JM (1986) Increasing compliance by improving the deal: The that's-not-all technique. Journal of Personality and Social Psychology 51(2):277–283.

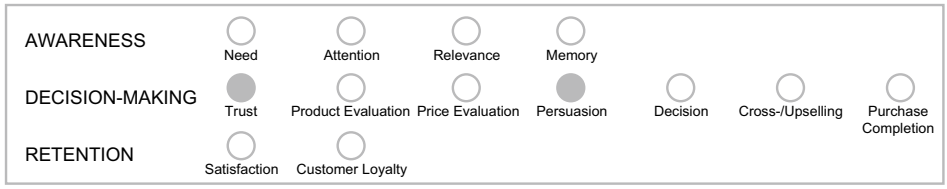
**See also:** Door-in-the-Face Technique, Anchoring, Reciprocity.

**My Notes**

### 5.4.20 Unity

In 2016, behavioral economics author Robert Cialdini concluded that his established six principles of persuasion – after resisting harsh criticism for around 30 years – did indeed have a gap. To close these, he added a seventh principle: Unity.

Unity is the intersection of identity-forming characteristics that creates a sense of group belonging between the influencer and the influenced. The sooner we perceive people as part of "us" (as a social group), the more likely they will affect and convince us. Group membership is perceived as desirable for evolutionary reasons. It gives us support and security, creates a feeling of connectedness, and, as a heuristic, simplifies our decisions (see: Social Proof). An extreme example is belonging to a family or the same gene pool, which even puts our lives at risk to help other group members. Slightly less drastic: In questionnaire completion experiments, the response rate was 20% for non-family members, compared to 97% for family members. Unity is stronger the more feature overlaps we detect. If there is a high level of overlap, we accept the person as "one of us" who stands up for our interests and speaks for us.




**Implementation in e-commerce**

- In the context of influencer marketing and social media, the Unity pattern suggests a mass or majority opinion with customer group advocates. This carries the risk of a reality distortion ("filter bubble"). Combined with modern targeting options, the feeling of belonging to the majority opinion can lead to strong conflicts and – apart from the web marketing context – is jointly responsible for strengthening populist movements in politics.
- The use of users' language is also a Unity application: certain groups recognize the affiliation of their members using a specific vocabulary (e.g., youth slang, Star Trek quotes, etc.).
- Marketers should use family associations: "I also gave this recommendation to my family."
- Customer Co-Creation: Customers and vendors become the joint manufacturer of a product or idea (see: Labor Love Effect). The decisive factor here is which terms are used: "participation" or "advice" activate the pattern much more strongly than "opinion" or "expectations" because they emphasize the common aspect.




**Source:** Cialdini R (2016) Pre-Suasion: A Revolutionary Way to Influence and Persuade. Simon & Schuster: New York.

**See also:** Not-Invented-Here Syndrome, Authority, Social Proof, Labor Love Effect, Aesthetics Heuristic.

**My Notes**

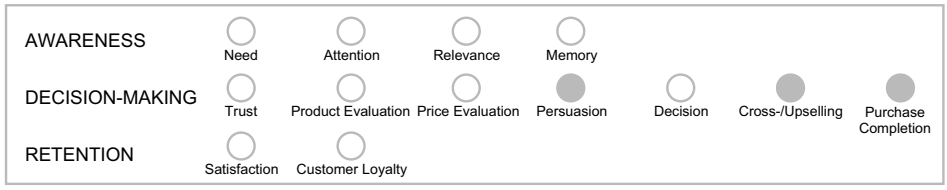
### 5.4.21 Time Inconsistency

People do not consistently evaluate periods regarding renunciation or purchase decisions. Many of us encounter a good example every year in December: New Year's resolutions. In the long term (i.e., in a few weeks or the new year), we resolve to live healthier, exercise more, and give up tobacco and alcohol. However, in the short term (now in December), treating yourself to a quick reward by watching TV on the couch with a bag of chips seems more suitable. As the new year dawns, the short-term temptations are often still too strong. Our decisions about the date of the first of January have changed depending on the time gap, so they are inconsistent.

One explanation for the time inconsistency is that people charge different interest rates for different behavior. In Thaler's experiment (1981), students were asked what they would have to be offered to give up US$15 today. The responses revealed widely differing interest claims depending on the period for which the waiver was requested: $20 was charged for one month (equivalent to an annual interest rate of 345%), $50 (120% annual interest) for one year, and US$100 (19% annual interest) for ten years. The longer the period, the lower the waiver interest. That is why it is so easy to make good intentions for a distant period (low-interest rate). At the same time, the immediate renunciation (sports instead of lying on the couch) is subject to a considerably higher interest rate. In addition to New Year's resolutions, the bank overdraft rate, for example, also works according to this principle, which is absurdly high from an economic point of view and yet is used by bank customers with inconsistent time behavior. A similar approach to explaining this effect is that impulsive influences such as hunger, greed, lust, or pain influence our decisions. Simply put, human impatience leads to inconsistent time behavior. Anyone who has ever stood at a supermarket checkout with children and had to fight through the notorious "whining zone" packed with sweet temptations will be able to confirm this.

The famous marshmallow test wonderfully illustrates the effect (Mischel et al. 1972), which is also used several times in advertising as supposed evidence of particularly irresistible products.

The fact that people tend to make absurdly high-interest demands for small amounts or behave irrationally here can indicate that, in these cases, decisions are made freely based on gut feeling (i.e., purely System 1-based). In contrast, the weight of the decision is correspondingly higher for larger sums. As a result, System 2 is activated, and we make much more realistic demands.




**Implementation in e-commerce**

- If the value of small, quick rewards is systematically overestimated, the performance of fulfillment should be used as a selling point: "Ordered by noon, shipped today."
- If necessary, same-day delivery should also be offered as a delivery option, as this is also considered a quick reward with a high-interest rate.
- In the case of digital products, it is also possible to advertise with "order and use immediately." In the case of products that exist both digitally and physically, when ordering the physical variant (e.g., book, CD, Blu-Ray), the digital version can be provided in excerpts or completely immediately.




**Source:** Thaler R (1981) Some empirical evidence on dynamic inconsistency. Economics Letters 8(3):201–207.

**See also:** Hyperbolic Discounting, Money Illusion.

**My Notes**

## 5.5 Decision

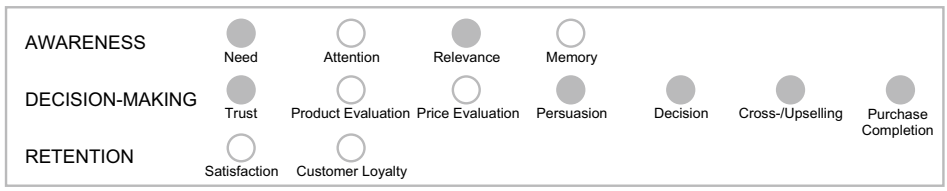
### 5.5.1 Autonomy Bias

Also: Fluency

Autonomy Bias is based on the Self-Determination Theory developed by Ryan and Deci to understand intrinsic motivation (Ryan & Deci 2000). Accordingly, one's motivation always depends on how much the three basic psychological needs of competence, social integration, and autonomy are satisfied.

So, we have an innate need to make our own decisions and implement them self-determinedly. A high degree of autonomy is associated with a sense of security, reduced stress, and increased intrinsic motivation. Conversely, it undermines our motivation when we feel forced or controlled by others (see: Reactance). Autonomy can also be derived evolutionarily: Individuals who align their actions exclusively with their parents, friends, and competitors have less chance of passing on their genome.

In management and marketing, Autonomy Bias generates increased intrinsic motivation among employees or customers through actual or perceived freedoms. The decisive factor is always the extent of freedom: in many cases, complete autonomy also leads to feeling lost in the face of an endless space of options (see: Paradox of Choice). In addition, as social beings, we are usually not interested in acting against the interests and decisions of our group – it is no coincidence that the group of social behavior patterns is one of the largest.




**Implementation in e-commerce**

- Especially in the case of products that are directly related to one's autonomy (e.g., retirement provision, financial products or investments, real estate), it is highly likely that customers will be cautious not to be influenced from outside. Accordingly, it makes sense to take up this need, to emphasize the existing freedom of decision and, above all, to refrain from clumsy attempts at influence (this is, of course, universally true!).
- Another convincing way to make complex decisions is to fully open up the option space and gradually reduce it. This creates an impression of competence and trustworthiness (see: Authority) and corresponds to the goal of System 1 to make good decisions with as little cognitive effort as possible.
- Taking advantage of Autonomy Bias and emphasizing decision-making complexity will lead to greater authorities' involvement in decision-making and a preference for proven

behaviors. Using the example of the well-known savings bank advertising: "How should we best meet the complex market changes?" – "We do it with the flags."

- A digital business model of its own based, among other things, on Autonomy Bias, is so-called curated shopping (in all its manifestations): A qualified expert who knows my needs makes a pre-selection, but the final decision-making power remains with me.
- In the ordering process, e-commerce managers should respect the need for autonomy, avoid non-changeable form fields or shopping carts, and offer sufficient choice in delivery and payment methods.




**Source:** Ryan RM, Deci EL (2000) Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being. American Psychologist 55(1):68–78.

**See also:** Overjustification Effect, Self-Efficacy, Evoking Freedom, Buyers Remorse, Reactance, Hobson's +1 Choice Effect.

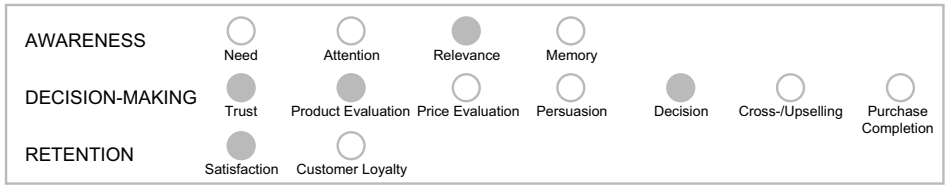
**My Notes**

### 5.5.2 Cognitive Ease

Also: Fluency

Cognitive Ease describes a state that feels good, in which you know no worries and consider the tasks ahead of you to be easily feasible. System 1 is clearly recognizable as the leading decision-making system in this state. As a result, information is checked superficially for intuitive plausibility, and decisions are made based on heuristics.

Cognitive Ease can have various causes, e.g., the task at hand is simple, we know comparable situations well, we have been primed for simplicity, or we are simply in a good mood. The severity of this condition determines how much time and effort one is willing to invest in a task. If the level of cognitive ease decreases (for example, because a task becomes increasingly complex and requires more mental resources), System 2 is activated. As a result, the perception of the task shifts: we become more rational, vigilant, and critical, leading to a decrease in trust in the website, confidence, and enjoyment of using it. It is, therefore, desirable that users have a high level of cognitive ease, as they are more receptive to using behavior patterns.




**Implementation in e-commerce**

- A simple basis for Cognitive Ease is the stringent adherence to UX conventions and learned processes: Known processes create a perceived simplicity and fluidity of interaction. System 1 remains the dominant system.
- Consequently, it must be ensured that no usability obstacles slow cognitive ease. This is especially true before the go-live of new pages and functions. At these times, comprehensive usability labs and bugability analyses should be carried out. But even in live operation, it is worthwhile to regularly test whether the most important pages can be used fluently and barrier-free.
- When using discounts, familiar representations should be used. We are used to easy-to-process price information such as "- 50%" or "only $9.99" – but not "12.4% on the old price of $21.79", which is perceived as a complex arithmetic operation and immediately activates System 2.
- Product names should be easy to remember and pronounce or, if possible, not raise questions about their meaning (see: Availability Heuristic).




**Source:** Tversky A, Kahneman D (1974) Judgment under uncertainty: Heuristics and biases. Science 185(4157):1124–1131.

**See also:** Availability Heuristic, Joy and Fun, Curse of Knowledge, Priming.

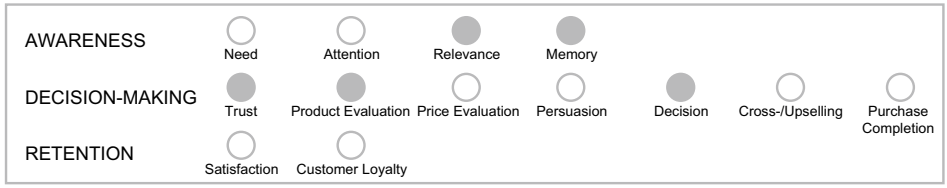
**My Notes**

### 5.5.3 Halo Effect

Also: Salience Effect

The Halo Effect describes how a dominant trait can outshine the perception of all other traits. For example, beautiful people are often assessed as competent, although these attributes are not correlated. The effect is caused by the brain's inability to pay the same attention to all stimuli. At this point, "inability" means nothing more than that an energy-saving heuristic is used. More concretely, on the one hand, there is an influence of the Primacy Effect (i.e., the characteristic perceived at the beginning due to its dominance shapes the further evaluation) and, on the other hand, there is a connection with Confirmation Bias (i.e., we specifically look for evidence that supports our initial assessment).

The Halo Effect also works negatively and is occasionally called the "Devil Effect." This means ensuring that a potential Halo feature has a positive connotation among all users. Otherwise, the conversion will improve for one target group and decrease for another.




**Implementation in e-commerce**

- Spill-over effects can be used with the Halo Effect: Positive experiences with a brand's product lead to these properties being attributed to other products of the same brands without being checked. Example: Satisfied iPhone customers are also very likely to be interested in the Apple Watch – although the inherent interest in (smart) watches may not even be there.
- Companies can specifically look for positive aspects that can be set as a halo: Famous bestsellers from the past, an award as a test winner, or a generally high level of credibility in other segments. This is well implemented at Red Bull, where the occupied topic of "action and energy" dominates the product's possible health risks and taste deficits. Nespresso provides another example: For years, the coffee brand has hardly presented the attributes of the actual product in its advertising but has benefited from the attractiveness of the testimonial (in the person of George Clooney, who has been awarded several times as "Sexiest Man Alive") and created desirability with the help of the Halo Effect.
- If it's appropriate to the context, good-looking people should be shown on the site. As described above, this can increase the credibility of the site. It is essential not to conflict with the Authenticity pattern, i.e., the images must still look credible and authentic.




**Source:** Asch SE (1946) Forming impressions of personality. The Journal of Abnormal and Social Psychology 41(3):258–290.

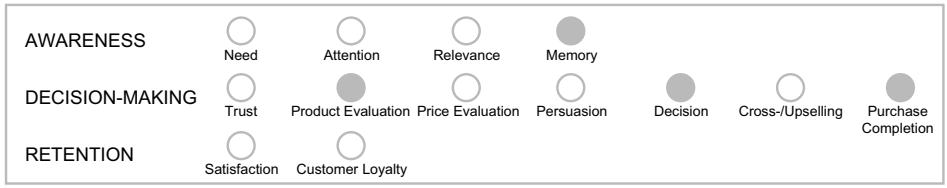
**See also:** Confirmation Bias, Primacy-Effect, Authenticity, Aesthetics Heuristic, Neomania.

**My Notes**

### 5.5.4 Hick's Law

There is a strong correlation between the number of choices and response time. Each additional alternative must be evaluated and weighed against the other options, which costs valuable cognitive resources and time. As a rule of thumb, it can be assumed that if the possibilities are doubled, the response time increases by 150 ms (Kent 1998). Other models assume a logarithmic relationship. Contrary to what one might think at first glance, this connection applies not only to System 2-dominated processes but also to the decision-making effort, which increases with intuitive considerations (System 1).

Regardless of this, Hick's Law is so stable that it is also used as an intelligence test. It is assumed that the shorter the reaction time with an increasing number of choices, the more intelligent a test person is. Important: Hick's Law has only limited applicability in complex decision-making processes that require reading much text and is not suitable as a predictor of decision speed.




**Implementation in e-commerce**

- The Hick's Law is one of the fundamental laws of user experience optimization. A low response time is used as an evaluation benchmark for good UX. The law is also based on the design principle KISS ("keep it short and simple"), which has not lost its validity to this day. It is essential to understand that KISS is not about eliminating processes but about disassembling and designing them so they can be edited with minimal effort. Otherwise, neither Hick's Law nor the KISS principle would be usable in complex use cases.
- Larger decisions and tasks are better divided into small units ("decision chunks"). For example, a multi-level primary navigation can be more suitable than a mega flyout in which hundreds of navigation points are shown in one plane. The same applies to payment and many other selection processes, such as the identification process for contractual products.




**Source:** Hick WE (1952) On the rate of gain of information. Quarterly Journal of Experimental Psychology 4(1):11–26.

**See also:** Paradox of Choice, Hobson's +1 Choice Effect, Ockham's Razor.

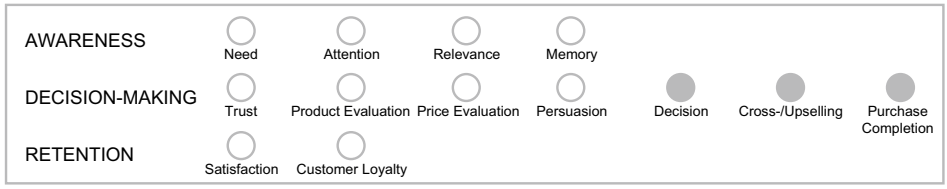
**My Notes**

### 5.5.5 Hobson's +1 Choice Effect

To understand the effect, you first need to know Hobson's Choice: This is a choice with only one option, which you either choose or don't ("take it or leave it"). In the case of Hobson's +1 Choice, a second option is added in the decision room.

Barry Schwartz's work on the "Paradox of Choice" has shown that the more alternatives there are, the worse a decision gets. However, this effect usually only takes effect from three available options. With fewer options, it is worth offering additional alternatives, which is what Hobson's +1 Choice Effect takes advantage of.

Why is that? In Hobson's Choice, we use all our mental resources to decide whether to buy a product. The "Buy" option gets as much attention as the "Don't Buy" option. According to the Status Quo Bias, there is a tendency not to buy a product. In Hobson's + 1 Choice, we use the same amount of mental resources to evaluate whether option 1 or 2 is better. Accordingly, the (not shown) third option, "non-purchase," receives considerably less attention.




**Implementation in e-commerce**

- At critical points (i.e., with a high abandonment rate) in the buying process, it may be a good idea to offer two call-to-actions with different labels but with the same link target (e.g., "buy now" and "add to cart"). This has been identified as very effective across contexts in the context of many A/B tests. Ideally, the button names correspond to the needs of the most important target groups. For a known goal-oriented visitor (e.g., a returning user), it may make sense to hide the additional button again.
- If you offer a "save" or notepad function on your product pages, you create an alternative to the buy button without leading the user out of the purchase process.




**Source:** Schwartz B (2004b) The tyranny of choice. Scientific American 290(4):70–75.

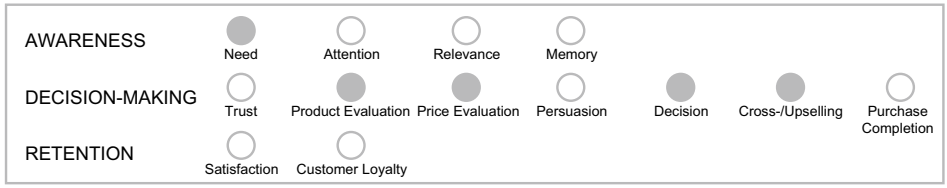
**See also:** Status Quo Bias, Decoy Effect, Paradox of Choice, Hick's Law, Autonomy Bias.

**My Notes**

### 5.5.6 Loss Aversion

Loss Aversion derives directly from Kahneman and Tversky's prospect theory. It states that people behave irrationally when uncertainties play a role in decision-making. According to this, the pain of losing something is perceived as roughly twice as high as the joy of receiving a surprise. Suppose people with a 50:50 probability can either win $100 or lose $100 in a game. In that case, the majority do not want to participate in that game because the expected value is irrationally perceived as negative. In fact, it is exactly 0, so a rational person would have to be indifferent to participate.

Loss Aversion also leads to the well-known sunk costs phenomenon. For example, if we sit in the cinema and accidentally picked a lousy movie, we stay seated instead of walking and spending time with a nicer activity. Why? In such situations, we convince ourselves that the effort (financial, time, physical) should not have been in vain. However, this has already been incurred and will not be reimbursed. Instead of looking back, we should look ahead and immediately stop the activity with negative marginal utility. However, realizing losses creates discomfort and pain due to Loss Aversion, so we prefer to endure a lousy movie instead of a more refreshing activity (see: Status Quo Bias).




**Implementation in e-commerce**

- Practical applications are temporarily limited special offers or vouchers that expire early. Customers will take loss precautions (i.e., buy) to not lose access to the offer. In Marketing, this is often referred to as triggering "fear of missing out" (FOMO).
- Critical inventory reports can be actively communicated: "Only 1 room left at this price!"
- Delivery cost flat rates: Delivery costs are costs that do not generate any physical value and are, therefore, particularly reluctant to be paid – in other words, they create much pain because a noticeable gain hardly offsets the pain. As a flat rate, this pain is significantly alleviated. According to the same principle, the preference for card payment, chips in the casino, or club money on vacation (in each case instead of cash) is justified.
- The pre-assignment of a high-quality product on a comparison site is also an option. This makes customers aware of losing product components or features that would not be included in a cheaper alternative. As a result, more users will opt for the higher-quality product.
- The voucher field should instead be hidden to avoid generating loss aversion for users without a voucher code. This also reduces the proportion of customers who redeem freely

available vouchers after they have only come up with the idea of searching for a code through the voucher field.




**Source:** Kahneman D, Tversky A (1979) Prospect theory: An analysis of decision under risk. Econometrica 47(2):263–291.

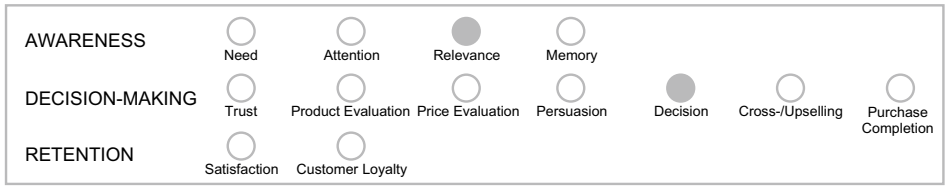
**See also:** Scarcity, Status Quo Bias, Endowment Effect, House Money Effect, Neomania, Pennies-a-day Effect, Precision Heuristic, Zero Risk Bias, Mayfly Effect.

**My Notes**

### 5.5.7 Mental Accounting

People sort financial transactions, such as online purchases, into various mental fictitious accounts (e.g., rent, food, clothing, vacations). Such mental accounts exist for all income and expenditure items. So, instead of looking at the full context of a decision, we only include a categorized excerpt. Mental Accounting is, thus, first and foremost, a complexity-reducing heuristic: It is not necessary to assess all the consequences of a decision, but only the influences within the respective mental account. However, depending on how these accounts are structured, we can come up with entirely different decisions (e.g., buy or not buy). If a mental account is still well-funded, we decide to buy it. If, on the other hand, more considerable expenses have recently been sorted into an account, we decide against it due to the lack of funds. This indicates that the assignment to mental accounts occurs regardless of whether the resulting conclusions are economically rational.

Richard Thaler, the "father" of this behavior pattern, likes to tell the following anecdote to illustrate how mental accounts work: When the Swiss franc was at an all-time high against the dollar, Thaler gave a lecture in Switzerland. He then spent a week there with his wife to see the region. The two sweetened this holiday with the thought that the lecture fee would cover the astronomical costs for the holiday well. If he had received the same fee a week earlier for a lecture in New York, he would not have enjoyed his vacation in Switzerland nearly as much. The high costs of the vacation and the lecture fee were thus encoded into the same mental account, which was therefore still balanced despite the unfavorable exchange rate.




**Implementation in e-commerce**

- E-commerce companies should ask themselves for each product: In which mental account should customers sort this product? Ideally, it is possible to classify the product in a comprehensive account that is always adequately covered or to place it in a niche where there are hardly any alternative scenarios (e.g., competitors). This can be achieved, for example, by presenting the product as a world-first or a real innovation.
- Products with different use cases (e.g., smartphones or laptops) can be sorted into separate mental accounts (e.g., work and entertainment). Depending on the coverage of the accounts, the allocation will decide on evaluating the value for money. The same applies to products at the intersection of two accounts: Animal health insurance can be placed in the "Pet Supplies" or "Insurance" accounts, for example.

- Whether discounts work (e.g., for a newsletter subscription) depends on the mental account to which the respective product is assigned. For example, many people have a "minimal account" that includes small purchases. If a product is sorted here, there is a high probability that discounts (even if they involve effort) will be claimed. In the example, newsletter subscriptions should be advertised for low-priced products.




**Sources:** Thaler RH (1985) Mental accounting and consumer choice. Marketing Science 4(3):199–214; Thaler RH (1999) Mental accounting matters. Journal of Behavioral decision making 12(3):183–206.

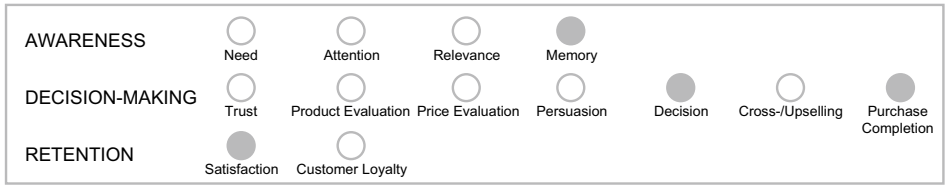
**See also:** House Money Effect, Hyperbolic Discounting, Moral Licensing.

**My Notes**

### 5.5.8 Peak-End Rule

Similar to the Primacy Effect and the Recency Effect, the Peak-End Rule also assumes that not all partial experiences with the same weight shape the overall experience but that individual impressions dominate the big picture. The rule shows that the strongest and the last emotional experiences (positive or negative) have a powerful and lasting impact on the overall impression. The peak is the point that deviates most from the norm or the expected value.

Carmon and Kahneman (1995) conducted a series of experiments. They found that participants who were mostly dissatisfied during an experience but briefly satisfied at the end rated the entire experience as positive. In another experiment, participants had the choice of holding their hand in icy water for 60 s or holding their hand in icy water for 60 s and then another 30 s in slightly less cold water. The majority irrationally opted for the second option with 90 s because the end of the alternative had a more positive ending.




**Implementation in e-commerce**

- If the "peak" decisively defines the customers' overall impression, the need to create positive surprises in the ordering process is derived from this. If this positive moment is not given, there is a latent danger that a partial experience that deviates from the norm downwards will lead to a negative overall evaluation. E-commerce companies, therefore, have a constant obligation to innovate to inspire customers again and again. For the sake of completeness, it should be mentioned that under certain circumstances, an experience that is merely expected can also function as a positive peak if comparable experiences in the past were usually perceived as negative (e.g., a smooth and lean check-out process).
- The formative endpoint can be achieved with fast fulfillment or attractive packaging, as well as small gifts as additions to the package.




**Source:** Carmon Z, Kahneman D (1995) The experienced utility of queuing: experience profiles and retrospective evaluations of simulated queues, Duke University working paper, Durham.

**See also:** Primacy Effect, Recency Effect, Halo Effect.

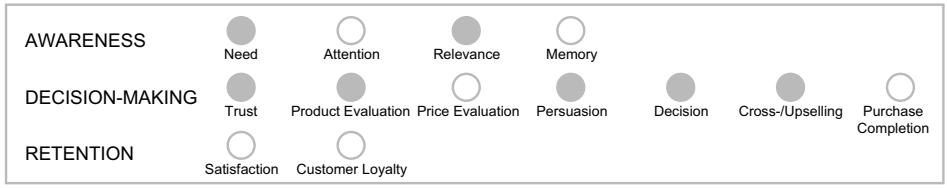
**My Notes**

### 5.5.9 Priming

One of the most central assumptions of behavioral economics is our brain's associative mode of action: With every experience we have, a substantial neural representation is activated. If we read the word "moon," associations arise for "night," "wolf," or "white." So "moon" is the 'prime' (literally: preparation, priming, paving) for a thought of the color white. Priming thus describes nothing more than the deliberate confrontation with certain impulses that let us mentally pre-select a specific action. At the moment of decision, this pre-selected option can be chosen with less cognitive effort.

Priming is largely universal: people primed for aging move more slowly, people primed for money are more selfish (see: Dollar Eyes Effect), and people primed for an unpleasant experience get the increased need to wash themselves. Another interesting priming experiment was conducted by Williams and Bargh (2008): they showed that evaluating a person unknown to us directly depends on what drink we are currently enjoying. If we drink a hot coffee, we are more likely to develop warm feelings towards strangers. On the other hand, we are much "cooler" with iced coffee. In this experiment, the two fundamental characteristics of Priming also become clear: We distinguish between positive (i.e., the prime increases the probability of showing a specific behavior) and negative Priming (i.e., the prime reduces the likelihood of delivering a particular behavior).

Priming has become a real hype in recent years because of its effectiveness. This prompted Nobel laureate Daniel Kahneman to write an open letter to the scientific community urging them to review research findings more intensively before publication to "restore the good reputation" of the mechanism. At this point, it is essential to mention that some key studies on priming have not been replicated recently. This has triggered a severe earthquake in the behavioral science community and other social sciences (the so-called "replication crisis").




**Implementation in e-commerce**

- Every purchase involves subconscious emotions. Through an empathetic examination of one's target group, these emotions can be identified and integrated in a targeted manner during the customer journey. An example: A leading insurance company has discovered in in-depth analyses that in animal health insurance, many customers are disproportionately moved by grief and feelings of loss. The experience of having lost a beloved animal leads to an increased need for insurance. After objectifying the tonality of the page and partially

replacing pictures of playing puppies with diseased animals, the activation potential of the site increased measurably.

- Beware of too obvious primes: If the effect is used too aggressively, so-called reverse priming effects are possible, i.e., the effect is reversed (psychological backfire).
- The entire buying process should be analyzed for unwanted negative primes. These can be, for example, minus signs, asterisks, and footnotes, or even colors and images that contradict the desired Prime.
- Priming is also used to build brand associations. Due to its highly visible involvement in action sports, RedBull (mind you: a drink with nine (!) sugar cubes per can) is associated with sportiness and dynamism.
- Ultimately, with influencer marketing, an entire advertising discipline is based on the Priming effect: Beautiful people present a product in beautiful surroundings that subsequently also appears desirable to us consumers.




**Source:** Bargh J, Pietromonaco P (1982) Automatic information processing and social perception: The influence of trait information presented outside of conscious awareness on impression formation. Journal of Personality and Social Psychology 43(3):437–449.

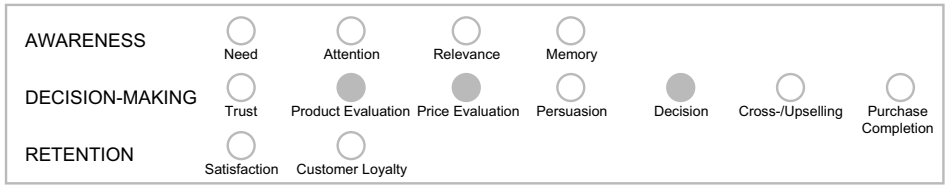
**See also:** Cognitive Ease, Availability Heuristic, Dollar Eyes Effect, Anchoring, Framing.

**My Notes**

### 5.5.10 Scarcity

Everything scarce has a magical attraction for us. If something is limited, we fear it may no longer be available if our decision-making process takes too long (see: Loss Aversion). As a result of the shortage, we are making more decisions based on heuristics and are showing a positive attitude towards buying. It's not for nothing that many commercials end with "Hurry! Offer will end soon!". In addition, we process the scarcity of a product as a sign of its desirability (see: Social Proof).

Originally, the Scarcity principle was proven using an experiment in which the participants were asked to choose a biscuit from one of two cans: one full and one almost empty. The biscuits in the empty tin were much more sought after and appreciated by the test subjects – as if they had gotten hold of something special. Interestingly, the participants who opted for biscuits from the almost empty can (i.e., were receptive to the effect of the pattern) made their decision much faster. The scarcity of the product created pressure to act and decisiveness.




**Implementation in e-commerce**

- Scarcity is usually based on a product's time and/or quantity limitation, e.g., limited editions.
- Inventory messages: "Only a few copies available due to high demand" can be combined well with patterns from the social proof tools: "12 people are looking at this product right now". The effect can be further enhanced if high-stock information is written in a small font.
- Customer group exclusives (e.g., for students, retirees, Facebook fans, Twitter / X followers) or channel-exclusive offers ("Only available online!") are also effective applications.
- Appointment offers can only be offered for the distant future at the beginning. Appointments then appear scarce, and the business is therefore particularly desirable (often used by celebrity hairdressers).
- In textbook form, Scarcity is also used as part of Amazon's flash sales (Cyber Monday, Black Friday, Prime Day, etc.). Here, new products with limited units come onto the market every hour; the shortage is quantitative and temporal. In this way, daily sales are generated (with supposedly high discounts) that would otherwise be generated in several weeks.

- Rolex uses the effect at almost every product launch and has already educated its customers that a sell-out within a few hours can always be assumed. Customers respond to this with pre-orders, attempts to bribe concessionaires, and inflated prices on the secondary market. Hardly any of them suspect that the initial number of units available could be deliberately kept low.




**Source:** Worchel S, Lee J, Adewole A (1975) Effects of supply and demand on ratings of object value. Journal of personality and social psychology 32(5):906–914.

**See also:** Loss Aversion, Social Proof, Reverse Psychology.

**My Notes**

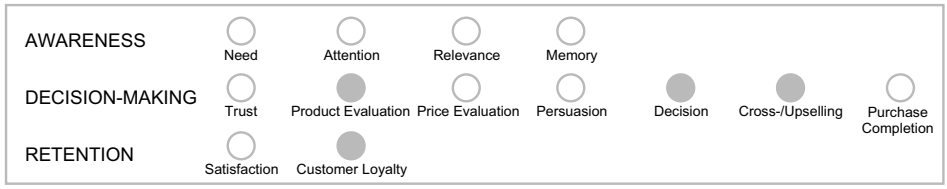
## 5.6 Cross-/up-Selling

### 5.6.1 Diderot Effect

The Diderot Effect describes how, after purchasing an item, people can come under pressure to make further follow-up purchases to create a coherent overall picture. The effect's name goes back to an essay by the French writer Denis Diderot, who described in 1772 why he mourned his old dressing gown: his furnishings no longer fit the new coat, and he would probably have to buy everything else new. So, he plunges into debt, renovates his house, and renews all the furniture – only to discover that he was the master of his old house coat but is now the slave of his new house coat.

The effect thus describes the distortion that we exaggerate the importance of new things (see: Recency Effect) and irrationally make far more severe decisions instead of revising the initial decision (see: Confirmation Bias). For example, if someone buys a pair of new shoes for $100, they might feel compelled to buy a matching coat, belt, bag, and trousers (for a total of $1′000). A considerable decision or investment is therefore triggered by a small decision in the context of striving for consistency (see: Commitment and Consistency).

The well-known "Sunk-Cost Fallacy" is similar, i.e., people are urged to continue a project to justify the initial start-up costs (or not to have to write them off). This is true even if the project is no longer profitable. Example: Accessories for a drone continue to be purchased, even though you have never flown with the device again after initial euphoria. In this way, one can maintain the impression that the initial decision (purchase of the drone) was not a mistake and thus avoid dissonance.




**Implementation in e-commerce**

- For retailers, products like Diderot's dressing gown are true saviors. They are perfect as an introduction to a long-term and profitable customer relationship. In e-commerce, shopping cart analyses can identify items often purchased with others and increase the probability of subsequent purchases. Very often, first-time purchases are on-demand purchases. Sensible measures are strengthening the assortment context around these articles and bundling-oriented cross-selling.
- The entry-level product should be attractively priced and, in exceptional cases, can even be sold below cost. In addition to a suitable "supplementary product world," a direct possibility of addressing customers after the initial purchase is also a prerequisite.




**Source:** McCracken F (1988) Diderot unities and the Diderot effect. In: McCracken G (ed.) Culture and Consumption: New Approaches to the Symbolic Character of Consumer Goods and Activities. Indiana University Press, Bloomington/Indianapolis, pp. 118–129.

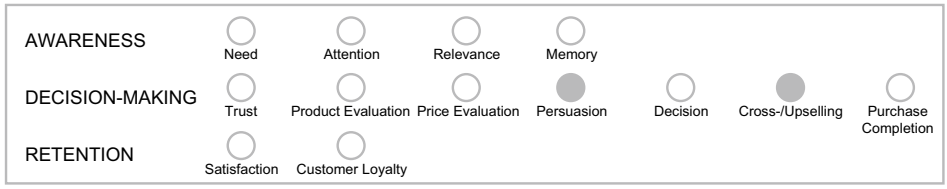
**See also:** Recency Effect, Zeigarnik Effect, Confirmation Bias, Commitment and Consistency, Functional Fixation, Mayfly Effect.

**My Notes**

### 5.6.2 Low Ball Effect

The Low Ball Effect starts similarly to the Foot-in-the-Door technique: A small request is formulated, which is not rejected by most. After the approval has been given, however, further additional costs are disclosed. These can be financial (e.g., shipping costs, handling fees) or non-financial costs (e.g., long delivery time, mandatory collection at the store). Despite this unpleasant surprise, the prices are accepted more often if you consent initially than if the undesirable constraints had been communicated from the outset.

Another difference to the Foot-in-the-Door Technique is the lack of a time lag between the two requests: the second request is communicated immediately after the first request has been approved. This triggers the Commitment and Consistency Pattern ("In for a penny, in for a pound").




**Implementation in e-commerce**

- In e-commerce, according to the pattern, unpleasant constraints should only be communicated when users have already committed to achieving the goal (e.g., in the form of a significant workload). A well-known example is the photo book for only $ 10. After the user elaborately designs it, shipping costs of $ 9.99 or image processing fees are displayed. Astonishingly, cross-selling is particularly successful for users who accept the constraint: For example, the offer to order another copy for $ 7.99 is gladly accepted because it relativizes the (already accepted) high shipping costs.
- Similar application scenarios are bait offers. For example, a very cheap toolbox is advertised but is already out of stock in the shop. Instead, a higher-priced alternative product is displayed, which then benefits from the affection for the original product and, despite the adverse ancillary conditions (higher price than the original product), is bought more often than with direct marketing.
- It is obvious that such approaches should only be used very discreetly and should never become the system of an online shop. Therefore, the pattern is not particularly well suited for building customer loyalty but is suitable for business models with low purchase frequency.



**Source:** Cialdini RB et al. (1978) Low-ball procedure for producing compliance: commitment then cost. Journal of Personality and Social Psychology, 36(5):463–476.

**See also:** Foot-in-the-Door Technique, Commitment and Consistency, Labor Love Effect.

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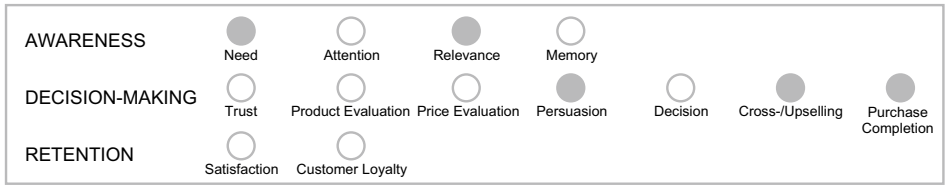
## 5.7 Purchase Completion

### 5.7.1 Action Bias

Action Bias, as a counter-effect to the Status Quo Bias, refers to the tendency of people to act actively even if the action is unlikely to achieve anything or is even harmful. This actionism is particularly effective in unclear situations where the best alternative course of action cannot be determined.

The effect was initially identified based on goalkeepers who rarely stop in the middle of a penalty but, in about 95% of cases, dive into one of the two corners. This does not make sense because the shooters shoot penalties relatively evenly across the width of the goal. Standing still would not reduce the chances of saving the penalty but possibly even increase it if you pulled in the enlarged area with arms spread out in both directions. Goalkeepers, however, succumb to the human tendency to intervene actively in the action rather than passively waiting.

On the one hand, this can be explained by the Overconfidence Effect, which makes us believe excessively that we can anticipate and cope with the situation. In addition, there is an evolutionary biological explanatory approach: Early humans usually only had the option of flight or fight. However, both possibilities imply active action, so doing nothing is not intended evolutionarily. A third explanation lies in the social evaluation of an action or non-action: Those who actively act have at least visibly demonstrated their will – those who do nothing at all, on the other hand, are quickly considered apathetic and indecisive. Consequently, this correlation is more pronounced in younger people, for whom social profiling has not yet been completed, than in older people.




**Implementation in e-commerce**

- The Action Bias can be used to promote decision-making in the context of the designation of buttons. These should be formulated literally as real call-to-actions, i.e., calls to action such as “vote now” or “order immediately.” Although this insight is part of the basics of online conception, it is surprisingly often neglected.
- In the content, formulations such as “Don’t wait any longer!”, “Take the decision into your own hands!” or “Act now!” can reinforce the urge to act and trigger corresponding behavior.




**Source:** Bar-Eli M, Azar OH, Ritov I, Keidar-Levin Y, Schein G (2007) Action bias among elite soccer goalkeepers: The case of penalty kicks. Journal of economic psychology 28(5):606–621.

**See also:** Status Quo Bias, Overconfidence Effect.

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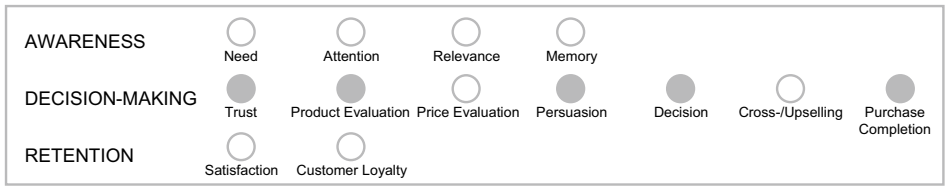
### 5.7.2 Cheering

Also: Spurring

If you want to understand the motivation to buy in e-commerce, we differentiate between hygiene and motivational factors – Frederick Herzberg's two-factor theory expresses this. Hygiene factors are, for example, good usability and the technical functioning of a shop, which enable us to buy a product in the first place. In the context of intuitive purchasing decisions, motivational factors are the aspects that ultimately decide between purchase and abandonment.

One of the most effective known motivational factors is positive feedback or positive reinforcement – a true classic of behavioral research. What began with the famous conditioning experiments of Burrhus Frederic Skinner on rats is now a popular and proven (albeit not entirely uncontroversial) principle in childrearing and sports training: The positive stimuli of our environment (especially from authority figures) have a strong influence on our behavior. The teacher's diligence card, the coach's praise for the best performance, or the appreciation of parents for "good behavior" motivates us to repeat the behaviors that have been highlighted and to behave consistently. It should not go unmentioned at this point that this type of reward works similarly to a drug: we need more and more of it, the effect is constantly waning, and without it, we feel powerless. It undermines intrinsic motivation. Therefore, this approach is hardly used in modern, needs-oriented parenting styles.

But back to the digital context: There are certainly approaches to using positive feedback to "plug motivation holes" – for example, in the ordering process. This is where behavior patterns such as Cheering come into play. It refers to approaches that motivate users to persevere in a sometimes long and cognitively exhausting process.




**Implementation in e-commerce**

- The most subtle form of Cheering is feedback: an action by the user triggers a reaction from the system (e.g., the online shop). In this way, we convey to the user that his action has been registered and that he has done something right.
- For example, selected products can stand out in color in a comparison table, or non-selected products are grayed out.
- The same applies to visual confirmations that clarify that a product has been successfully added to the shopping cart – which is not yet well implemented in many online shops.

- Another example of motivational feedback is form fields representing a green check mark or a green frame when the user's input matches the required format (e.g., for the date of birth "MM.DD.YY").
- In the broadest sense, an empoweringly designed progress bar can also be an example of cheering (see: Zeigarnik Effect).

If users have an increased degree of insecurity, cheering can be used less neutrally (or as mere feedback) and take on a more empowering or encouraging character. It must make sense in the context and pay off to the user's actual needs (especially confirmation reduction of uncertainty).

- The critical moment, when customers add a product to their shopping cart, can be supported with an empowering message such as "Good choice" or "Congratulations, you have found the right product."
- This form of cheering can be perfectly combined with other behavior patterns, e.g., with the Bandwagon Effect ("Awesome! This good choice has already been made by 426 customers before you") or Scarcity (Airbnb uses something like "This is a rare find. This accommodation is usually already fully booked").




**Source:** Lo LYS, Lin SW, Hsu LY (2016) Motivation for online impulse buying: A two-factor theory perspective. International Journal of Information Management 36(5):759–772.

**See also:** Commitment and Consistency, Self-Efficacy, Framing, Liking, Confirmation Bias, Zeigarnik Effect.

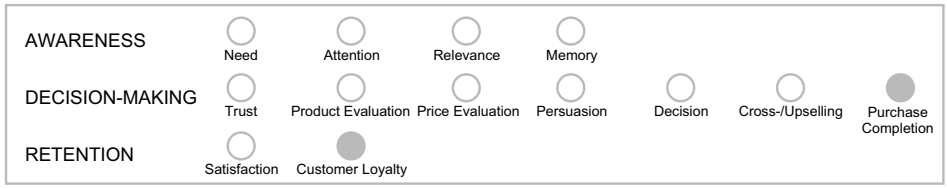
**My Notes**

### 5.7.3 Commitment and Consistency

Commitment and Consistency describes the striving for consistency between one's actions on the one hand and one's values and self-image on the other. Consistency is considered a desirable social trait and indicates rationality, trustworthiness, and stability. In addition, the principle is a mental shortcut in the decision-making process. Previous decisions are used as the basis for current tasks, and the same point of view is applied again as a heuristic.

In practical terms, once we have decided on an idea or a goal, either orally or in writing, we tend to maintain and defend this decision. The deep anchoring in everyday life can also be seen in the expressions "If you said A, you must also say B" and "If you start something, you finish it."

What is essential about the effect of the pattern is that the commitment to a cause was made voluntarily, actively, and publicly. The effect is reduced if one or more of these factors are absent. On the other hand, the effect increases if the commitment has also been made in writing.




**Implementation in e-commerce**

- In late steps in application or ordering processes with a high abandonment rate, it can be made clear that the decision has already been made, and users should remain consistent. This pattern can also be used effectively in the order confirmation to reduce the return rate.
- Getting a small commitment from users to the shop or brand can be the door opener for a long-term customer relationship because the casually made original decision (e.g., download an audiobook, newsletter subscription, purchase of a cheap sale item) for the provider should not be revised later (see: Foot-in-the-Door Technique).
- A wish-list function can also use this pattern. This way, a small, non-binding decision is easily made, which cognitively prepares the later purchase decision.




**Source:** Cialdini RB (1984) Influence: the psychology of persuasion. Harper Collins: New York.

**See also:** Foot-in-the-Door Technique, Confirmation Bias, Zeigarnik Effect, Mere Agreement, Diderot Effect, Functional Fixation, Moral Licensing, Cheering, Mayfly Effect.

**My Notes**

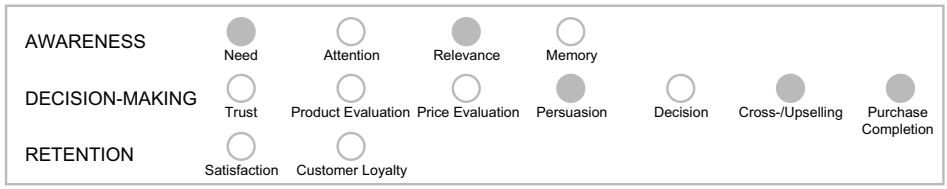
### 5.7.4 Confirmation Bias

Also: Choice Supportive Bias, My Side Bias, Inertia

After an opinion or decision has been made, people tend to focus only on the positive aspects of that decision and interpret all the facts in the spirit of the existing decision. All decision-relevant factors are implicitly weighted so that it feels like the best possible decision. System 1 defends the decision. As a result, there is no guilty conscience, no cognitive dissonance, and no necessary activation of System 2. Without System 2, the need for reversal is not even examined. Daniel Kahneman (2011, p. 81) summarizes the bias: "The confirmatory bias of System 1 favors uncritical acceptance of suggestions and exaggeration of the likelihood of extreme and improbable events."

There is always the risk of a wrong decision, as arguments that could refute one's opinion are distorted or ignored. The weighting can also be adjusted as a strategy for dealing with opposing views. If one weighs arguments that match one's thesis more heavily, one also speaks of "My Side Bias." Incidentally, the Choice Supportive or Confirmation Bias is the basis for the broad success of superstition, populism, charlatanry, conspiracy theories, and general resistance to advice. Those who want to get rich quickly are likelier to believe those who promise quick riches than a realistic critic. As a result, such people are more likely to participate in a Ponzi scheme, for example.

In a nutshell: People want to believe and confirm. This also explains the "clustering illusion," which refers to the phenomenon in which we recognize evidence for our opinion in everything instead of acknowledging chance.




**Implementation in e-commerce**

- Statements in advertising campaigns or product descriptions can be formulated with a certain amount of room for interpretation. In this way, interested parties can find themselves in the statement regardless of their existing opinion and feel addressed or confirmed.
- For shopping cart abandoners, it should be emphasized that the decision to add the product to the shopping cart has already been made. This can be achieved by once again mentioning all the emotional and functional advantages of the order in the abandonment address.
- It's a good idea to use user-centered methods to understand customers' deep-seated beliefs. In the next step, these target group segments can be targeted with a message that matches their convictions. Thus, no independent conviction is necessary. This works for a wide range of target groups – from climate change deniers to vegans.




**Source:** Forer BR (1949) The fallacy of personal validation: A classroom demonstration of gullibility. Journal of Abnormal and Social Psychology 44:118–123.

**See also:** False Consensus Effect, Availability Heuristics, Belief Bias, Cognitive Dissonance, Commitment and Consistency, Cheering.

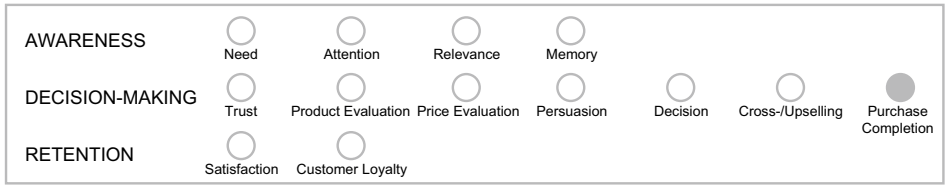
**My Notes**

### 5.7.5 Hindsight Bias

Also: Retrospective Error, Knew-It-All-Along Effect

Hindsight Bias describes the phenomenon that people who know the outcome of an event are convinced that they could have predicted the outcome. In short: "I've always known it!". It is also called the "Knew-It-All-Along Effect" (Hertwig et al. 1997).

It leads us to find explanations where none can be found. Nothing seems to be random or arbitrary anymore. This is particularly explosive in the case of legal issues, such as the assessment of sworn experts as to whether accidents could have been foreseen or avoided. Appraisers who know the event's outcome are far more likely to find a logical explanation. Another example is significant stock market crashes, which, in retrospect, are regularly glorified as predictable.




**Implementation in e-commerce**

- The Hindsight Bias has only limited applicability for interface design. Nevertheless, you should know it for conversion optimization. This is because it contributes to the fact that you may not get the most out of the essential tool you have for optimization – A/B testing: Test results are often not analyzed in detail enough due to Hindsight Bias if they seem plausible and, therefore, predictable. This applies, in particular, to results that prove the already implemented variant and are consequently not considered valuable. Thus, you should define an evaluation and interpretation process for A/B tests that both positive and negative results go through. This ensures that Hindsight Bias is minimized and A/B testing is used optimally.




**Source:** Pohl RF (2004) Hindsight bias. In: Pohl RF (ed.) Cognitive illusions: A handbook on fallacies and biases in thinking, judgement and memory. Psychology Press, Hove, pp. 363–378.

**See also:** Confirmation Bias, Illusion of Control.

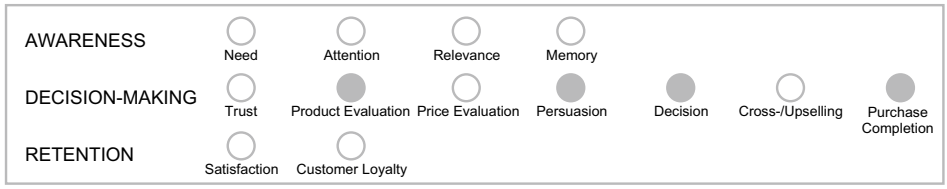
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### 5.7.6 Illusion of Control

Illusion of Control refers to the mistaken belief of people that they can influence the outcome of random experiments. For example, it can often be observed that people roll dice with more force when they want to roll a high number in a dice game. Many people also believe they can influence a completely random coin toss.

The effect was discovered in a famous experiment: subjects were presented with two cards face down, a winning card and a losing card. The probability of receiving a win was assessed significantly higher by the participants if they could choose a card themselves than if the card was given to them by the game master. The effect can be further enhanced by so-called "skill cues," which are signs of skills usually needed in comparable situations (e.g., selecting, researching, deciding).

Further experiments have shown that we feel in control when the process to which we have contributed leads to success. These can be professional basketball players who are told how to make a free throw (Ellsberg 1961) – but also websites that respond to user input and send positive feedback.




**Implementation in e-commerce**

- If users believe they have control over uncontrollable processes within the buying process, you shouldn't take that belief away from them. Critical are, therefore, situations in which the lack of control becomes apparent to users, e.g., in case of technical errors and interruptions. Generally, control-confirming elements positively affect the course of the customer journey and reduce the abandonment rate. The reason is simple: those who believe they can control a process feel less subjective risk and are willing to take an objectively higher risk. Sentences such as "You decide how you pay" or "You have full control over the ordering process at all times" support System 1 and have activating potential.
- In the ordering process, fictitious decisions that would not be necessary can be queried since a choice between several options is also a form of perceived control. However, this approach clashes with UX conventions, according to which pages must be as decision-lean as possible and should therefore be used cautiously.
- Important information and frequently required data should be omnipresent, e.g., the shopping cart (e.g., as a sticky element at the top of the page) or the contact options (e.g., as a flag on the right edge of the screen).

- For long listings (e.g., when hundreds of products are offered in a category), "More" buttons or page numbers make more sense than infinite scrolling (i.e., the page automatically reloads products as soon as you reach the bottom of the list) to let the user take control.
- Filters set are requirements of the user or instructions for action for the shop. They should, therefore, be applied to all products and persist when the page is reloaded.




**Source:** Langer EJ (1975) The illusion of control. Journal of personality and social psychology 32(2):311–328.

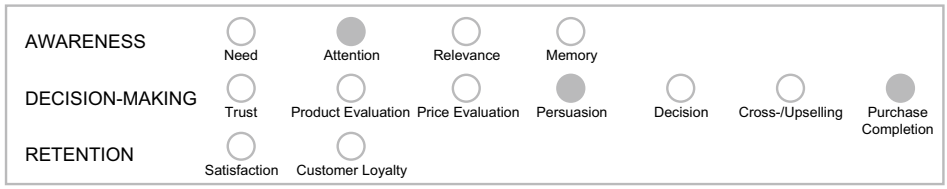
**See also:** Overconfidence, Self-Efficacy, Hindsight Bias.

**My Notes**

### 5.7.7 Motivating Uncertainty Effect

Research into behavior at risk has been intensively pursued since the 1970s. Most studies conclude that we have an aversion to uncertainty and ambiguity: we tend to avoid decisions with uncertain consequences (see: Ambiguity Aversion). Current findings, on the other hand, suggest that uncertainty can also be a strong motivator. In several experiments, subjects who did not know the extent of a reward showed higher motivation and perseverance in completing a task than participants who were told the reward amount right at the experiment's beginning. The basis for this Motivating Uncertainty Effect was laid by the famous experiments with the Skinner box in the 1950s: Here, it was shown that laboratory mice reacted particularly strongly to a random reward.

On the one hand, this can be explained by the tension we feel when working on the task. On the other hand, we attribute our wishes and ideas to the vacuum, hoping (regardless of the probability) for an exceptionally high reward. This can be further reinforced with appropriate framing of the situation.




**Implementation in e-commerce**

- In the context of newsletters with special offers and discounted products, showing a range of savings (e.g., 20–70%) is a good idea. The uncertainty or hope for the highest possible discount motivates users to call up the products in the shop and increases the click-through rate of the newsletter accordingly.
- When the page is called up, a message can also be displayed announcing a voucher code of an unknown amount in the ordering process. Users who have worked their way from the product page to the shopping cart to the ordering process then succumb to the Zeigarnik Effect and the Loss Aversion Pattern and glorify the value of the coupon to justify the effort made so far.




**Source:** Shen L, Fishbach A, Hsee CK (2014) The motivating-uncertainty effect: Uncertainty increases resource investment in the process of reward pursuit. Journal of Consumer Research 41(5):1301–1315.

**See also:** Ambiguity Aversion, Status Quo Bias, Framing, Curiosity, Joy and Fun.

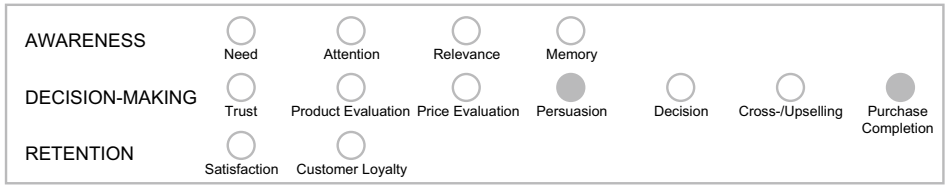
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### 5.7.8 Overjustification Effect

The effect describes that the excessive justification of a desired action leads to the intrinsic motivation of this action being displaced by extrinsic motivation. However, if an action is only extrinsically motivated, people will no longer perform it as soon as the extrinsic motivational incentive disappears. In other words, an extrinsic incentive (such as a reward) increases motivation in the short term. However, as soon as the incentive disappears, the motivation drops significantly below the original level. However, external stimuli are a valuable and suitable motivational tool when people are not intrinsically motivated.

This can be explained as follows: You don't expect a reward for an activity you enjoy. If, contrary to expectations, a reward is given, a cognitive reassessment of the situation takes place. The activity is equated with situations for which one usually receives a reward. Most of the time, these are situations in which people are not intrinsically motivated (e.g., heavy physical work only for appropriate pay). This equation deprives the activity of its intrinsic motivation and corrupts it.

In several application contexts, research has been carried out into how rewards develop the maximum benefit for motivation. The findings are largely unanimous that unexpected rewards have the most significant leverage – whether in the design of incentive systems for employees in the e-commerce team or for the activation of users on the website.




**Implementation in e-commerce**

- When users are already intrinsically motivated to show a specific behavior, extrinsic motivation should not be used in addition. A strong driver of intrinsic motivation is, for example, curiosity. Others are fun, involvement, inquisitiveness, sympathy, or one's self-image. So, if curiosity led to the purchase process in the past, extrinsic amplifiers (e.g., a voucher code) should be avoided.
- Rewards should not be fully predictable for users if intrinsic motivation is present. If it makes sense to rely on extrinsic rewards additionally, these should be played out surprisingly.
- As a rule of thumb, the effect of an extrinsic stimulus is usually more positive for repetitive tasks than for complex tasks (i.e., higher demands on attention or creativity). In the case of the latter, intrinsic motivation should rather be supported, e.g., by emphasizing the positive effect of the action on the user (autonomy, personal realization, security, etc.).




**Sources:** Bem DJ (1967) Self-perception. An alternative interpretation of cognitive dissonance phenomena. Psychological Review 74:536–537; Deci EL, Koestner R, Ryan RM (1999) A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. Psychological Bulletin 125(6):627–668.

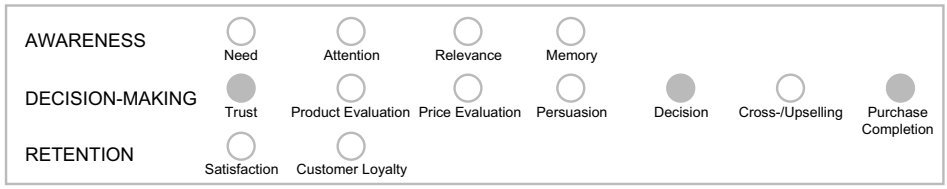
**See also:** Self-Efficacy, Joy and Fun, Curiosity, Cognitive Dissonance, Autonomy Bias.

**My Notes**

### 5.7.9 Response Efficacy

We are more willing to take action if we believe that this action will lead to the desired result. What at first glance sounds similar to the Self-Efficacy pattern has an entirely different twist: While Self-Efficacy asks how competent we feel to act ("Will I manage to do this?"), Response Efficacy is about the consequences of the action ("If I do it, does it bring me to the desired goal?"). Uncertainty about the consequences of action, therefore, undermines our willingness to act.

An intention to act always arises from emotional and cognitive components. Response efficacy is one of the most powerful cognitive influences. It also has a mediator function for emotional decision-making aspects. This is stronger in the case of emotions with positive connotations (such as pride or joy) than with negative ones (such as fear or compulsion).




**Implementation in e-commerce**

- A target group analysis reveals customers' specific objectives when purchasing a product. These objectives should then be addressed communicatively. In the example of a retirement provision contract: Do customers conclude this contract to have security in old age or as a well-interest-bearing investment?
- It should then be checked whether positive or negative framing leverages behavior more (e.g., "Calculate how much you can save with this product" vs. "Calculate how much money you're missing out on without this product").
- The message that buying a product leads to the desired result can be reinforced with testimonials, statistics, or awards and seals of approval.




**Source:** Lewis IM, Watson B, White KM (2010) Response efficacy: The key to minimizing rejection and maximizing acceptance of emotion-based anti-speeding messages. Accident Analysis & Prevention 42(2):459–467.

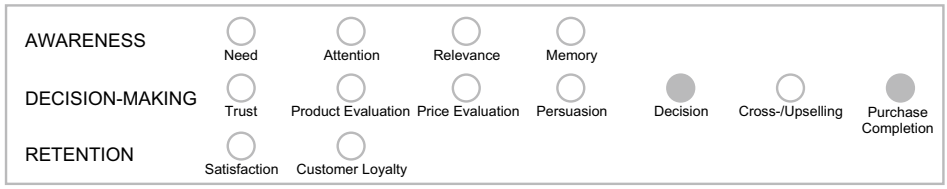
**See also:** Self-Efficacy, Social Proof, Trust Bias.

**My Notes**

### 5.7.10 Self-Efficacy

Albert Bandura's concept of self-efficacy refers to the expectation of a person to successfully carry out an action based on their competencies. It is not the actual (objective) ability that is decisive but the (distorted) self-assessment of one's competencies.

This pattern becomes behaviorally effective when a sense of achievement strengthens the belief in one's abilities. In this case, the motivation to complete the task increases. Motivation, on the other hand, decreases when failures raise doubts about one's competence. As a result, there is an increased intention to abandon to escape the "pain" of recognizing one's inability. Self-efficacy is distinguished from response efficacy, which is not about the competence to act but the consequence of action.




**Implementation in e-commerce**

- The website should constantly give the user affirmative feedback to do everything right (e.g., with inline validation when entering addresses or bank details).
- Information about the user's current position in the ordering process should also be provided transparently (e.g., with pictograms). Steps already completed can be presented as successes to support self-efficacy.
- The pattern can also be used in the onsite search: Intelligent auto-suggest can anticipate search suggestions and pre-empt input problems (e.g., exotic synonyms, spelling mistakes, regional idioms) not to jeopardize the user's self-efficacy.
- Other well-suited tools are FAQs, how-to pages with tutorials, explanatory videos, intermediate conclusions, positively worded hints and error messages, or social proof (seeing that others have mastered a task also generates motivation as a "vicarious experience").




**Source:** Bandura A (1977) Self-efficacy: toward a unifying theory of behavioral change. Psychological Review 84(2):191–215.

**See also:** Overconfidence, Social Proof, Bandwagon Effect, Response Efficacy, Autonomy Bias, Cheering.

**My Notes**

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# Behavior Patterns for the Customer Retention Phase

6


**Summary**

The purchase process explicitly does not end with clicking the buy button or delivering the ordered goods. Because existing customer care has a significantly better "return on marketing investment" than new customer acquisition, the first purchase should be understood as the starting point of a long-term customer relationship. The post-purchase or customer loyalty phase contains two sections in which a total of nine primary behavior patterns are classified. **Satisfaction** with the buying experience and the purchased product is an essential prerequisite for customer loyalty and can be psychologically supported with appropriate behavior patterns. If this is the case, there is an opportunity to retain customers long-term and build **customer loyalty** (Fig. 6.1).


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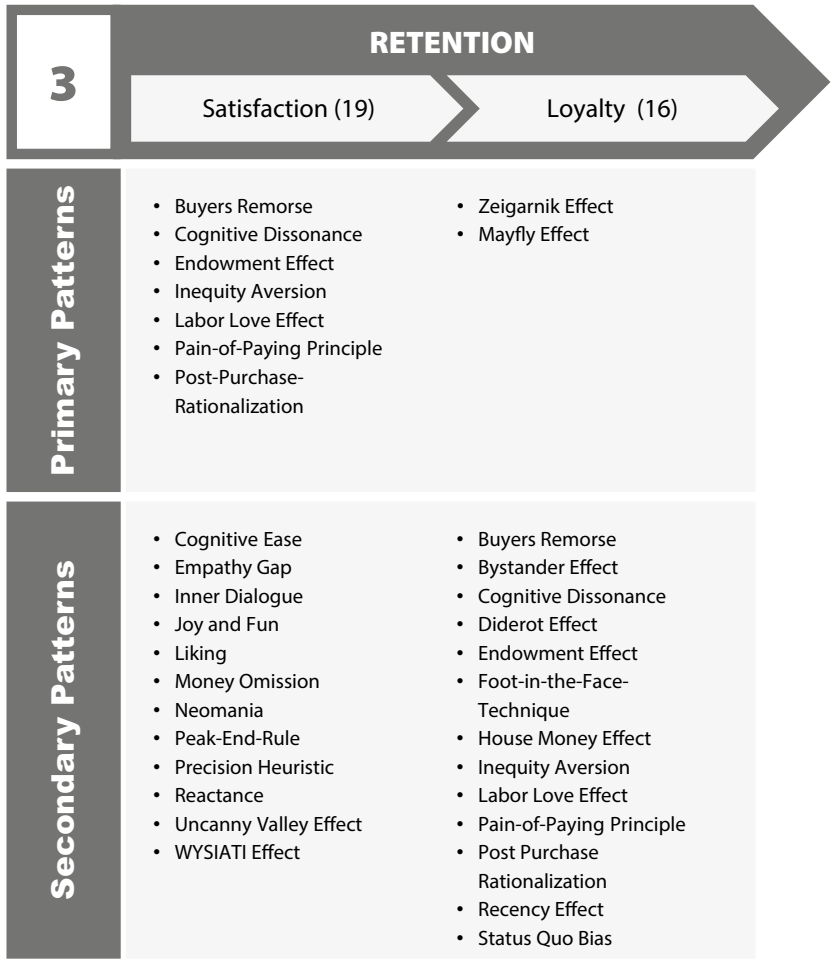



**Fig. 6.1** Overview of primary and secondary patterns for the customer retention phase

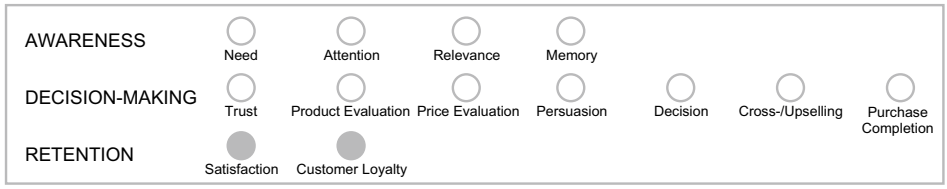
## 6.1 Satisfaction

### 6.1.1 Buyer's Remorse

Also: Buying Remorse, Post-Purchase Dissonance, Choice Closure

Buyer's Remorse is the counter-effect of Post-Purchase Rationalization: While Post-Purchase Rationalization justifies a purchase decision retroactively, Buyer's Remorse creates doubts about the correctness of the decision. This post-purchase dissonance describes a discrepancy between customer expectations and actual satisfaction.

The effect is amplified when customers have been pressured into making a decision or "bribed" with short-term incentives. The consequence can be high return rates. This makes it clear once again that the use of behavior patterns does not have to be geared towards a quick sale but towards long-term customer benefit.




**Implementation in e-commerce**

- The effect can be reduced with congratulations on the excellent deal or decision.
- The instruments of post-purchase rationalization also include summarizing the original motivation of the customer and arguments for the superiority of the shop.
- The assurance that you will still be there for customers after the purchase is also effective.
- An example from the insurance industry: ERGO (one of Europe's largest insurance companies) offers supplementary dental insurance with the positioning "Insure when it is actually already too late". In this way, post-purchase dissonance is immediately avoided, as the sensible use of the product is already ensured. As a result, the disadvantage of significantly higher premiums is no longer significant.
- The effect can be anticipated by offering social sharing buttons on the thank you page (confirmation page after completing the ordering process). As soon as users use these buttons, a decision becomes socially visible and thus much more difficult to reverse, as the risk of social devaluation would accompany the inconsistent behavior (see: Commitment and Consistency).
- Another option is to complete the decision with a "physical act. In this way, we help the brain accept the decision-making process's finality. This can be achieved by closing the order window or a note on the thank you page "You can close this window now".




**Source:** Festinger L (1962) A theory of cognitive dissonance (Vol. 2). Stanford university press, Palo Alto.

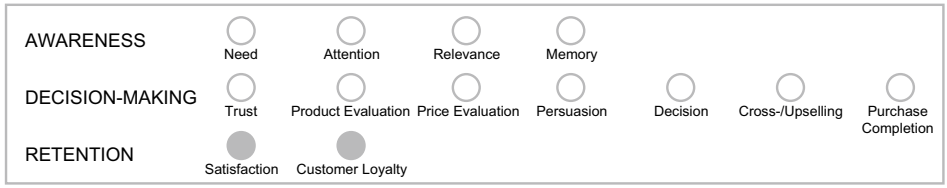
**See also:** Commitment and Consistency, Post-Purchase Rationalization, Cognitive Dissonance, Autonomy Bias.

**My Notes**

### 6.1.2 Cognitive Dissonance

Conflicts between our attitudes, our beliefs, and our behavior create dissonance – an unpleasant feeling. We try to dissolve this feeling of tension and restore balance. To do this, we have three strategies: adaptation of our behavior, adaptation of our beliefs, and adaptation of our perception. Thus, smokers can dissolve their sense of dissonance by quitting smoking (behavior change), trivializing smoking (change of belief), or celebrating the joy of smoking (change in perception).

A vivid example is provided by the fable "The Fox and the Grapes" by the Greek poet Aesop: After the fox has not managed to reach the grapes hanging above him after several attempts, he says to himself that he actually does not want them anyway because they are not yet ripe enough for him.




**Implementation in e-commerce**

- In e-commerce, cognitive dissonance usually occurs immediately after the order has been placed or when a product is used for the first time. In this context, after-sales communication is of great importance. If dissonance is felt, providers should provide a template for changing beliefs or perceptions to reduce the risk of behavior change (= return). This can be achieved, for example, by summarizing the product USPs and the utility value in the order confirmation email (see: Buyer's Remorse).
- A post-purchase survey can also reduce dissonance. For example, if you ask, "Why did you buy this product?" and provide several good reasons as possible answers, you get users to reaffirm their decision. Questionnaires can, therefore, not only record behavior but also influence it.




**Source:** Festinger L (1957) A theory of cognitive dissonance. Stanford University Press, Stanford.

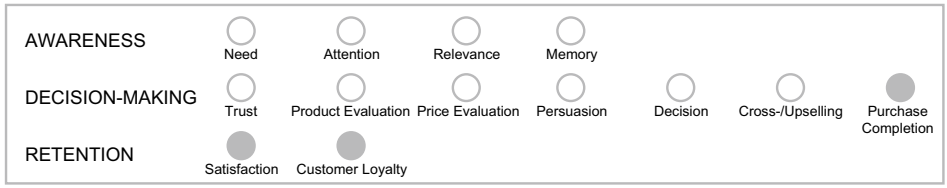
**See also:** Confirmation Bias, Buyer's Remorse, False Consensus, Post-Purchase Rationalization.

**My Notes**

### 6.1.3 Endowment Effect

People tend to consider a good more valuable if they own it. Therefore, appreciation for an object is not objective but depends on the current ownership status. This effect often leads to complicated negotiations in business relationships. It can reduce the overall transaction volume because the seller calls up significantly higher prices than the buyer's willingness to pay. The endowment effect is more substantial the less often you buy or sell an object; it is, therefore, relatively weak for everyday goods.

The mode of action is similar to the Status Quo Bias: one tries to maintain the current state as much as possible. Prospect Theory can also be used as an explanatory approach with Loss Aversion when people weigh the loss of an object through the sale higher than the profit in the form of a market price.




**Implementation in e-commerce**

- Store owners can offer long exchange periods to get users used to ownership and increase the product's perceived value. Extended exchange periods and a reduction in the return rate are, therefore, not a paradox, as the examples of Zalando, About You, and IKEA show.
- After the purchase, the Endowment Effect also applies to new customers. It, therefore, makes sense to implement social sharing buttons at the end of the ordering process (e.g., on the thank you page). Since users consider the product they have just purchased to be very valuable, the probability of willingness to use it is comparatively high.
- The situation is similar with customer reviews. Due to the Endowment Effect, these also tend to be sold immediately after purchase. The leverage would be even more substantial after receiving the product, but there is often a lack of suitable touchpoints to the customer where the reviews could be requested.
- In addition, after the purchase, all interactions should be referred to as the customer's product ("your product"). This again triggers the rewarding idea of ownership of the customers.
- The effect can also be observed within the e-commerce organization. As a professional e-commerce company, you and your employees should only be subject to the Endowment Effect to a minimal extent. However, product or category managers can also tend to emotionally adopt "their" products or products from "their" category. This identification is absolutely desirable to a certain extent but must not lead to pricing moving away from market realities and negatively influencing sales.




**Sources:** Thaler RH (1980) Toward a Positive Theory of Consumer Choice. Journal of Economic Behavior and Organization 1(1):39–60; Kahneman D, Knetsch JL, Thaler RH (1990). Experimental tests of the endowment and the Coase theorem. Journal of political economy 98(6):1325–1348.

**See also:** Status Quo Bias, Zeigarnik Effect, Loss Aversion.

**My Notes**

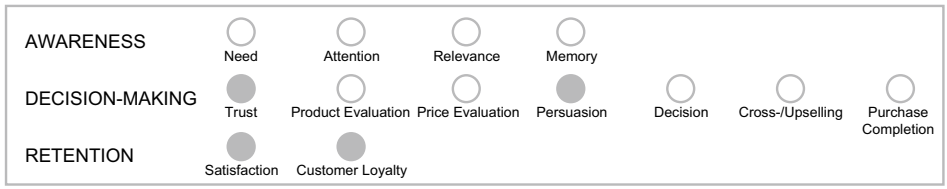
### 6.1.4 Inequity Aversion

Also: Inequality Aversion, Unfairness Aversion

Inequity Aversion is the preference for fairness and equality within a social structure. Curiously, fairness is usually interpreted as an absolute and not a relative value: For example, if two people (one of whom is wealthy and one very poor) find a $100 bill on the street, they usually consider a 50:50 split fair. The fact that the banknote has a much greater benefit for the poor person is ignored. It is also interesting to note that people are sensitive to both inequalities that are negative for them and positive for them.

The inequity aversion was shown memorably in the context of the dictator game. Here, a single participant (the dictator) decides how a reward will be divided between him and other participants. More than 50% of the subjects in the role of the dictator decide to give up at least part of the amount, which would be incompatible with maximizing the benefits of homo economicus. One stage of expansion is the ultimatum game. It also includes the rule that the other participant can veto it, resulting in both players coming away empty-handed. This veto is usually used when there is a low level of participation. The participants, therefore, prefer to receive nothing at all instead of a little if the other side also suffers a loss. So, they are willing to pay for maintaining fairness. This can also be proven neurologically: Low offers to the second player activate brain regions in the dictator responsible for feelings such as pain and disgust. Fairness is, therefore, not only strategically motivated but also emotional.

The effect was later demonstrated with capuchin monkeys, suggesting an evolutionary sense of fairness and that it is not just a social construct. It is also worth noting that our reaction to unfairness is stronger when the deceived people are close to us (e.g., the farmer from our village).




**Implementation in e-commerce**

- Fairness and social norms can be used in returns behavior: For example, it has been shown that a parcel insert with an allusion to social norms can significantly reduce returns behavior (Pfrang et al. 2015). Suppose customers are confronted with selfishness and point out ecological (environmental impact of transport) and social consequences (return costs must be passed on to the customer community). In that case, the often unhesitating returns are questioned. In addition to the decreasing return rate, this also positively affects follow-up orders.

- The effect can also be used in pricing: Documenting, for example, fairness towards all members of the supply chain can significantly improve willingness to pay and the evaluation of value for money. This is successfully used, for example, in "Fair Trade" coffee, where it is shown that the final sale price is necessary to secure an income for the coffee farmers.




**Source:** Fehr E, Schmidt KM (1999) A theory of fairness, competition, and cooperation. The Quarterly Journal of Economics 114(3):817–868.

**See also:** Unity, Bystander Effect, Reverse Psychology.

**My Notes**

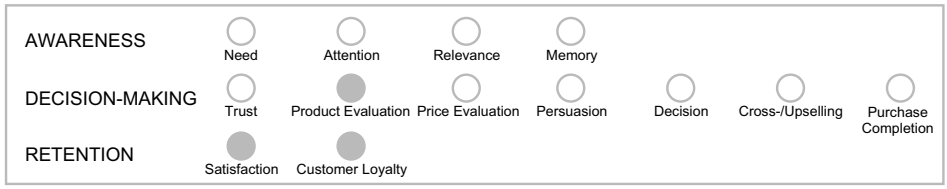
### 6.1.5 Labor Love Effect

Also: Egg Theory, IKEA Effect

The Labor Love Effect describes the cognitive bias that people like things more (love) to which they themselves have contributed or invested effort (labor). The effect was discovered and sometimes even named after the Swedish furniture store, which outsources a significant part of the added value (especially the logistics and construction of the furniture) to the customers and thus not only saves a lot of costs but surprisingly also increases the appreciation for its products. Quantitatively, this reaches almost the same level as individual carpentry. Conversely, this means that making it easier for customers does not necessarily mean that the experience will also be better. Thus, the phenomenon is similar to the Endowment Effect, which describes how people systematically overestimate the value of their own products.

In addition, there is a second aspect: The effort invested makes us blind to the perspective of others who evaluate the product (without their own invested effort) quite differently or less. However, this only applies if the results of our efforts are crowned with success. To stay with the example: If we fail to set up a shelf, there is no affection for the furniture, but the responsibility for the failure falls on the manufacturer.

The alternative name "Egg Effect" dates back to the 1950s, when consumer goods manufacturer Pillsbury realized that a baking mix that required adding water and an egg was far more desirable than the same product that only needed water. Apparently, a cake symbolizes a special event that you want to give to your loved ones. However, a cake only deserves this status if it has been crafted into it. In addition to the well-known IKEA example, one of the discoverers of the effect cites an extreme example to illustrate it – presumably, it is not meant seriously: Dan Ariely considers children to be "unsaleable" mainly because their parents have simply invested an extremely large amount of work in them.




**Implementation in e-commerce**

- Customer co-creation: Customers can be involved in creating the product, for example, in the context of personalization or mass customization approaches (e.g., designing their own shoes). In this example, in addition to the conversion-increasing effect, the effect also contributes to reducing the return rate.
- In addition, the effect can be used for products that are difficult to find or only available locally (high search or procurement effort) and for limited products.

- As we have seen in the IKEA example, the final assembly of the product by the customer is also an option for implementation. This also enables a smaller pack size and lower logistics costs.
- Cooking boxes (ingredients by courier, you have to cook yourself) are an excellent example of a direct transfer of the IKEA principle and are consequently enjoying increasing popularity.
- Finally, customer surveys can also be used as a means of persuasion: having products tested by customers and obtaining feedback increases the willingness to participate and the appreciation for the product and the provider.




**Source:** Norton MI, Mochon D, Ariely D (2011) The 'IKEA Effect': When Labor Leads to Love. Harvard Business School Marketing Unit Working Paper No. 11–091. https://doi.org/10.2139/ssrn.1777100.

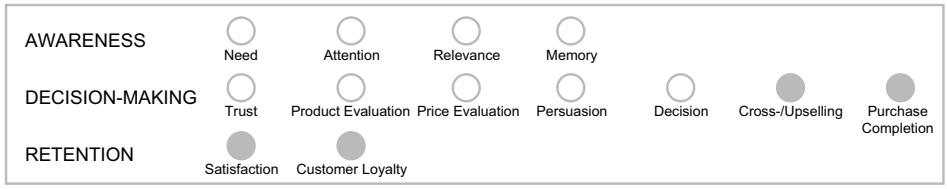
**See also:** Endowment Effect, Not-Invented-Here Syndrome.

**My Notes**

### 6.1.6 Pain-of-Paying Principle

There is a close psychological connection between the pleasure of consumption and the payment for it. The Pain-of-Paying Principle describes that we are reluctant to part with our money because it has much more than just the function of a medium of exchange psychologically (see: Dollar Eyes Effect). Every time you spend money, therefore, triggers the questioning of the meaningfulness of the respective purchase decision, which carries the risk of post-purchase dissonance (see: Buyer's Remorse) or purchase abandonment. For example, a ticking taximeter noticeably reduces the pleasure of taking a taxi. Neurologically, spending money activates the regions that are also associated with pain and disgust. This correlation is more pronounced in the case of cash than in the case of payment by invoice or credit card. Regardless, the amount of expenses is closely related to the pain felt.

Prelec and Loewenstein, who described this effect for the first time, also discovered that the use of products that have already been paid for is perceived almost as "free". They refer to this as "prospective accounting" (1998).




**Implementation in e-commerce**

- The pain of paying can be reduced if it does not coincide with the purchase decision. In the case of purchase on account, this takes place much later (often after the end of the regular return period), and it is also hyperbolically discounted and is, therefore, weaker from today's point of view (see: Hyperbolic Discounting).
- Because of the pattern, payment methods with a high degree of immediacy and transparency are still struggling in e-commerce and will probably also have a niche existence in the future.
- The "pain of paying" is significantly reduced if customers have stored their payment information in the customer account and can skip the pain-causing step in the ordering process. This is well implemented with Amazon's 1-click order and a deep integration of PayPal. The transaction, perceived as more pleasant as a result, increases customer loyalty to the provider and creates lock-in-like situations.
- Bundling: If several products can be added to the shopping cart with just one click, the pain is correspondingly lower (e.g., with "shop the look"). In addition, products are then regarded as an elementary component of a superordinate unit that you no longer want to dissolve.

- Multi-channel companies with their own store network can score points with free pick-up on-site (click-and-collect). Several positive aspects come together: The absolute payment amount decreases, and thus, the pain of paying is almost the same. In addition, the word "free" triggers the perception of value for money. Ultimately, there are cross-selling opportunities when picking up on-site, so the pick-up counter should never be placed in the checkout area or even outside the store.
- Return guarantees and refund promises should be prominently communicated, even if they only fail within legal requirements. Attention: Compliance with legal requirements must not be presented as an exceptional goodwill gesture. In many industries, however, the possibility of withdrawal is currently hardly communicated, thus setting the threshold for the purchase artificially high.
- The verbalized form of money is currency symbols. It may make sense to forego the price units ("14.99" instead of "14.99 Dollar" or "$ 14.99") to present the payment less explicitly. This approach is becoming more common in the US, especially in high-end e-commerce environments.




**Source:** Prelec D, Loewenstein G (1998) The red and the black: Mental accounting of savings and debt. Marketing Science 17(1):4–28.

**See also:** Dollar Eyes Effect, Hyperbolic Discounting, Cognitive Dissonance, Buyer's Remorse, Zero-Price-Effect, Mayfly Effect.

**My Notes**

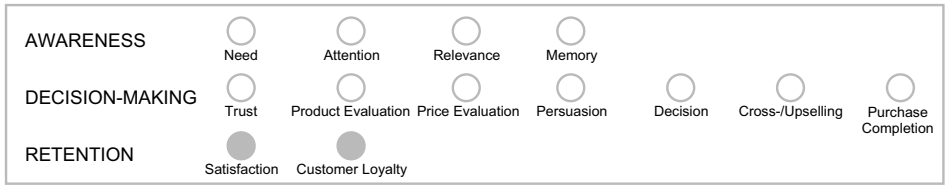
### 6.1.7 Post-Purchase Rationalization

Also: Buyer's Stockholm Syndrome

We unconsciously attribute unreasonably high importance or quality to products after purchase. In this way, we want to convince ourselves of the correctness of the purchase decision and weaken the feeling of having bought a superfluous or overpriced product. This is intended to provide a commitment to the original decision and avoid post-purchase dissonance (see: Buyer's Remorse). Therefore, the pattern is considered a particular manifestation of Confirmation Bias.

The purchase decision is, therefore, often made emotionally, after which a rational justification must be provided immediately. This is done by adjusting the weights of the distinctive features. An example: "Of course, the iPhone is more expensive than the Android Phone; it has a compass app. This can be vital in certain situations!".

Because of this mildly schizophrenic buying behavior, the pattern is also known as Buyer's Stockholm Syndrome – in reference to Stockholm Syndrome, which describes why victims of hostage-taking sometimes form an intimate relationship with their captors.




**Implementation in e-commerce**

- Customers can be helped after the purchase and given a rational justification to avoid or reduce this post-purchase dissonance (instead of simply: "Thank you for your order").
- The justification does not have to be explicitly formulated. The advantages of the product or the shop can very well be implicitly formulated as values from the customer's point of view. For example, the online florist Bloomy Days wrote for a long time on its thank you page, "Look forward to the most beautiful and freshest flowers that we can find on the market and send to you today"). This was probably a much stronger post-purchase rationalization than simply talking about "high quality" and "fast shipping".
- Texts on the thank you page or in order confirmation emails should always be written in post-purchase language. This leaves no doubt that the decision has already been made and that a return is not an option. This can also be achieved by describing which processes (e.g., in logistics) have been initiated and are already running immediately after the order has been sent.



**Source:** Cohen JB, Goldberg ME (1970) The dissonance model in post-decision product evaluation. Journal of Marketing Research 7(3):315–321.

**See also:** Confirmation Bias, Buyer's Remorse, Cognitive Dissonance.

**My Notes**

## 6.2 Customer Loyalty

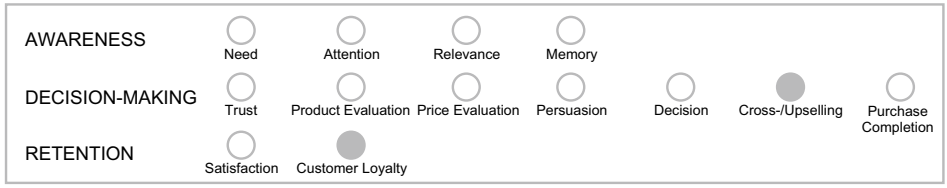
### 6.2.1 Zeigarnik Effect

Also: Endowed Progress Effect, Completion

The Zeigarnik Effect assumes that it is human nature to want to finish things once they have started. As long as this goal is not achieved, we feel an unpleasant dissonance (see: Cognitive Dissonance) that builds tension. People tend to reduce these tensions and not abandon a task that has been started. This can also be seen in the resources we invest in reminding us of tasks: Tasks that have already been completed are much more difficult to remember than open to-dos, as fewer resources are allocated.

The waiter, who is able to remember complex orders and pass them on to the kitchen, can be used as an example. However, after the guests of a table have paid and the process is completed, it is much more difficult for him to remember the food consumed.

In a well-known experiment, Nunes and Drèze issued loyalty cards to car wash customers. Half the customers needed eight stamps to get a car wash for free. The other customers needed ten stamps, two of which were given to them when the loyalty card was issued. The first two stamps created a feeling of incompleteness and the need to complete the card, which led to a significant increase in loyalty from this group of customers.




**Implementation in e-commerce**

- The use of "cliffhangers" is conceivable: This works well in guiding users from one step of the buying process to the next. Example: "On the next page, you will find a coupon code to save a surprising amount."
- Progress bars visualize the imperfection of the task during the ordering process and motivate completion. The first step should be "gifted" so that the progress bar starts right at the second step. At Zalando, for example, the first step stands for "register", which has already been done.
- Due to its increased clarity, a single-page ordering process can lead to the fact that not the individual steps but the entire ordering process is perceived as a task that needs to be done. However, this should be urgently tested before the roll-out, as clarity usually suffers compared to a multi-page check-out.
- The Zeigarnik Effect also supports the clear communication of the completion of the ordering process: Users must be completely clear when they can classify the "online

shopping" task as completed and mentally discard it. If this is not the case, emerging dissonance can lead to an increased return rate.

- Customer loyalty program: A simple points system proves its worth instead of a complicated discount scale. Even before the first purchase, customers should be given the first points, e.g., for the first visit or creating a customer account. Many digital loyalty programs (e.g., at Gearbest, Rakuten, or Bestsecret) work according to this principle.
- Creation of a customer profile: The first data required for the transaction (e.g., name and e-mail address) are already stored, and a progress bar shows the first "successes" in completing the profile. With this presentation, customers are much more likely to be willing to provide further information that is valuable for later contact, such as date of birth, marital status, and telephone number.
- Cross-selling: If you present a product as part of a bundle that belongs together, customers do not want to split up this bundle and do not want to finish shopping until it has been entirely placed in the shopping cart.




**Source:** Zeigarnik B (1938) On finished and unfinished tasks. A source book of Gestalt Psychology 1:300–314; Nunes JC, Drèze X (2006) The endowed progress effect: How artificial advancement increases effort. Journal of Consumer Research 32(4):504–512.

**See also:** Endowment Effect, Commitment and Consistency, Cognitive Dissonance, Zero Risk Bias, Cheering.

**My Notes**

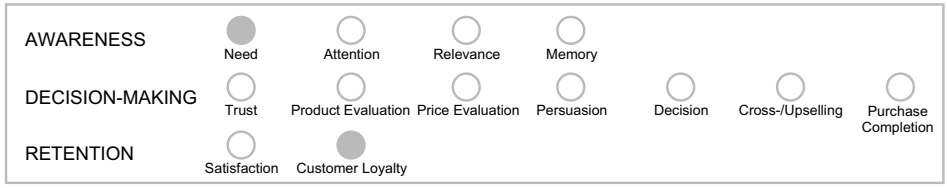
### 6.2.2 Mayfly Effect

Also: Psychology of Consumption.

The goal of sustainably thinking e-commerce managers is to persuade users to buy and ensure that the purchased products are used regularly. On the one hand, this is a question of professional honor, and on the other hand, it also makes sense from a business point of view – especially in the context of contract and subscription products. As is well known, acquiring new customers is considerably more expensive than reactivating existing customers.

The Mayfly Effect describes that we usually only use many (contract) products for a very short time. Anyone who has already signed a gym contract should now feel caught. Various studies confirm, show, and explain the connection with the sunk cost phenomenon: When we spend money on something, we use it – just to avoid the feeling of having made a wrong decision. This is true even if we no longer consider the use of the product to be beneficial. However, this effect wears off quickly. If the contract was signed some time ago, sunk costs no longer motivate us.

So, how do you manage not to let customers become a flash in the pan and achieve sustained product use? Regular reminders are a surprisingly simple yet effective remedy. Even if it seems counterintuitive, reminding customers of the commitment made prolongs the effect of the sunk cost effect, and the use remains at a high level. If, on the other hand, we use a product regularly, the probability of doing so beyond the termination date increases considerably (see: Status Quo Bias, Endowment Effect).




**Implementation in e-commerce**

- Sending regular e-mails reminding recipients of the booked contract (flat rate for delivery cost, magazine subscription, gym membership, software license, etc.) has been proven to increase product usage and customer loyalty.
- In such loyalty campaigns, whether you are addressing more action-oriented or status-oriented customers should be clarified in advance. Status-oriented customers generally find it difficult to change an existing state; the sunk cost effect is much stronger and longer for them. For action-oriented customers, it is essential to lead the existing actionism into active product use and continuously build up ownership effects (see: Endowment Effect).
- The "pain of paying" should be reduced in parallel by automating payments. Processes that require the customer's active involvement (e.g., bank transfer) are to be avoided.

- A day-based representation of the payment amount has a similar effect. Presenting a price of $79 per month for a gym as $2.63 per day" is perceived as cheaper due to the lower reference point (see: Anchoring). In addition, with odd amounts, we are hardly able to intuitively infer monthly costs and process time in a non-linear manner (see: Hyperbolic Discounting), which also makes the perception of costs more bearable.




**Source:** Van Putten M, Zeelenberg M, Van Dijk E (2010) Who throws good money after bad? Action vs. state orientation moderates the sunk cost fallacy. Judgment and Decision Making 5(1):33–36.

**See also:** Pain of Paying Principle, Loss Aversion, Diderot Effect, Commitment and Consistency, Precision Heuristic.

**My Notes**

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# Operational Work with Behavior Patterns

# 7


**Summary**

A robust library of behavior patterns can only develop power with the proper operational application. For this reason, it is essential to know the specifics of its use in e-commerce. With this knowledge, practical triggers can be developed in the next step, which initiates the corresponding behavior patterns on the digital surface. For a stimulus to work, the ability and motivation are essential prerequisites on the customer side. Generally, "the more, the better" does not apply when using behavior patterns. Instead, the selection should be made carefully and based on well-founded hypotheses. To ensure that the conversion does not deteriorate with the use (this is also possible if misused), an optimization project must always be accompanied by experimental methods (such as A/B testing). Especially in the context of personalized websites and the dynamic selection of behavior patterns for specific user groups, this involves a complexity that should not be underestimated – especially in the context of new legal circumstances, as a guest article by Joachim Stalph (Managing Partner, elaboratum) shows. Despite all the conversion focus, customer needs must never be neglected. An "alignment of interests" between company and customer requires ethical and moral responsibility on the one hand. On the other hand, this is in the interest of a sustainable and profitable customer relationship.



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## 7.1 Advantages and Disadvantages of Using Behavior Patterns in a Digital Context

Behavior patterns, especially in the digital sector, have advantages and disadvantages compared to other application contexts. This will be briefly discussed at the beginning of this chapter to enable a differentiated view of the implementation.

**Advantages**

- **Immediacy:** The effect of behavior patterns on conversion and sales is much more direct in the digital context than in brick-and-mortar retail, for example. When confronted with the respective triggers, users are only clicks or seconds away from the purchase so that the triggered impulse can become fully behaviorally relevant. In brick-and-mortar retail, there is a significantly longer chain of effects (both locally and temporally), which entails the risk of relativizing or weakening the behavioral impulses.
- **Measurability:** The knowledge of the effect of behavior patterns can not be built up as well in a non-experimental setting as in the digital field. Only here does a user leave masses of clues and traces that allow his decision-making process to be reconstructed step by step. With the help of session IDs, cookies, and cluster approaches, almost the entire customer journey can be traced – often even over several visits, days, and end devices (see also ► Sect. 7.5).
- **Freedom from social influences:** Unlike brick-and-mortar retail, online shopping usually takes place without any direct influence from other people. Neither sellers nor accompanying persons "disturb" the activation potential of the behavior patterns. This makes them all the more effective in this context and makes it possible to assign the effect more clearly to a specific polluter.

**Disadvantages**

- **Validity:** Many of the behavior patterns have so far only been proven in an experimental context. Often, the database of the pointed and often particularly effective patterns is unsatisfactory. Therefore, it is not always possible to say unequivocally whether an observed anomaly is a universal, psychologically or neuroscientifically explainable pattern or statistical coincidence. In practice, the only thing that helps is trial-and-error or testing, testing, testing.
- **Transferability:** Since many patterns were initially discovered and described in other contexts (e.g., social interactions, brick-and-mortar retail, direct selling, fundraising), the question of transferability to online retail often arises (see also the following point "Transparency"). However, since the industry influences are usually not more potent than other contextual factors of the impact, transferability can be ensured with tests that are necessary anyway. In addition, strict attention was paid to digital applicability when selecting the listed PsyConversion® patterns.

- **Transparency:** Patterns are often used in e-commerce, but mostly "behind closed doors". Conversion optimization is challenging, with only a few opening their books and openly sharing valuable insights. The seemingly trivial question of whether a measured uplift is large or small is often difficult to answer without reference values. All readers are recommended to join existing experience exchange groups or to create their own formats. The exchange with like-minded people with whom one is not in direct competition is a considerable accelerator and efficiency driver of working with behavior patterns since the behavioral patterns on which the test results are based are usually universal and can be transferred excellently. In this way, it is possible to set the right priorities in testing and not get lost in replicating multiple contexts.

## 7.2 Conception of Triggers to Activate Behavioral Patterns

If we are honest, the crucial question in e-commerce practice for most users of this book is not which psychological or neurological explanatory patterns theoretically underlie the behavior of users. Instead, it is: How can I trigger desired behavior in my website visitors in a targeted manner? To achieve this, two things are needed: a profound knowledge of the psychological behavioral patterns of the users and the ability to develop triggers that activate the respective patterns. The former was taught in chapters 4, 5, and 6; the latter will be addressed in the following section.

A commonly used and easily accessible model for the relationship between triggers and behavior is the Fogg Behavior Model (Fogg, 2009; www.behaviormodel.org). Despite or perhaps because of its simplicity, it is often regarded as a central point of reference for persuasive design. It contains reference points to many significant theories, such as Bandura's Social Cognitive Theory and Self-Efficacy (1977, 1989), Fishbein and Ajzen's Theory of Reasoned Action or Planned Behavior (1975, 1991) or Petty's and Cacioppo's Elaboration Likelihood Model (1986). The behavioral model describes that three aspects must be present at the same time for people to act:

1. Motivation (there must be a reason to do something)
2. Ability (there must be the ability to do this)
3. Trigger (there must be an impulse or stimulus that makes people do something)

One of the three factors is missing if a desired behavior fails to materialize. The list also serves as a checklist for activating behavior patterns, which user experience professionals

and conversion optimizers can use to determine where the bottleneck is located. The connection can be expressed in a simple formula:

**Behavior/Action = Motivation * Ability * Trigger (B = MAT).**

In the model, there are three core motivators (→ motivation), six simplifying framework conditions (→ ability), and three types of stimuli (→ triggers). "**Motivation**" includes the three motivators feelings (in the model: "sensation"), expectations ("anticipation"), and belonging ("belonging"), which are understood as linear dimensions with pairs of opposites at the extreme positions:

- Feelings: Pleasure /Pain
- Expectations: Hope/ Fear
- Belonging: social acceptance/social rejection

The fact that users must be able to act at all (i.e., "**Ability**" is sufficiently present) to do just that seems trivial at first glance. However, user experience developers and designers often expect too much "ability" from users, especially in this area, which can lead to the failure of digital projects. To eliminate the bottleneck of users' capability, there are two strategies:

- Training to develop their skills: Fogg (2009) refers to this path as "the hard path" and advises avoiding it whenever possible due to the considerable effort involved.
- "Simplicity": The more manageable tasks are to master, the higher the average user's ability. Simplicity is the "minimum satisfactory solution at the lowest cost". According to Fogg (2009), this simplicity is a precisely decomposable construct consisting of six elements, the weighting of which can vary depending on the user:
  - Time: Completing tasks must take up as little of our time resources as possible.
  - Money: Accomplishing tasks must cost as little money as possible.
  - Physical effort: Completing tasks requires as little physical effort as possible.
  - Brain cycles: Accomplishing tasks must generate as little cognitive effort (thinking) as possible.
  - Social Deviance: Accomplishing tasks must comply with our society's generally accepted social rules.
  - Non-routine: Accomplishing tasks must be compatible with existing routines and habits.

Each user has their simplicity profile: teenagers, for example, have more time and less money than adults in the middle of their working lives. The actual ability also depends on the most limited resource – and this is perhaps the most critical insight for the development of simple digital services. So, if there is a bottleneck in one of these six factors,

it no longer matters how extensive the other five factors may be. It is essential to consider this when modeling the simplicity for your core target group.

There is an exchange relationship between the ability and the motivation: If the ability is low, the motivation must be correspondingly higher to trigger a behavior. Alternatively, to put it another way: With high motivation, people are willing to do hard or strenuous things.

The third and final factor in the motivation model is **triggers.** They are also referred to as "cue", "prompt", "call-to-action", or "request" in other models. There are three different types of triggers, which are suitable for activating the user differently depending on the use case:

- "Facilitator" ( high motivation/low ability): Motivated users who find a task complex, however, can be activated with a trigger that is geared towards the actual simplicity of the task (see: Self-Efficacy). These types of triggers are prevalent in social networks or, for example, during software updates.
- "Signal" (high motivation/high ability): The ideal case for shop operators. To motivate users to take an action, a simple signal is enough that an action is now appropriate. Accordingly, the trigger should not be aimed at the simplicity of the action or the user's motivation to not fizzle out ineffectively due to the already high severity of these factors.
- "Spark" (low motivation/high ability): With a high ability, a spark can be used as a trigger, increasing users' core motivation.

All three types of triggers have one thing in common: their effect is significantly influenced by the **timing** of the placement. For e-commerce companies, this is a recommendation to develop a precise understanding of how users use their product and when motivation and ability are sufficiently available. If one of these factors is absent (e.g., lack of time capability because users are very busy during the day), the trigger will fizzle out without effect. The same applies to the end device or channel through which the trigger is played out (e.g., push notification in the app, newsletter mailing). The following applies: The more direct the access, the higher the control over the time of perception of the trigger. For example, a push message on a smartphone is usually noticed within seconds, while an e-mail newsletter can go unnoticed for days.

## 7.3 Scope of Use of Behavior Patterns

In chap. 3, it was explained that using some patterns can also be associated with risks. The Door-in-the-Face Technique can nip the delicate plant of a customer relationship in the bud, Threat can scare customers away in the long term, and negative Framing may lead to customers permanently associating bad associations with a product –just a few

examples. From this alone, one can derive that behavior patterns should only be used **consciously, thoughtfully, and tested**. In addition, there is the risk of negatively influencing the user experience, as every trigger in the interface can interfere as an additional element and prevent flow. It can be helpful to use the Fogg Behavior Model (► Sect. 7.2) to keep in mind and to be aware of what bad triggers can trigger in users: If, for example, motivation is lacking, an ability-oriented impulse is annoying. If the essential ability to act is missing and powerfully activating triggers are sent, you create moments of frustration among users. "**Visual noise**", i.e., intrusive or disruptive external stimuli that take up scarce attention resources from users, must be avoided at all costs. Typical examples are pop-up banners, customer surveys in the shop, chatbots activated without being asked, or newsletter advertising – all of them usually do not represent the reason for the visit and disrupt the purchase process. In the meantime, it has been proven that such interruptions increase mental stress and cause fatigue (Youyou et al. 2015) – neither of which is particularly conducive to conversion.

Again and again, the phenomenon of "**psychological backfire**" comes up (Petty & Cacioppo 1986). This means that behavior patterns can also trigger the opposite of the desired effect: If someone tries too hard to gain our trust, it fuels mistrust. We know similar effects, for example, from the reactance pattern. In the meantime, large meta-studies have shown (Xie et al. 2017) that the backfire effect is rare. While it has been widely acknowledged that people are reluctant to adopt viewpoints that contradict their current beliefs (see: Status Quo Bias and Belief Bias), there has been little evidence of a reverse effect. Nevertheless, it should be noted that it is certainly not one of the most effective negotiation and persuasion tactics to confront people directly with opposing points of view.

Between the patterns, there are mutual **reinforcement or cannibalization processes.** It is well known that these interactions exist. However, how and to what extent they influence the effect is unclear. This can only be determined with a clean test setup and simultaneously makes it clear that each additional pattern can have a long sequence of consequences and should, therefore, be implemented cautiously.

When using behavior patterns, it is essential always to keep an eye on the ultimate conversion goal, not just the next step of the customer journey. The stubborn but stupid claim "Sex sells!" in online marketing, for example, leads at best to an increase in attention but usually not to a measurable increase in sales. The abandonment point of the users is only pushed further back by acquiring such unqualified traffic (with the negative consequence that the measured conversion rate collapses in the purchase process). In addition, we have learned from the example of the behavior patterns "Authenticity" and the "Aesthetics Heuristic" that such representations per se can only appeal to a limited group of users.

E-commerce organizations with high professionalism in quantitative testing can set up a multidimensional test matrix with pattern/target group/customer journey as a model for

determining such cross-effects and visualizing possible consequences as a constant appeal to use behavior patterns sparingly.

## 7.4 Personalization for Maximized Effectiveness

To uncover the full potential of behavior patterns, it must be ensured that the patterns that generate the desired willingness to act are activated in users. However, this is precisely where the most complex challenge for companies lies. The size and diversity of the target groups with their different needs, goals, possibilities, and preferences make it clear that a "one fits all" strategy does not unleash the full impact potential of the patterns.

According to current knowledge, the best answer to this challenging situation is the real-time personalization of the content played. It stands for the renunciation of the watering can principle and pursues the goal of directing the right information and triggers to a specific target group at the right moment and the right channel. For the personalized approach, the first step is to **create segments**, e.g., based on socio-demographic characteristics such as gender (consequence: e.g., showing only women's clothing), age (consequence: e.g., presenting mobile phones with senior-friendly features on the homepage) or region of residence (consequence: e.g., offer same-day delivery in large cities). However, it is also possible to create a psychographic and needs-oriented typology of customers, e.g., with the Limbic® Types of the Nymphenburg Group or the dimensions of needs according to Hassenzahl (Diefenbach et al. 2014). This form of segmentation is usually more promising than the aforementioned socio-demographic segmentation because it reveals how users think and feel. In practice, a third form of segmentation is prevalent: it is based on easy-to-collect **primary movement data and historical behavioral characteristics.** It is often perceived as superior to the other two segmentation approaches because it can make real-time recommendations based on actual behavioral data. However, their forecast quality is proportional to the amount of data available: An anonymous first-time visitor with an activated ad blocker can hardly make a valid statement based on data, while a loyal regular customer who rarely deletes his cookies can be classified excellently. However, since the measurable behavior on a website can only be used vaguely to determine the actual emotional needs, a combination of the different segmentation approaches is the ideal solution. Data-based segmentation is beneficial in later customer journey phases when users have left sufficient data traces. On the other hand, at the beginning of the customer journey, psychographic approaches often generate more precise profiles.

#### The Pitfalls of Socio-Demographic Segmentation

The following well-known example shows how imprecise socio-demographic segmentation can be (no matter how many criteria they contain): A segment consists of men in their late 60 s, born and raised in England, wealthy and professionally successful, with a penchant for holidays in the Alps. At first glance, this sounds like a meaningful basis for uniformly and needs-based addressing users with these features. However, the description applies to both Prince Charles and Ozzy Osbourne.

The two are socio-demographic twins but ultimately different characters. It is highly doubtful that an online shop could successfully sell its products to both people with this segmentation. This makes it clear that for meaningful segmentation as the basis for a personalized approach, it is usually not age or gender but rather behavior, values, and preferences that are decisive.

Especially for socio-demographic and psychographic segmentation, the question always arises: How is it possible to find out which **user is behind the IP address** currently entering an online shop? Unfortunately, this user does not introduce himself to the shop owner with a meaningful profile. Nevertheless, you can gain much knowledge about him. The **digital** footprint that each Internet user leaves behind when surfing reveals, for example, with which device and from which location he visits the site, whether his software is up to date, which page he was on before, and whether this is his first visit.

However, segmentation based on this data usually does not make sense, as it says nothing about the users' needs. It is more promising to identify statistical correlations of psychographic user characteristics with these characteristics from the digital footprint. For this purpose, a model must be established that establishes assumption- and **data-based correlation relationships** between digital footprint characteristics and user segmentation characteristics (see the guest article in ► sec. 7.5). The latter can be excellently developed and defined with **user-centered methods**. There are also good opportunities within the framework of quantitative approaches (e.g., the popular Myers-Briggs type indicator or the "Big Five" of personality traits).

Suppose you want to take the Big Five personality test and determine how the five focal points of openness, conscientiousness, extraversion, agreeableness, and neuroticism are developed. In that case, you can do so free at https://bigfive-test.com/.

The assumptions in the correlation model are then checked and adjusted until a probability of error defined in advance as acceptable is reached. From this moment on, it can be said with a correspondingly high degree of probability: Anyone who visits the site at 11 p.m. with a current iPhone from a comparison portal is probably a user from the "Performer" segment (see Limbic® Types), who can be activated particularly strongly by a combination of the behavior patterns "Curiosity" and "Illusion of Control". Such personality modeling is already astonishingly valid today and can sometimes describe a user's personality more accurately than their friends and family (Youyou et al. 2015).

As soon as this assignment has been made, the personalization system automatically plays out the **individually optimized page variant** with the appropriate orientation of the triggers to activate the respective conversion-relevant behavior patterns. In the case of the above-mentioned performer type with a strong dominance orientation (essential values are, e.g., performance, status, efficiency, assertion), these could be, for example, the patterns of overconfidence, self-efficacy, and action bias (see Sect. 3.6).

The underlying model should be **designed as a self-learning system**: With each user interaction, the conversion of the controlled variant is measured. In this way, a large

number of measuring points are created in a short time, which allows the system to be continuously and iteratively improved. This is also necessary because user groups are by no means static constructs. They are subject to constant change and usually have a short half-life. This makes it clear again that conversion optimization may have a starting point but certainly not an endpoint.

What must be taken into account with all personalization approaches is that they carry the **risk of reactance.** If users consciously notice the individualized control of the content, they can feel limited in their freedom and autonomy. Some companies that rely heavily on retargeting are already receiving similar negative feedback today. With this form of online advertising, an interested visitor is given a cookie in an online shop without being noticed ("Has visited shop XYZ, interested in product 123"), and from now on, he will be followed at every turn by advertisements for this product on the Internet. As a result, the website operators' otherwise hidden tracking and targeting options suddenly become visible, which is often perceived as a creepy moment and an intrusion into the user's privacy.

Retargeting is the perfect keyword to move on to the following guest article by digital marketing and data strategy expert Joachim Stalph, as this marketing channel is severely affected by the changed legislation. Before that, however, a small conclusion on personalization: It must not be intrusive or creepy for users. On the contrary, it is designed as a service that provides users with suitable products more quickly, reduces complexity, and cleans up the confusing range of irrelevant information.

## 7.5 Guest Article: Customer-Driven Marketing

***The author of the following guest article is Joachim Stalph, data strategist, managing partner at elaboratum, and expert in the field of data-based user approach. In it, he describes the external shock that advertisers in the digital industry face due to the changed legal framework. Stalph believes this shock will lead to a stringent focus on customer needs in online marketing. Their effectiveness is hardly conceivable without behavioral economic principles. The article thus provides a valuable addition to the question of how the PsyConversion® principles can be scalable in the context of a changed technological setup.***

### A New Era: Customer-Centric Customer Journeys

The introduction of the new General Data Protection Regulation in the European Union and in similar forms all over the world and the changed technical settings of the browsers eliminate the setting of third-party cookies and thus pose significant challenges for advertisers. Data collected on other sites may soon no longer be used. Minor adjustments to tried-and-tested marketing strategies and changes in tracking are, therefore, not enough. The new situation triggers a fundamental rethinking of the entire structure of data-driven marketing. First and

foremost is data sovereignty: What data (= information about customers) does a company possess, and in which structure and quality?

Until now, data sovereignty in advertising companies has been left to marketing agencies. The user approach was, therefore, predominantly a task of the marketers and the corresponding technology providers. However, now it is a matter of **regaining data sovereignty** and bringing it into line with the use and analysis of your own collected data (such as transaction data, search queries, data from forms, and CRM data). In addition, existing silos in the user approach must be dismantled. In short, if you want to market successfully in the future, you should **bury data-driven marketing and rely on customer-driven marketing**. This "re-engineering" includes the following disciplines:

- Customer centricity and identity management
- Customer Intelligence
- PsyConversion®-optimized customer journeys
- Measurability
- Technologies

**Identity Management: Focus on the User**

The cross-channel user approach along the entire customer journey has been considered ideal for many years. However, reality still lags behind this picture. We often find a linear, siloed, channel-specific, and episodic user approach. Third-party cookies have characterized the customer approach in recent years; data has been increasingly used by advertisers who collect user information on other sites with the cookies. This led to a vigorous campaign-centricity.

The decisive factor for concrete measures was the user's current (website) behavior, detached from the customer journey phase.

The focus on first-party data now provokes a paradigm shift, as shown in Fig. 7.1: The customer and his customer journey become the new maxim. Considering many individual factors, a holistic customer experience is to be created. The goal: a tailor-made coordination of all channels and a customer approach according to the dramaturgy of the storytelling.

**Build Comprehensive Customer Profiles Without Cookies**

Without cookies, there are no customer identities; without identities, there is no practical customer approach. Is there a risk of a relapse into the watering can principle? If you take the proper measures, you can counteract this. To do this, advertising companies, advertisers, or publishers must try to retain users and arouse their willingness to use data. The key to success: the creation of all-encompassing customer profiles. This is the only way to meet customer expectations and use marketing measures efficiently.

| | campaign centricity | customer centricity |
|---|---|---|
| Metrics | campaign performance<br>• ad impressions<br>• clicks<br>• conversions | customer quality<br>• CLV<br>• retention |
| Data | static use | dynamic use |
| Personalisation | static | dynamic |
| Customer Journeys | silo-like, triggered campaigns | co-orchestrated |
| Optimization | channel optimization | customer experience |

**Fig. 7.1** Key differences between campaign centricity and customer centricity in marketing. (Source: own illustration)

However, creating customer profiles poses challenges: Information from different channels and data sources must be aggregated, normalized, and unified in compliance with data protection regulations. The data collected differs between:

- personalized data: data left by the user (login, registration, purchase process)
- anonymized data: cookie-based information without personal reference

**Identity resolution management** is the order of the day. This refers to the process of creating customer profiles, taking into account all data sets. This process requires interdisciplinary cooperation among the advertising companies. This is because the points of contact between a customer and a company occur via several channels and departments, as shown in Fig. 7.2.

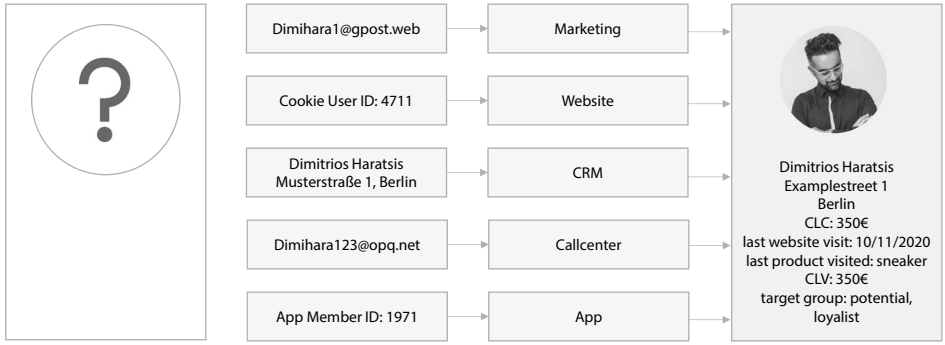



**Fig. 7.2** Personalized and anonymized data are aggregated into a profile. (Source: own representation)

| | BASIC | AVERAGE | ADVANCED |
|---|---|---|---|
| Data source | website | website + database | website + database + AI |
| Information | onsite data | onsite data + customer quality | onsite data + customer quality + propensity |
| Models | RFME | RFME enriched with CLV, retention, churn | RFME + customer quality + predictions/ recommendations |
| | = data-driven marketing | = customer-driven Marketing | |

**Fig. 7.3** Maturity model of Customer Intelligence (source: own illustration)

If this data is combined into a singular profile, there are many advantages:

- Basics for the creation of a multi-level, multi-channel approach
- Increase in sales through the efficient channel use
- Improved brand image by avoiding unnecessary communication
- Increased conversion rate through storytelling approaches.

During creating customer identities, comprehensive knowledge about the customer is built up. Two databases constitute the appropriate basis for this:

- Web data: all the data left by the customer when visiting their website
- Database data: all data that the customer leaves in the company's systems (data warehouse, CRM, Point of Sale)

By linking these databases, customers can be classified and segmented well. Not only affinities and socio-demographics are considered, but also attributes of customer quality (Fig. 7.3).

**Behavioral Economics in Segmentation**

The linking of the two databases makes it possible to enrich the profiles or segments with the characteristics that, according to current findings in behavioral economics and behavioral psychology, are responsible for > 95% of all decisions: those of the needs level.

The **interdependence between psychology and marketing technology** is as straightforward as unknown (Fig. 7.4).

Even though the example in the previous section has already been strained, it illustrates the data dilemma well: Ozzy Osbourne and Prince Charles have identical socio-demographic characteristics, but they could not be more different in their level of personality and needs. A purely data-based approach logic is, therefore, almost bound to fail. The needs dimension is, therefore, the decisive and, above all, hardly considered a new criterion for the successful

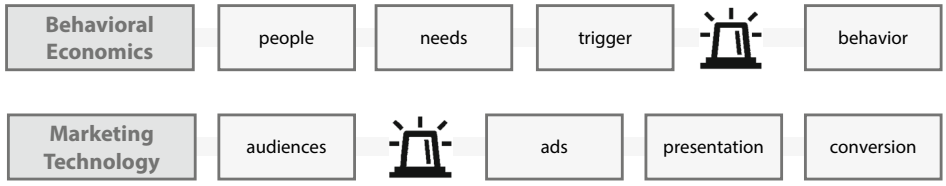



**Fig. 7.4** Interdependence between psychology and marketing technology. (Source: own presentation)

design of customer journeys. On the one hand, the fact that it is not yet on everyone's lips can be explained by the fact that the research findings are only gradually being implemented. On the other hand, it is due to the complexity of opening up this level, which should not be underestimated.

There are two ways to implement a psychologically sound approach in customer-centric marketing: conceptual and automated.

1. The **conceptual** approach is a data-based, empathetic approach. Experts such as customer journey managers with extensive customer knowledge and psychological competence add psychographic characteristics to the existing target groups or segments. In the next step, several strictly target group-specific customer journeys are developed, which run parallel to each other (usually up to the checkout).
2. The **automated process** is currently still at an early stage. Innovative start-ups are already showing that machine learning algorithms can be used to deliver psychological triggers very effectively. However, psychological knowledge is also required here in the conception of the triggers, and the interaction of the systems must be clarified.

Whatever your path, the chain of effects of human need trigger conversion can effectively affect the interplay between data and psychology.

Behavioral economics will play an increasingly important role in building customer intelligence in the future. In addition, forecasts of future customer behavior will also be part of every customer profile. Forecasts (conversion probability, churn probability) and recommendations (Next Best Product, Next Best Channel) further sharpen the customer profile.

True customer intelligence, therefore, means **merging three disciplines as the basis for intelligent segmentation** (Fig. 7.5):

- Identity Resolution: The generation of data protection-compliant, consolidated user identities

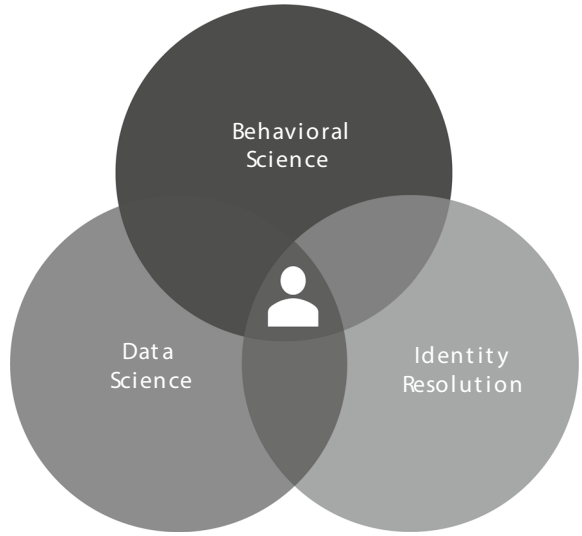



**Fig. 7.5** Customer intelligence as an intersection of three disciplines. (Source: own presentation)

- Data Science: The statistical analysis and interpretation of existing data, as well as the derivation of patterns and trends for the development of propensity models (predictive analytics)
- Behavioral Science: The qualitative enrichment of the collected data with psychographic characteristics (fundamental needs and relevant decision-making patterns)

As a result, advertisers are getting closer and closer to the demand for a holistic and consistent user experience.

**Personalized Customer Journeys**

The consideration of the new dimensions in user segmentation will have a significant influence on personalization measures and thus bring about the **next evolutionary stage in user language**: user- or needs-centered customer journeys (hyper-personalization, see Fig. 7.6). The new era of customer journeys will take greater account of user engagement and intent. It will bring a new, agile, and dynamic approach to customer journeys.

The often simultaneous and inflationary user approach via many channels is ending.

**Measurability as a Success Factor**

Due to the multitude of data sources, we can use today to get an idea of "our" user and his interaction with us, our brand, or our products - the opposite often happens: we do not know where to look first.

This "**data overkill**" prevents us from finding the right answers to our (business-relevant) questions. In practice, we are presented with key figures at regular intervals that, in the best

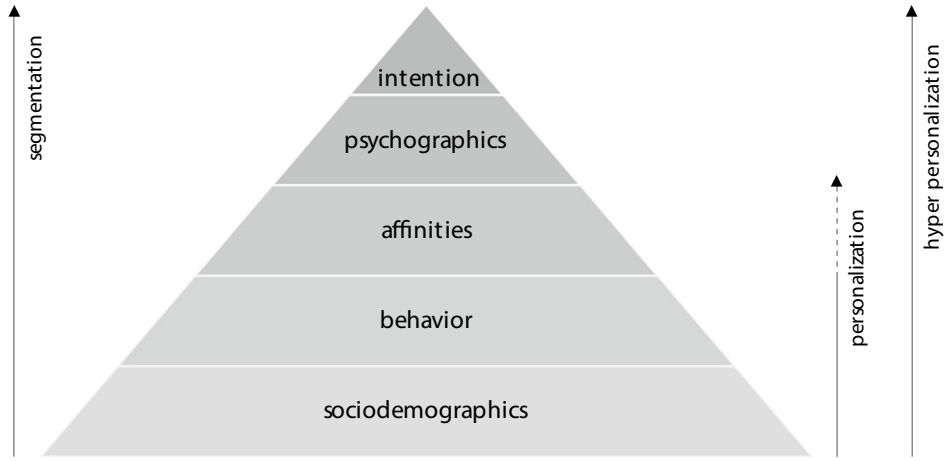



**Fig. 7.6** Step-by-step model of personalization. (Source: own representation)

case, show a positive upward trend but do not enable us to make the right decisions. What we need now is **a new KPI framework**: agile, dynamic, and enticing action.

It is essential to say goodbye to a macroscopic view and focus on a microscopic one. Moreover, to analyze the user and not the respective campaign. In conclusion, the campaign-centric maxim "CPx" (Cost per Lead, Cost per Order, Cost per Action), which has been popular until now, will soon be an anachronism.

The new maxim is user-centered key figures. What stage of the customer journey is the user at? How is his involvement changing, how is his transaction intention changing? Where are decision bottlenecks or negative experiences in the interaction with our company? The new approach requires the creation of a new framework to map customer experience and engagement schematically.

Until now, the customer journey analysis was the schematic basis for creating key figures. The AIDA funnel (**A**wareness, **I**nterest, **D**esire, **A**ction) was the compass for monitoring success. The goal of the "awareness" phase is to build new reach. Here we look at metrics such as "New Visitors", "Engagement Rate" or "Impression Share". In the transaction-heavy "Desire/Action" phase, "Conversion Rate", "Cart Abandonment" or even the "Average Order Value" provide us with information about the success of the journey section.

However, we are dealing with two challenges: Firstly, the metrics just mentioned are highly processed key figures based on historical data and less customer-centric metrics. On the other hand, we will no longer be able to collect or evaluate much of this data, which mostly comes from third-party providers, in the future.

One solution to this is the perspective of customer-driven marketing. By focusing on the customer himself and no longer "only" on the phases of the buying process, we achieve a new quality of measuring success. To do this, we need to regain sovereignty over our

customer data and merge it into a "**single source of truth"** so we become independent of third parties and can aggregate holistic customer intelligence.

This enables us to view the customer journey no longer as a linear buying process but as a dynamic interaction (with the possibility of hyper-personalization). New data dimensions, such as psychographics and intention, can help understand users better and contextualize the journey more appropriately. In this way, we can focus even more on the user himself and use new metrics to measure success.

As is well known, a historical customer lifetime value provides information about previous purchasing behavior. However, it is much more exciting to analyze what buying behavior will look like in the future (**predictive analytics**). In this way, measures taken within a customer journey can be better evaluated. Predicted conversions and a predicted average shopping cart give us an estimate of how our customer will behave in the coming period X. Is this true, or, better yet, could the behavior be positively influenced? How valuable is this customer for our company (RLV; Residual Lifetime Value)? "Customer Lifespan" is becoming the new dynamic benchmark in measuring the success of customer loyalty. Customers (segments) can also be evaluated in a different quality by probabilities of being "alive" (clustering). In this way, in combination, e.g., allocate the marketing budget more efficiently to segments that might otherwise have migrated, etc.

Finally, it can be said that the new approach to customer-driven marketing opens up the possibility of incorporating dynamic and customer-centric metrics into measuring a company's success. The consequence is better budget allocation and increased ROI via contextualized personalization of the customer journey.

**Technologies in Focus**

Data-driven marketing will be replaced by customer-driven marketing in the future. As a result of this revolution, the focus is finally on the customer instead of third-party data. **Marketing orchestration instead of marketing automation** is the new maxim.

This change requires basic technical prerequisites. To date, setting up a data infrastructure to implement a customer-driven user approach has been opportunistic and not coordinated. Depending on the responsibilities in companies, several technologies with overlapping functionalities were often in use. This created a fragmented tool landscape. This produces additional costs and does not do justice to the goal of co-orchestrating all channels to create a successful user experience.

For successful customer-driven marketing, a re-evaluation is more than up-to-date and necessary. The prerequisite for this is the definition of a clear objective. Only then can the needs be determined and the delimitation and selection of technological partners carried out. **Customer Data Platforms** (CDPs) will be central to the new architecture. They are the further development of existing CRM solutions and the nucleus of the future data infrastructure. It is around them that customer understanding is built.

**Result**

After the cookie apocalypse, it is now all about building customer-driven marketing. The customer himself, not the mere data, is the new focus. Identity resolution management is the first and decisive step. In the future, the customer approach will be based on central and holistic customer intelligence. Tailor-made customer experiences are the logical consequence and the guarantee of success of the new data normal.

When building a comprehensive understanding of the customer (customer intelligence), we still rely primarily on interaction and transaction data. This allows the modeling of segments, but they are incomplete or not all-encompassing. The level of needs is missing. Even if, from today's perspective, this is intended to be progressive with regard to the tool landscape and the degree of use of data-based insights in companies, the enrichment of segments with findings from behavioral economics and behavioral psychology will herald a new era of segmentation and lay a new foundation in customer-driven marketing.

Developing a scalable data infrastructure tailored to the individual parts is the cornerstone of success. The technologies used must meet this goal and then be evaluated. **Marketing co-orchestration is preceded by technology co-orchestration**. Acting strategically in tactical times is the only way out of the recently ignited ("identity") crisis in digital marketing.

## 7.6 Impact Measurement with A/B Testing

Of course, conversion optimization is not complete with the pattern-based relaunch of a page. The effect of the revision must be accompanied and evaluated by A/B testing in every project. In an A/B test, some visitors to a website (= A) are directed to another variant (= B) of the same page, which can be enriched with a particular behavior pattern, for example. The test aims always to find the "better" variant (based on many possible key figures). Since an A/B test can measure the actual behavior of real users in a real need situation, it is often the tool of choice for analyzing the effect of behavior patterns. If the number of visitors is low, you should limit yourself to simple **split tests**. This means that the default state of a page is tested against a variant of the same page that has changed in a characteristic. In the case of correspondingly higher traffic, **multivariate tests** can also be considered, which allow the interdependencies between several changed characteristics to be investigated.

In short, effective conversion optimization only works with testing. In addition, A/B testing also has massive **internal process advantages.** It (almost) always generates clear insights: Variant A sells more, variant B sells less, or also: the topic we are currently thinking about intensively does not influence user behavior. As a result, tasteful discussions about the ideal wording or design can be nipped in the bud, and irrelevant discussions about the key figures can be shortened. If there are signs of a protracted debate that cannot lead to a generally accepted result, it makes sense to quickly bring an A/B test into play and outsource the decision based on data. In principle, however, political

trench warfare and emotional points of view should only be the sole reason for a test in explicit exceptional cases to make the best possible use of resources.

A/B testing should be an **iterative analysis, evaluation, and optimization process**. This should be set up as a cross-device process (i.e., across multiple hardware devices) because the perception of a website on a smartphone will be very different from that of the same page on a desktop computer or laptop. This is not only due to the responsive or adaptive transfer of the content and the resulting completely different appearance (see: Aesthetics Heuristic) but also to the different environmental contexts and needs situations. A visit via a smartphone with a mobile data connection, for example, indicates that the user is not at home and may, therefore, have a more local search interest. This fact is highly relevant for e-commerce and multi-channel companies because the conversion rate of a customer journey with a location reference is much higher. Google (2016) puts the proportion of local searches that lead to a purchase at around 28%.

However, A/B testing, as the most essential tool for building actual user knowledge, also has its **limits.** Not all effects can be isolated and evaluated in a simple A/B test. A pattern like the Aesthetics Heuristic affects the entire website design and all touchpoints along the customer journey. If, for such methodological reasons, no empirical evidence for the effectiveness of a measure can be found, one should rely (as a heuristic, so to speak) on recognized guidelines, UX conventions, and, if necessary, out-of-context research results.

## 7.7 Limitations of Working with Behavior Patterns

In ► Sec. 7.1 Some disadvantages of using behavior patterns in e-commerce have already been mentioned. In addition, there are several limitations that you should be aware of when working with patterns.

A **demystification** of the topic seems particularly important. Behavior patterns are a promising tool for unlocking previously untapped conversion potential, but they are not a silver bullet. If a product is not competitive or overpriced, if the webshop is not technically error-free, if there are supply bottlenecks or problems with fulfillment–all these (and many more) points will lead to the effect of the use of behavior patterns drastically reducing or no longer being measurable at all. This instrument is undoubtedly of great interest to the vast majority of e-commerce professionals but can only be fully effective in shops with "homework done".

As e-commerce becomes increasingly globalized, target groups and their needs are becoming increasingly difficult to quantify in a uniform manner. When working with behavior patterns, there is a strong **cultural influence** on their effectiveness. Likely, a trigger for American internet users will not be caught to the same extent by an Arab or Chinese user. A striking example is the different forces that patterns such as Authority or Social Proof unfold in individualistic and collectivist societies.

In describing the decision-making process and explaining the behavior patterns, **findings from neuroeconomics** have often been incorporated, which tries to expand economic teaching with measurement methods and models from the neurosciences. However, this approach is not devoid of criticism. In order to be able to classify the representations correctly and to evaluate the relevance of the research results for oneself, this criticism must be named. Some of the most common objections are therefore presented below.

- **Insignificance:** It is often argued that it is entirely irrelevant for economic analysis which processes occur in the brain during decisions – only the measured or observed result, i.e., the actual behavior, is essential. Whether the amygdala, the neocortex, or the hippocampus became active for this purpose is irrelevant (Beck 2014). In addition, it is questionable whether looking into the black box brain brings profound new insights for e-commerce practice. For example, von Bernheim (2008) mentions in a very differentiated view that one does not need a deep knowledge of string theory to be a research-strong physicist.
- **Inaccuracy:** Functional magnetic resonance imaging is the most essential tool of neuroeconomics. However, this method only measures the blood flow to different brain areas. The currently smallest measurable unit (1 voxel) usually covers several million nerve cells. Thus, an effect in the brain can only be roughly located, especially since measurements are often delayed (Rosenfeld 2005). This means that magnetic resonance imaging is still relatively imprecise in its informative value.
- **Irreliability:** In a meta-analysis of 48 published neuroscience papers, Button et al. (2013) found that the studies could only prove the claimed effects with a probability of 20%, primarily due to too few cases or weak effects. The reliability (reliability of scientific measurements or non-random proportion of observed effects) is therefore not sufficiently given.

Two insights can be derived from this: On the one hand, it becomes clear that neuroscience can hardly refute economic theory – because it makes no statements about the brain's processes. In addition, economic models must not be misunderstood 1:1 as models of the brain. So there is no **battle of the sciences**, no clash of beliefs, no better or worse theory. Each discipline can enrich the other; none can be complete in their own right. On the other hand, these limitations underline the necessity of structured support through A/B testing (see ► Sect. 7.6). Limitations of the significance of the effectiveness of behavior patterns are nothing more than uncertainties about the conversion lever of an optimization measure. This uncertainty can only be eliminated by collecting your own primary data based on real user behavior.

Finally, it should be mentioned as a limitation that many of the patterns explain how decisions are made. However, the decisions made will not always lead to **actual behavior**. This puts the binding nature of the effect of patterns into perspective and once again emphasizes the need for extensive testing.

## 7.8 Ethical-Moral Discussion

> To be persuasive, we must be believable; to be believable, we must be credible; to be credible, we must be truthful (Edward Murrow, journalist legend and media activist).

A critical section is intended to conclude this book (why one should save the most important things for the end, the Recency Effect in chap. 4 shown): The ethical-moral discussion of the use of behavior patterns, in a sense, the **"disclaimer"** to the book. It has already been mentioned at various points in the book: You must always be aware that knowledge of behavior patterns is a powerful tool. Behavior patterns are subconscious processes in which (without explicit training) there is little possibility of perceiving or correcting them as such. In this way, you can meet customers where they are particularly susceptible to external influences. It is obvious that this is accompanied by the **possibility of manipulation** – especially given the sometimes astonishing effectiveness.

Since the beginning of the renunciation of the belief in "homo economicus", there has been a passionate **discussion about the freedom of will** in decision-making situations. After about three decades of being predominantly based on philosophical and psychological arguments, neuroscientific arguments have been increasingly added in recent years. These indicate that a large part of the decisions are made based on implicit processes and that at no time does a conscious rational filter pass through (see Sect. 2.2). So, in the end, are we not free to decide at all? Moreover, are we no longer responsible for our decisions? Indeed, this a thought with extreme implications; just think of our legal system, which to this day holds almost every offender fully accountable for his actions – regardless of whether he committed his actions consciously or intuitively. Fortunately, it is not quite that dramatic. After all, important decisions are made in the **interaction of both decision-making systems**. The processes in intuitive System 1 (amygdala-dependent) can be controlled by processes in rational System 2 (prefrontal-dependent) within a particular corridor. Deliberate decision-making approaches of System 2, thus overwriting spontaneous decision-making impulses of System 1. A good example is enduring pain during medical treatment in anticipation of a later cure. Decisive for the actual decision are the balance of power between the two decision-making systems and, above all, **one's willpower** – defined as a combination of determination and self-discipline (Burns & Bechara 2007). In a nutshell, a part of our brain wants to make decisions we have never consciously thought through. However, whether this part comes into play and we follow the impulse is something we can influence ourselves to a large extent. However, this requires effort and cognitive resources that are scarce and limited.

It is now known that this can also be mapped neurologically: The **glucose content** in the **blood** plays a vital role in enforcing willpower. The volitional resource is physiologically consumed by self-control, complex decisions, and the regulation of emotions and stress (Gailliot & Baumeister 2007). Glucose is, therefore, not only the energy source for

our muscles but also for our thinking and decision-making processes. A low glucose content in the blood leads to decisions being made more in System 1 and based on heuristics or behavior patterns, while an increased glucose content is more likely to hand over control to System 2. Masicampo and Baumeister (2008) impressively demonstrate this using the example of the Decoy Effect with a glass of glucose-containing lemonade.

Experts in user experience and conversion optimization are responsible for guiding people to their desired goals with minimal effort but not manipulating them. The fact that persuasiveness cannot and must not be anything dishonorable, immoral, or dubious but must be based on the needs of people is by no means a new insight. Aristotle formulated it well over 2000 years ago. His three **principles of persuasion** form the bracket of the conversion term used here:

1. Logos: Persuasion with logic – use of facts, statistics, quotes, expert opinions, customer reviews.
2. Pathos: Persuasion with emotions – use of emotional stories, outbursts, descriptions of events, language, and images.
3. Ethos: Persuasion with character – use of competence, non-partisanship, objectivity, morality, and good intentions.

The use of behavior patterns should, therefore, always bring a benefit to customers. For example, a well-designed time-saving effect, which can break down a significant decision into several small decisions and, on top of that, maintains the user's motivation to finish the process that has begun, reduces the decision-making effort. This "**control without manipulation"** sounds trivial, but in practice, the demarcation and implementation are often precisely the opposite. When sales targets are defined, e-commerce managers almost inevitably find themselves in a conflict of objectives between a mainly quantitative target agreement and their own professional ethics. Identifying where goal-oriented control ends and systematic manipulation begins is difficult. The fundamental law of working with behavior patterns is that the use must serve the mutual interest.

This "**alignment of interests**" is essential: it asks companies to only consider measures that are also in the users' interest. In some application areas, it is said somewhat less concretely that the measures must not run counter to the users' interests. What sounds trivial is often almost impossible in operational practice. This usually involves the **organizational structures and working models**, which are only organized along the customer journey in highly exceptional cases. In most cases, responsibilities and roles are organized vertically in silos: One team is responsible for media purchasing, another for search engine optimization, another takes care of the conception of landing pages and product pages, and another team supervises the ordering process. In this way, there can be no end-to-end orientation towards customer needs, and the probability increases that it will

be controlled in isolation according to departmental goals or micro-conversions – the "alignment of interests" is gone.

**Example**

In order to promote the conclusion of newsletter subscriptions with relevant content for users, one can rely on users' loss aversion and status quo bias. As a reminder, Prospect Theory has explained to us that possible losses outweigh possible gains (see Sect. 2.5). For example, if you change the headline of the order form from a profit-oriented one (e.g., "Sign up here and always receive important news") to a loss-oriented phrase (e.g., "Sign up here so you do not miss any more important news"), you will tend to see higher subscriber numbers. From an economic point of view, these two options are identical: Users can still freely decide whether they want to receive the newsletter or not – there is no compulsion. Thus, the use of loss aversion would be sensible and honest.

The situation is different if the default option is set from "deselected by default" (= no receipt of the newsletter) to "selected by default" (= receipt of the newsletter) without adequately pointing this out. The same applies if the selection logic is reversed, e.g., "Click here if you do not want to receive newsletters." In these cases, users might become newsletter subscribers against their will. The use of the pattern would not only be morally reprehensible but also legally illegitimate. Nevertheless, such approaches can be found repeatedly in a globalized internet where jurisprudence reaches its limits. Also, from an economic standpoint and with regard to customer satisfaction, loyalty, and willingness to recommend, it certainly makes no sense to force unwanted advertising on users. Suppose you feel taken by surprise, cheated, or not taken seriously; not only do you not develop any motivation to buy that case, but you are even permanently demotivated, i.e., you develop an explicitly negative attitude towards the website. Manipulative behavior patterns are, therefore, one of the most significant demotivation factors.◀

A guide to the fine line between justifiable and dishonest influences can be examples of **companies crossing this red line**. The portal "darkpatterns.org" and the associated Twitter-/X- account "@darkpatterns" have created a constantly growing (predominantly American) collection of them. Their "Hall of Shame" includes plaintiff names such as Amazon, LinkedIn, Citibank, or Google, which are said to have used behavior patterns that do not lead users to their actual goal but develop dishonest manipulation mechanisms.

So that you do not run the risk of finding your company in such lists, it is advisable to develop your own **Code of Conduct**. On the one hand, this draws attention to the need for discussion in the organization. On the other hand, it creates a generally accepted consensus to which all employees must submit. Over several projects, the following example

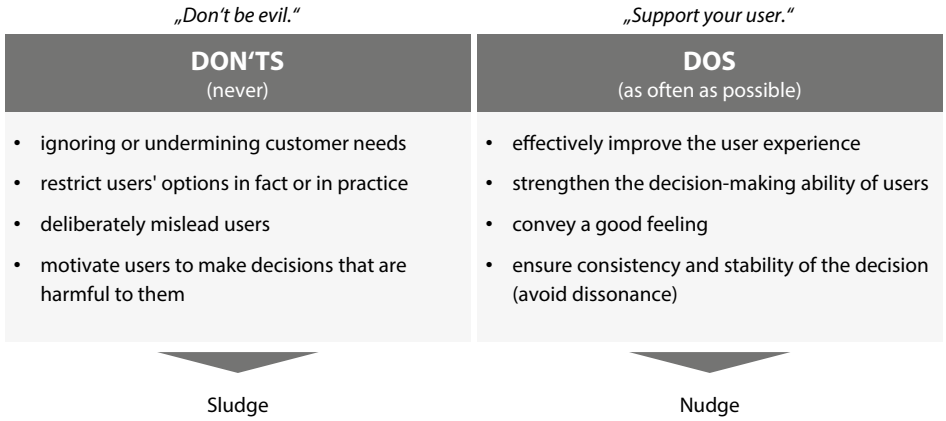



**Fig. 7.7** Example of a "Code of Conduct" for working with behavior patterns. (Source: own illustration)

(Fig. 7.7) a Code of Conduct, which can serve as a starting point for the development of its own Code of Conduct.

**Result**

Behavior patterns are a **powerful tool** with highly versatile applications. You can use patterns to understand your customers better and give them smoother shopping experiences. However, you can also use it to manipulate them (at least to some extent). The same discussion can also be excellently conducted about a kitchen knife, which can be both a dangerous weapon and an indispensable utensil for a magical 5-course meal. What matters is what these tools are used for by their users. In other words, it is up to you to use behavior patterns correctly and honestly. The American term for working with behavior patterns is revealing. It is often referred to as the **"Art of Persuasion".** So, persuasion is an art, not a weapon. Conviction does not assume that you can "hunt" a customer like a victim that others can control but that you understand and respect their needs.

There should, therefore, be agreement on the overarching objective: to create a **sustainable win–win situation** for online shoppers and e-commerce companies. The goals of the company must correspond with the goals of the customers. For customers, achieving their goals means receiving the best product on the market. If your shop offers this product, customer orientation really dictates that you convince your visitors to buy by all decent means. If you do not, others do – and your designated customers will only find the second-best solution to their problem. This sustainability is anything but altruistic customer orientation: it is the basis of successful customer lifecycle management. Only sustainably satisfied customers return daily follow-up purchases and recommend you to others.

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P. Spreer, *PsyConversion®*, https://doi.org/10.1007/978-3-658-44593-5